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(54) **PROPHYLACTIC OR THERAPEUTIC
COMPOSITION FOR GRAFT-VERSUS-HOST
DISEASE**

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(57)

ABSTRACT

The present invention provides a prophylactic or therapeutic composition for graft-versus-host disease (GVHD). There is provided a prophylactic or therapeutic composition for GVHD, which comprises bacteria belonging to any genus selected from the group consisting of the following genera: *Blautia*, *Clostridium*, unclassified Clostridiales, *Actinomyces*, *Parabacteroides*, *Lachnoclostridium*, *Bacteroides*, *Faecalibacterium*, unclassified Lachnospiraceae, *Roseburia*, *Ruminococcus*, unclassified Firmicutes, *Dorea*, *Phascolarctobacterium*, *Sutterella*, *Megamonas*, *Collinsella*, *Eubacterium*, and *Coprococcus*, etc., or any combination of bacteria belonging to these genera.

Specification includes a Sequence Listing.

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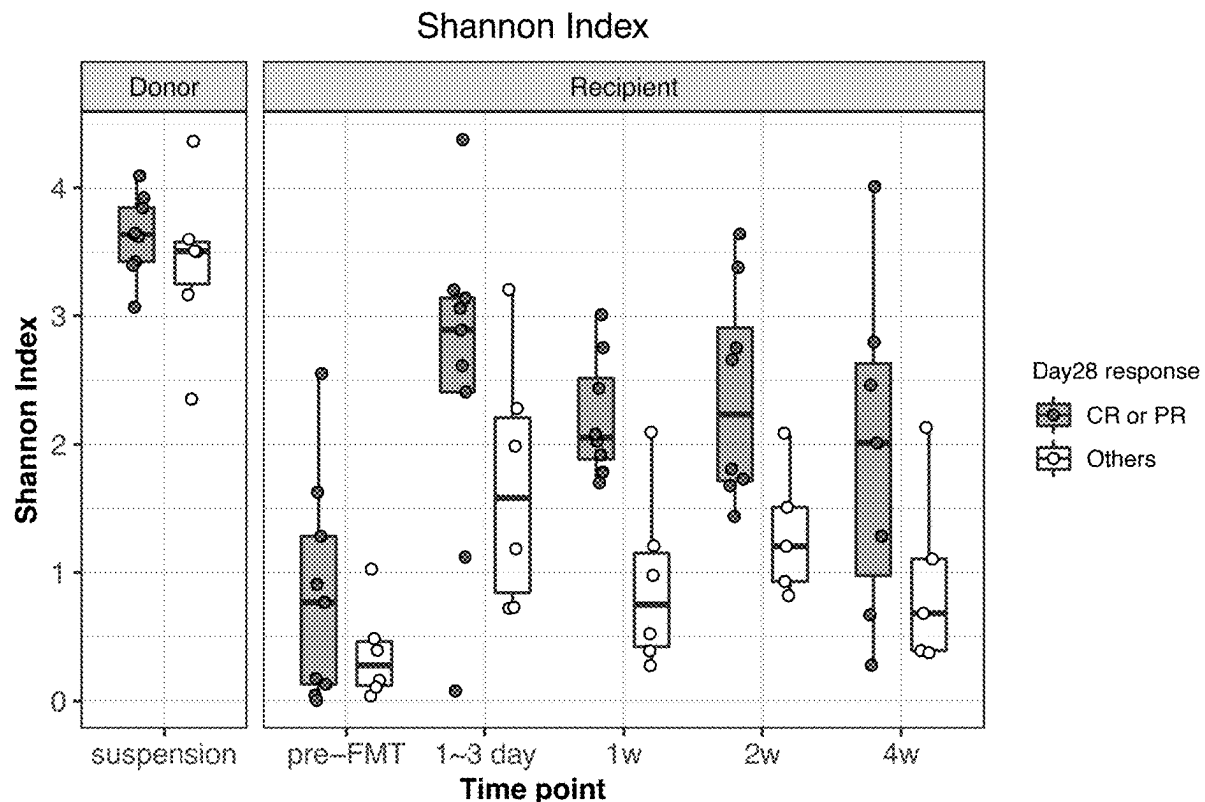


Fig. 1

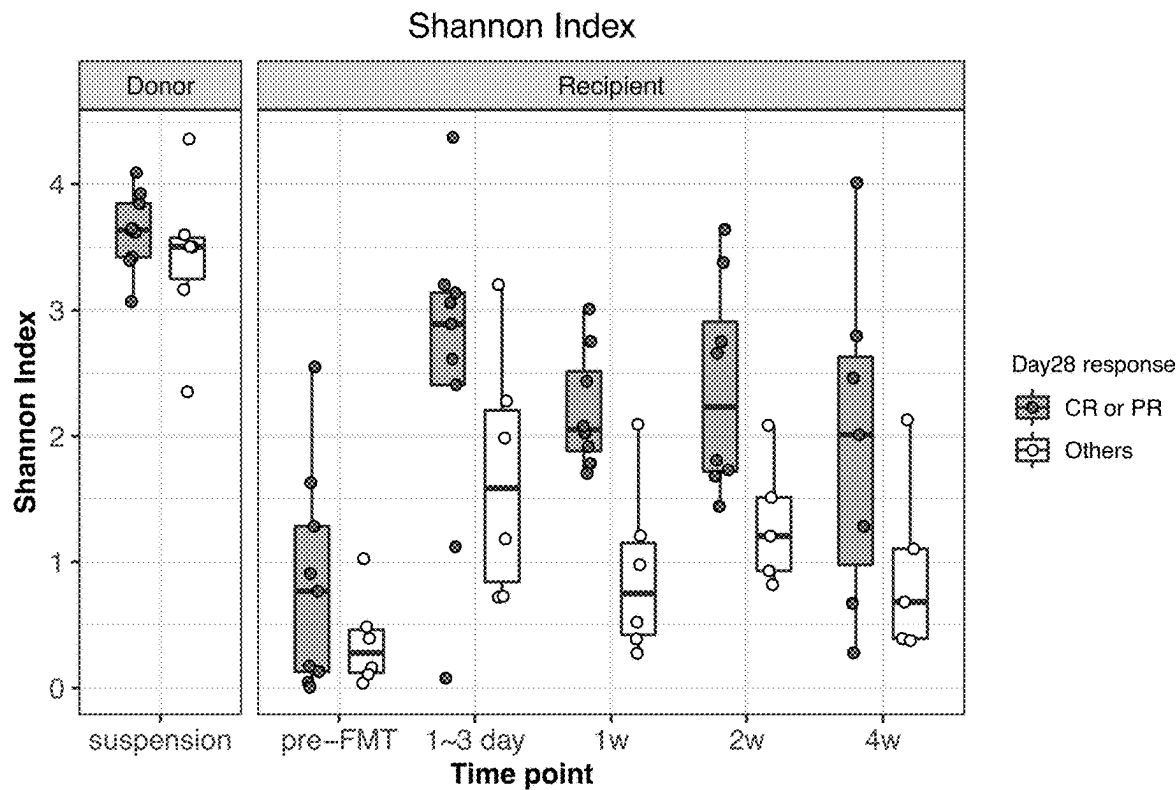


FIG. 2A

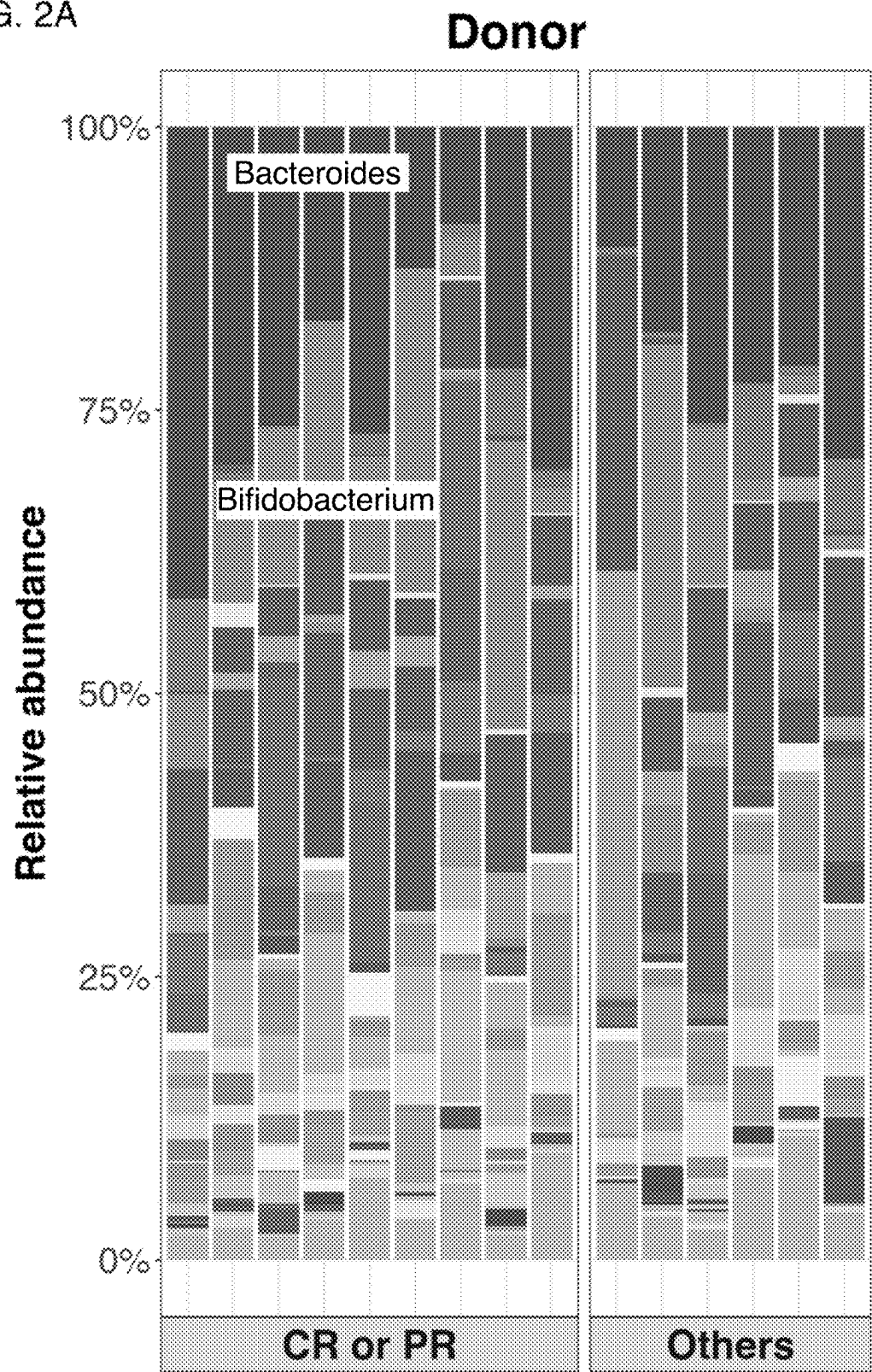


FIG. 2B

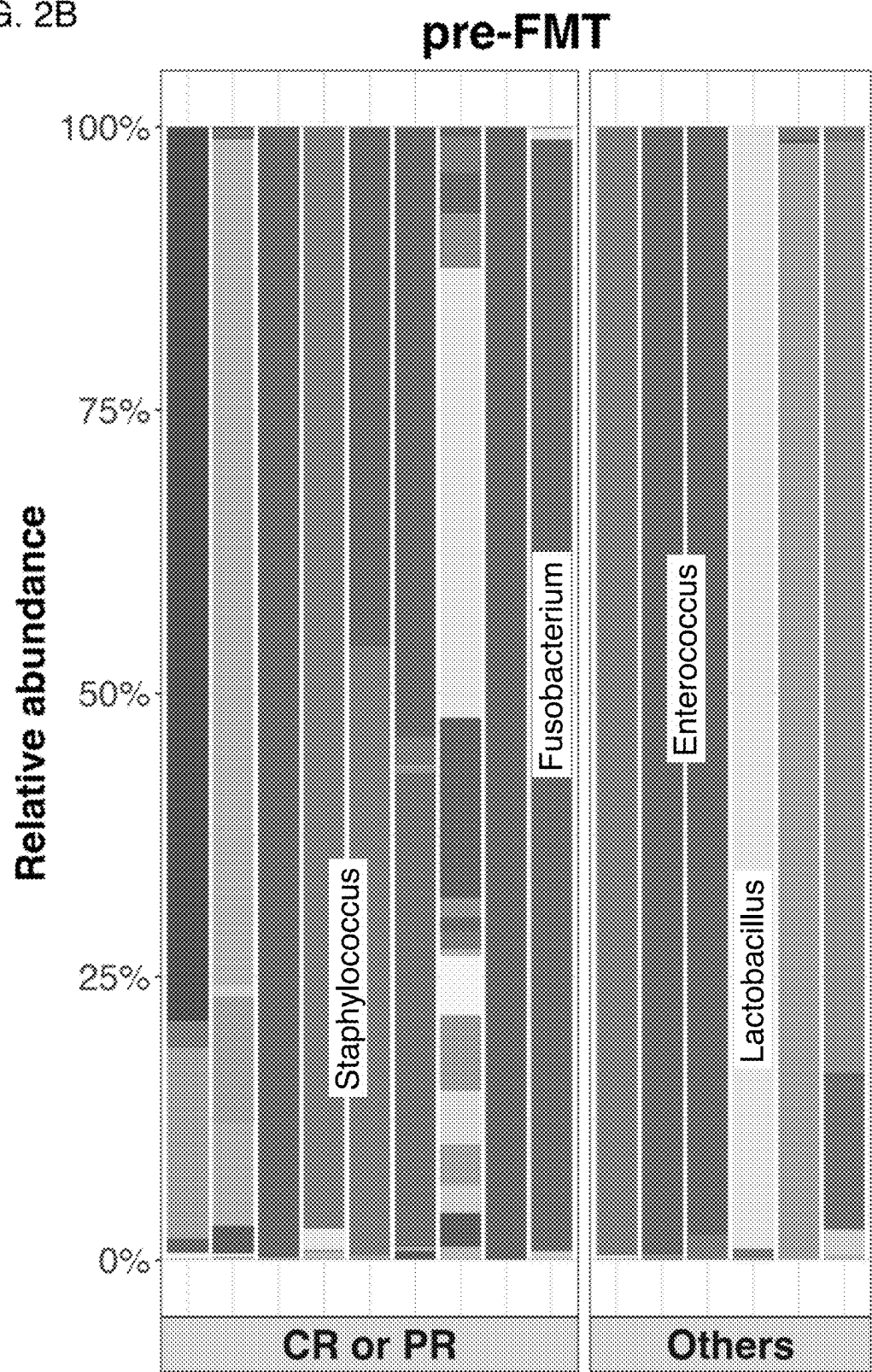


FIG. 2C

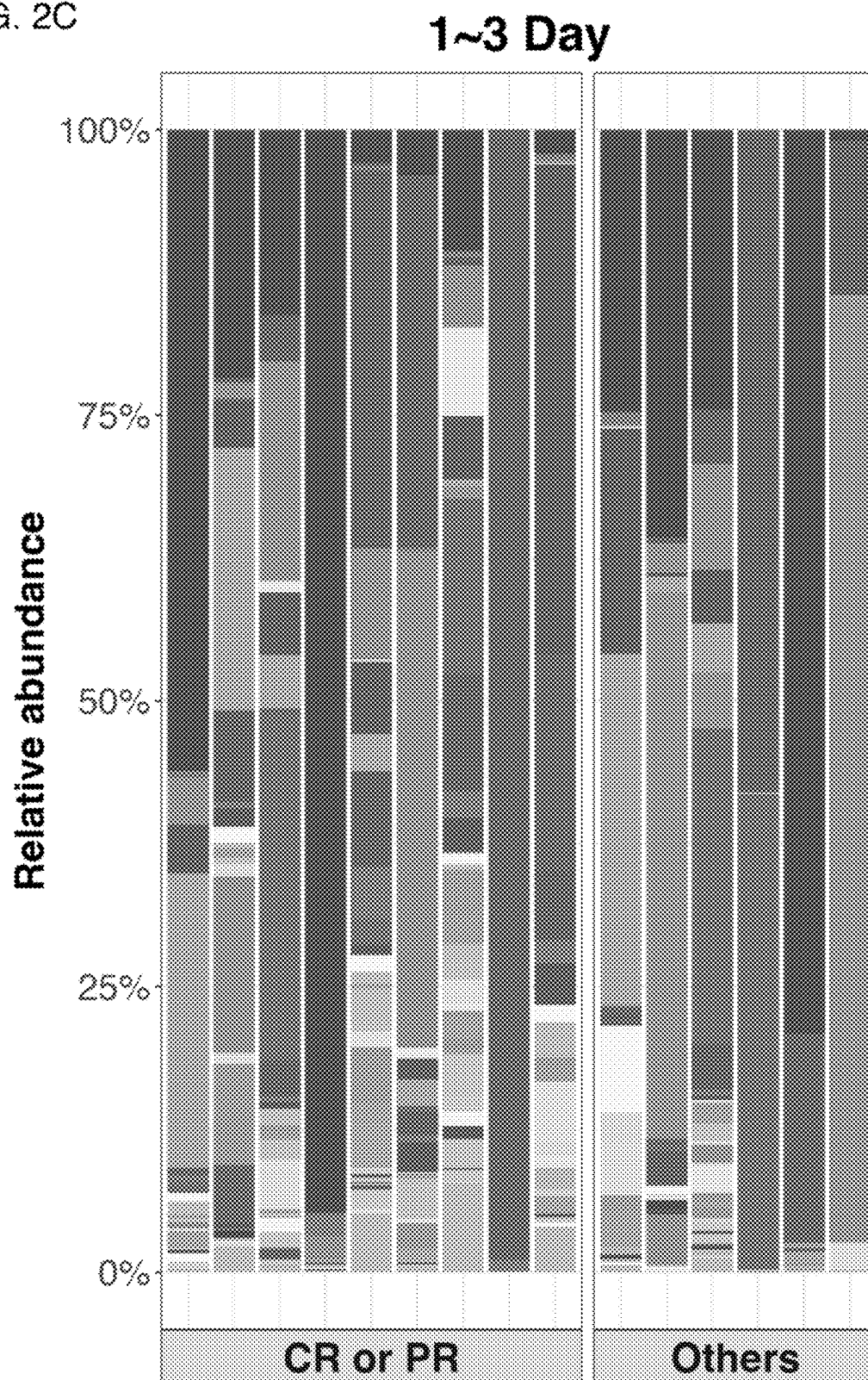


FIG. 2D

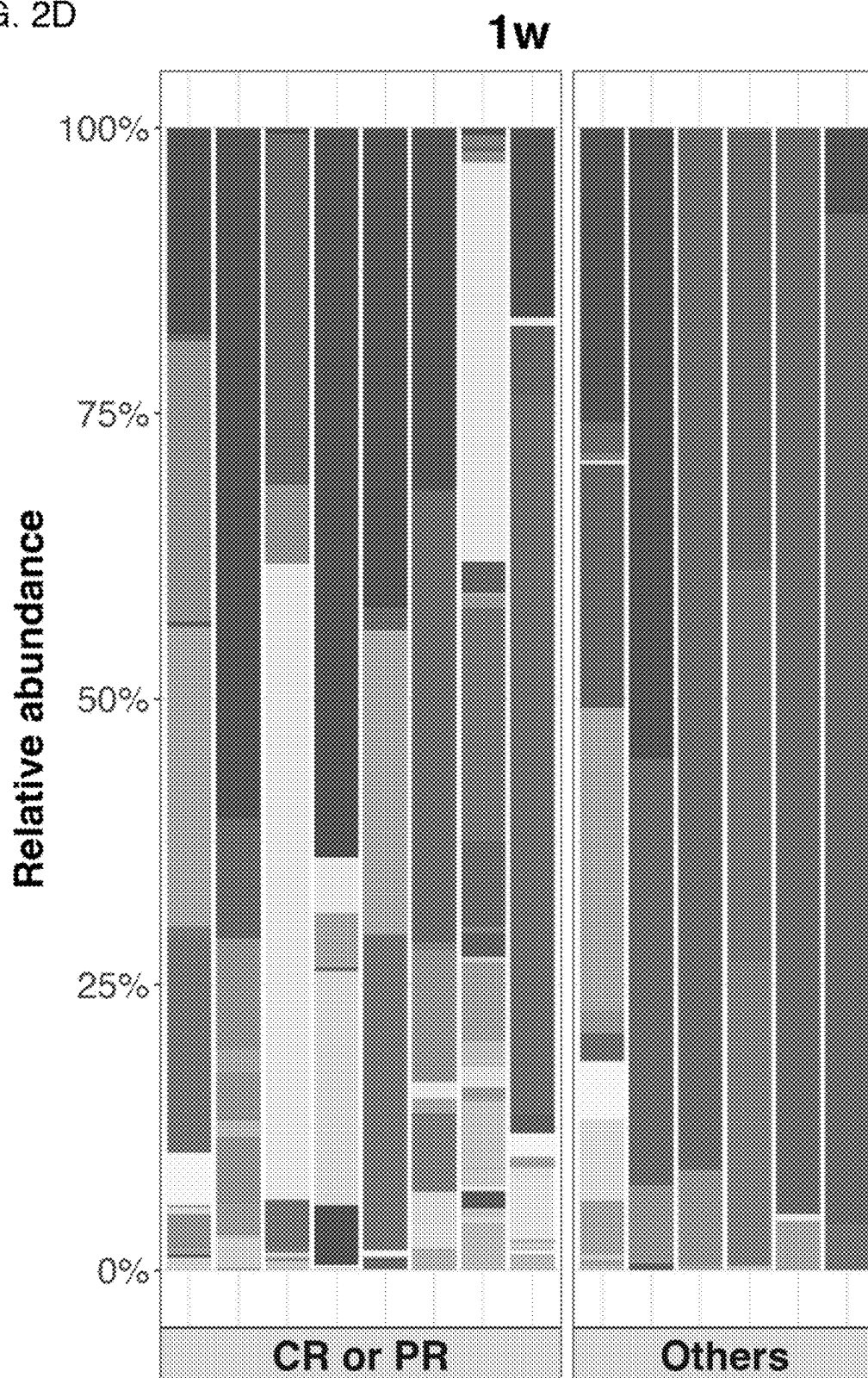


FIG. 2E

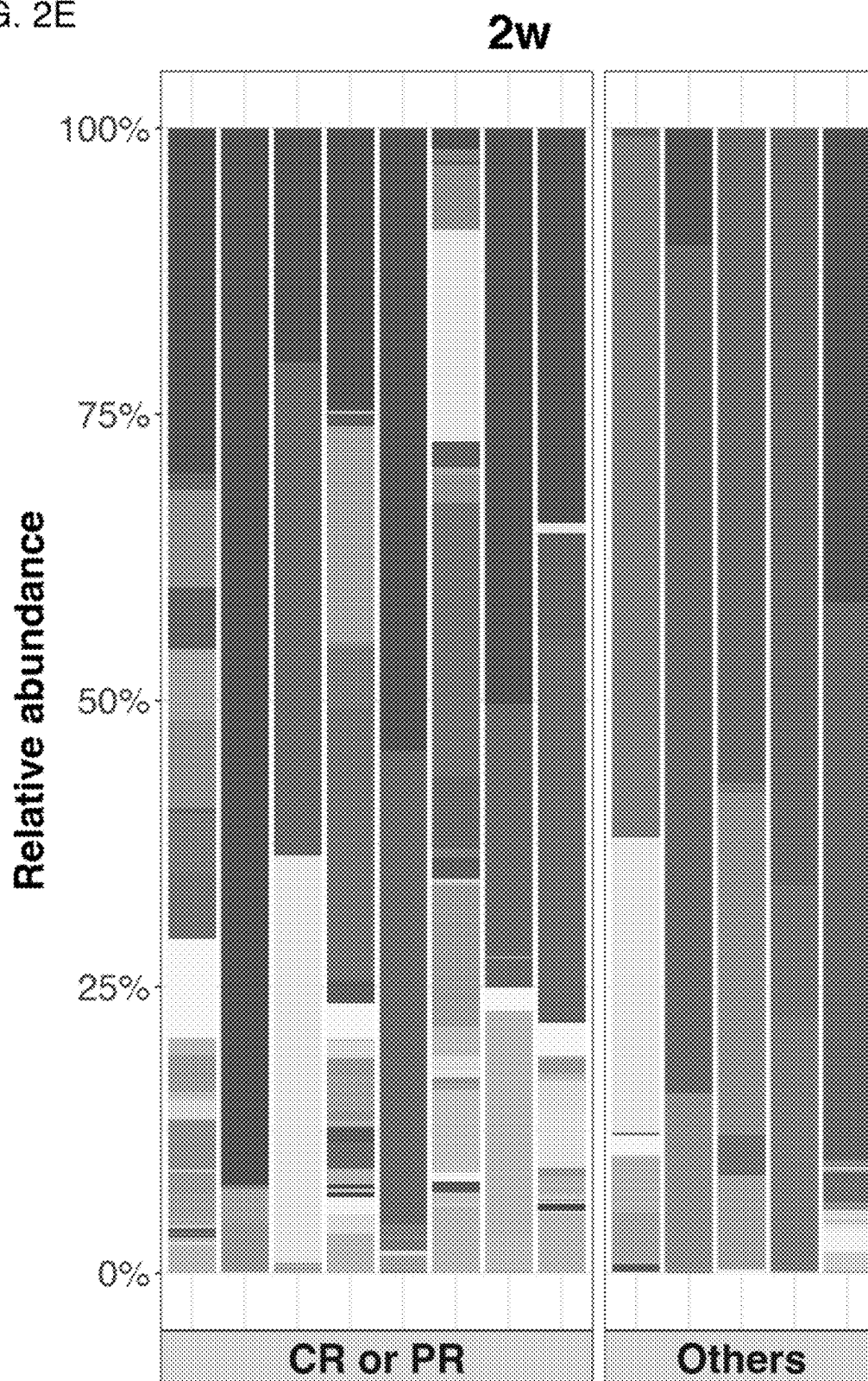


FIG. 2F

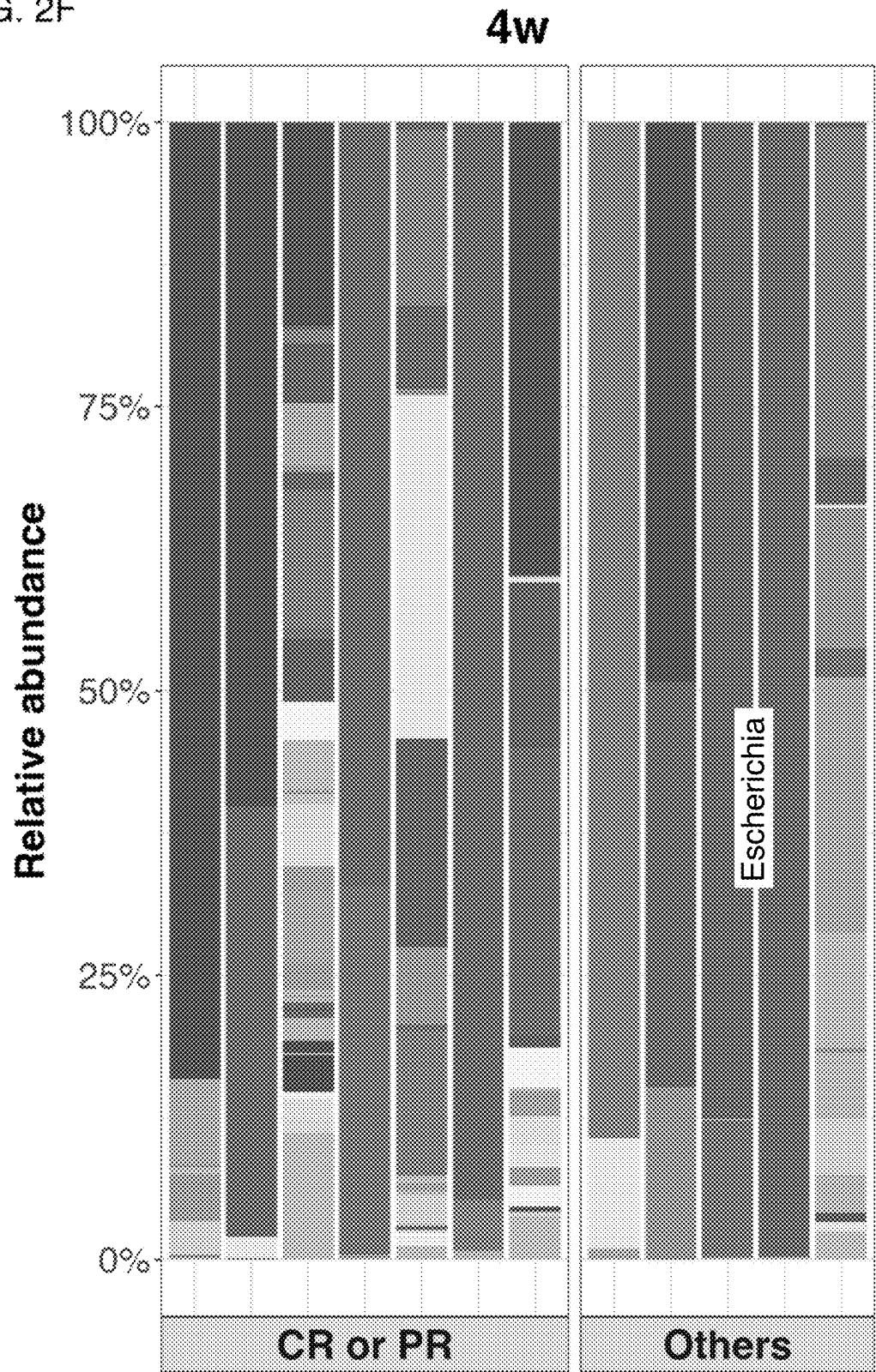


FIG. 2G

■ Bacteroides	■ Alistipes	■ unclassified Ruminococcaceae
■ Enterococcus	■ Megasphaera	■ Peptostreptococcus
■ Staphylococcus	■ Erysipelatoclostridium	■ Eggerthella
■ Escherichia	■ Anaerostipes	■ Faecalimonas
■ Bifidobacterium	■ Phascolarctobacterium	■ Paraprevotella
■ Lactobacillus	■ Robinsoniella	■ Faecalitalea
■ Blautia	■ Sutterella	■ unclassified Clostridiaceae
■ Parabacteroides	■ Fusicatenibacter	■ Butyricicoccus
■ Streptococcus	■ Abiotrophia	■ Faecalicatena
■ Fusobacterium	■ Veillonella	■ Haemophilus
■ Prevotella	■ Prevotellamassilia	■ Akkermansia
■ Klebsiella	■ Rothia	■ Campylobacter
■ Faecalibacterium	■ Schaalia	■ Lachnospira
■ Clostridium	■ Eubacterium	■ Dialister
■ Corynebacterium	■ Coprococcus	■ Finegoldia
■ Megamonas	■ Holdemanella	■ Coprobacter
■ unclassified Clostridiales	■ Parasutterella	■ unclassified Erysipelotrichaceae
■ Okadaella	■ Barnesiella	■ Peptoniphilus
■ Lachnoclostridium	■ Actinomyces	■ Romboutsia
■ unclassified Lachnospiraceae	■ Catenibacterium	■ Lachnobacterium
■ Roseburia	■ Drancourtella	■ Paeniclostridium
■ Ruminococcus	■ Acidaminococcus	■ Olsenella
■ unclassified Firmicutes	■ Bifilophila	■ Others
■ Collinsella	■ Ruthenibacterium	
■ Dorea	■ Flavonifractor	

Fig. 3

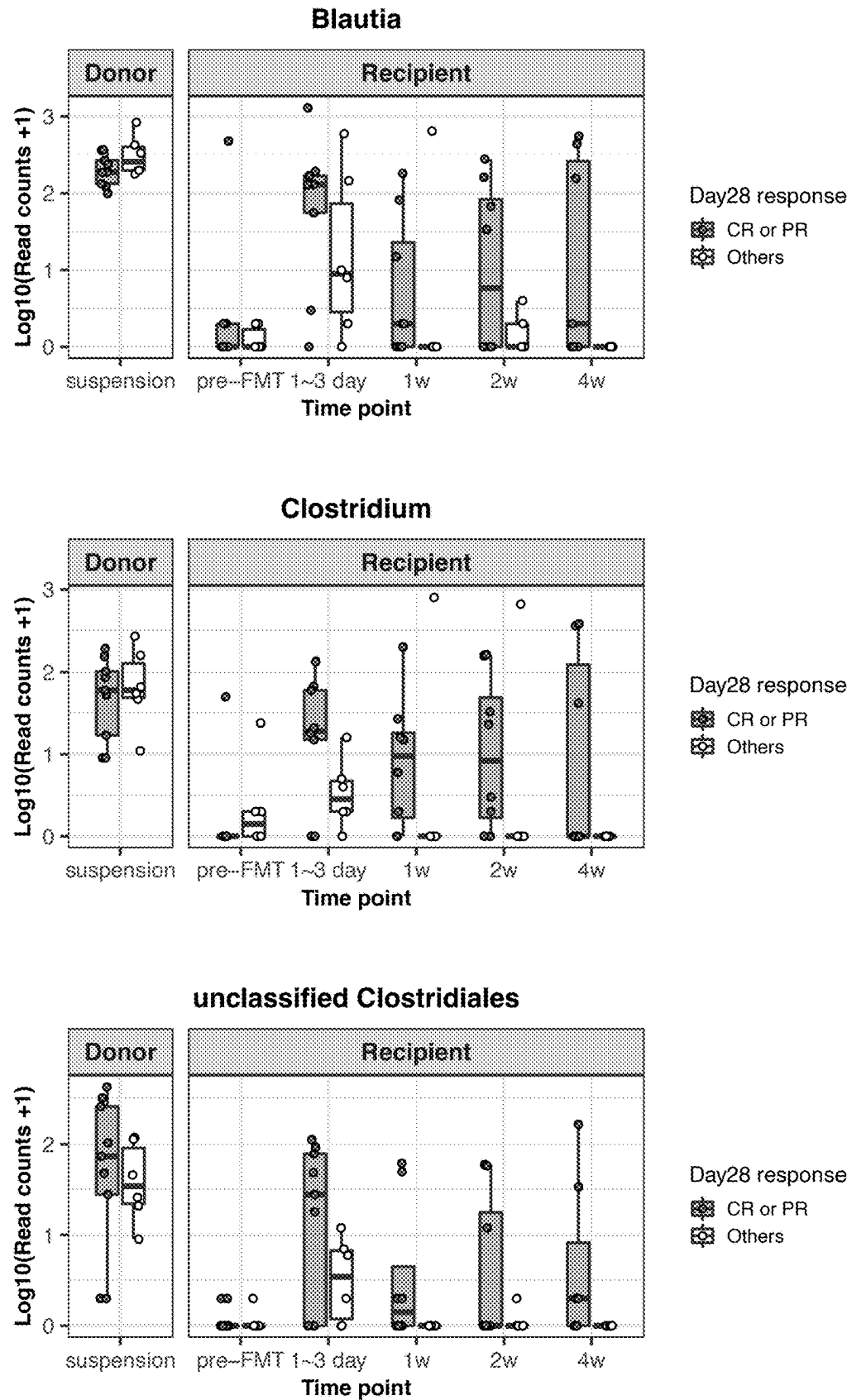


Fig. 4

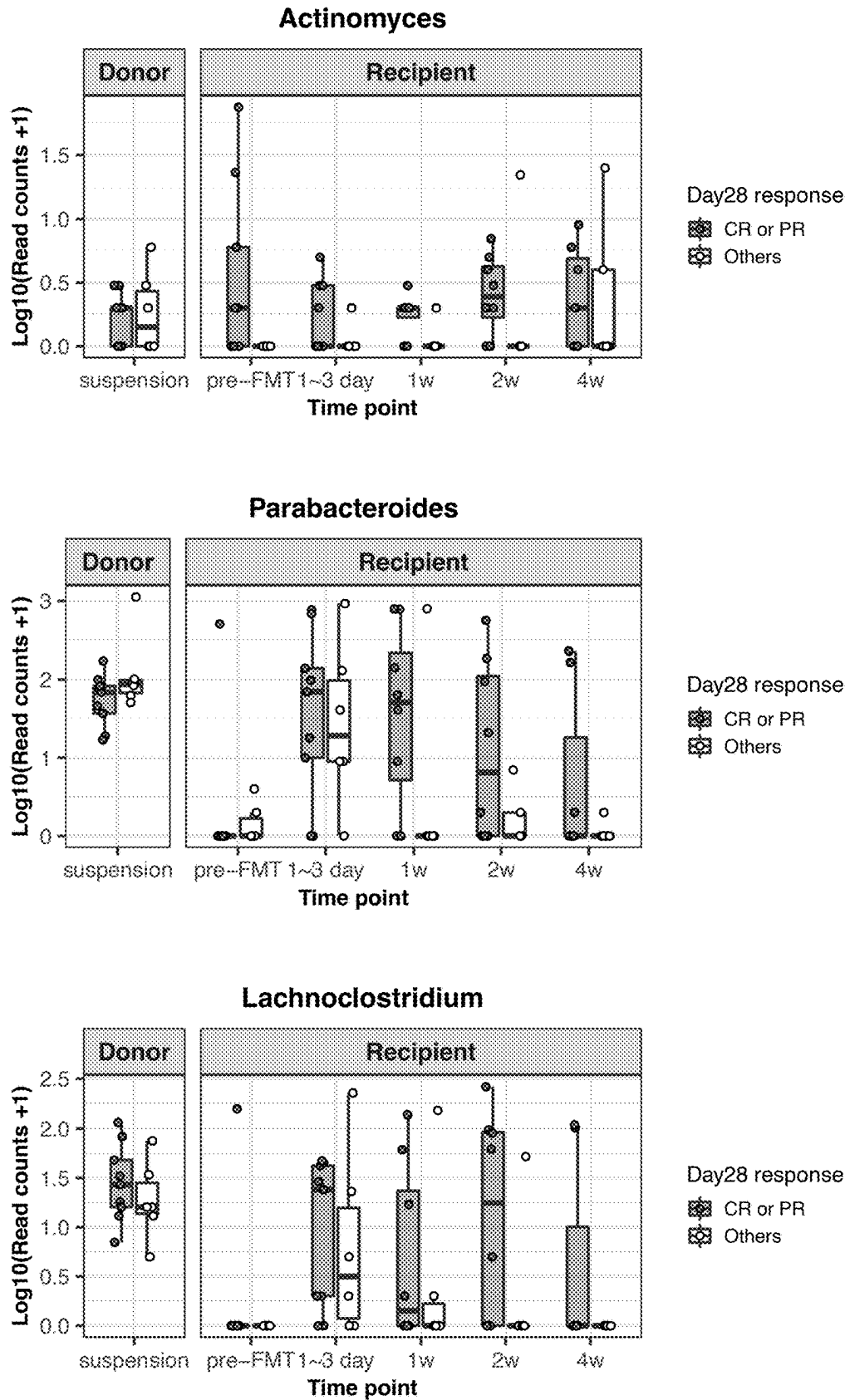


Fig. 5

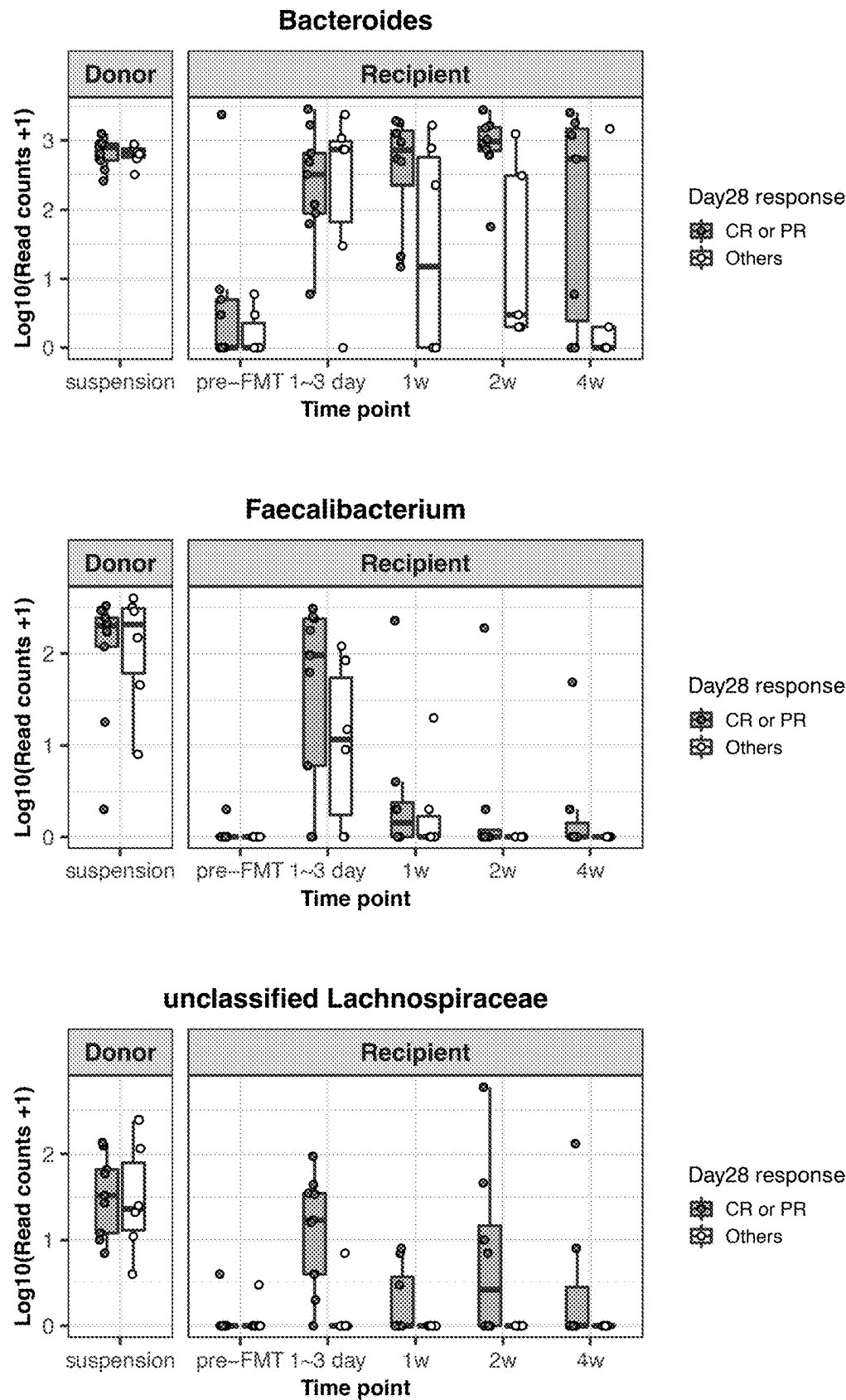


Fig. 6

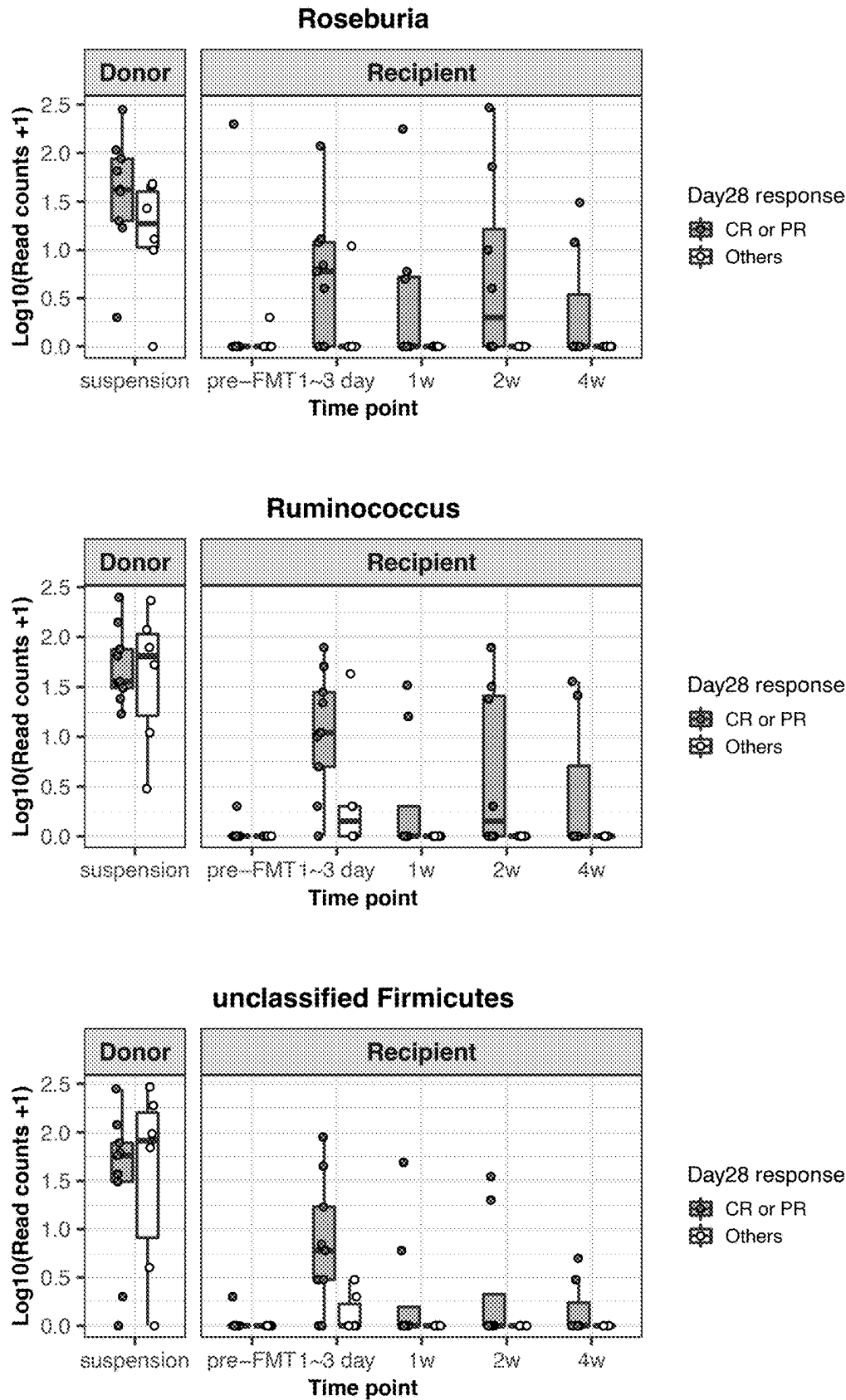


Fig. 7

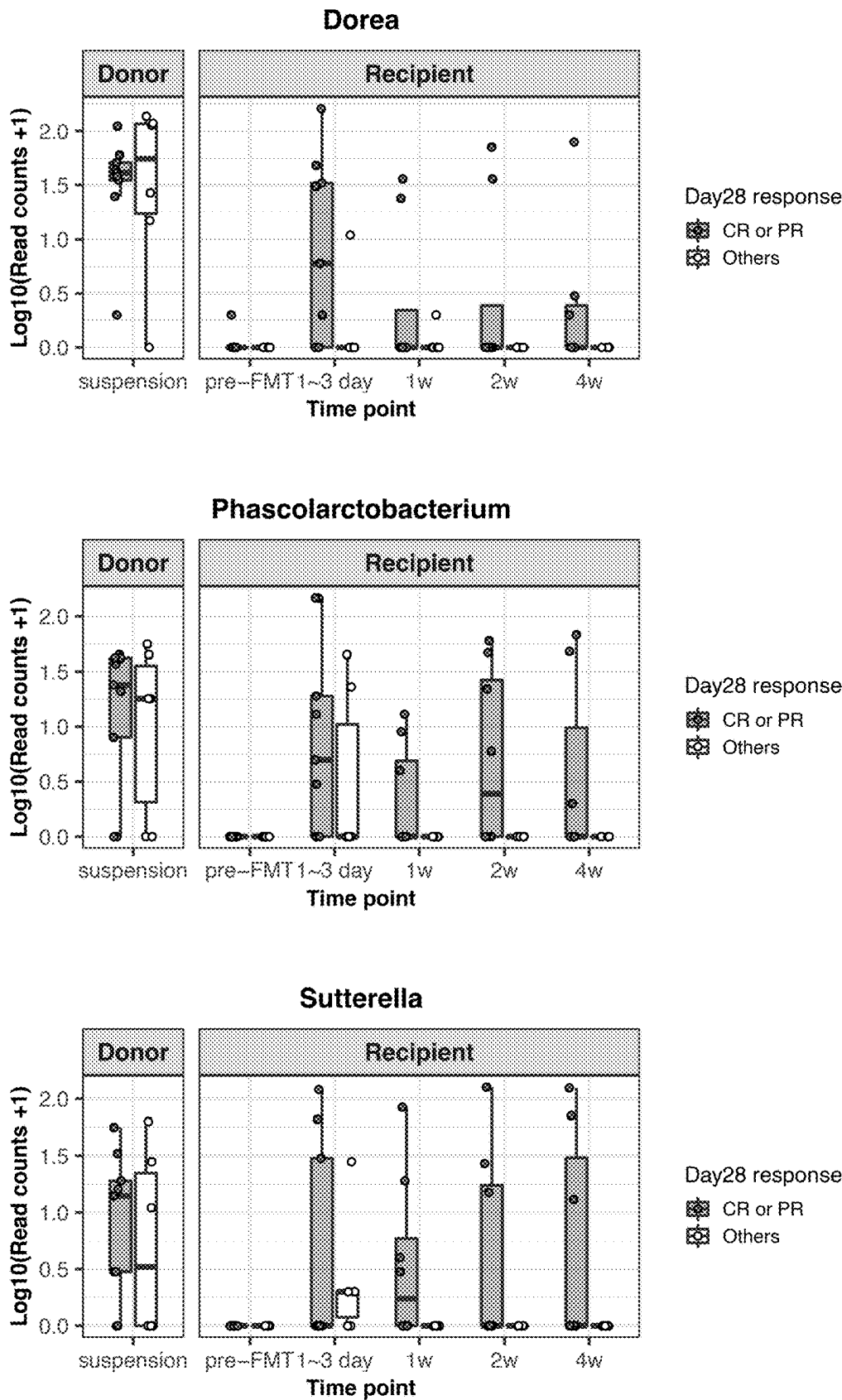


Fig. 8

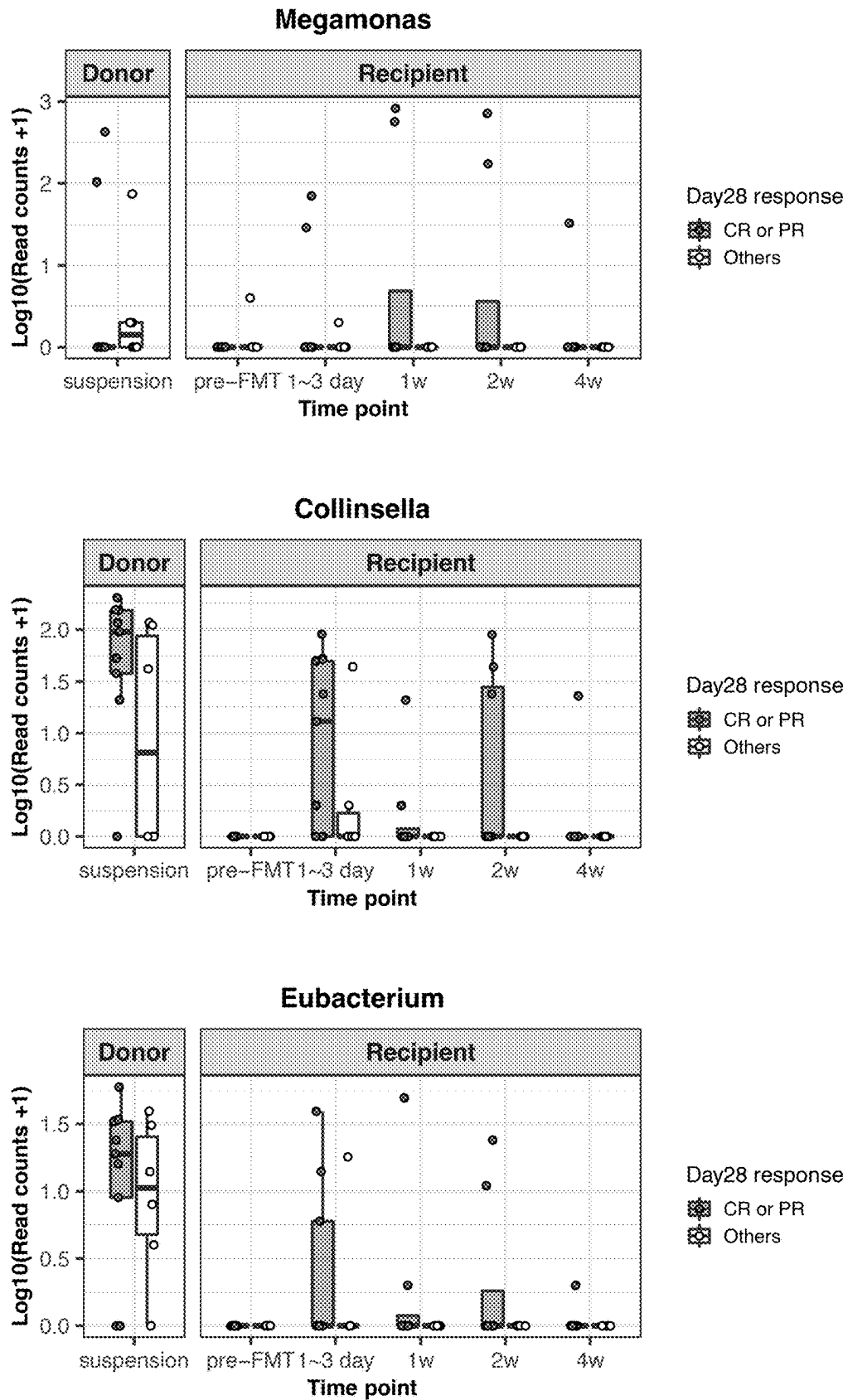


Fig. 9

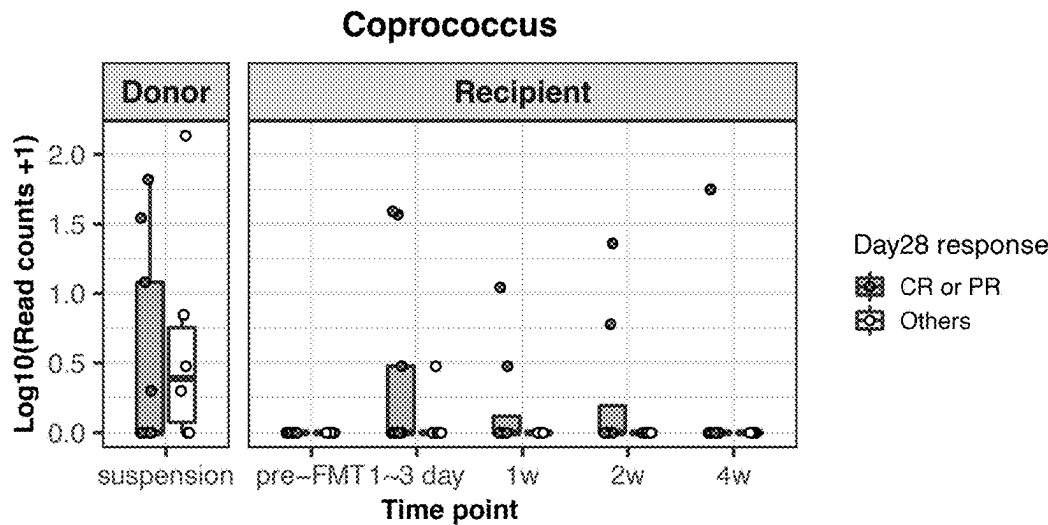


Fig. 10

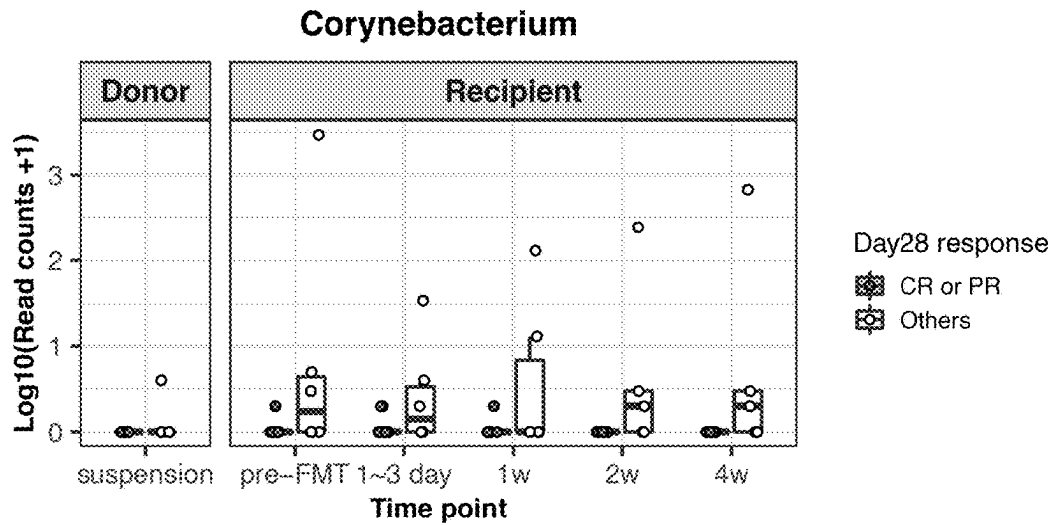


Fig. 11

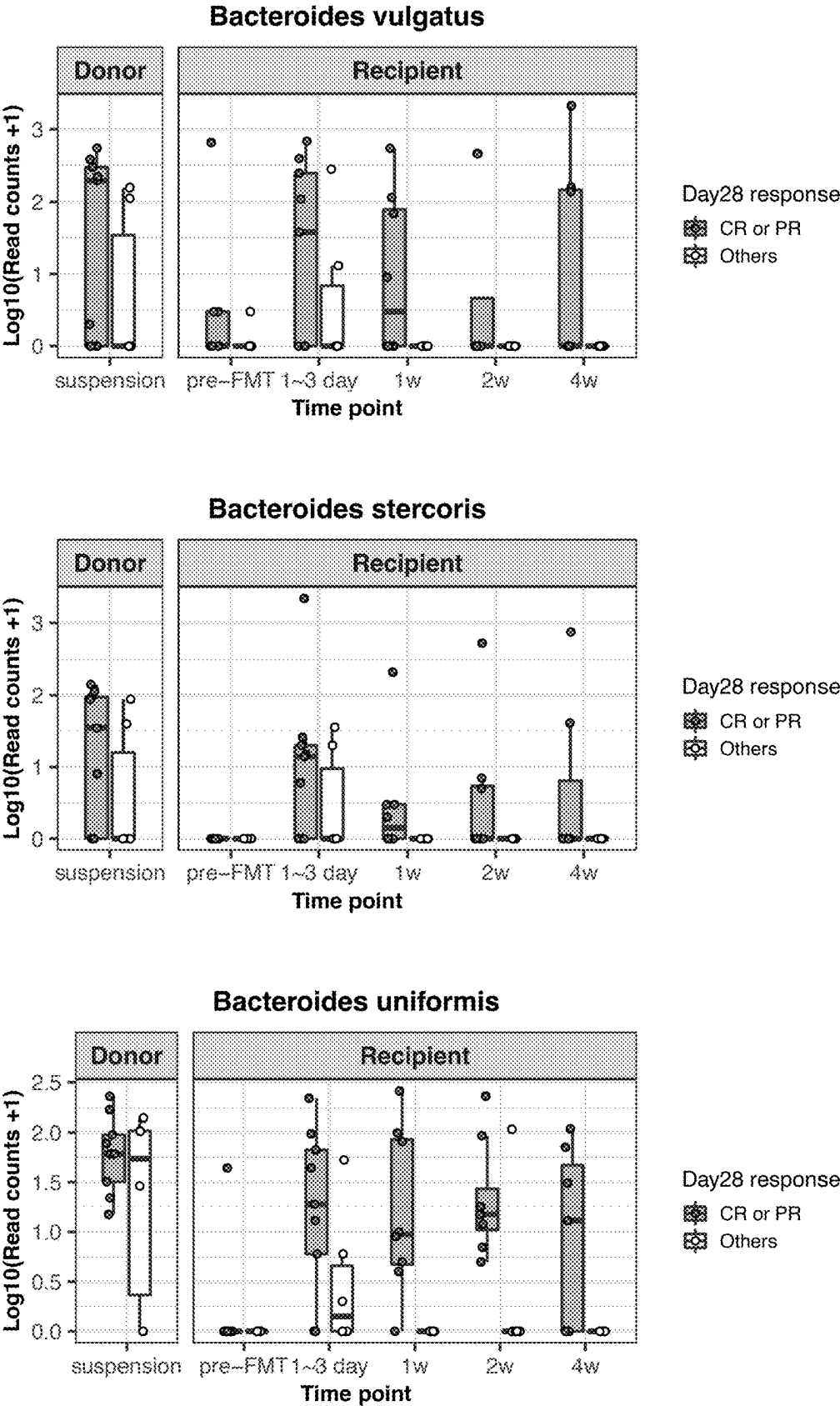


Fig. 12

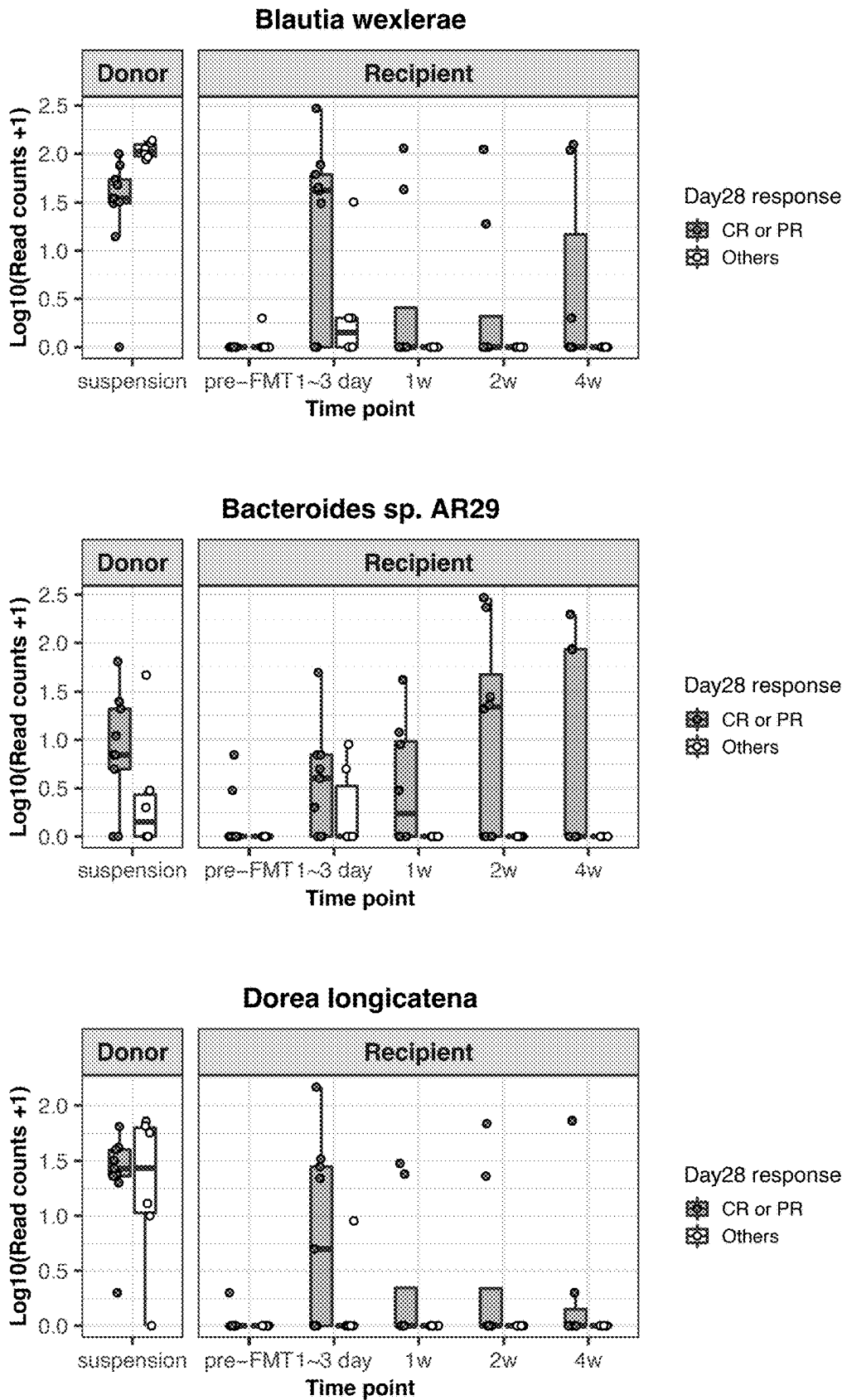


Fig. 13

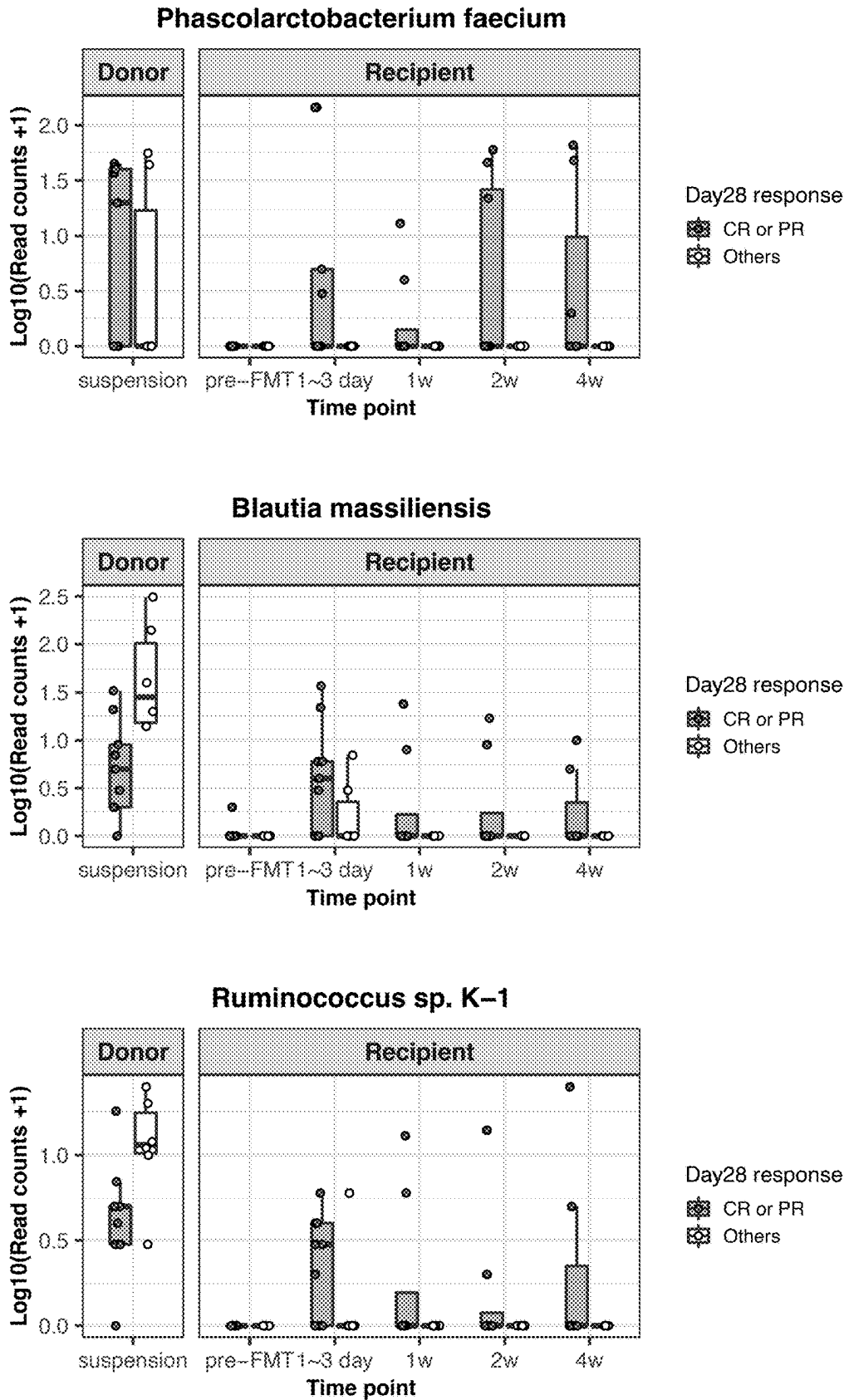


Fig. 14

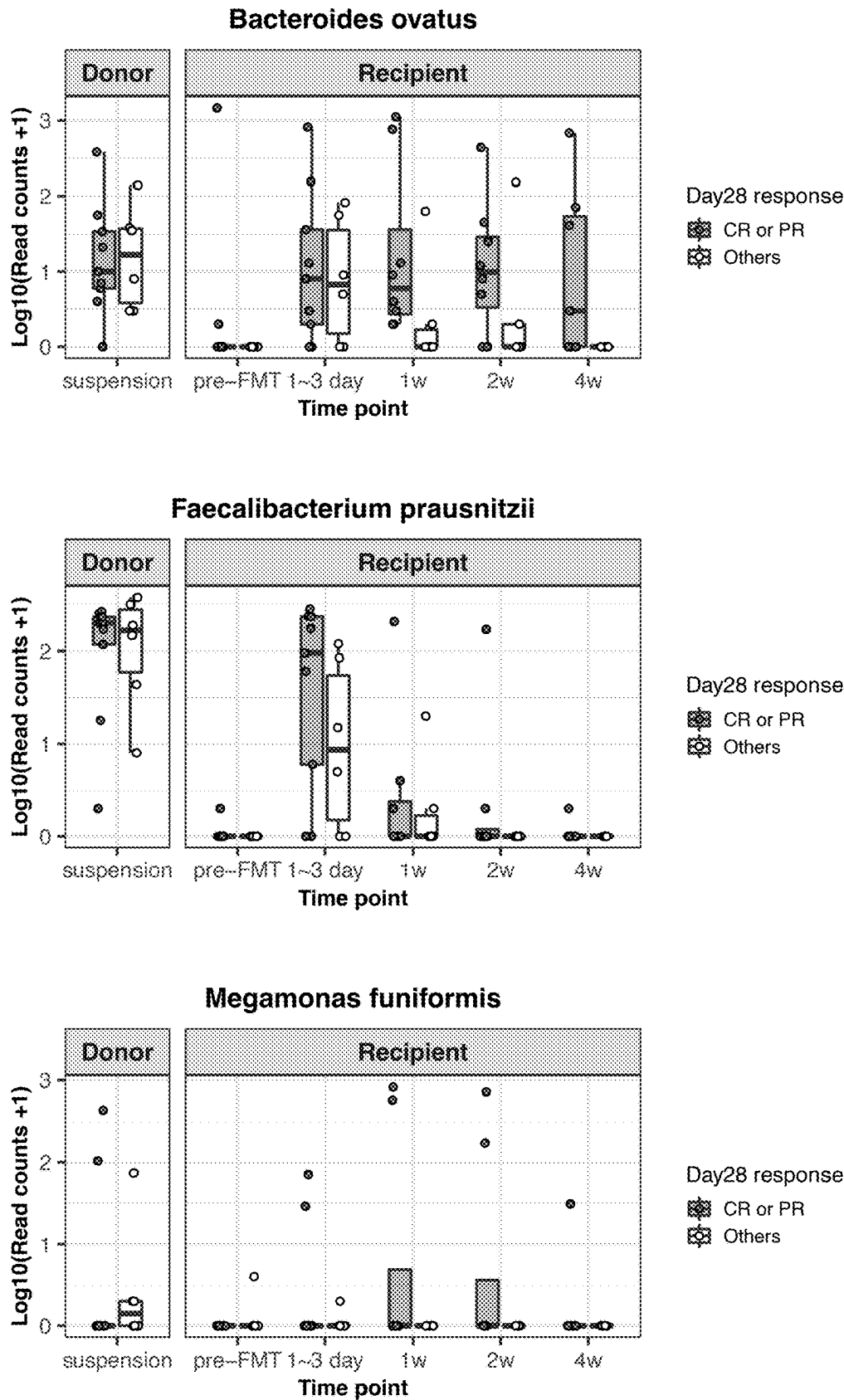


Fig. 15

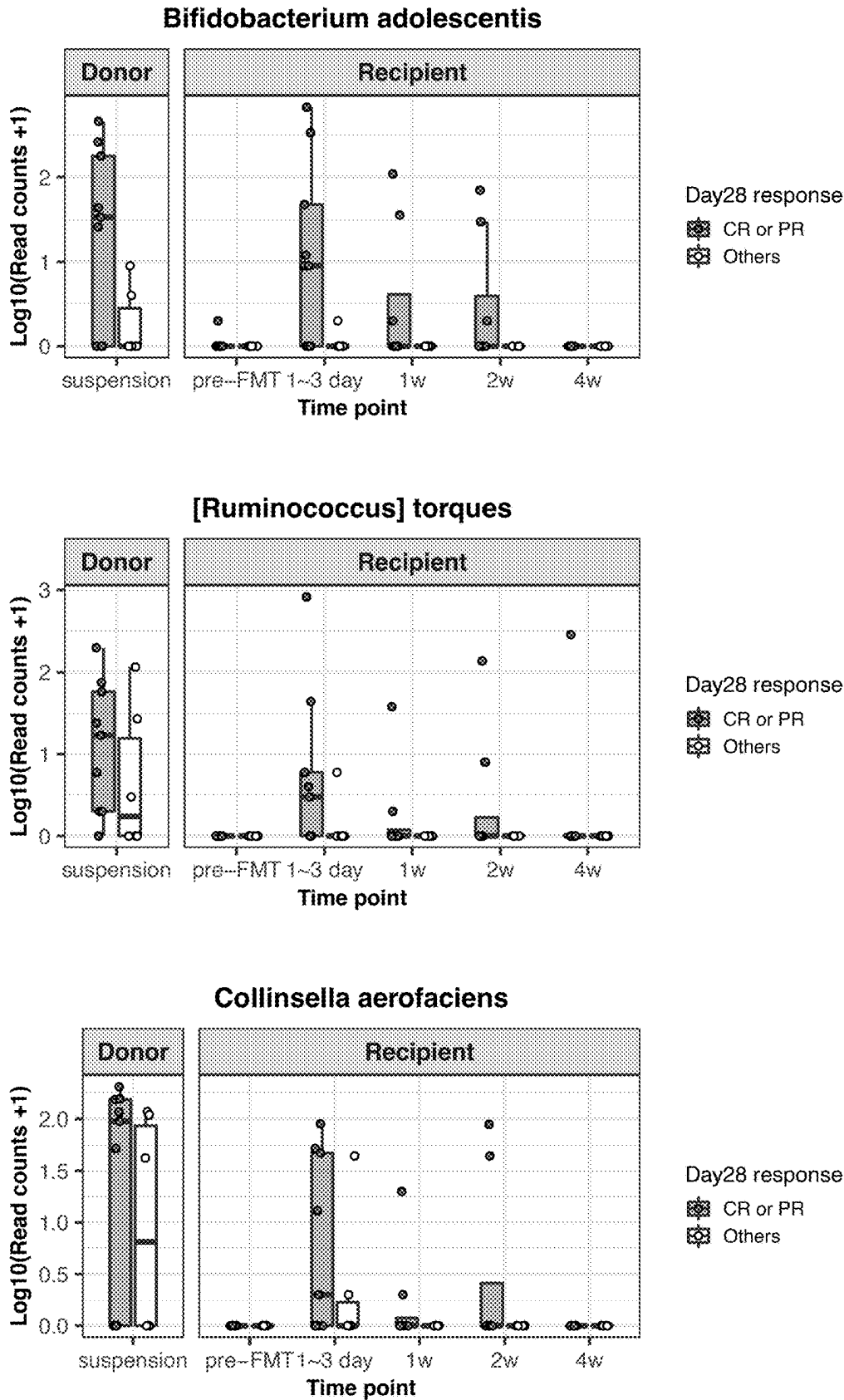


Fig. 16

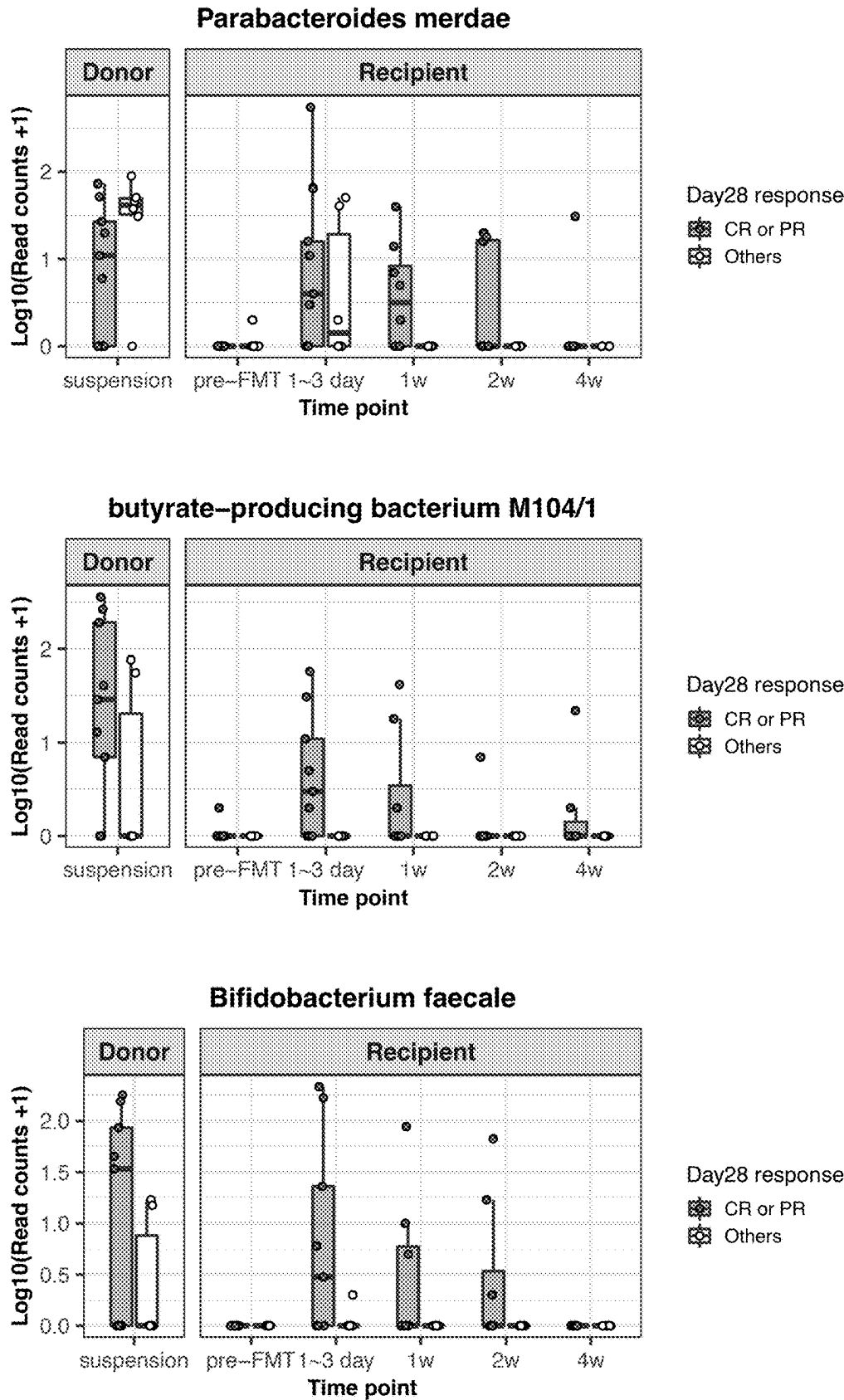


Fig. 17

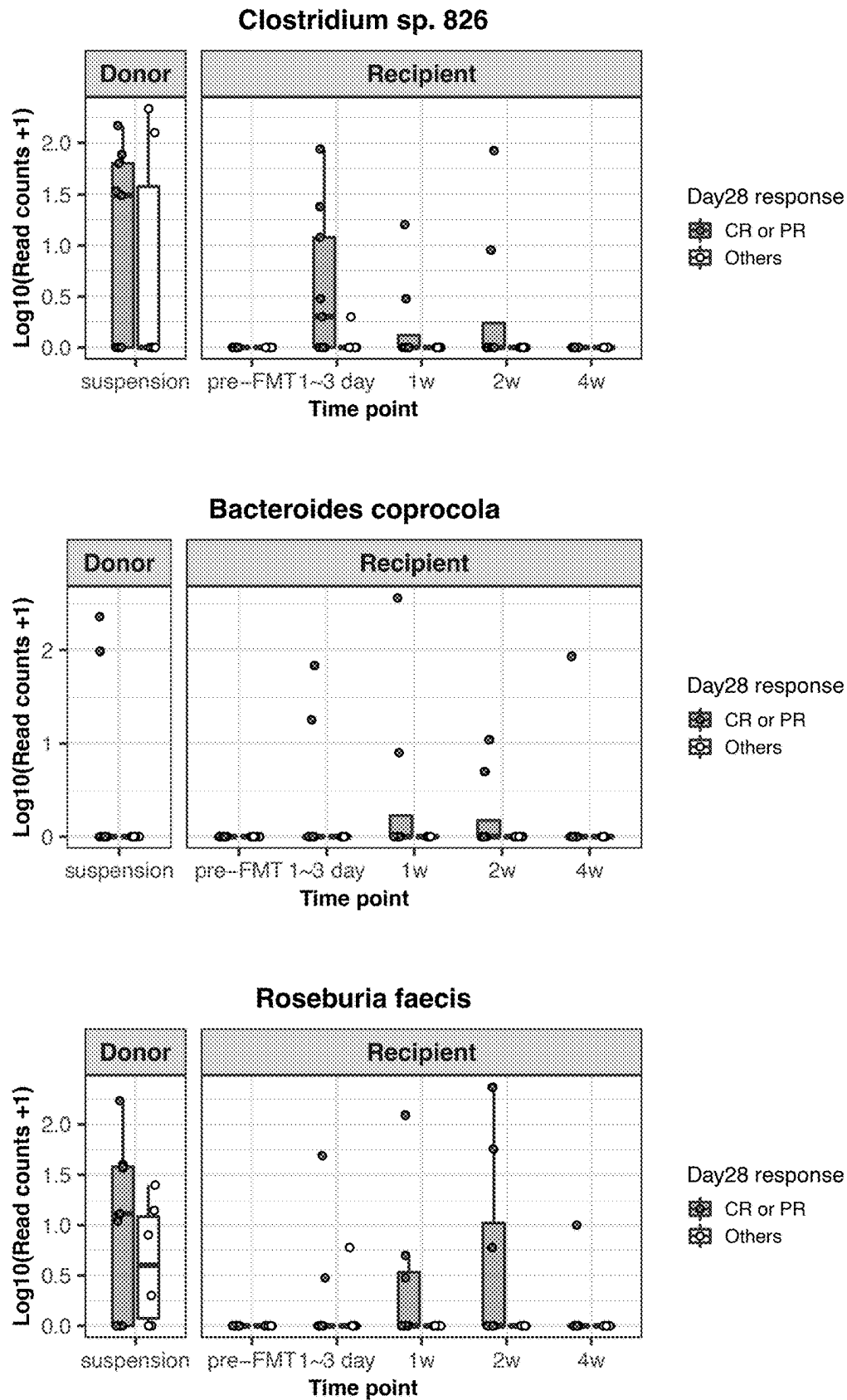


Fig. 18

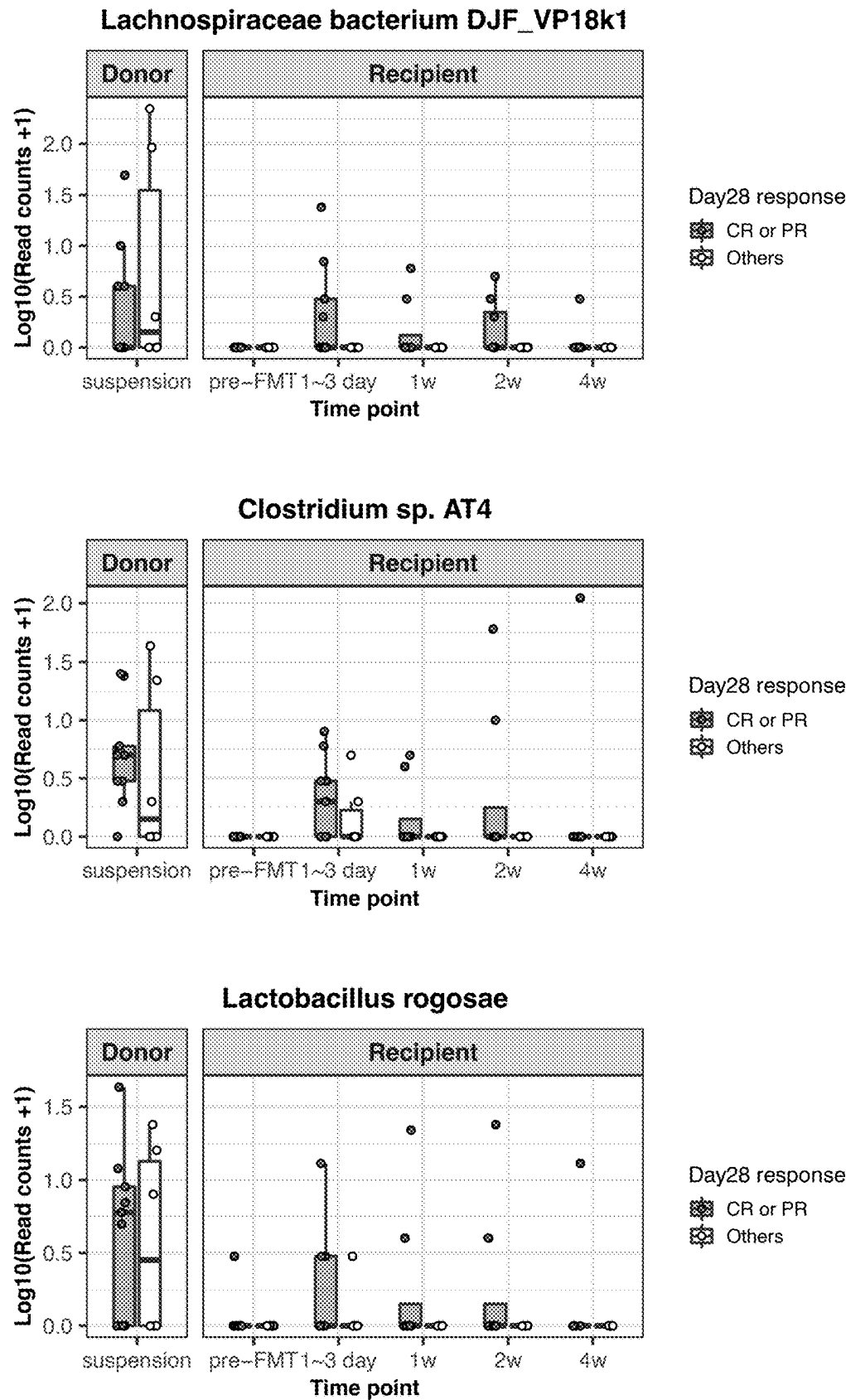
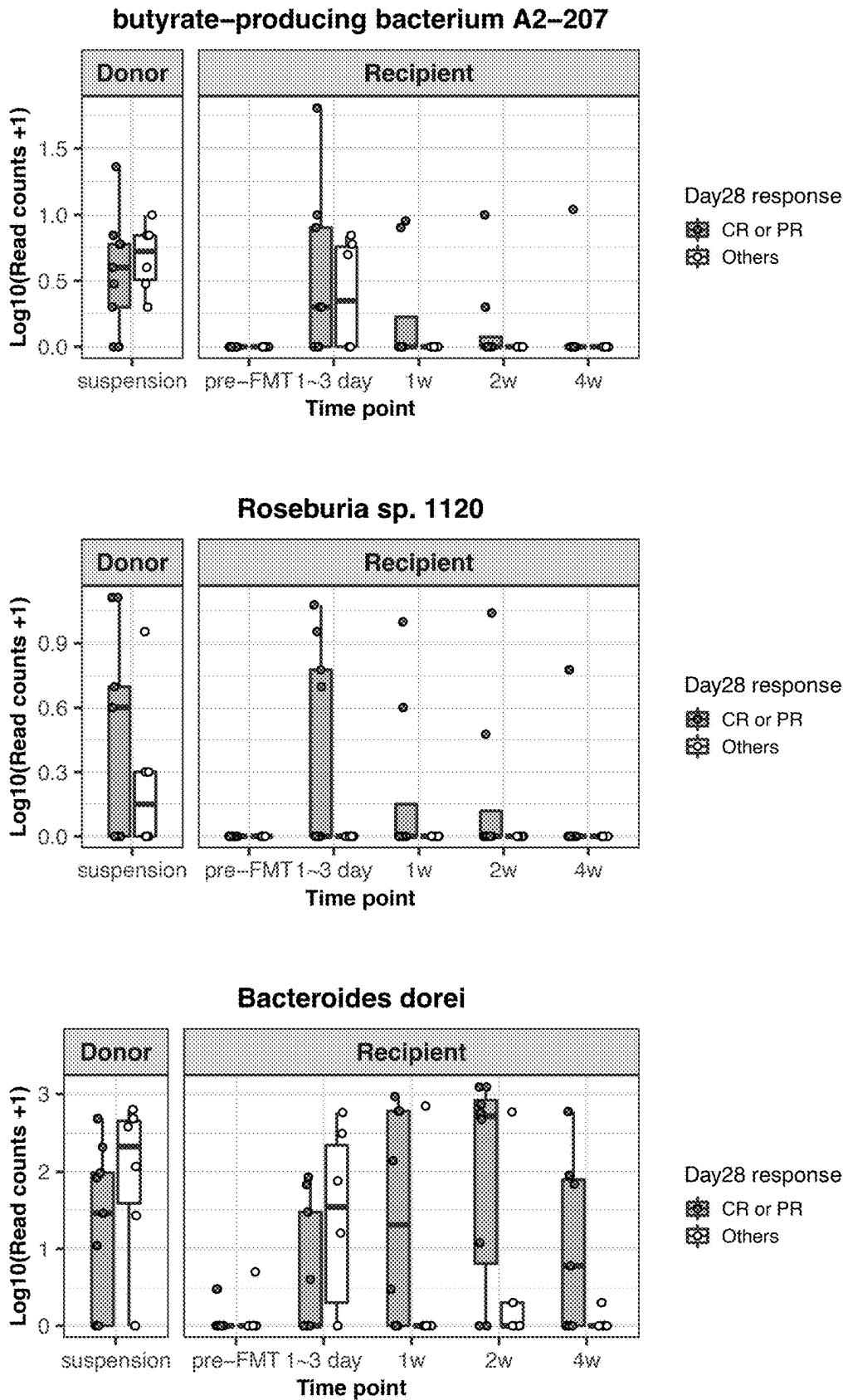


Fig. 19



PROPHYLACTIC OR THERAPEUTIC COMPOSITION FOR GRAFT-VERSUS-HOST DISEASE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a Divisional of copending application Ser. No. 17/779,286, filed on May 24, 2022, which is the National Phase under 35 U.S.C. § 371 of International Application No. PCT/JP2020/043883, filed on Nov. 25, 2020, which claims the benefit under 35 U.S.C. §§ 119(a) to Patent Application No. 2019-213436, filed in Japan on Nov. 26, 2019, all of which are hereby expressly incorporated by reference into the present application.

REFERENCE TO ELECTRONIC SEQUENCE LISTING

[0002] This application contains a Sequence Listing which has been submitted electronically in .XML format and is hereby incorporated by reference in its entirety. Said .XML copy, created on May 2, 2025, is named "4456-0295PUS2.xml" and is 419,095 bytes in size. The sequence listing contained in this .XML file is part of the specification and is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0003] The present invention relates to a composition comprising a microbiota, and more particularly relates to a prophylactic or therapeutic composition for graft-versus-host disease (GVHD).

BACKGROUND ART

[0004] Allogeneic hematopoietic stem cell transplantation (allo-HSCT) has been widely used as a radical therapy for various blood diseases, but acute GVHD associated with allo-HSCT is comparable to recurrence and infection among serious complications. Agents used in initial therapy (primary therapy) for this GVHD are adrenocorticosteroid hormones (steroids), but only about half of them are confirmed to be effective (Blood. 2007; 109(10): 4119-4126. (Non-patent Document 1)), and no secondary therapy has been established.

[0005] Enterobacteria and their metabolites have been widely known to play important roles in inflammatory suppression and immunomodulation in the gut, and recent reports have suggested the applicability of fecal microbiota transplantation (FMT) to GVHD following allo-HSCT (Blood. 2014; 124(7): 1174-1182. (Non-patent Document 2)). Moreover, as to FMT, the inventions disclosed in JP 2013-537531 A (Patent Document 1), JP 2016-501852 A (Patent Document 2) and Japanese Patent No. 6408092 have been known.

[0006] However, the relation between GVHD and FMT is not clear.

PRIOR ART DOCUMENTS

Patent Documents

- [0007] Patent Document 1: JP 2013-537531 A
- [0008] Patent Document 2: JP 2016-501852 A
- [0009] Patent Document 3: Japanese Patent No. 6408092

Non-Patent Documents

- [0010] Non-patent Document 1: Blood. 2007; 109(10): 4119-4126.
- [0011] Non-patent Document 2: Blood. 2014; 124(7): 1174-1182.
- [0012] Non-patent Document 3: Bone Marrow Transplantation 1995; 15, 825-828.

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

[0013] The object of the present invention is to provide a prophylactic or therapeutic composition particularly for acute gut GVHD.

Means to Solve the Problem

[0014] As a result of extensive and intensive efforts made to solve the problem stated above, the inventors of the present invention have succeeded in preventing or treating GVHD by transplantation of a composition comprising a feces-derived microbiota, and have identified bacteria effective for the prevention or treatment of GVHD from a feces-derived microbiota, thereby completing the present invention.

[0015] Namely, the present invention is as follows.

[0016] (1)

[0017] A prophylactic or therapeutic composition for GVHD, which comprises bacteria belonging to any genus selected from the group consisting of the following genera:

- [0018] *Blautia*,
- [0019] *Clostridium*,
- [0020] unclassified Clostridiales,
- [0021] *Actinomyces*,
- [0022] *Parabacteroides*,
- [0023] *Lachnoclostridium*,
- [0024] *Bacteroides*,
- [0025] *Faecalibacterium*,
- [0026] unclassified Lachnospiraceae,
- [0027] *Roseburia*,
- [0028] *Ruminococcus*,
- [0029] unclassified Firmicutes,
- [0030] *Dorea*,
- [0031] *Phascolarctobacterium*,
- [0032] *Sutterella*,
- [0033] *Megamonas*,
- [0034] *Collinsella*,
- [0035] *Eubacterium*,
- [0036] *Coprococcus*,
- [0037] *Schaalia*,
- [0038] *Alistipes*,
- [0039] *Bifidobacterium*,
- [0040] *Lactobacillus*,
- [0041] *Veillonella*,
- [0042] *Anaerostipes*,
- [0043] *Bilophila*,
- [0044] *Butyrivibrio*,
- [0045] *Barnesiella*,
- [0046] *Fusicatenibacter*,
- [0047] *Flavonifractor*,
- [0048] unclassified Ruminococcaceae,
- [0049] unclassified Clostridiaceae,
- [0050] *Faecalicatena*,

- [0051] *Prevotella*,
 [0052] *Megasphaera*,
 [0053] *Robinsoniella*,
 [0054] *Faecalitalea*,
 [0055] *Lachnospira*, and
 [0056] *Romboutsia*,
 [0057] or any combination of bacteria belonging to these genera.
 [0058] (2)
 [0059] The composition according to (1) above, which comprises bacteria belonging to any genus selected from the group consisting of the following genera:
 [0060] *Blautia*,
 [0061] *Clostridium*,
 [0062] unclassified Clostridiales,
 [0063] *Actinomyces*,
 [0064] *Parabacteroides*,
 [0065] *Lachnoclostridium*,
 [0066] *Bacteroides*,
 [0067] *Faecalibacterium*,
 [0068] unclassified Lachnospiraceae,
 [0069] *Roseburia*,
 [0070] *Ruminococcus*,
 [0071] unclassified Firmicutes,
 [0072] *Dorea*,
 [0073] *Phascolarctobacterium*,
 [0074] *Sutterella*,
 [0075] *Megamonas*,
 [0076] *Collinsella*,
 [0077] *Eubacterium*, and
 [0078] *Coprococcus*,
 [0079] or any combination of bacteria belonging to these genera.
 [0080] (3)
 [0081] The composition according to (1) above, which comprises bacteria belonging to any genus selected from the group consisting of the following genera:
 [0082] *Blautia*,
 [0083] *Clostridium*,
 [0084] unclassified Clostridiales,
 [0085] *Actinomyces*,
 [0086] *Parabacteroides*,
 [0087] *Lachnoclostridium*,
 [0088] *Bacteroides*,
 [0089] unclassified Lachnospiraceae,
 [0090] *Roseburia*,
 [0091] *Ruminococcus*,
 [0092] *Phascolarctobacterium*,
 [0093] *Faecalibacterium*,
 [0094] *Schaalia*,
 [0095] *Alistipes*,
 [0096] *Bifidobacterium*,
 [0097] *Lactobacillus*,
 [0098] *Veillonella*,
 [0099] *Dorea*,
 [0100] unclassified Firmicutes,
 [0101] *Anaerostipes*,
 [0102] *Collinsella*,
 [0103] *Butyrivibrio*,
 [0104] *Barnesiella*, and
 [0105] *Fusicatenibacter*,
 [0106] or any combination of bacteria belonging to these genera.
 [0107] (4)
 [0108] The composition according to (3) above, which comprises bacteria belonging to any genus selected from the group consisting of the following genera:
 [0109] *Blautia*,
 [0110] *Clostridium*,
 [0111] unclassified Clostridiales,
 [0112] *Actinomyces*,
 [0113] *Parabacteroides*,
 [0114] *Lachnoclostridium*, and
 [0115] *Bacteroides*,
 [0116] or any combination of bacteria belonging to these genera.
 [0117] (5)
 [0118] The composition according to (1) above, which comprises bacteria belonging to any genus selected from the group consisting of the following genera:
 [0119] *Faecalibacterium*,
 [0120] unclassified Clostridiales,
 [0121] unclassified Lachnospiraceae,
 [0122] *Roseburia*,
 [0123] *Ruminococcus*,
 [0124] unclassified Firmicutes,
 [0125] *Dorea*,
 [0126] *Phascolarctobacterium*,
 [0127] *Sutterella*,
 [0128] *Bacteroides*,
 [0129] *Blautia*,
 [0130] *Parabacteroides*,
 [0131] *Megamonas*,
 [0132] *Collinsella*,
 [0133] *Eubacterium*,
 [0134] *Coprococcus*,
 [0135] *Anaerostipes*,
 [0136] *Bilophila*,
 [0137] *Flavonifractor*,
 [0138] unclassified Ruminococcaceae,
 [0139] unclassified Clostridiaceae,
 [0140] *Butyrivibrio*,
 [0141] *Faecalicatena*,
 [0142] *Prevotella*,
 [0143] *Clostridium*,
 [0144] *Lachnoclostridium*,
 [0145] *Alistipes*,
 [0146] *Megasphaera*,
 [0147] *Robinsoniella*,
 [0148] *Fusicatenibacter*,
 [0149] *Barnesiella*,
 [0150] *Faecalitalea*,
 [0151] *Lachnospira*, and
 [0152] *Romboutsia*,
 [0153] or any combination of bacteria belonging to these genera.
 [0154] (6)
 [0155] The composition according to (5) above, which comprises bacteria belonging to any genus selected from the group consisting of the following genera:
 [0156] *Faecalibacterium*,
 [0157] unclassified Clostridiales,

- [0158] unclassified Lachnospiraceae,
 [0159] *Roseburia*,
 [0160] *Ruminococcus*,
 [0161] unclassified Firmicutes,
 [0162] *Dorea*,
 [0163] *Phascolarctobacterium*,
 [0164] *Sutterella*,
 [0165] *Bacteroides*,
 [0166] *Blautia*,
 [0167] *Parabacteroides*,
 [0168] *Megamonas*,
 [0169] *Collinsella*,
 [0170] *Eubacterium*, and
 [0171] *Coprococcus*,
 [0172] or any combination of bacteria belonging to these genera.
 [0173] (7)
 [0174] A prophylactic or therapeutic composition for GVHD, which comprises bacteria comprising DNA consisting of a nucleotide sequence sharing 94% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 1 to 120, or any combination of these bacteria.
 [0175] (8)
 [0176] The composition according to (7) above, which comprises bacteria comprising DNA consisting of a nucleotide sequence sharing 94% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 121 to 152, or any combination of these bacteria.
 [0177] (9)
 [0178] The composition according to (7) above, which comprises bacteria comprising DNA consisting of a nucleotide sequence sharing 94% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 153 to 198, or any combination of these bacteria.
 [0179] (10)
 [0180] The composition according to (9) above, which comprises bacteria comprising DNA consisting of a nucleotide sequence sharing 94% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 199 to 201, or any combination of these bacteria.
 [0181] (11)
 [0182] The composition according to (7) above, which comprises bacteria comprising DNA consisting of a nucleotide sequence sharing 94% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 202 to 312, or any combination of these bacteria.
 [0183] (12)
 [0184] The composition according to (11) above, which comprises bacteria comprising DNA consisting of a nucleotide sequence sharing 94% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 313 to 344, or any combination of these bacteria.
 [0185] (13)
 [0186] The composition according to (1) above, wherein the bacteria comprise at least one selected from the group consisting of:
 [0187] *Bacteroides vulgatus*,
 [0188] *Bacteroides stercoris*,
 [0189] *Bacteroides uniformis*,
 [0190] *Blautia wexlerae*,
 [0191] *Bacteroides* sp. AR29,
 [0192] *Dorea longicatena*,
 [0193] *Phascolarctobacterium faecium*,
 [0194] *Blautia massiliensis*,
 [0195] *Ruminococcus* sp. K-1,
 [0196] *Bacteroides ovatus*,
 [0197] *Faecalibacterium prausnitzii*,
 [0198] *Megamonas funiformis*,
 [0199] *Bifidobacterium adolescentis*,
 [0200] [*Ruminococcus*] *torques*,
 [0201] *Collinsella aerofaciens*,
 [0202] *Parabacteroides merdae*,
 [0203] butyrate-producing bacterium M104/1,
 [0204] *Bifidobacterium faecale*,
 [0205] *Clostridium* sp. 826,
 [0206] *Bacteroides coprocola*,
 [0207] *Roseburia faecis*,
 [0208] Lachnospiraceae bacterium DJF_VP18k1,
 [0209] *Clostridium* sp. AT4,
 [0210] *Lactobacillus rogosae*,
 [0211] butyrate-producing bacterium A2-207,
 [0212] *Roseburia* sp. 1120,
 [0213] *Bacteroides dorei*,
 [0214] *Bifidobacterium pseudocatenulatum*,
 [0215] *Megasphaera massiliensis*,
 [0216] *Bacteroides massiliensis*,
 [0217] *Anaerostipes hadrus*,
 [0218] *Alistipes putredinis*,
 [0219] *Parabacteroides* sp. D25,
 [0220] *Roseburia inulinivorans*,
 [0221] *Bilophila* sp. 4130,
 [0222] *Flavonifractor plautii*,
 [0223] *Eubacterium ventriosum*,
 [0224] Clostridiales bacterium 80/3,
 [0225] *Butyricicoccus faecihominis*,
 [0226] *Bacteroides cellulosilyticus*,
 [0227] *Coprococcus catus*,
 [0228] *Parabacteroides johnsonii*,
 [0229] Lachnospiraceae bacterium DJF_RP14,
 [0230] [*Ruminococcus*] *gnavus*,
 [0231] *Bifidobacterium longum*,
 [0232] *Bacteroides xylanisolvens*,
 [0233] *Prevotella stercora*,
 [0234] *Bacteroides plebeius*,
 [0235] Firmicutes bacterium DJF_VR50,
 [0236] *Sutterella wadsworthensis*,
 [0237] Lachnospiraceae bacterium 1_1_57FAA,
 [0238] *Blautia obeum*,
 [0239] *Barnesiella intestinihominis*,
 [0240] [*Eubacterium*] *eligens*,
 [0241] *Coprococcus comes*,
 [0242] *Ruminococcus* sp. DJF_VR70k1,
 [0243] butyrate-producing bacterium A2-175,
 [0244] *Faecalitalea cylindroides*,
 [0245] *Blautia* sp. YHC-4,
 [0246] [*Clostridium*] *glycyrrhizinilyticum*,
 [0247] *Ruminococcus* sp. 653,
 [0248] *Ruminococcus lactaris*,
 [0249] butyrate-producing bacterium SM6/1,
 [0250] butyrate-producing bacterium SS3/4,
 [0251] Clostridiaceae bacterium DJF LS40,
 [0252] *Bacteroides* sp. 1130,

- [0253] *Bacteroides nordii*,
- [0254] *Fusicatenibacter saccharivorans*,
- [0255] butyrate-producing bacterium SL7/1,
- [0256] [*Clostridium*] *scindens*,
- [0257] *Parabacteroides distasonis*,
- [0258] *Schaalia odontolytica*,
- [0259] *Bacteroides faecis*,
- [0260] *Bacteroides fragilis*,
- [0261] [*Clostridium*] *bolteae*, and
- [0262] *Clostridium* sp. HGF2.
- [0263] (14)
- [0264] The composition according to (13) above, wherein the bacteria comprise at least one selected from the group consisting of:
- [0265] *Bacteroides vulgatus*,
- [0266] *Bacteroides stercoris*,
- [0267] *Bacteroides uniformis*,
- [0268] *Blautia wexlerae*,
- [0269] *Bacteroides* sp. AR29,
- [0270] *Dorea longicatena*,
- [0271] *Phascolarctobacterium faecium*,
- [0272] *Blautia massiliensis*,
- [0273] *Ruminococcus* sp. K-1,
- [0274] *Bacteroides ovatus*,
- [0275] *Faecalibacterium prausnitzii*,
- [0276] *Megamonas funiformis*,
- [0277] *Bifidobacterium adolescentis*,
- [0278] [*Ruminococcus*] *torques*,
- [0279] *Collinsella aerofaciens*,
- [0280] *Parabacteroides merdae*,
- [0281] butyrate-producing bacterium M104/1,
- [0282] *Bifidobacterium faecale*,
- [0283] *Clostridium* sp. 826,
- [0284] *Bacteroides coprocola*,
- [0285] *Roseburia faecis*,
- [0286] Lachnospiraceae bacterium DJF_VP18k1,
- [0287] *Clostridium* sp. AT4,
- [0288] *Lactobacillus rogosae*,
- [0289] butyrate-producing bacterium A2-207,
- [0290] *Roseburia* sp. 1120, and
- [0291] *Bacteroides dorei*.
- [0292] (15)
- [0293] The composition according to (13) above, wherein the bacteria comprise at least one selected from the group consisting of:
- [0294] *Bacteroides vulgatus*,
- [0295] *Bacteroides stercoris*,
- [0296] *Bacteroides uniformis*,
- [0297] *Blautia wexlerae*,
- [0298] *Bacteroides* sp. AR29,
- [0299] *Dorea longicatena*,
- [0300] *Phascolarctobacterium faecium*,
- [0301] *Blautia massiliensis*,
- [0302] *Ruminococcus* sp. K-1, and
- [0303] *Bacteroides ovatus*.
- [0304] (16)
- [0305] The composition according to any one of (1) to (15) above, wherein the GVHD is steroid-refractory or steroid-dependent GVHD.
- [0306] (17)
- [0307] The composition according to any one of (1) to (15) above, wherein the GVHD is acute gut GVHD.

- [0308] (18)
- [0309] A capsule formulation for the prevention or treatment of GVHD, which comprises the composition according to any one of (1) to (17) above.
- [0310] (19)
- [0311] A therapeutic method for GVHD, which comprises administering the composition according to any one of (1) to (17) above or the capsule formulation according to (18) above to a GVHD patient.
- [0312] (20)
- [0313] A prophylactic method for GVHD, which comprises administering the composition according to any one of (1) to (17) above or the capsule formulation according to (18) above either before or after or both before and after hematopoietic stem cell transplantation to a patient who is a subject of the hematopoietic stem cell transplantation.
- [0314] (21)
- [0315] A method for preparing a suspension for fecal microbiota transplantation, which comprises confirming the presence of the bacteria shown in any one of (1) to (15) above in feces collected from a human subject, and suspending the feces confirmed for the presence of the bacteria into an aqueous medium.
- [0316] (22)
- [0317] A method for preparing a bacterial mixture for fecal microbiota transplantation, which comprises confirming the presence of the bacteria shown in any one of (1) to (15) above in feces collected from a human subject, separating the bacteria from the feces confirmed for the presence of the bacteria, and mixing the separated bacteria.
- [0318] (23)
- [0319] A method for preparing a formulation for fecal microbiota transplantation, which comprises confirming the presence of the bacteria shown in any one of (1) to (15) above in feces collected from a human subject, separating the bacteria from the feces confirmed for the presence of the bacteria, and formulating the separated bacteria.
- [0320] (24)
- [0321] The method according to (23) above, wherein the formulation is a capsule formulation.
- [0322] (25)
- [0323] The method according to any one of (21) to (24) above, wherein the presence of the bacteria is confirmed by 16S rRNA gene analysis.

Effects of the Invention

[0324] The present invention enables the prevention or treatment of acute gut GVHD.

BRIEF DESCRIPTION OF THE DRAWINGS

[0325] The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

[0326] FIG. 1 shows the results of a diversity evaluation by the Shannon index.

[0327] FIGS. 2A-2G show the relative proportions of fecal microbiota at the genus level before and after FMT.

[0328] FIG. 3 shows the results of microbiota analysis based on the 16S rRNA gene in bacteria belonging to each genus.

[0329] FIG. 4 shows the results of microbiota analysis based on the 16S rRNA gene in bacteria belonging to each genus.

[0330] FIG. 5 shows the results of microbiota analysis based on the 16S rRNA gene in bacteria belonging to each genus.

[0331] FIG. 6 shows the results of microbiota analysis based on the 16S rRNA gene in bacteria belonging to each genus.

[0332] FIG. 7 shows the results of microbiota analysis based on the 16S rRNA gene in bacteria belonging to each genus.

[0333] FIG. 8 shows the results of microbiota analysis based on the 16S rRNA gene in bacteria belonging to each genus.

[0334] FIG. 9 shows the results of microbiota analysis based on the 16S rRNA gene in bacteria belonging to each genus.

[0335] FIG. 10 shows the results of microbiota analysis based on the 16S rRNA gene in bacteria belonging to the genus *Corynebacterium*.

[0336] FIG. 11 shows the results of microbiota analysis based on the 16S rRNA gene in each species.

[0337] FIG. 12 shows the results of microbiota analysis based on the 16S rRNA gene in each species.

[0338] FIG. 13 shows the results of microbiota analysis based on the 16S rRNA gene in each species.

[0339] FIG. 14 shows the results of microbiota analysis based on the 16S rRNA gene in each species.

[0340] FIG. 15 shows the results of microbiota analysis based on the 16S rRNA gene in each species.

[0341] FIG. 16 shows the results of microbiota analysis based on the 16S rRNA gene in each species.

[0342] FIG. 17 shows the results of microbiota analysis based on the 16S rRNA gene in each species.

[0343] FIG. 18 shows the results of microbiota analysis based on the 16S rRNA gene in each species.

[0344] FIG. 19 shows the results of microbiota analysis based on the 16S rRNA gene in each species.

DESCRIPTION OF EMBODIMENTS

[0345] In the expectation that intervention in the gut microbiota would lead to novel prophylactic and therapeutic methods for GVHD following hematopoietic stem cell transplantation, the inventors of the present invention have attempted to perform fecal microbiota transplantation (FMT) on GVHD patients. FMT is a therapy designed to administer a fecal suspension from a normal subject into the gut of a patient, thereby allowing high volume administration of a normal microbiota, and this therapy has been attempted for use in diseases which are deemed to be associated with dysbiosis of the gut microbiota.

[0346] Further, the inventors of the present invention have performed gut microbiota analysis before and after FMT on patients who had developed GVHD following hematopoietic stem cell transplantation, whereby bacteria useful for the prevention and/or treatment of GVHD have been identified. As a result of the analysis, it has been found that there are clear differences in the distribution of the engrafted microbiota between patients whose GVHD has gotten better and the other patients. Further, the engrafted gut microbiota in

patients whose GVHD has gotten better were picked up as bacteria useful for the treatment of GVHD, and then grouped under unique indicators such as heterogeneity in abundance due to the therapeutic effect of FMT in recipients, as analyzed by two group comparison (between CR or PR group and Others group), the length of the engraftment period in recipients, etc.

As a result, the inventors of the present invention have succeeded in identifying bacteria useful for the prevention and/or treatment of GVHD.

1. FMT

[0347] Since a fecal microbiota is contained in feces per se or a treated product of feces, a treated product of feces can be used as the composition of the present invention. Such a treated product of feces includes not only a suspension of collected feces in an appropriate solvent (e.g., physiological saline, buffer), but also a filtrate of this suspension passed through an appropriate sieve, gauze, filter or the like (e.g., pore size: 0.1 mm to 0.5 mm), or a precipitate of this suspension obtained after centrifugation. Further, these compositions may be frozen in a freezer or with liquid nitrogen, or may be subjected to lyophilization or spray drying. In the above case where feces are prepared into a suspension with a solvent, feces may be suspended with 1 to 20 ml of liquid per gram of feces. In a case where the prepared suspension is centrifuged to isolate bacteria, the bacteria may be suspended again in 0.2 to 1 mL of liquid per gram of bacteria and then provided for use.

[0348] For freezing or lyophilization, cryoprotectants and/or lyoprotectants may be added, as exemplified by various sugars (e.g., sucrose, fructose, lactose, mannitol), glycerol, polyethylene glycol (PEG), trehalose, glycine, glucose, dextran, erythritol and so on.

[0349] In the present invention, the collected feces or a treated product thereof may be stored for 6 to 10 hours after collection or treatment of the feces. The storage temperature is not limited in any way, but cold storage (e.g., 4° C.) is preferred for this purpose.

[0350] The thus prepared composition is used as a material for FMT. The prepared FMT material is preferably stored under anaerobic conditions (e.g., in an anaerobic unit, in an anaerobic bag) until use. In this case, cold storage (e.g., 4° C.) is also preferred.

[0351] In the present invention, a composition comprising a fecal microbiota (i.e., an untreated or treated fecal material) is transplanted between different individuals, for example, between humans or between animals. A composition for use in FMT may be transplanted to the same individual whose feces were collected, or may be configured such that a fecal microbiota collected from one individual is transplanted to another individual.

[0352] The disease to be targeted is GVHD, as exemplified by GVHD following hematopoietic stem cell transplantation. GVHD includes, but is not limited to, acute gut GVHD.

[0353] Transplantation may be performed in any manner, either by oral or parenteral administration. Examples include transplantation via a gastroduodenal tube, internal use in the form of capsules or the like, administration of suppositories, transplantation into the colon through a colon fibroscope or high pressure enema, etc.

[0354] The amount used for single transplantation is 150 ml to 300 ml in the case of liquid form, which is given once

a day. Depending on the condition of a recipient, transplantation may be repeated every 4 days to 2 weeks, twice to 4 times in total.

[0355] In this way, the composition of the present invention enables the treatment of GVHD when transplanted (administered) to GVHD patients.

[0356] For assessment of the therapeutic effect, the results of the maximum effect at 4 weeks after the final administration and within the observation period are used to evaluate the response rate (CR+PR), which is the sum of complete response (CR) and partial response (PR). A comparative analysis is made between microbiota in patients (group) assessed as CR or PR and microbiota in patients (group) assessed as the group of others. Aliquots of the donor fecal preparation and the patient's feces (e.g., collected before FMT, and at 1 to 3 days and 1, 2 and 4 weeks after the final FMT) are used for analysis of gut microbiota at each time point.

2. Heterogeneity in the Abundance of Bacteria Between CR or PR Group and Others Group

[0357] In the present invention, the following two methods are used for narrow down search based on heterogeneity in the abundance of bacteria between CR or PR group and Others group.

Method 1. Narrow Down Search with Median

[0358] (1) At each time point of fecal collection, the average of relative abundance of the respective genera or OTUs (operational taxonomic units) in samples of the same group is calculated for the CR or PR group (hereinafter referred to as "CR or PR group") and the group of others (hereinafter referred to as "Others group"), and genera or OTUs in which a greater one of the averages of the "CR or PR group" and the "Others group" is 0.1% or more are used for analysis.

[0359] (2) At each time point, the median of relative abundance of the genera or OTUs in samples of the same group is calculated for each of the CR or PR group and the Others group.

[0360] (3) The median Log 2 fold change (CR or PR group/Others group) is calculated. The Log 2 fold change intended here refers to a logarithmic value (base 2) of the ratio obtained by dividing the median of the CR or PR group by the median of the Others group. Such a median may include 0; and hence 0.0001 is added to each median before calculation of the Log 2 fold change.

[0361] (4) Genera or OTUs showing a Log 2 fold change of 0.5 or more are assumed to be abundant in the CR or PR group, while those showing a Log 2 fold change of -0.5 or less are assumed to be abundant in the Others group. However, genera or OTUs showing a negative Log 2 Fold change for the ratio of median between the donor and the recipient before FMT in the CR or PR group, or genera or OTUs for which the carrying rate of bacteria belonging thereto is 0 in the donor are excluded, because they cannot be determined to be transplanted through FMT from the donor.

[0362] (5) At the time points of fecal collection after fecal transplantation, the number of times when the Log 2 fold change was 0.5 or more or was -0.5 or less is counted.

[0363] (6) In step 5, cases where the number of times when the Log 2 fold change is 0.5 or more exceeds 1

are assumed to be preferred, and bacteria showing a higher number of times are assessed to give more contribution to the effect of FMT. If the number of times counted exceeds more than half of the number of fecal collection after transplantation, such bacteria are assessed to give more contribution to the effect of FMT. However, if the number of times when the Log 2 fold change is -0.5 or less exceeds the number of times when the Log 2 fold change is 0.5 or more, such bacteria are excluded because they are not determined to be abundant in the CR or PR group.

Method 2. Narrow Down Search for Bacteria with Mean and Carrying Rate

[0364] (1) At each time point of fecal collection, the average of relative abundance of the respective genera or OTUs in samples of the same group is calculated for the CR or PR group (hereinafter referred to as "CR or PR group") and the group of others (hereinafter referred to as "Others group"), and genera or OTUs in which a greater one of the averages of the "CR or PR group" and the "Others group" is 0.1% or more are used for analysis.

[0365] (2) At each time point, the mean of relative abundance of the genera or OTUs in samples of the same group is calculated for each of the CR or PR group and the Others group. Further, at each time point, the carrying rate of genera or OTUs (i.e., the proportion of donors or recipients in which the relative abundance of these genera or OTUs is greater than zero) in each group is calculated.

[0366] (3) The mean Log 2 fold change (CR or PR group/Others group) is calculated. The Log 2 fold change intended here refers to a logarithmic value (base 2) of the ratio obtained by dividing the mean of the CR or PR group by the mean of the Others group. Such a mean may include 0; and hence 0.0001 is added to each mean before calculation of the Log 2 fold change.

[0367] (4) At the time points of fecal collection after fecal transplantation, the number of times when the Log 2 fold change was 0.5 or more and the carrying rate in the CR or PR group was 20% or more is counted. Genera or OTUs showing a higher number of times were assumed to be more abundant in the CR or PR group. However, in (4), genera or OTUs showing a negative Log 2 Fold change of the ratio of mean between the donor and the recipient before FMT in the CR or PR group, or genera or OTUs for which the carrying rate of bacteria belonging thereto is 0 in the donor are excluded, because they cannot be determined to be transplanted through FMT from the donor.

[0368] (5) At the time points of fecal collection after fecal transplantation, the number of times when the Log 2 fold change was -0.5 or less and the carrying rate in the Others group was 20% or more is counted. Genera or OTUs showing a higher number of times are assumed to be more abundant in the Others group.

[0369] (6) In (4), cases where the number of times counted exceeds 1 are assumed to be preferred, and bacteria showing a higher number of times are assessed to give more contribution to the effect of FMT. If the number of times counted exceeds more than half of the number of fecal collection after transplantation, such bacteria are assessed to give more contribution to the effect of FMT. However, if the number of times in (5)

exceeds the number of times in (4), such bacteria are excluded because they are not determined to be abundant in the CR or PR group.

3. Microbiota Constituting a Composition

3-1. Definition by Sequence

[0370] The bacteria extracted by narrow down search using Methods 1 and 2 are bacteria which have DNA shearing 94% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 1 to 120 in the sequence of their 16S rRNA gene (e.g., v1-v2 region).

[0371] As used herein, the expression “94% or more homology” is intended to mean being, for example, 94% or more, 95% or more, 96% or more, 97% or more, 98% or more, 99% or more, 99.5% or more, 99.8% or more, 99.9% or more, or 100% homologous (the same applies hereinafter).

[0372] More preferred are bacteria which have DNA sharing 94% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 121 to 152.

[0373] In another embodiment of the present invention, the bacteria extracted by narrow down search using Methods 1 and 2 are bacteria which have DNA sharing 97% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 1 to 120 in the sequence of their 16S rRNA gene (e.g., v1-v2 region).

[0374] More preferred are bacteria which have DNA sharing 97% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 121 to 152.

[0375] In the present invention, bacteria determined to be preferred by Method 1 (narrow down search with median) are bacteria which have DNA sharing 94% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 153 to 198 in the sequence of the v1-v2 region in their 16S rRNA gene.

[0376] More preferred are bacteria which have DNA sharing 94% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 199 to 201.

[0377] In yet another embodiment of the present invention, bacteria determined to be preferred by Method 1 (narrow down search with median) are bacteria which have DNA sharing 97% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 153 to 198 in the sequence of the v1-v2 region in their 16S rRNA gene.

[0378] More preferred are bacteria which have DNA sharing 97% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 199 to 201.

[0379] In the present invention, bacteria determined to be preferred by Method 2 (narrow down search with mean and carrying rate) are bacteria which have DNA sharing 94% or

more homology with any of the nucleotide sequences shown in SEQ ID NOs: 202 to 312 in the sequence of the v1-v2 region in their 16S rRNA gene.

[0380] More preferred are bacteria which have DNA sharing 94% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 313 to 344.

[0381] In yet another embodiment of the present invention, bacteria determined to be preferred by Method 2 (narrow down search with mean and carrying rate) are bacteria which have DNA sharing 97% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 202 to 312 in the sequence of the v1-v2 region in their 16S rRNA gene.

[0382] More preferred are bacteria which have DNA sharing 97% or more homology with any of the nucleotide sequences shown in SEQ ID NOs: 313 to 344.

3-2. Definition by Genus Name

[0383] In the present invention, bacteria determined to be preferred by Method 1 (narrow down search with median) are those belonging to the genera given below. Bacteria belonging to “unclassified taxonomy names” (e.g., unclassified Clostridiales) are bacteria which have DNA sharing 94% or more homology with the nucleotide sequence of 16S rRNA gene in bacteria found under the species names shown in Table 1A below.

- [0384]** *Blautia*
- [0385]** *Clostridium*
- [0386]** unclassified Clostridiales
- [0387]** *Actinomyces*
- [0388]** *Parabacteroides*
- [0389]** *Lachnoclostridium*
- [0390]** *Bacteroides*
- [0391]** unclassified Lachnospiraceae
- [0392]** *Roseburia*
- [0393]** *Ruminococcus*
- [0394]** *Phascolarctobacterium*
- [0395]** *Faecalibacterium*
- [0396]** *Schaalia*
- [0397]** *Alistipes*
- [0398]** *Bifidobacterium*
- [0399]** *Lactobacillus*
- [0400]** *Veillonella*
- [0401]** *Dorea*
- [0402]** unclassified Firmicutes
- [0403]** *Anaerostipes*
- [0404]** *Collinsella*
- [0405]** *Bilophila*
- [0406]** *Butyricicoccus*
- [0407]** *Barnesiella*
- [0408]** *Fusicatenibacter*

TABLE 1A

Genus group ID	Species subgroup ID	Species	NCBI norank group
unclassified Clostridiaceae	Clostridiaceae bacterium	Clostridiaceae bacterium DJF_B063	unclassified Clostridiaceae
		Clostridiaceae bacterium bSSV31	unclassified Clostridiaceae
		Clostridiaceae bacterium MS3	unclassified Clostridiaceae
		Clostridiaceae bacterium DJF_LS40	unclassified Clostridiaceae
		Clostridiaceae bacterium END-2	unclassified Clostridiaceae
unclassified Clostridiales	butyrate-producing bacterium	butyrate-producing bacterium A2-175	unclassified Clostridiales
		butyrate-producing bacterium A2-207	unclassified Clostridiales
		butyrate-producing bacterium ART55/1	unclassified Clostridiales
		butyrate-producing bacterium L1-93	unclassified Clostridiales
		butyrate-producing bacterium M104/1	unclassified Clostridiales

TABLE 1A-continued

Genus group ID	Species subgroup ID	Species	NCBI norank group
		butyrate-producing bacterium M21/2	unclassified Clostridiales
		butyrate-producing bacterium SL6/1/1	unclassified Clostridiales
		butyrate-producing bacterium SL7/1	unclassified Clostridiales
		butyrate-producing bacterium SM6/1	unclassified Clostridiales
		butyrate-producing bacterium SS3/4	unclassified Clostridiales
		butyrate-producing bacterium YE53	unclassified Clostridiales
	Clostridiales bacterium	Clostridiales bacterium 1_7_47FAA	unclassified Clostridiales
		Clostridiales bacterium 10-3b	unclassified Clostridiales
		Clostridiales bacterium 21-4c	unclassified Clostridiales
		Clostridiales bacterium 24-4c	unclassified Clostridiales
		Clostridiales bacterium 30-4c	unclassified Clostridiales
		Clostridiales bacterium 80/3	unclassified Clostridiales
		Clostridiales bacterium A2-162	unclassified Clostridiales
		Clostridiales bacterium CIEAF 029	unclassified Clostridiales
		Clostridiales bacterium DJF_VP48	unclassified Clostridiales
	Elbe River snow isolate	Elbe River snow isolate Iso15_5	unclassified Clostridiales
unclassified Firmicutes	Firmicutes bacterium	Firmicutes bacterium DJF_VR50	unclassified Firmicutes
unclassified Lachnospiraceae	—	Firmicutes bacterium DJF_VP44	unclassified Firmicutes
		[<i>Eubacterium</i>] <i>rectale</i>	unclassified Lachnospiraceae
	Lachnospiraceae bacterium	Lachnospiraceae bacterium 1_1_57FAA	unclassified Lachnospiraceae
		Lachnospiraceae bacterium 1_4_56FAA	unclassified Lachnospiraceae
		Lachnospiraceae bacterium 2_1_46FAA	unclassified Lachnospiraceae
		Lachnospiraceae bacterium 3_1_46FAA	unclassified Lachnospiraceae
		Lachnospiraceae bacterium 8_1_57FAA	unclassified Lachnospiraceae
		Lachnospiraceae bacterium 9_1_43BFAA	unclassified Lachnospiraceae
		Lachnospiraceae bacterium DJF_RP14	unclassified Lachnospiraceae
		Lachnospiraceae bacterium DJF_RP61	unclassified Lachnospiraceae
		Lachnospiraceae bacterium DJF_VP08k1	unclassified Lachnospiraceae
		Lachnospiraceae bacterium DJF_VP18k1	unclassified Lachnospiraceae
		Lachnospiraceae bacterium TF01-11	unclassified Lachnospiraceae
unclassified Ruminococcaceae	—	[<i>Clostridium</i>] <i>leptum</i>	Ruminococcaceae incertae sedis
	—	[<i>Clostridium</i>] <i>methylopentosum</i>	Ruminococcaceae incertae sedis
	—	[<i>Eubacterium</i>] <i>siraeum</i>	Ruminococcaceae incertae sedis
	Ruminococcaceae bacterium	Ruminococcaceae bacterium D16	unclassified Ruminococcaceae
		Ruminococcaceae bacterium LM158	unclassified Ruminococcaceae
unclassified Erysipelotrichaceae	Erysipelotrichaceae bacterium	Erysipelotrichaceae bacterium 3_1_53	unclassified Erysipelotrichaceae
		Erysipelotrichaceae bacterium 5_2_54FAA	unclassified Erysipelotrichaceae

[0409] Among those listed above, bacteria determined to be more preferred in terms of the length of the engraftment period in recipients are those belonging to the following genera.

- [0410] *Blautia*
- [0411] *Clostridium*
- [0412] unclassified Clostridiales
- [0413] *Actinomyces*
- [0414] *Parabacteroides*
- [0415] *Lachnoclostridium*
- [0416] *Bacteroides*

[0417] Likewise, in the present invention, bacteria determined to be preferred by Method 2 (narrow down search with mean and carrying rate) are those belonging to the genera given below. Bacteria belonging to “unclassified taxonomy names” are bacteria which have DNA sharing 94% or more homology with the nucleotide sequence of 16S rRNA gene in bacteria found under the species names shown in Table 1A.

- [0418] *Faecalibacterium*
- [0419] unclassified Clostridiales
- [0420] unclassified Lachnospiraceae
- [0421] *Roseburia*
- [0422] *Ruminococcus*
- [0423] unclassified Firmicutes
- [0424] *Dorea*
- [0425] *Phascolarctobacterium*
- [0426] *Sutterella*

- [0427] *Bacteroides*
- [0428] *Blautia*
- [0429] *Parabacteroides*
- [0430] *Megamonas*
- [0431] *Collinsella*
- [0432] *Eubacterium*
- [0433] *Coproccoccus*
- [0434] *Anaerostipes*
- [0435] *Bilophila*
- [0436] *Flavonifractor*
- [0437] unclassified Ruminococcaceae
- [0438] unclassified Clostridiaceae
- [0439] *Butyricicoccus*
- [0440] *Faecalicatena*
- [0441] *Prevotella*
- [0442] *Clostridium*
- [0443] *Lachnoclostridium*
- [0444] *Alistipes*
- [0445] *Megasphaera*
- [0446] *Robinsoniella*
- [0447] *Fusicatenibacter*
- [0448] *Barnesiella*
- [0449] *Faecalitalea*
- [0450] *Lachnospira*
- [0451] *Romboutsia*

[0452] Among those listed above, bacteria determined to be more preferred in terms of the length of the engraftment period in recipients are those belonging to the following genera.

- [0453] *Faecalibacterium*
- [0454] unclassified Clostridiales
- [0455] unclassified Lachnospiraceae
- [0456] *Roseburia*
- [0457] *Ruminococcus*
- [0458] unclassified Firmicutes
- [0459] *Dorea*
- [0460] *Phascolarctobacterium*
- [0461] *Sutterella*
- [0462] *Bacteroides*
- [0463] *Blautia*
- [0464] *Parabacteroides*
- [0465] *Megamonas*
- [0466] *Collinsella*
- [0467] *Eubacterium*
- [0468] *Coprococcus*

[0469] The genera of bacteria extracted by the above two methods for narrow down search are as listed below. Bacteria belonging to “unclassified taxonomy names” are bacteria which have DNA sharing 94% or more homology with the nucleotide sequence of 16S rRNA gene in bacteria found under the species names shown in Table TA.

[0470] *Blautia*, *Clostridium*, unclassified Clostridiales, *Actinomyces*, *Parabacteroides*, *Lachnoclostridium*, *Bacteroides*, *Faecalibacterium*, unclassified Lachnospiraceae, *Roseburia*, *Ruminococcus*, unclassified Firmicutes, *Dorea*, *Phascolarctobacterium*, *Sutterella*, *Megamonas*, *Collinsella*, *Eubacterium*, *Coprococcus*, *Schaalia*, *Alistipes*, *Bifidobacterium*, *Lactobacillus*, *Veillonella*, *Anaerostipes*, *Bilophila*, *Butyricoccus*, *Barnesiella*, *Fusicatenibacter*, *Flavonifractor*, unclassified Ruminococcaceae, unclassified Clostridiaceae, *Faecalicatena*, *Prevotella*, *Megasphaera*, *Robinsoniella*, *Faecalitalea*, *Lachnospira*, *Romboutsia*

[0471] Thus, the composition of the present invention comprises bacteria belonging to any genus selected from the group consisting of the following genera: *Blautia*, *Clostridium*, unclassified Clostridiales, *Actinomyces*, *Parabacteroides*, *Lachnoclostridium*, *Bacteroides*, *Faecalibacterium*, unclassified Lachnospiraceae, *Roseburia*, *Ruminococcus*, unclassified Firmicutes, *Dorea*, *Phascolarctobacterium*, *Sutterella*, *Megamonas*, *Collinsella*, *Eubacterium*, *Coprococcus*, *Schaalia*, *Alistipes*, *Bifidobacterium*, *Lactobacillus*, *Veillonella*, *Anaerostipes*, *Bilophila*, *Butyricoccus*, *Barnesiella*, *Fusicatenibacter*, *Flavonifractor*, unclassified Ruminococcaceae, unclassified Clostridiaceae, *Faecalicatena*, *Prevotella*, *Megasphaera*, *Robinsoniella*, *Faecalitalea*, *Lachnospira*, and *Romboutsia*.

[0472] More preferably, the composition of the present invention comprises bacteria belonging to any genus selected from the group consisting of the following genera: *Blautia*, *Clostridium*, unclassified Clostridiales, *Actinomy-*

ces, *Parabacteroides*, *Lachnoclostridium*, *Bacteroides*, *Faecalibacterium*, unclassified Lachnospiraceae, *Roseburia*, *Ruminococcus*, unclassified Firmicutes, *Dorea*, *Phascolarctobacterium*, *Sutterella*, *Megamonas*, *Collinsella*, *Eubacterium*, and *Coprococcus*.

3-3. Definition by Species Name

[0473] In the present invention, Method 1 (narrow down search with median) and Method 2 (narrow down search with mean and carrying rate) mentioned above can also be applied to define preferred bacterial species. For definition of species names, a narrow down search was made for bacterial species belonging to the genus names defined in the section “3-2. Definition by genus name.”

[0474] In the present invention, bacteria determined to be preferred by Method 1 (narrow down search with median) are of the following species.

TABLE 1B

Narrow down search with median	
<i>Bacteroides uniformis</i>	
<i>Bacteroides</i> sp. AR29	
<i>Bacteroides ovatus</i>	
<i>Bacteroides dorei</i>	
<i>Bacteroides vulgatus</i>	
<i>Parabacteroides distasonis</i>	
<i>Bacteroides stercoris</i>	
<i>Schaalia odontolytica</i>	
<i>Parabacteroides merdae</i>	
<i>Bifidobacterium longum</i>	
<i>Bacteroides faecis</i>	
<i>Dorea longicatena</i>	
<i>Ruminococcus</i> sp. K-1	
<i>Blautia wexlerae</i>	
<i>Blautia massiliensis</i>	
<i>Bacteroides fragilis</i>	
[<i>Clostridium</i>] <i>bolteae</i>	
<i>Bifidobacterium adolescentis</i>	
[<i>Ruminococcus</i>] <i>torques</i>	
butyrate-producing bacterium M104/1	
<i>Bifidobacterium faecale</i>	
<i>Anaerostipes hadrus</i>	
<i>Clostridium</i> sp. 826	
<i>Collinsella aerofaciens</i>	
<i>Clostridium</i> sp. HGF2	
<i>Clostridium</i> sp. AT4	
<i>Bilophila</i> sp. 4_1_30	
<i>Faecalibacterium prausnitzii</i>	
[<i>Ruminococcus</i>] <i>gnavus</i>	
<i>Roseburia inulinivorans</i>	
<i>Bifidobacterium pseudocatenulatum</i>	
<i>Butyricoccus faecihominis</i>	
Firmicutes bacterium DJF_VR50	
Lachnospiraceae bacterium 1_1_57FAA	
<i>Blautia obeum</i>	
<i>Ruminococcus</i> sp. DJF_VR70k1	

[0475] Likewise, in the present invention, bacteria determined to be preferred by Method 2 (narrow down search with mean and carrying rate) are of the following species.

TABLE 1C

Narrow down search with mean and carrying rate	
<i>Bacteroides vulgatus</i>	<i>Eubacterium ventriosum</i>
<i>Bacteroides stercoris</i>	Clostridiales bacterium 80/3
<i>Bacteroides uniformis</i>	<i>Butyricoccus faecihominis</i>
<i>Blautia wexlerae</i>	<i>Bacteroides cellulosilyticus</i>
<i>Bacteroides</i> sp. AR29	<i>Coprococcus catus</i>

TABLE 1C-continued

Narrow down search with mean and carrying rate	
<i>Dorea longicatena</i>	<i>Parabacteroides johnsonii</i>
<i>Phascolarctobacterium faecium</i>	Lachnospiraceae bacterium DJF_RP14
<i>Blautia massiliensis</i>	[<i>Ruminococcus</i>] <i>gnavus</i>
<i>Ruminococcus</i> sp. K-1	<i>Bifidobacterium longum</i>
<i>Faecalibacterium prausnitzii</i>	<i>Bacteroides xylanisolvens</i>
<i>Megamonas funiformis</i>	<i>Prevotella stercorea</i>
<i>Bifidobacterium adolescentis</i>	<i>Bacteroides plebeius</i>
[<i>Ruminococcus</i>] <i>torques</i>	Firmicutes bacterium DJF_VR50
<i>Collinsella aerofaciens</i>	<i>Sutterella wadsworthensis</i>
<i>Parabacteroides merdae</i>	Lachnospiraceae bacterium 1_1_57FAA
butyrate-producing bacterium M104/1	<i>Blautia obeum</i>
<i>Bifidobacterium faecale</i>	<i>Barnesiella intestinihominis</i>
<i>Clostridium</i> sp. 826	[<i>Eubacterium</i>] <i>eligenis</i>
<i>Bacteroides coprocola</i>	<i>Coprococcus comes</i>
<i>Roseburia faecis</i>	<i>Ruminococcus</i> sp. DJF_VR70k1
Lachnospiraceae bacterium DJF_VP18k1	butyrate-producing bacterium A2-175
<i>Clostridium</i> sp. AT4	<i>Faecalitalea cylindroides</i>
<i>Lactobacillus rogosae</i>	<i>Blautia</i> sp. YHC-4
butyrate-producing bacterium A2-207	[<i>Clostridium</i>] <i>glycyrrhizinilyticum</i>
<i>Roseburia</i> sp. 1120	<i>Ruminococcus</i> sp. 653
<i>Bacteroides dorei</i>	<i>Ruminococcus lactaris</i>
<i>Bifidobacterium pseudocatenulatum</i>	butyrate-producing bacterium SM6/1
<i>Megasphaera massiliensis</i>	butyrate-producing bacterium SS3/4
<i>Bacteroides massiliensis</i>	Clostridiaceae bacterium DJF_LS40
<i>Anaerostipes hadrus</i>	<i>Bacteroides</i> sp. 1_1_30
<i>Alistipes putredinis</i>	<i>Bacteroides nordii</i>
<i>Parabacteroides</i> sp. D25	<i>Fusicatenibacter saccharivorans</i>
<i>Roseburia inulinivorans</i>	butyrate-producing bacterium SL7/1
<i>Bilophila</i> sp. 4_1_30	[<i>Clostridium</i>] <i>scindens</i>
<i>Flavonifractor plautii</i>	

[0476] Species extracted by the above two methods for narrow down search are summarized as follows.

TABLE 1D

Species	Species
<i>Blautia wexlerae</i>	<i>Roseburia faecis</i>
<i>Blautia massiliensis</i>	<i>Roseburia</i> sp. 1120
[<i>Ruminococcus</i>] <i>torques</i>	<i>Roseburia inulinivorans</i>
[<i>Ruminococcus</i>] <i>gnavus</i>	<i>Ruminococcus</i> sp. K-1
<i>Blautia obeum</i>	<i>Ruminococcus</i> sp. DJF_VR70k1
<i>Blautia</i> sp. YHC-4	<i>Ruminococcus</i> sp. 653
<i>Clostridium</i> sp. 826	<i>Ruminococcus lactaris</i>
<i>Clostridium</i> sp. AT4	Firmicutes bacterium DJF_VR50
<i>Clostridium</i> sp. HGF2	<i>Dorea longicatena</i>
butyrate-producing bacterium M104/1	<i>Phascolarctobacterium faecium</i>
butyrate-producing bacterium A2-207	<i>Sutterella wadsworthensis</i>
Clostridiales bacterium 80/3	<i>Megamonas funiformis</i>
butyrate-producing bacterium A2-175	<i>Collinsella aerofaciens</i>
butyrate-producing bacterium SM6/1	<i>Eubacterium ventriosum</i>
butyrate-producing bacterium SS3/4	[<i>Eubacterium</i>] <i>eligenis</i>
butyrate-producing bacterium SL7/1	<i>Coprococcus catus</i>
<i>Parabacteroides merdae</i>	<i>Coprococcus comes</i>
<i>Parabacteroides</i> sp. D25	<i>Schaalia odontolytica</i>
<i>Parabacteroides johnsonii</i>	<i>Alistipes putredinis</i>
<i>Parabacteroides distasonis</i>	<i>Bifidobacterium adolescentis</i>
[<i>Clostridium</i>] <i>glycyrrhizinilyticum</i>	<i>Bifidobacterium faecale</i>
[<i>Clostridium</i>] <i>scindens</i>	<i>Bifidobacterium pseudocatenulatum</i>
[<i>Clostridium</i>] <i>bolteae</i>	<i>Bifidobacterium longum</i>
<i>Bacteroides vulgatus</i>	<i>Lactobacillus rogosae</i>
<i>Bacteroides stercoris</i>	
<i>Bacteroides uniformis</i>	<i>Anaerostipes hadrus</i>
<i>Bacteroides</i> sp. AR29	<i>Bilophila</i> sp. 4_1_30
<i>Bacteroides ovatus</i>	<i>Butyrivibrio faecihominis</i>
<i>Bacteroides coprocola</i>	<i>Barnesiella intestinihominis</i>
<i>Bacteroides dorei</i>	<i>Fusicatenibacter saccharivorans</i>
<i>Bacteroides massiliensis</i>	<i>Flavonifractor plautii</i>
<i>Bacteroides cellulosilyticus</i>	
<i>Bacteroides xylanisolvens</i>	Clostridiaceae bacterium DJF_LS40

TABLE 1D-continued

Species	Species
<i>Bacteroides plebeius</i>	<i>Prevotella stercorea</i>
<i>Bacteroides</i> sp. 1_1_30	<i>Megasphaera massiliensis</i>
<i>Bacteroides nordii</i>	
<i>Bacteroides faecis</i>	<i>Faecalitalea cylindroides</i>
<i>Bacteroides fragilis</i>	
<i>Faecalibacterium prausnitzii</i>	
Lachnospiraceae bacterium DJF_VP18k1	
Lachnospiraceae bacterium DJF_RP14	
Lachnospiraceae bacterium 1_1_57FAA	

[0477] In the present invention, the results of the narrow down search for species indicate that 76 species shown in Table 1D (referred to as Group A) are abundant in CR or PR. Among them, the following 27 species (referred to as Group B) are more preferred. Among these species falling within Group B, the following 10 species (referred to as Group C) are most preferred.

Group B (27 Species):

- [0478] *Bacteroides vulgatus*
- [0479] *Bacteroides stercoris*
- [0480] *Bacteroides uniformis*
- [0481] *Blautia wexlerae*
- [0482] *Bacteroides* sp. AR29
- [0483] *Dorea longicatena*
- [0484] *Phascolarctobacterium faecium*
- [0485] *Blautia massiliensis*
- [0486] *Ruminococcus* sp. K-1
- [0487] *Bacteroides ovatus*
- [0488] *Faecalibacterium prausnitzii*
- [0489] *Megamonas funiformis*
- [0490] *Bifidobacterium adolescentis*
- [0491] [*Ruminococcus*] *torques*
- [0492] *Collinsella aerofaciens*
- [0493] *Parabacteroides merdae*
- [0494] butyrate-producing bacterium M104/1
- [0495] *Bifidobacterium faecale*
- [0496] *Clostridium* sp. 826
- [0497] *Bacteroides coprococla*
- [0498] *Roseburia faecis*
- [0499] Lachnospiraceae bacterium DJF_VP18k1
- [0500] *Clostridium* sp. AT4
- [0501] *Lactobacillus rogosae*
- [0502] butyrate-producing bacterium A2-207
- [0503] *Roseburia* sp. 1120
- [0504] *Bacteroides dorei*

Group C:

- [0505] *Bacteroides vulgatus*
- [0506] *Bacteroides stercoris*
- [0507] *Bacteroides uniformis*
- [0508] *Blautia wexlerae*
- [0509] *Bacteroides* sp. AR29
- [0510] *Dorea longicatena*
- [0511] *Phascolarctobacterium faecium*
- [0512] *Blautia massiliensis*
- [0513] *Ruminococcus* sp. K-1
- [0514] *Bacteroides ovatus*

[0515] In the composition of the present invention, bacteria belonging to the genera listed above may be contained

alone, or some of these bacteria may be contained in combination. When some of these bacteria are contained in combination, up to 40 types of bacteria are preferred, up to 30 type of bacteria are more preferred, and up to 20 types of bacteria are most preferred. It is possible to select any combination of up to 20 types of bacteria, for example, any combination of 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3 or 2 types of bacteria.

[0516] Moreover, the composition of the present invention may be in any powder, solid or liquid form in order that it is used as appropriate in allo-HSCT or can be used for a long period of time. Such a powder, solid or liquid form may also be formulated into a capsule formulation. The composition of the present invention when formulated into a capsule formulation is advantageous in that it is possible to avoid complications such as hemorrhage caused by tube insertion and/or a colon fiberscope, etc., and it is also possible to reduce the burdens on patients during implementation.

[0517] Further, the composition of the present invention may comprise either live bacteria or dead bacteria, or may comprise a mixture of live and dead bacteria.

[0518] Furthermore, the composition of the present invention may be configured to comprise at least one selected from pH stabilizers, acidifiers, antiseptics, vitamins, minerals, nutritional supplements, prebiotics and probiotics.

[0519] Moreover, the composition of the present invention may be administered either before or after or both before and after hematopoietic stem cell transplantation to a patient who is a subject of this transplantation, whereby GVHD can be prevented or treated.

[0520] GVHD may be exemplified by steroid-refractory or steroid-dependent GVHD, or acute gut GVHD.

[0521] In the context of the term “treatment” or “therapeutic” as used herein, the degree of suppression is not limited in any way as long as the development of GVHD can be suppressed. Thus, the “treatment” includes both complete response (CR) and partial response (PR). Complete response (CR) means the disappearance of all gut GVHD-related symptoms, while partial response (PR) means at least one down staging of gut GVHD. In the present invention, the treatment is assumed to be effective (i.e., GVHD has been treated) when reaching PR or CR in steroid-resistant cases or when succeeding in 40% or more steroid reduction in steroid-dependent cases as compared to before treatment.

[0522] Likewise, the term “prevention” or “prophylactic” is intended to include all of the following meanings: to suppress the development of GVHD before it occurs, to prevent the condition of already developed GVHD from becoming worse, and to prevent the recurrence of resolved GVHD.

[0523] In the present invention, there is provided a therapeutic method for GVHD, which comprises administering the above composition or capsule formulation to a GVHD patient.

[0524] The present invention further provides a prophylactic method for GVHD, which comprises administering the above composition or capsule formulation either before or after or both before and after hematopoietic stem cell transplantation to a patient who is a subject of the hematopoietic stem cell transplantation.

4. Bacterial Mixture and Formulation

[0525] The present invention provides a method for preparing a suspension for fecal microbiota transplantation, which comprises confirming the presence of the above bacteria in feces collected from a human subject, and suspending the feces confirmed for the presence of the bacteria into a solvent (e.g., physiological saline, buffer).

[0526] Moreover, the present invention provides a method for preparing a suspension for fecal microbiota transplantation, which comprises confirming the presence of the above bacteria in feces collected from a human subject, obtaining a microbiota from the feces confirmed for the presence of the bacteria, and suspending the microbiota into a solvent.

[0527] Further, the present invention provides a method for preparing a bacterial mixture for fecal microbiota transplantation, which comprises confirming the presence of the above bacteria in feces collected from a human subject, separating the bacteria from the feces confirmed for the presence of the bacteria, and mixing the separated bacteria. In the separation of bacteria, bacteria contained in a sample may be of a single type or may be of two or more types.

[0528] In yet another embodiment of the present invention, there is provided a method for preparing a formulation for fecal microbiota transplantation, which comprises confirming the presence of the above bacteria in feces collected from a human subject, and formulating a microbiota from the feces confirmed for the presence of the bacteria, or the bacteria separated from the feces confirmed for the presence of the bacteria.

[0529] Formulation techniques are exemplified by encapsulation into capsules.

[0530] Feces may be collected according to the procedures described in the section “1. FMT” above.

[0531] Techniques used to confirm the presence of bacteria are not limited in any way, and examples include 16S rRNA gene analysis, qPCR or Microarray using the DNA sequences of regions specific to bacteria of interest, or MALDI-TOF MS, etc.

[0532] Once the presence of bacteria has been confirmed by 16S rRNA gene analysis or other techniques, it can be assumed that the bacteria are contained in the donor feces. A diluted suspension of such feces is subjected to various culture conditions, and single colonies are collected to isolate the bacteria.

[0533] The formulation of the present invention may be in any dosage form as long as it comprises the above bacteria of the present invention, but preferred is a formulation for oral administration.

[0534] In the case of preparing an oral solid formulation, a base drug may be supplemented with an excipient and optionally with a binder, a disintegrant, a lubricant, a coloring agent, a corrective and so on, and then formulated in

a standard manner into tablets, coated tablets, granules, fine granules, powders, capsules, etc.

[0535] Capsules used for this purpose include, for example, acid-resistant capsules. Such acid-resistant capsules may be exemplified by DR Caps® (Capsugel).

[0536] Examples of an excipient available for use include lactose, corn starch, sucrose, glucose, sorbit, crystalline cellulose, silicon dioxide and so on. Examples of a binder available for use include polyvinyl alcohol, ethyl cellulose, methyl cellulose, gum arabic, hydroxypropyl cellulose, hydroxypropyl methyl cellulose and so on. Examples of a lubricant available for use include magnesium stearate, talc, silica and so on. Examples of a coloring agent available for use include those permitted to be added to medicaments. Examples of a corrective available for use include cocoa powder, menthol, aromatic acids, peppermint oil, borneol, cinnamon powder and so on.

[0537] In the present invention, a capsule formulation is preferred among the above formulations.

[0538] Bacteria contained in the formulation may be set to, for example, 1 to 2000 mg, preferably 10 to 500 mg, and more preferably 100 to 250 mg by dry weight, and may be administered in an amount of 150 to 200 mg per day, preferably 175 mg per day, depending on the patient's body weight and symptoms, etc.

[0539] The bacteria intended in the present invention may be used either as a medicament or as a quasi drug.

EXAMPLES

[0540] The present invention will be further described in more detail by way of the following illustrative examples. However, the scope of the present invention is not limited by these examples.

Example 1

FMT for Screening of Preferred Microbiota

[Subjects and Methods]

Subjects of FMT

[0541] FMT was performed on 15 cases of steroid-refractory or steroid-dependent acute gut GVHD.

[0542] Not only the gut, but also the skin and the liver can be target organs of acute GVHD, but this study was targeted at the gut in terms of the fact that acute gut GVHD is more severe than acute GVHD in the other organs. A steroid-refractory case refers to a case where in spite of treatment with adequate steroid dose (1 mg/kg or more of prednisolone), the clinical condition of a patient remains unchanged from day 5 after initiation of the treatment, while a steroid-dependent case refers to a case where a patient initially responds to steroid treatment, but becomes worse with reduction in steroid dose, so that the steroid dose is required to be increased again (i.e., a case where steroid dose is difficult to reduce).

[0543] In addition to the above cases, this study was also targeted at cases where acute gut GVHD progressed on day 3 after initiation of steroid treatment.

[0544] Even in the case of patients complicated with other gut lesions, cases where acute gut GVHD was considered to be the leading cause of diarrhea were available for entry.

[0545] Exclusion criteria were set as shown below.

- [0546] 1) GVHD responsive to steroid treatment
- [0547] 2) GVHD progressive after primary steroid treatment (regardless of affected organ)
- [0548] 3) Cases with uncontrollable infections
- [0549] 4) Cases where diarrhea is deemed to result from any cause other than GVHD

Selection of FMT Donors

[0550] Donor candidates were selected from the spouses or relatives within the fourth degree of relationship of patients. These candidates were between 20 and 64 years of age and were selected from those who did not have any infection risks as shown below.

- [0551] 1) Not have gotten a new tattoo or pierced within the last three months
- [0552] 2) Not have had sexual contact with a new partner within the last three months
- [0553] 3) Not have received a blood transfusion within the last three months
- [0554] 4) Not have had a travel history to tropical regions within the last three months
- [0555] 5) Not have used any antibiotics within the last one month
- [0556] 6) Not have a previous history of malignant disease or inflammatory bowel disease
- [0557] 7) Not have any digestive symptoms such as diarrhea on the day of FMT (as confirmed on the day of FMT)
- [0558] 8) Not have hypertension, diabetes, hyperlipidemia, cardiac diseases (e.g., myocardial infarction, angina pectoris, heart failure), apoplexy, or allergic diseases (e.g., asthma, atopic dermatitis, food allergy) (except for pollinosis)

[0559] If donor candidates have no problem with the above items, they were subjected to blood collection and fecal examination. Blood examination was made to check HIV, human T-lymphotropic virus type I (HTLV-1), hepatitis A, B and C, syphilis, cytomegalovirus (CMV) and Epstein-Barr virus (EBV). Likewise, fecal examination was made to check the presence or absence of parasites, *Clostridium difficile*, *Cryptosporidium*, *Giardia*, Microsporidia, *Entamoeba histolytica*, *Cyclospora*, *Isospora*, *Dientamoeba fragilis*, *Blastocystis hominis*, *Schistosoma* and other pathogenic bacteria, etc. With regard to CMV and EBV, patients showing a past infection pattern of CMV or EBV were determined to have no problem.

FMT Method

[0560] On the day of FMT, feces were collected from the donors and stored at 4° C. under anaerobic conditions until use.

[0561] At the timing of inserting a gastroduodenal tube into each patient, feces were started to be prepared. For preparation of feces, feces were first weighed, and sterile physiological saline was added in a volume of 200 to 300 mL depending on the weight of feces, followed by thoroughly stirring to give a uniform mixture. This mixture was passed once through a metal sieve to remove large undigested matters, and then passed twice through sterile gauze to prepared a suspension. FMT was performed as soon as patients were ready.

[0562] The suspension was filled into 50 mL syringes, and the fecal suspension was administered through the gas-

trooduodenal tube. The administration rate was set not to exceed 30 seconds per 50 mL. After the entire suspension was introduced, 50 mL of physiological saline was used to wash the inside of the tube, and the tube was then removed to complete the administration.

[0563] When no FMT-related adverse event of grade 3 (as defined in Non-patent Document 3) or higher was observed, additional administration was allowed once within 4 to 14 days after the first FMT while checking its effect. In this case, feces were collected from the same donor as used for the first administration. Fecal transplantation was performed within 12 hours (within 8 hours, if possible) after fecal collection from the donor.

[0564] Patients who were fasting due to their gut GVHD were allowed to ingest supplements containing dietary fiber, oligosaccharides and so on (GFO®: Otsuka Pharmaceutical Co., Ltd., Japan) as prebiotics from the day before initiation of FMT. Moreover, the administration of antibiotics was stopped for 24 hours before and after FMT, if possible.

[0565] Cases newly developed or showing progression of grade 1 (as defined in Non-patent Document 3) or higher within one week after FMT were all regarded as adverse events, and evaluated by the National Cancer Institute Common Terminology Criteria for Adverse Events (CTC-AE) version 4.0.

[0566] The therapeutic effect was evaluated at 4 weeks after the final FMT. Criteria for therapeutic effect assessment are as shown below.

- [0567] 1) Complete response (CR): disappearance of all gut GVHD-related symptoms
- [0568] 2) Partial response (PR): at least one down staging of gut GVHD
- [0569] 3) Progression (PG): progression of at least one stage (as defined in Non-patent Document 3)
- [0570] 4) No change (NC): variation within the same stage (as defined in Non-patent Document 3)
- [0571] 5) Not determined (ND): early death (within 7 days) after treatment or any case difficult to evaluate due to symptoms other than GVHD

[0572] FMT was assessed to be effective when reaching PR or CR in steroid-resistant cases or when succeeding in 40% or more steroid reduction in steroid-dependent cases as compared to before treatment.

Analysis of Gut Microbiota (Meta 16S Analysis)

[0573] Aliquots of the prepared fecal preparation and the patient's feces (collected before FMT, and at 1 to 3 days and 1, 2 and 4 weeks after the final FMT) were used for analysis of gut microbiota.

[0574] Bacterial DNA was extracted from each sample in a standard manner. This DNA was amplified by PCR (polymerase chain reaction) with primers (27Fmod: 5'-agrgttt-gatymtggtcag-3' (SEQ ID NO: 345), 338R: 5'-tgctgctcccgtaggagt-3' (SEQ ID NO: 346)) designed to cover the variable region v1-v2 in the 16S rRNA gene, followed by 250 bp paired-end sequencing on MiSeq® (illumina). After Read 1 and Read 2 of the resulting sequence were merged, quality checking was conducted and 3,000 sequence data reads which passed this quality checking were used for analysis. The resulting reads were sorted in descending order by the abundance of the same sequence, and reads at the same abundance were sorted in descending order by the average of quality. The sorted sequences were subjected to operational taxonomic units (OTUs) clustering

using the UCLUST algorithm with a homology threshold of 97%. Moreover, glsearch was used to perform homology search against 16S rRNA gene databases (RefSeq, RDP, GRD, CORE) to thereby identify bacterial species. The homology threshold in genus identification was set to 94%. The homology threshold in species identification was set to 97%.

[0575] Moreover, the Shannon index was used for a diversity evaluation. It should be noted that taxonomy names in this analysis were in accordance with taxdmp.zip on the FTP site of the NCBI Taxonomy Database as of May 7, 2019.

Intergroup Comparison of Gut Microbiota

[0576] The CR or PR group and the Others group were subjected to the two methods described in the sections “Method 1. Narrow down search with median” and “Method 2. Narrow down search for bacteria with mean and carrying rate” to extract enterobacteria abundant in the CR or PR group.

RESULTS AND DISCUSSION

[0577] In the therapeutic effect assessment using the results obtained at 4 weeks after the final administration, 6 of the 15 cases were assessed as CR and 3 of the 15 cases were assessed as PR, thus resulting in a response rate of 60%. Thus, this example demonstrated the effectiveness of FMT as a therapeutic or prophylactic strategy for GVHD.

[0578] The time course of changes in the diversity of fecal microbiota before and after FMT is as shown in FIG. 1. The

CR or PR group and the Others group both showed an improvement in a diversity after FMT. The a diversity was gradually reduced over 4 weeks after FMT, but this reduction was suppressed in the CR or PR group as compared to the Others group.

[0579] Graphs representing the relative proportions of fecal microbiota at the genus level before and after FMT are shown in FIG. 2. Before FMT, the diversity of the microbiota was reduced (i.e., dysbiosis), and bacteria such as *Staphylococcus* and *Enterococcus* were dominant. After FMT, the diversity of the microbiota was recovered, so that the microbiota was altered to include various bacteria such as *Bacteroides* and *Bifidobacterium*. In the Others group, cases where *Escherichia* was dominant after 4 weeks were also observed. Moreover, for the respective genera, the time courses of their changes are shown in FIGS. 3 to 9. The genera shown in FIGS. 3 to 9 were all confirmed to be abundant in the CR or PR group in comparison with the Others group. Moreover, *Corynebacterium* shown in FIG. 10 was confirmed to be abundant in the Others group at all the time points.

[0580] The results of two group comparison obtained by the two methods described in the sections “Method 1. Narrow down search with median” and “Method 2. Narrow down search for bacteria with mean and carrying rate” are shown below. The results at the OTU level are shown in Table 2 and Table 3, while the results at the genus level are shown in Table 4 and Table 5. These tables represent bacteria showing at least one time when the Log 2 Fold change was 0.5 or more.

TABLE 2

Results calculated with median for Log2 Fold change at the OTU level											
OTU	The number of times based on the value of Log2FC		Log2 FC (Donor/pre-FMT)	Log2 Fold Change		Doner prevalence					
	over 0.5	under -0.5		Donor	pre-FMT	1-3 day	1 w	2 w	4 w	CR or PR	Others
OTU00031	4	0	6.71	4.12	0.00	5.47	4.87	5.57	5.36	100	50
OTU00040	3	0	4.39	2.98	0.00	3.46	2.12	6.15	0.00	77.8	50
OTU00002	3	1	6.56	-3.13	0.00	-7.19	7.87	10.77	4.14	66.7	83.3
OTU00003	2	0	-6.08	0.00	0.23	4.79	0.00	2.12	0.00	0	16.7
OTU00005	2	0	9.33	9.33	0.00	6.96	3.84	0.00	0.00	66.7	33.3
OTU00020	2	0	4.95	-0.40	0.00	0.21	4.14	4.70	0.00	77.8	83.3
OTU00024	2	0	6.79	6.79	0.00	5.47	1.42	0.00	0.00	66.7	33.3
OTU00026	2	0	7.45	0.67	0.00	6.34	4.50	0.00	0.00	100	83.3
OTU00053	2	0	4.14	-2.67	0.00	1.52	1.42	0.00	0.00	55.6	83.3
OTU00043	2	0	-2.94	0.00	2.94	0.00	0.00	4.00	2.94	0	16.7
OTU00070	2	0	0.00	0.00	0.00	0.00	2.58	2.58	0.00	44.4	0
OTU00058	2	0	2.12	0.00	0.00	0.00	0.00	2.94	2.12	77.8	66.7
OTU00008	2	0	6.34	0.22	0.00	-0.66	5.36	2.12	0.00	88.9	83.3
OTU00010	2	0	2.94	0.82	0.00	-3.22	0.00	3.45	2.12	55.6	50
OTU00021	1	0	4.39	4.39	0.00	0.00	0.00	5.30	0.00	66.7	33.3
OTU00009	1	0	8.31	2.59	0.00	5.92	0.00	0.00	0.00	88.9	66.7
OTU00033	1	0	6.84	-1.62	0.00	5.69	0.00	0.00	0.00	88.9	100
OTU00103	1	0	8.22	3.19	0.00	7.00	0.00	0.00	0.00	77.8	50
OTU00032	1	0	6.75	0.67	0.00	2.12	0.00	0.00	0.00	88.9	100
OTU00034	1	0	5.76	3.65	0.00	2.94	0.00	0.00	0.00	88.9	50
OTU00072	1	0	7.55	-0.50	0.00	2.94	0.00	0.00	0.00	77.8	66.7
OTU00027	1	0	6.71	6.71	0.00	4.79	0.00	0.00	0.00	66.7	16.7
OTU01091	1	0	5.36	0.00	0.00	3.40	0.00	0.00	0.00	88.9	100
OTU00057	1	0	8.30	3.34	0.00	2.12	0.00	0.00	0.00	66.7	50
OTU00048	1	0	6.56	6.56	0.00	2.94	0.00	0.00	0.00	77.8	33.3
OTU00085	1	0	2.12	-0.47	0.00	-0.47	0.00	6.22	0.00	77.8	66.7
OTU00035	1	0	6.56	6.56	0.00	2.94	0.00	0.00	0.00	55.6	33.3
OTU00045	1	0	6.66	6.66	0.00	2.12	0.00	0.00	0.00	55.6	33.3
OTU00068	1	0	2.12	-0.47	0.00	2.12	0.00	0.00	0.00	55.6	66.7
OTU00147	1	0	6.22	0.21	0.00	3.46	0.00	0.00	0.00	88.9	83.3
OTU00080	1	0	3.84	-2.77	0.00	3.46	0.00	0.00	0.00	88.9	100

TABLE 2-continued

Results calculated with median for Log2 Fold change at the OTU level											
OTU	The number of times based on the value of Log2FC		Log2 FC (Donor/pre-FMT)	Log2 Fold Change		Doner prevalence					
	over 0.5	under -0.5		Doner	pre-FMT	1~3 day	1 w	2 w	4 w	CR or PR	Others
OTU00054	1	0	5.93	-0.50	0.00	2.94	0.00	0.00	0.00	66.7	83.3
OTU00062	1	0	6.28	2.01	0.00	5.67	0.00	0.00	0.00	88.9	66.7
OTU00126	1	0	4.79	0.19	0.00	2.12	0.00	0.00	0.00	66.7	50
OTU00106	1	0	6.01	2.79	0.00	5.35	0.00	0.00	0.00	88.9	50
OTU00125	1	0	4.95	4.95	0.00	2.12	0.00	0.00	0.00	66.7	16.7
OTU00067	1	0	3.84	2.43	0.00	2.12	0.00	0.00	0.00	88.9	50
OTU00164	1	0	4.14	-1.71	0.00	3.46	0.00	0.00	0.00	55.6	83.3
OTU00111	1	0	4.39	0.93	0.00	2.94	0.00	0.00	0.00	66.7	50
OTU00079	1	0	0.00	0.00	0.00	2.12	0.00	0.00	0.00	44.4	16.7
OTU00109	1	0	3.84	0.90	0.00	2.12	0.00	0.00	0.00	66.7	50
OTU00179	1	0	4.95	3.54	0.00	2.12	0.00	0.00	0.00	77.8	50
OTU00113	1	0	3.46	1.34	0.00	2.12	0.00	0.00	0.00	77.8	50
OTU00112	1	0	0.00	-1.42	0.00	0.00	0.00	1.42	0.00	11.1	50
OTU00233	1	0	3.84	-0.30	0.00	3.84	0.00	0.00	0.00	88.9	66.7
OTU00197	1	0	2.12	0.00	0.00	2.12	0.00	0.00	0.00	77.8	66.7
OTU00194	1	0	2.12	-0.47	0.00	0.70	0.00	0.00	0.00	66.7	83.3
OTU00762	1	0	2.12	0.70	0.00	2.12	0.00	0.00	0.00	66.7	50
OTU00028	1	1	2.94	-1.06	0.00	-1.84	1.42	0.00	0.00	66.7	83.3
OTU00011	1	2	2.12	-1.11	-1.42	-4.27	5.30	0.00	-8.66	66.7	66.7

TABLE 3

Results calculate with mean for Log2 Fold change at the OTU level									
OTU	The number of times based on the value of Log2 FC		Log2 FC (Doner/ preval-)	Log2 FC after FMT	CR or PR prevalence after FMT				
	Log2 FC >= 0.5 & CR or PR preval-	Log2 FC <= -0.5 & Others preval-			1~3 day	1 w	2 w	4 w	1~3 day
OTU00005	4	0	1.31	1.74	8.26	8.58	10.16	55.6	50
OTU00020	4	0	-1.55	2.61	4.14	0.83	8.53	66.7	87.5
OTU00024	4	0	7.46	4.87	6.49	7.48	8.53	66.7	50
OTU00031	4	0	3.56	5.10	7.42	1.17	6.72	77.8	87.5
OTU00033	4	0	7.15	3.39	6.04	5.76	6.81	66.7	25
OTU00040	4	0	3.68	1.77	4.75	7.95	7.46	66.7	50
OTU00051	4	0	6.07	6.78	2.86	5.73	5.78	44.4	25
OTU00147	4	0	6.12	3.95	4.50	5.24	5.12	55.6	25
OTU00080	4	0	4.37	2.35	3.75	3.46	2.85	66.7	25
OTU00060	4	0	-3.39	1.52	5.94	3.29	2.75	33.3	37.5
OTU00059	4	0	2.23	1.69	6.27	5.51	2.75	22.2	25
OTU00015	3	0	7.61	4.57	9.18	8.53	3.93	22.2	25
OTU00032	3	0	2.60	2.73	3.88	3.80	7.14	55.6	12.5
OTU00034	3	0	7.00	6.35	4.04	5.87	7.01	55.6	25
OTU00027	3	0	7.80	7.04	5.71	5.42	0.00	66.7	37.5
OTU00057	3	0	8.01	1.97	3.16	5.56	0.00	55.6	25
OTU00048	3	0	7.92	5.29	4.68	1.81	3.52	66.7	37.5
OTU00053	3	0	5.64	2.06	3.46	4.48	3.93	66.7	50
OTU00401	3	0	-3.72	4.02	7.15	4.72	1.54	33.3	37.5
OTU00035	3	0	7.41	6.42	5.36	5.01	0.00	55.6	37.5
OTU00045	3	0	7.02	4.90	3.01	5.28	0.00	55.6	25
OTU00030	3	0	6.91	5.01	7.27	2.77	5.37	22.2	25

TABLE 3-continued

Results calculate with mean for Log2 Fold change at the OTU level									
OTU00063	3	0	6.81	2.84	5.74	6.85	2.40	22.2	37.5
OTU00068	3	0	1.25	1.29	6.35	4.27	7.72	55.6	12.5
OTU00070	3	0	-3.45	0.56	2.42	2.35	-0.27	44.4	62.5
OTU00092	3	0	4.60	3.64	1.97	1.97	0.97	44.4	25
OTU00125	3	0	5.49	3.41	5.43	4.30	3.69	55.6	37.5
OTU00067	3	0	4.65	0.95	1.97	4.87	5.74	55.6	25
OTU00064	3	0	4.71	2.97	3.29	2.58	0.00	55.6	25
OTU00111	3	0	5.30	4.63	3.16	4.04	0.00	77.8	25
OTU00073	3	0	0.00	1.31	1.33	6.63	2.53	22.2	25
OTU00109	3	0	5.37	1.66	4.48	3.29	3.26	55.6	37.5
OTU00084	3	0	-0.32	1.55	2.77	2.86	1.54	33.3	25
OTU01483	3	0	1.36	4.29	4.36	2.94	3.26	22.2	25
OTU00148	3	0	4.07	1.71	3.46	3.56	2.76	33.3	25
OTU00180	3	0	3.25	1.33	1.97	1.42	1.54	33.3	25
OTU00231	3	0	3.72	3.46	2.58	2.48	1.76	44.4	25
OTU00002	3	1	7.32	-2.99	1.28	2.20	7.74	44.4	62.5
OTU00004	2	0	-6.54	5.64	9.62	8.68	8.49	22.2	12.5
OTU00009	2	0	9.12	1.74	7.17	6.01	0.00	77.8	37.5
OTU00050	2	0	-0.80	2.63	9.12	7.44	-0.28	22.2	12.5
OTU00103	2	0	7.59	5.42	5.03	5.20	0.00	55.6	25
OTU00049	2	0	5.68	7.48	6.77	5.80	0.00	22.2	25
OTU00041	2	0	6.68	4.77	6.17	6.20	5.80	33.3	12.5
OTU00114	2	0	7.28	2.55	0.50	3.22	1.54	22.2	12.5
OTU00054	2	0	5.88	-0.62	5.64	5.43	4.06	55.6	25
OTU01754	2	0	5.12	2.91	3.92	3.88	0.00	22.2	25
OTU00091	2	0	4.85	-0.94	4.63	4.75	2.85	44.4	25
OTU00078	2	0	5.10	3.00	5.39	3.35	0.00	33.3	25
OTU01289	2	0	2.23	4.08	3.51	3.96	5.16	11.1	25
OTU00156	2	0	3.95	0.36	3.56	4.72	0.00	44.4	25
OTU00179	2	0	5.57	3.25	1.81	2.86	0.00	55.6	12.5
OTU00427	2	0	3.01	0.80	3.03	5.14	4.14	11.1	25
OTU00113	2	0	3.54	-0.50	-3.61	5.01	3.11	66.7	25
OTU00083	2	0	4.11	4.20	0.00	3.29	4.95	44.4	0
OTU00145	2	0	3.68	-0.16	0.00	4.00	4.46	44.4	0
OTU00140	2	0	3.72	3.25	1.42	3.40	4.02	44.4	12.5
OTU00138	2	0	5.21	2.12	1.17	2.37	0.00	33.3	12.5
OTU00175	2	0	4.69	3.01	1.17	0.00	0.00	22.2	25
OTU00197	2	0	3.07	1.93	2.12	1.81	1.76	55.6	25
OTU00512	2	0	3.72	1.60	0.50	3.80	0.00	44.4	12.5
OTU01408	2	0	2.54	-0.84	1.81	0.87	2.94	22.2	25
OTU00194	2	0	3.31	0.96	1.17	1.17	0.56	55.6	12.5
OTU00201	2	0	1.31	1.31	1.17	1.81	4.71	11.1	25
OTU00144	2	0	1.31	1.85	1.81	3.56	3.64	33.3	12.5
OTU00442	2	0	0.43	1.69	2.12	3.56	2.27	22.2	12.5
OTU00008	2	1	0.30	-1.15	0.29	8.12	6.95	77.8	50
OTU00014	2	1	-7.58	1.85	3.78	1.42	-3.41	33.3	25
OTU00026	2	1	7.55	1.01	2.08	-2.96	0.00	88.9	62.5
OTU00047	2	1	5.93	-2.12	0.16	4.21	5.25	44.4	12.5
OTU00065	2	1	0.00	2.34	1.62	3.61	-2.02	22.2	12.5
OTU00001	2	2	-8.10	2.29	-1.74	-0.56	1.91	44.4	37.5
OTU00017	2	2	-1.61	1.76	2.01	-3.90	-3.09	33.3	37.5
OTU00085	2	2	-1.23	-1.60	-1.85	1.94	5.67	66.7	37.5
OTU00052	2	2	-5.92	3.00	2.01	-1.34	-3.59	44.4	37.5
OTU00112	2	2	-4.66	1.69	1.17	-1.66	-1.06	33.3	37.5
OTU00042	1	0	8.22	0.01	7.92	8.10	0.00	22.2	25
OTU00072	1	0	7.40	4.76	4.36	3.60	0.00	66.7	12.5
OTU00066	1	0	5.46	-2.42	1.81	4.58	2.40	22.2	12.5
OTU00055	1	0	0.01	-0.16	-5.80	6.04	6.52	33.3	12.5
OTU00044	1	0	6.71	7.61	3.09	0.00	0.00	22.2	12.5
OTU00038	1	0	6.93	4.83	2.37	2.86	0.00	44.4	12.5
OTU00037	1	0	5.21	1.54	5.15	5.73	5.88	22.2	12.5
OTU00158	1	0	4.95	-0.46	-6.98	3.40	4.42	55.6	0
OTU00062	1	0	6.35	5.39	0.87	3.92	3.69	66.7	12.5
OTU00126	1	0	5.82	5.98	4.39	3.46	0.00	66.7	12.5
OTU00106	1	0	6.25	5.53	0.50	1.62	0.00	77.8	12.5
OTU00223	1	0	5.97	5.83	0.50	0.00	0.56	33.3	12.5
OTU00130	1	0	4.54	3.68	2.68	3.71	0.00	22.2	12.5
OTU00108	1	0	4.63	3.56	1.42	0.50	0.00	22.2	12.5
OTU00135	1	0	4.63	-2.68	5.93	0.00	0.00	11.1	25
OTU00088	1	0	4.14	1.72	0.00	0.00	0.00	22.2	0
OTU00079	1	0	4.47	4.87	1.17	1.81	2.94	55.6	12.5
OTU00773	1	0	-2.32	2.00	0.00	1.62	0.00	33.3	0
OTU00227	1	0	-3.63	0.69	4.60	4.75	3.40	11.1	12.5
OTU00081	1	0	-4.42	-0.64	6.11	0.00	0.56	0	25
OTU00134	1	0	4.88	0.32	4.14	2.77	0.00	11.1	25

TABLE 3-continued

Results calculate with mean for Log2 Fold change at the OTU level									
OTU00120	1	0	5.28	3.41	0.50	2.68	0.00	33.3	12.5
OTU00115	1	0	-1.85	0.56	1.17	3.71	1.52	44.4	12.5
OTU00143	1	0	2.79	3.68	0.50	5.30	3.98	33.3	12.5
OTU00199	1	0	4.41	1.31	0.00	0.00	0.00	22.2	0
OTU00141	1	0	3.64	1.85	1.81	4.79	0.00	33.3	12.5
OTU00233	1	0	4.63	2.69	0.87	2.25	0.00	77.8	12.5
OTU00217	1	0	3.41	2.77	2.86	3.66	0.00	33.3	12.5
OTU00131	1	0	4.99	4.17	0.50	0.00	0.00	22.2	12.5
OTU00116	1	0	4.07	1.00	0.00	0.00	0.56	33.3	0
OTU00090	1	0	2.71	3.41	0.00	4.68	3.11	33.3	0
OTU00100	1	0	4.85	3.98	0.00	0.00	0.00	22.2	0
OTU00168	1	0	5.22	2.44	0.00	0.00	0.00	22.2	0
OTU00166	1	0	3.84	3.19	0.00	2.37	4.36	33.3	0
OTU00389	1	0	4.60	3.01	0.00	2.68	0.00	22.2	0
OTU00258	1	0	3.41	4.08	0.50	1.62	2.85	22.2	12.5
OTU00946	1	0	3.84	4.29	1.42	2.12	0.00	33.3	12.5
OTU00117	1	0	0.00	0.80	2.68	4.33	3.64	22.2	12.5
OTU00226	1	0	4.17	2.72	0.50	0.87	1.54	44.4	12.5
OTU00762	1	0	3.41	3.01	1.17	1.42	0.00	55.6	12.5
OTU00576	1	0	2.94	1.46	1.62	1.62	0.00	22.2	12.5
OTU00153	1	0	-2.94	0.97	1.97	-0.35	0.06	22.2	12.5
OTU00191	1	0	2.74	2.43	0.00	1.42	0.00	22.2	0
OTU00171	1	0	3.60	1.21	0.00	1.42	3.33	44.4	0
OTU00215	1	0	0.00	0.45	2.12	4.53	2.27	11.1	25
OTU00295	1	0	3.36	3.76	0.50	0.50	0.00	33.3	12.5
OTU00235	1	0	4.11	3.07	0.50	0.00	0.00	22.2	12.5
OTU00146	1	0	2.63	4.29	0.00	0.00	0.00	22.2	0
OTU00169	1	0	0.80	0.16	4.30	1.81	1.95	11.1	25
OTU00159	1	0	1.08	2.34	0.00	4.36	0.00	11.1	0
OTU00160	1	0	-1.06	3.01	3.66	1.42	0.00	22.2	12.5
OTU00246	1	0	2.12	1.03	0.00	0.00	0.00	22.2	0
OTU00361	1	0	3.60	1.85	1.42	1.17	0.00	33.3	12.5
OTU00357	1	0	1.51	1.08	0.50	0.50	0.00	33.3	12.5
OTU01249	1	0	1.31	0.00	2.48	3.51	1.95	0	12.5
OTU00400	1	0	0.45	-1.53	0.50	0.87	0.00	11.1	12.5
OTU00167	1	0	3.80	0.80	0.00	0.00	0.00	22.2	0
OTU00013	1	1	-2.63	-3.63	0.87	2.77	2.27	11.1	12.5
OTU01091	1	1	5.87	-2.02	-7.18	3.26	3.69	55.6	0
OTU00058	1	1	-3.75	-0.28	0.12	0.60	-1.89	22.2	37.5
OTU00086	1	1	-2.53	-3.87	-5.40	5.19	0.00	11.1	0
OTU00136	1	1	0.20	-3.60	2.47	-2.81	1.28	0	25
OTU00082	1	1	1.31	-2.80	4.33	4.24	4.91	22.2	12.5
OTU00255	1	1	0.80	-2.29	0.00	-2.15	1.28	0	0
OTU00021	1	2	5.41	-7.01	-2.93	1.18	-4.67	44.4	37.5
OTU00006	1	2	0.20	0.80	-3.64	-1.21	1.95	11.1	12.5
OTU00010	1	2	3.68	7.48	-2.55	-1.65	6.23	22.2	25
OTU00043	1	2	-7.05	3.68	-0.23	-1.41	-1.55	33.3	37.5
OTU00003	1	3	-10.76	1.10	-0.56	-4.44	-5.42	66.7	37.5
OTU00019	1	3	1.31	-3.38	0.52	-5.33	-7.92	11.1	37.5

OTU	2 w	4 w	Others prevalence		Doner prevalence		CR or PR	Others
			after FMT day	1 w	2 w	4 w		
OTU00005	25	42.9	33.3	0	0	0	68.7	33.3
OTU00020	62.5	42.9	66.7	16.7	40	0	77.8	83.3
OTU00024	37.5	28.6	33.3	0	0	0	66.7	33.3
OTU00031	100	57.1	33.3	0	20	0	100	50
OTU00033	25	42.9	50	0	0	0	88.9	100
OTU00040	62.5	42.9	33.3	0	0	0	77.8	50
OTU00051	37.5	42.9	0	0	0	0	55.6	33.3
OTU00147	25	28.6	16.7	0	0	0	88.9	83.3
OTU00080	25	28.6	33.3	0	0	0	88.9	100
OTU00060	25	28.6	33.3	0	0	0	22.2	50
OTU00059	25	28.6	0	0	0	0	22.2	0
OTU00015	25	14.3	16.7	0	0	0	22.2	50
OTU00032	25	28.6	16.7	0	0	20	88.9	100
OTU00034	25	14.3	16.7	0	0	0	88.9	50
OTU00027	37.5	0	16.7	0	0	0	66.7	16.7
OTU00057	25	0	33.3	0	0	0	66.7	50
OTU00048	12.5	28.6	0	0	0	0	77.8	33.3
OTU00053	37.5	14.3	50	0	0	0	55.6	83.3
OTU00401	37.5	14.3	16.7	0	0	0	44.4	50

TABLE 3-continued

Results calculate with mean for Log2 Fold change at the OTU level								
OTU00035	37.5	0	16.7	0	0	0	55.6	33.3
OTU00045	25	0	16.7	0	0	0	55.6	33.3
OTU00030	25	14.3	0	0	0	0	22.2	0
OTU00063	37.5	14.3	16.7	0	0	0	66.7	66.7
OTU00068	37.5	28.6	33.3	0	0	0	55.6	66.7
OTU00070	62.5	28.6	33.3	33.3	20	40	44.4	0
OTU00092	37.5	14.3	0	0	0	0	44.4	50
OTU00125	37.5	14.3	0	0	0	0	66.7	16.7
OTU00067	25	14.3	33.3	0	0	0	88.9	50
OTU00064	25	0	33.3	0	0	0	55.6	83.3
OTU00111	25	0	0	0	0	0	66.7	50
OTU00073	25	14.3	0	16.7	0	0	22.2	33.3
OTU00109	37.5	14.3	16.7	0	0	0	66.7	50
OTU00084	25	14.3	16.7	0	0	0	22.2	16.7
OTU01483	25	14.3	0	0	0	0	33.3	33.3
OTU00148	25	14.3	16.7	0	0	0	66.7	50
OTU00180	25	14.3	50	0	0	0	66.7	100
OTU00231	25	14.3	0	0	0	0	55.6	50
OTU00002	75	57.1	66.7	16.7	40	40	66.7	83.3
OTU00004	25	14.3	16.7	0	0	0	22.2	16.7
OTU00009	12.5	0	50	0	0	0	88.9	66.7
OTU00050	25	14.3	0	0	20	40	0	0
OTU00103	12.5	0	33.3	16.7	0	0	77.8	50
OTU00049	12.5	0	0	0	0	0	22.2	0
OTU00041	25	14.3	0	0	0	0	33.3	16.7
OTU00114	25	14.3	16.7	0	0	0	33.3	50
OTU00054	25	14.3	16.7	0	0	0	66.7	83.3
OTU01754	12.5	0	16.7	0	0	0	22.2	16.7
OTU00091	25	14.3	16.7	0	0	0	66.7	33.3
OTU00078	12.5	0	16.7	0	0	0	44.4	16.7
OTU01289	25	14.3	0	0	0	0	33.3	33.3
OTU00156	25	0	16.7	0	0	0	33.3	66.7
OTU00179	25	0	0	0	0	0	77.8	50
OTU00427	12.5	28.6	0	16.7	0	0	22.2	0
OTU00113	25	14.3	33.3	0	0	0	77.8	50
OTU00083	25	14.3	0	0	0	0	55.6	16.7
OTU00145	37.5	28.6	50	0	0	0	100	66.7
OTU00140	25	14.3	0	0	0	0	77.8	50
OTU00138	25	0	0	0	0	0	77.8	50
OTU00175	0	0	0	0	0	0	55.6	16.7
OTU00197	12.5	14.3	33.3	0	0	0	77.8	66.7
OTU00512	25	0	16.7	0	0	0	66.7	33.3
OTU01408	25	14.3	16.7	0	0	0	66.7	83.3
OTU00194	25	14.3	50	0	0	0	66.7	83.3
OTU00201	25	14.3	0	0	0	0	11.1	0
OTU00144	12.5	28.6	0	0	0	0	33.3	16.7
OTU00442	25	14.3	0	0	0	0	22.2	0
OTU00008	62.5	42.9	66.7	16.7	0	0	88.9	83.3
OTU00014	12.5	28.6	0	16.7	0	20	33.3	50
OTU00026	37.5	0	33.3	16.7	20	0	100	83.3
OTU00047	37.5	28.6	33.3	16.7	40	0	33.3	50
OTU00065	25	28.6	0	0	0	20	55.6	50
OTU00001	37.5	42.9	33.3	33.3	20	40	44.4	0
OTU00017	12.5	28.6	33.3	33.3	40	40	11.1	0
OTU00085	50	28.6	66.7	33.3	20	0	77.8	66.7
OTU00052	37.5	42.9	33.3	16.7	40	45	33.3	50
OTU00112	50	42.9	0	0	20	20	11.1	50
OTU00042	12.5	0	16.7	0	0	0	22.2	16.7
OTU00072	12.5	0	16.7	0	0	0	77.8	66.7
OTU00066	25	14.3	16.7	0	0	0	22.2	33.3
OTU00055	25	14.3	33.3	16.7	0	0	33.3	50
OTU00044	0	0	0	0	0	0	22.2	33.3
OTU00038	12.5	0	33.3	0	0	0	66.7	66.7
OTU00037	12.5	14.3	50	0	0	0	33.3	50
OTU00158	25	14.3	66.7	16.7	0	0	88.9	100
OTU00062	12.5	14.3	16.7	0	0	0	88.9	66.7
OTU00126	12.5	0	0	0	0	0	66.7	50
OTU00106	12.5	0	0	0	0	0	88.9	50
OTU00223	0	14.3	0	0	0	0	44.4	16.7
OTU00130	12.5	0	0	0	0	0	33.3	33.3
OTU00108	12.5	0	16.7	0	0	0	22.2	16.7
OTU00135	0	0	16.7	0	0	0	33.3	50
OTU00088	0	0	16.7	0	0	0	44.4	33.3
OTU00079	12.5	14.3	16.7	0	0	0	44.4	16.7
OTU00773	12.5	0	16.7	0	0	0	33.3	0

TABLE 3-continued

Results calculate with mean for Log2 Fold change at the OTU level								
OTU00227	25	14.3	16.7	0	0	0	11.1	16.7
OTU00081	0	143	16.7	0	0	0	0	0
OTU00134	12.5	0	16.7	0	0	0	44.4	50
OTU00120	12.5	0	16.7	0	0	0	33.3	33.3
OTU00115	12.5	14.3	16.7	0	0	0	55.6	50
OTU00143	12.5	14.3	0	0	0	0	22.2	16.7
OTU00199	0	0	0	0	0	0	44.4	50
OTU00141	12.5	0	0	0	0	0	55.6	33.3
OTU00233	12.5	0	16.7	0	0	0	88.9	66.7
OTU00217	12.5	0	16.7	0	0	0	44.4	50
OTU00131	0	0	0	0	0	0	55.6	33.3
OTU00116	0	14.3	0	0	0	0	33.3	50
OTU00090	12.5	34.3	0	0	0	0	33.3	16.7
OTU00100	0	0	0	0	0	0	22.2	16.7
OTU00168	0	0	0	0	0	0	33.3	0
OTU00166	12.5	14.3	0	0	0	0	44.4	0
OTU00389	12.5	0	0	0	0	0	33.3	16.7
OTU00258	12.5	14.3	0	0	0	0	33.3	33.3
OTU00946	12.5	0	0	0	0	0	33.3	0
OTU00117	12.5	14.3	0	0	0	0	0	0
OTU00226	12.5	14.3	0	0	0	0	66.7	50
OTU00762	12.5	0	0	0	0	0	66.7	50
OTU00576	12.5	0	16.7	0	0	0	55.6	83.3
OTU00153	12.5	28.6	33.3	0	40	20	0	0
OTU00191	12.5	0	16.7	0	0	0	66.7	83.3
OTU00171	12.5	14.3	16.7	0	0	0	66.7	66.7
OTU00215	12.5	14.3	0	0	0	0	0	0
OTU00295	12.5	0	0	0	0	0	55.6	16.7
OTU00235	0	0	0	0	0	0	44.4	0
OTU00146	0	0	0	0	0	0	33.3	0
OTU00169	12.5	14.3	16.7	0	0	0	11.1	0
OTU00159	25	0	0	0	0	0	11.1	0
OTU00160	12.5	0	0	0	0	0	11.1	16.7
OTU00246	0	0	0	0	0	0	22.2	33.3
OTU00361	12.5	0	0	0	0	0	55.6	33.3
OTU00357	12.5	0	0	0	0	0	44.4	66.7
OTU01249	25	14.3	0	0	0	0	11.1	0
OTU00400	25	0	16.7	0	0	0	11.1	16.7
OTU00167	0	0	0	0	0	0	33.3	0
OTU00013	25	14.3	33.3	0	0	0	0	0
OTU01091	25	14.3	50	16.7	20	0	88.9	100
OTU00058	62.5	71.4	16.7	16.7	20	20	77.8	66.7
OTU00086	25	0	50	16.7	0	0	11.1	50
OTU00136	0	14.3	16.7	50	40	0	22.2	33.3
OTU00082	25	14.3	33.3	0	0	0	22.2	33.3
OTU00255	12.5	28.6	16.7	02	20	0	22.2	16.7
OTU00021	50	28.6	33.3	16.7	20	20	66.7	33.3
OTU00006	25	42.9	0	50	60	20	11.1	16.7
OTU00010	62.5	57.1	50	16.7	20	0	55.6	50
OTU00043	75	57.1	16.7	16.7	20	40	0	16.7
OTU00003	62.5	28.6	33.3	33.3	04	40	0	16.7
OTU00019	25	0	50	33.3	0	20	33.3	33.3

TABLE 4

Results calculated with median for Log Fold change at the genus level											
Genus	The number of times based on the value of Log2 FC		Log2 FC (Doner/ pre-FMT)	Log2 Fold Change		1~3				Doner prevalence	
	over 0.5	under -0.5		Doner	pre-FMT	day	1 w	2 w	4 w	CR or PR	Others
<i>Blautia</i>	4	0	9.278	-0.499	0.000	3.950	2.115	5.807	2.115	100	100
<i>Staphylococcus</i>	4	0	-1.122	0.670	1.035	5.470	1.415	2.939	2.115	88.9	100
<i>Clostridium</i>	3	0	7.627	-0.012	-1.415	2.992	5.030	5.358	0.000	100	100
undefined_	3	0	7.933	1.054	0.000	3.048	1.415	0.000	2.115	100	100
<i>Clostridiales</i>											
<i>Actinomyces</i>	3	0	0.000	0.700	2.115	0.000	2.115	2.585	2.115	66.7	50
<i>Parabacteroides</i>	3	0	7.831	-0.337	0.000	1.512	7.432	5.170	0.000	100	100

TABLE 4-continued

Results calculated with median for Log Fold change at the genus level											
Genus	The number of times based on the value of Log2 FC		Log2 FC (Doner/ pre-FMT)	Log2 Fold Change		1~3				Doner prevalence	
	over 0.5	under -0.5		Doner	pre-FMT	day	1 w	2 w	4 w	CR or PR	Others
<i>Lachnospiraceae</i>											
<i>Roseburia</i>	2	0	7.105	1.098	0.000	4.143	0.000	2.585	0.000	100	83.3
<i>Ruminococcus</i>	2	0	6.879	-0.876	0.000	3.687	0.000	1.415	0.000	100	100
<i>Phascolarctobacterium</i>	2	0	6.279	0.430	0.000	3.841	0.000	3.222	0.000	77.8	66.7
<i>Faecalibacterium</i>	2	0	9.383	-0.127	0.000	3.076	1.415	0.000	0.000	100	100
<i>Rothia</i>	2	0	-2.939	0.000	2.939	0.000	0.000	4.000	2.939	0	16.7
<i>Schaalia</i>	2	0	2.115	0.000	0.000	0.000	0.000	2.939	2.115	77.8	66.7
<i>Alistipes</i>	2	0	3.841	1.726	0.000	2.044	1.415	0.000	0.000	77.8	50
<i>Bifidobacterium</i>	2	1	10.051	2.565	0.000	5.768	7.196	-3.815	0.000	100	100
<i>Lactobacillus</i>	2	1	3.242	-0.217	2.115	0.354	6.794	3.459	-2.115	77.8	83.3
<i>Veillonella</i>	2	1	2.115	2.115	0.000	-2.585	2.939	2.585	0.000	55.6	33.3
<i>Dorea</i>	1	0	7.070	-0.782	0.000	4.143	0.000	0.000	0.000	100	83.3
undefined_Firmicutes	1	0	7.577	-0.531	0.000	4.143	0.000	0.000	0.000	88.9	83.3
<i>Anaerostipes</i>	1	0	6.279	-0.175	0.000	3.841	0.000	0.000	0.000	77.8	83.3
<i>Collinsella</i>	1	0	8.296	2.181	0.000	5.358	0.000	0.000	0.000	88.9	50
<i>Bilophila</i>	1	0	3.841	1.726	0.000	2.115	0.000	0.000	0.000	77.8	50
<i>Butyrivibrio</i>	1	0	2.939	0.354	0.000	0.700	0.000	0.000	0.000	77.8	83.3
<i>Barnesiella</i>	1	0	0.000	0.000	0.000	2.115	0.000	0.000	0.000	44.4	16.7
<i>Fusicatenibacter</i>	1	0	6.508	-0.174	0.000	2.115	0.000	0.000	0.000	66.7	66.7
<i>Sutterella</i>	1	1	5.470	1.327	0.000	-2.115	2.115	0.000	0.000	77.8	50
<i>Fusobacterium</i>	1	1	0.000	-4.503	0.000	-3.841	0.000	1.415	0.000	33.3	66.7
<i>Erysipelatoclostridium</i>	1	1	4.143	-0.557	0.000	-0.905	1.415	0.000	0.000	88.9	83.3
<i>Enterococcus</i>	1	2	-4.283	2.115	6.398	0.942	-11.204	-7.353	0.189	55.6	33.3
<i>Escherichia</i>	1	2	2.115	-1.344	-1.415	-4.273	5.298	0.000	-8.671	66.7	66.7

TABLE 5

Results calculated with mean for Log Fold change at the genus level									
Genus	The number of times based on the value of Log2 FC		Log2 FC (Doner/ pre-FMT)	Log2 FC after FMT				CR or PR prevalence after FMT	
	Log2 FC >= 0.5 & CR or PR prevalence	Log2 FC <= -0.5 & Others prevalence		1~3 day	1 w	2 w	4 w	1~3 day	1 w
<i>Fusobacterium</i>	4	0	-5.98	2.91	4.22	5.12	8.53	33.3	25
<i>Faecalibacterium</i>	4	0	8.75	1.75	3.02	6.32	4.60	77.8	50
unclassified Clostridiales	4	0	8.06	3.34	5.57	5.02	6.59	66.7	50
unclassified Lachnospiraceae	4	0	6.36	4.36	2.86	8.08	6.06	88.9	37.5
<i>Roseburia</i>	4	0	1.71	3.15	6.29	7.30	4.36	66.7	37.5
<i>Ruminococcus</i>	4	0	7.49	1.55	4.36	5.81	4.89	88.9	25
unclassified Firmicutes	4	0	7.48	4.53	4.53	4.53	1.95	77.8	25
<i>Dorea</i>	4	0	6.76	3.97	4.02	5.48	5.31	66.7	25
<i>Phascolarctobacterium</i>	4	0	6.31	1.70	3.40	5.80	5.80	66.7	37.5
<i>Sutterella</i>	4	0	5.69	2.18	5.50	6.12	6.62	33.3	50
<i>Bacteroides</i>	3	0	1.43	-0.25	0.98	1.90	1.55	100	100
<i>Lactobacillus</i>	3	0	-3.22	4.10	6.16	0.41	1.21	66.7	75
<i>Blautia</i>	3	0	2.02	0.91	-1.62	5.93	9.09	88.9	62.5
<i>Parabacteroides</i>	3	0	0.26	0.10	0.78	5.99	6.83	77.8	75
<i>Megamonas</i>	3	0	7.62	4.58	9.18	8.54	4.02	22.2	25
<i>Collinsella</i>	3	0	8.26	1.73	3.29	6.03	3.52	66.7	25
<i>Eubacterium</i>	3	0	6.13	1.06	4.42	3.88	0.56	33.3	25
<i>Coproccoccus</i>	3	0	5.40	3.79	2.58	3.61	4.77	33.3	25
<i>Anaerostipes</i>	2	0	6.50	-0.55	6.25	6.22	4.79	77.8	25
<i>Veillonella</i>	2	0	-0.31	0.13	2.36	2.42	-0.31	44.4	62.5
<i>Bilophila</i>	2	0	3.72	-0.50	3.61	5.03	3.11	66.7	25

TABLE 5-continued

Results calculated with mean for Log Fold change at the genus level									
<i>Flavonifractor</i>	2	0	3.68	-0.16	0.00	4.00	4.49	44.4	0
unclassified <i>Ruminococcaceae</i>	2	0	3.60	2.39	2.12	3.01	1.28	33.3	12.5
unclassified <i>Clostridiaceae</i>	2	0	3.95	4.34	0.00	0.87	1.54	44.4	0
<i>Butyrivibrio</i>	2	0	3.68	0.54	1.17	1.17	1.28	55.6	12.5
<i>Faecalicatena</i>	2	0	3.64	3.31	0.00	2.37	3.74	33.3	0
<i>Campylobacter</i>	2	0	-2.94	0.97	2.12	-0.35	0.82	22.2	12.5
<i>Prevotella</i>	2	1	7.81	0.12	2.20	1.37	-5.67	22.2	25
<i>Clostridium</i>	2	1	3.71	3.09	-2.03	-1.50	8.55	77.8	75
<i>Okadaella</i>	2	1	-7.58	1.85	3.81	1.62	-3.42	33.3	25
<i>Lachnoclostridium</i>	2	1	1.16	-1.03	0.08	2.61	6.65	77.8	50
<i>Alistipes</i>	2	1	6.37	-0.62	8.22	3.45	4.91	66.7	50
<i>Megasphaera</i>	2	1	6.01	2.72	6.81	2.17	-2.94	22.2	25
<i>Enterococcus</i>	2	2	-7.98	1.97	-2.71	-1.32	1.68	88.9	50
<i>Staphylococcus</i>	2	2	-3.03	1.38	0.74	-5.30	-3.07	88.9	50
<i>Bifidobacterium</i>	2	2	4.58	2.26	1.44	-2.33	-5.00	88.9	75
<i>Streptococcus</i>	2	2	0.08	-7.19	0.80	4.12	-1.22	66.7	50
<i>Abiotrophia</i>	2	2	-5.73	3.00	1.67	-1.81	-3.56	44.4	37.5
<i>Actinomyces</i>	2	2	-3.30	1.48	1.33	-0.89	-1.06	44.4	75
<i>Robinsoniella</i>	1	0	0.07	-0.15	-5.80	6.04	6.52	33.3	12.5
<i>Fusicatenibacter</i>	1	0	7.01	4.90	2.58	2.86	0.00	55.6	12.5
<i>Barnesiella</i>	1	0	4.49	4.88	1.17	1.97	3.03	55.6	12.5
<i>Drancourtella</i>	1	0	-1.87	0.56	1.17	3.75	1.54	44.4	12.5
<i>Faecalitalea</i>	1	0	4.85	3.98	0.00	0.00	0.00	22.2	0
<i>Akkermansia</i>	1	0	0.00	0.80	2.68	4.36	3.69	22.2	12.5
<i>Lachnospira</i>	1	0	3.91	0.80	2.77	0.87	1.54	11.1	12.5
<i>Coprobacter</i>	1	0	-1.06	3.01	3.75	1.42	0.00	22.2	12.5
<i>Romboutsia</i>	1	0	3.25	1.69	0.00	0.00	0.00	22.2	0
<i>Schaalia</i>	1	1	-3.75	-0.28	0.12	0.60	-1.91	22.2	37.5
<i>Parasutterella</i>	1	1	4.97	-3.25	6.07	0.00	0.00	11.1	25
<i>Ruthenibacterium</i>	1	1	1.31	-2.80	4.33	4.27	4.91	22.2	12.5
<i>Rothia</i>	1	2	-7.05	3.70	-0.23	-1.41	-1.45	33.3	37.5

Genus	CR or PR prevalence after FMT		Others prevalence after FMT				Donor prevalence	
	2 w	4 w	1~3 day	1 w	2 w	4 w	CR or PR	Others
<i>Fusobacterium</i>	50	28.6	66.7	33.3	20	0	33.3	66.7
<i>Faecalibacterium</i>	25	28.6	66.7	33.3	0	0	100	100
unclassified <i>Clostridiales</i>	37.5	57.1	66.7	0	20	0	100	100
unclassified <i>Lachnospiraceae</i>	50	28.6	16.7	0	0	0	100	100
<i>Roseburia</i>	50	28.6	16.7	0	0	0	100	83.3
<i>Ruminococcus</i>	50	28.6	50	0	0	0	100	100
unclassified <i>Firmicutes</i>	25	28.6	33.3	0	0	0	88.9	83.3
<i>Dorea</i>	25	42.9	16.7	16.7	0	0	100	83.3
<i>Phascolarctobacterium</i>	50	42.9	33.3	0	0	0	77.8	66.7
<i>Sutterella</i>	37.5	42.9	66.7	0	0	0	77.8	50
<i>Bacteroides</i>	100	71.4	83.3	50	100	40	100	100
<i>Lactobacillus</i>	62.5	42.9	66.7	33.3	40	60	77.8	83.3
<i>Blautia</i>	50	57.1	83.3	16.7	40	0	100	100
<i>Parabacteroides</i>	62.5	42.9	83.3	16.7	40	20	100	100
<i>Megamonas</i>	25	14.3	16.7	0	0	0	22.2	50
<i>Collinsella</i>	37.5	14.3	33.3	0	0	0	88.9	50
<i>Eubacterium</i>	25	14.3	16.7	0	0	0	77.8	83.3
<i>Coprococcus</i>	25	14.3	16.7	0	0	0	44.4	66.7
<i>Anaerostipes</i>	37.5	14.3	16.7	0	0	0	77.8	83.3
<i>Veillonella</i>	62.5	28.6	50	33.3	20	40	55.6	33.3
<i>Bilophila</i>	25	14.3	33.3	0	0	0	77.8	50
<i>Flavonifractor</i>	37.5	28.6	50	0	0	0	160	66.7
unclassified <i>Ruminococcaceae</i>	25	14.3	16.7	0	0	0	66.7	50
unclassified <i>Clostridiaceae</i>	12.5	28.6	0	0	0	0	77.8	16.7
<i>Butyrivibrio</i>	25	14.3	50	0	0	0	77.8	83.3
<i>Faecalicatena</i>	25	14.3	0	0	0	0	66.7	33.3
<i>Campylobacter</i>	12.5	28.6	33.3	0	40	20	0	0
<i>Prevotella</i>	25	0	33.3	50	60	20	22.2	16.7
<i>Clostridium</i>	75	42.9	83.3	16.7	20	0	100	100
<i>Okadaella</i>	12.5	28.6	0	16.7	0	20	33.3	50
<i>Lachnoclostridium</i>	62.5	28.6	66.7	33.3	20	0	100	100
<i>Alistipes</i>	37.5	14.3	50	0	20	0	77.8	50
<i>Megasphaera</i>	12.5	0	16.7	0	20	20	33.3	16.7
<i>Enterococcus</i>	62.5	85.7	66.7	100	80	80	55.6	33.3
<i>Staphylococcus</i>	75	71.4	33.3	33.3	40	40	88.9	100
<i>Bifidobacterium</i>	75	28.6	66.7	50	60	40	100	100
<i>Streptococcus</i>	37.5	28.6	83.3	66.7	60	20	88.9	100
<i>Abiotrophia</i>	37.5	42.9	33.3	16.7	40	40	33.3	50

TABLE 5-continued

Results calculated with mean for Log Fold change at the genus level								
<i>Actinomyces</i>	75	57.1	16.7	16.7	20	40	66.7	50
<i>Robinsoniella</i>	25	14.3	33.0	16.7	0	0	55.6	66.7
<i>Fusicatenibacter</i>	12.5	0	33.3	0	0	0	66.7	66.7
<i>Barnesiella</i>	12.5	14.3	16.7	0	0	0	44.4	16.7
<i>Drancourtella</i>	12.5	14.3	16.7	0	0	0	55.6	50
<i>Faecalitalea</i>	0	0	0	0	0	0	22.2	16.7
<i>Akkermansia</i>	12.5	14.3	0	0	0	0	0	0
<i>Lachnospira</i>	25	14.3	0	0	0	0	44.4	66.7
<i>Coproacter</i>	12.5	0	0	0	0	0	11.1	16.7
<i>Romboutsia</i>	0	0	0	0	0	0	55.6	50
<i>Schaalia</i>	62.5	71.4	16.7	16.7	20	20	77.8	66.7
<i>Parasutterella</i>	0	0	33.3	0	0	0	33.3	50
<i>Ruthenibacterium</i>	25	14.3	33.3	0	0	0	22.2	33.3
<i>Rothia</i>	75	57.1	16.7	16.7	20	40	0	16.7

[0581] Further, among species in the fecal microbiota before and after FMT, the time courses of changes in the species of the above Group B are shown in FIGS. 11 to 19. The species shown in FIGS. 11 to 19 were all confirmed to be abundant in the CR or PR group in comparison with the Others group.

Sequence Listing Free Text

[0582] SEQ ID NO: 345: synthetic DNA

[0583] SEQ ID NO: 346: synthetic DNA

SEQUENCE LISTING

Sequence total quantity: 346

SEQ ID NO: 1 moltype = DNA length = 314
 FEATURE Location/Qualifiers
 source 1..314
 mol_type = genomic DNA
 organism = Bacteroides sp.

SEQUENCE: 1
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 ctaagtttga tggcgaccgg cgcacgggtg agtaacacgt atccaacctg cccgatgactc 120
 ggggtagagc ttctgaaaga aagattaata cccgatggca tagttcttcc gcctggtaga 180
 actattaaag aatttcggtc atcgatgggg atgcgttcca ttaggttggt ggcggggtaa 240
 cggcccacca agccttcgat ggataggggt tctgagagga aggtccccc cattggaact 300
 gagacacggt ccaa 314

SEQ ID NO: 2 moltype = DNA length = 316
 FEATURE Location/Qualifiers
 source 1..316
 mol_type = genomic DNA
 organism = Bacteroides sp.

SEQUENCE: 2
 gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcattt cagtttgctt 60
 gcaaaactgga gatggcgacc ggcgcacggg tgagtaacac gtatccaacc tgccgataac 120
 tcggggatag cctttcgaaa gaaagattaa taccgatggg tataattaga cccgatggtc 180
 ttgttattaa agaatttcgg ttatcgatgg ggatgcgttc cattaggcag ttggtgaggt 240
 aacggctcac caaaccttcg atggataggg gttctgagag gaaggtcccc cacattggaa 300
 ctgagacacg gtccaa 316

SEQ ID NO: 3 moltype = DNA length = 317
 FEATURE Location/Qualifiers
 source 1..317
 mol_type = genomic DNA
 organism = Bacteroides sp.

SEQUENCE: 3
 gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatgg tcttagcttg 60
 ctaaggctga tggcgaccgg cgcacgggtg agtaacacgt atccaacctg cccgtctactc 120
 ttggccagcc ttctgaaagg aagattaatc caggatggga tcatgagttc acatgtccgc 180
 atgattaaag gtattttccg gtagacgatg gggatgcgtt ccattagata gtaggcgggg 240
 taacggccca ctagtcaac gatggatagg ggttctgaga ggaaggtccc ccacattgga 300
 actgagacac ggtccaa 317

SEQ ID NO: 4 moltype = DNA length = 315
 FEATURE Location/Qualifiers
 source 1..315
 mol_type = genomic DNA
 organism = Bacteroides sp.

-continued

SEQUENCE: 4
 gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatgg tcttagcttg 60
 ctaaggccga tggcgaccgg cgcacgggtg agtaaacagt atccaacctg ccgtctactc 120
 ttggacagcc ttctgaaagg aagattaata caagatggca tcatgagtcg gcatgttcac 180
 atgattaaag gtattccggt agacgatggg gatgcgttcc attagatagt aggcggggta 240
 acggcccacc tagctctcga tggatagggg ttctgagagg aaggtccccc acattggaac 300
 tgagacacgg tccaa 315

SEQ ID NO: 5 moltype = DNA length = 314
 FEATURE Location/Qualifiers
 source 1..314
 mol_type = genomic DNA
 organism = Bacteroides sp.

SEQUENCE: 5
 gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatca tcaaagcttg 60
 ctttgatgga tggcgaccgg cgcacgggtg agtaaacagt atccaacctg ccgacaacac 120
 tgggatagcc ttctgaaaga aagattaata cgggatggca tagttttccc gcatgggatg 180
 attattaaag aatttcggtt gtcgatgggg atgcgttcca ttaggcagtt ggcggggtaa 240
 cggcccacca aaccaacgat ggataggggt tctgagagga aggtccccc cattggaact 300
 gagacacggt ccaa 314

SEQ ID NO: 6 moltype = DNA length = 323
 FEATURE Location/Qualifiers
 source 1..323
 mol_type = genomic DNA
 organism = Blautia sp.

SEQUENCE: 6
 gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggaaatatt tcattgagac 60
 ttcggtggat ttgatctatt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
 ttatacaggg ggataacagt cagaaatggc tgctaatacc gcataagcgc acagagctgc 180
 atggctcagt gtgaaaaact ccggtgggtat aagatggacc cgcgttggat tagcttggtg 240
 gtggggtaac ggcccaccaa ggcgacgac catagccggc ctgagagggt gaacggccac 300
 attgggactg agacacggcc cag 323

SEQ ID NO: 7 moltype = DNA length = 323
 FEATURE Location/Qualifiers
 source 1..323
 mol_type = genomic DNA
 note = Phascolarctobacterium
 organism = unidentified

SEQUENCE: 7
 gacgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggagaatttt atttcggtag 60
 aattcttagt ggcgaacggg tgaagtaacgc gtaggcaacc tgccttttag acggggacaa 120
 cattccgaaa gtagtgctaa taccggatgt gatcatcgtg ccgcattggca ggatgaagaa 180
 agatggcctc tacaagtaag ctatcgctaa aggatgggcc tgcgtctgat tagctagttg 240
 gtagtgtaac ggactaccaa ggcgatgatc agtagccggt ctgagaggat gaacggccac 300
 attgggactg agacacggcc caa 323

SEQ ID NO: 8 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 organism = Dorea sp.

SEQUENCE: 8
 gatgaacgct ggcggcgctgc ttaacacatg caagtcgagc gaagcactta agtttgattc 60
 ttcggatgaa gacttttgtg actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc 120
 ctacatacag gggataacag ttgaaatga ctgctaatac cgcataagac caccggtaccg 180
 catggtacag tggtaaaaaa tctggtggta tgagatggac ccgcgtctga ttaggtagtt 240
 ggtggggtaa cggcctacca agccgacgat cagtagccga cctgagaggg tgaccggcca 300
 cattgggact gagacacggc ccag 324

SEQ ID NO: 9 moltype = DNA length = 323
 FEATURE Location/Qualifiers
 source 1..323
 mol_type = genomic DNA
 organism = Blautia sp.

SEQUENCE: 9
 gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggaattact ttattgaaac 60
 ttcggtcgat atgatttaat tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
 ttatacaggg ggataacagt cagaaatgac tgctaatacc gcataagcgc acagagctgc 180
 atggctcggg gtgaaaaact ccggtgggtat aagatggacc cgcgttggat tagctagttg 240
 gtgaggtaac ggcccaccaa ggcgacgatc catagccggc ctgagagggt gaacggccac 300
 attgggactg agacacggcc cag 323

SEQ ID NO: 10 moltype = DNA length = 316
 FEATURE Location/Qualifiers

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source          1..316
                 mol_type = genomic DNA
                 organism = Bacteroides sp.

SEQUENCE: 10
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcattt tagtttgctt 60
gcaactataa gatggcgacc ggcgcacggg tgagtaacac gtatccaacc tgccgataac 120
tcggggatag cctttcgaaa gaaagattaa tatccgatgg tatattttct ccgcattggtg 180
agaatattaa agaatttcgg ttatcgatgg ggatgcgttc cattagtttg ttggcgggggt 240
aacggcccac caagaccacg atggataggg gttctgagag gaagggtccc caccattggaa 300
ctgagacacg gtccaa                                     316

SEQ ID NO: 11      moltype = DNA length = 322
FEATURE           Location/Qualifiers
source            1..322
                 mol_type = genomic DNA
                 organism = Megamonas sp.

SEQUENCE: 11
gacgaacgct ggcggcgctgc ttaacacatg caagtcgaac ggggtgttta ttccggtaaa 60
caccaaagtgg cgaacgggtg agtaacgcgt aagcaatcta ccttcaagat ggggacaaca 120
cttcgaaaagg ggtgctaata ccgaatgaat gtaagagtat cgcattgagac acttactaaa 180
ggaggcctct gacataagct tccgcttgaa gatgagcttg cgtctgatta gctagttggt 240
gagggtaaaag gccacccaag gcgacgatca gtagccggtc tgagaggatg aacggccaca 300
ttgggactga gacacggccc ag                                     322

SEQ ID NO: 12      moltype = DNA length = 316
FEATURE           Location/Qualifiers
source            1..316
                 mol_type = genomic DNA
                 organism = Staphylococcus sp.

SEQUENCE: 12
gacgaacgct ggcggcgctgc ctaatacatg caagtagaac gctgaagaga ggagcttgct 60
ctttctggat gagttgcgaa cgggtgagta acgcgtaggt aacctgcctt gtagcggggg 120
ataactattg gaaacgatag ctaataccgc ataacaatgg atgacacatg tcatttattt 180
gaaaggggca attgctccac tacaagatgg acctgcgttg tattagctag taggtgaggt 240
aacggctcac ctaggcgacg atacatagcc gacctgagag ggtgatcggc cacactggga 300
ctgagacacg gcccag                                     316

SEQ ID NO: 13      moltype = DNA length = 324
FEATURE           Location/Qualifiers
source            1..324
                 mol_type = genomic DNA
                 organism = Blautia sp.

SEQUENCE: 13
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcacctt gatttgattc 60
ttcggatgaa gatcttggtg actgagtggt ggcaggggtg gtaacgcgtg ggtaacctgc 120
ctcatcacgg gggataacag ttagaatga ctgctaatac cgcataagac cacagcaccg 180
catggtgcag gggtaaaaac tccggtggta tgagatggac ccgcgtctga ttaggtagtt 240
ggtggggtaa cggcctacca agccgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa                                     324

SEQ ID NO: 14      moltype = DNA length = 307
FEATURE           Location/Qualifiers
source            1..307
                 mol_type = genomic DNA
                 organism = Bifidobacterium sp.

SEQUENCE: 14
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggatcccag gagcttgctc 60
ctgggtgaga gtggcgaacg ggtgagtaat gcgtgaccga cctgccccat acaccggaat 120
agctcctgga aacgggtggt aatgcccgat gctccagttg gatgcatgtc cttctgggaa 180
agattcatcg gtatgggatg gggtcgcgtc ctatcagctt gatggcgggg taacggccca 240
ccatggcttc gacgggtagc cggcctgaga gggcgaccgg ccacattggg actgagatac 300
ggcccag                                     307

SEQ ID NO: 15      moltype = DNA length = 303
FEATURE           Location/Qualifiers
source            1..303
                 mol_type = genomic DNA
                 organism = Collinsella sp.

SEQUENCE: 15
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggcacccctc tccggaggga 60
agcgaagtgg gaacggctga gtaacacgtg gagaacctgc cccctccccg gggatagccg 120
cccgaaggga cgggtaatac cggatacccc ggggtgccgc atggcacccc ggctaaagcc 180
ccgacgggag gggatggctc cgcggcccat caggtagacg gcgggggtgac ggcccaccgt 240
gcccgaacg ggtagccggg ttgagagacc gaccggccag attgggactg agacacggcc 300
cag                                     303

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SEQ ID NO: 16 moltype = DNA length = 326
 FEATURE Location/Qualifiers
 source 1..326
 mol_type = genomic DNA
 organism = Clostridium sp.

SEQUENCE: 16

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgaac	gaagcacttt	atttgatttc	60
cttcgggact	gattatcttg	tgactgagtg	gcggacgggt	gagtaacgcg	tgggtaacct	120
gccttgatca	gggggataac	agttggaaac	ggctgctaata	accgcataag	cgcacagcat	180
cgcatgatgc	agtgtgaaaa	actccgggtg	tataagatgg	acccgcgttg	gattagctag	240
ttggtagagg	aacggcccac	caaggcgacg	atccatagcc	gacctgagag	ggtgaccggc	300
cacattggga	ctgagacacg	gccccaa				326

SEQ ID NO: 17 moltype = DNA length = 316
 FEATURE Location/Qualifiers
 source 1..316
 mol_type = genomic DNA
 organism = Parabacteroides sp.

SEQUENCE: 17

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tacagattga	tggcgaccgg	cgcacgggtg	agtaacgcgt	atgcaactta	cctatcagag	120
ggggatagcc	cgggcgaagt	cggattaata	ccccataaaa	caggggtccc	gcatgggaat	180
atttgttaaa	gattcatcgc	tgatagatag	gcatgcgttc	cattagcgag	ttggcggggt	240
aacggcccac	caaaccgacg	atggataggg	gttctgagag	gaaggtcccc	cacattggta	300
ctgagacacg	gaccaa					316

SEQ ID NO: 18 moltype = DNA length = 307
 FEATURE Location/Qualifiers
 source 1..307
 mol_type = genomic DNA
 organism = Bifidobacterium sp.

SEQUENCE: 18

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ctgggtgaga	gtggcgaaacg	ggtgagtaat	gcgtgaccga	cctgccccat	acaccggaat	120
agctcctgga	aacgggtggt	aatgccggat	gctccagttg	accgcaggtg	cctctgggaa	180
agcttttgcg	gtatgggatg	gggtcgcgtc	ctatcagctt	gatggcgggg	taacggccca	240
ccatgggttc	gacgggtagc	cggcctgaga	gggcgaccgg	ccacattggg	actgagatac	300
ggccccag						307

SEQ ID NO: 19 moltype = DNA length = 302
 FEATURE Location/Qualifiers
 source 1..302
 mol_type = genomic DNA
 organism = Clostridium sp.

SEQUENCE: 19

gatgaacgct	ggcggcgctgc	ctaatacatg	caagtcgaac	gcttcacttc	ggtgaagagt	60
ggcgaaacggg	tgagtaatac	ataagtaacc	tggcatctac	agggggataa	ctgatggaaa	120
cgtcagctaa	gaccgcatag	gtgtagagat	cgcatagaac	ctatatgaaa	agtgtactcg	180
gactggtaga	tgatggactt	atggcgcat	agctgggttg	tagggtaacg	gcctaccaag	240
gcgacgatgc	gtagccgacc	tgagaggggtg	accggccaca	ctgggactga	gacacggccc	300
ag						302

SEQ ID NO: 20 moltype = DNA length = 315
 FEATURE Location/Qualifiers
 source 1..315
 mol_type = genomic DNA
 organism = Bacteroides sp.

SEQUENCE: 20

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ctaagtttga	tggcgaccgg	cgcacgggtg	agtaacacgt	atccaacctt	ccgtttactc	120
agggatagcc	tttcgaaaga	aagattaata	cctgatagta	tggtagatt	gcatgatagc	180
accattaaag	atttattggt	aaacgatggg	gatgcgttcc	attaggtagt	aggcggggta	240
acggccacc	tagccgacga	tgatagggg	ttctgagagg	aaggtcccc	acattggaac	300
tgagacacgg	tccaa					315

SEQ ID NO: 21 moltype = DNA length = 326
 FEATURE Location/Qualifiers
 source 1..326
 mol_type = genomic DNA
 organism = Roseburia sp.

SEQUENCE: 21

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cttcggaaat	gaagattttg	tgactgagtg	gcggacgggt	gagtaacgcg	tgggtaacct	120
gcctcataca	gggggataac	agttggaaac	gactgctaata	accgcataag	cgcacaggat	180
cgcatgatcc	gggtgtgaaa	actccgggtg	tatgggatgg	acccgcgtct	gattagccag	240
ttggcagggt	aacggcctac	caaagcgacg	atcagtagcc	gacctgagag	ggtgaccggc	300

-continued

cacattggga ctgagacacg gcccac 326

SEQ ID NO: 22 moltype = DNA length = 323
 FEATURE Location/Qualifiers
 source 1..323
 mol_type = genomic DNA
 note = Lachnospiraceae
 organism = unidentified

SEQUENCE: 22
 gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggaaacctt ttattgaagc 60
 ttccggcagat ttagattggt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
 ttatacaggg ggataaacagt cagaaatgac tgctaatacc gcataagcgc acaggaccgc 180
 atggtccggt gtgaaaaaac ccggtggtat aagatggacc cgcgttggat tagctagttg 240
 gtgagggtaac ggcccaccaa ggcgacgat catagccggc ctgagagggt gaacggccac 300
 attgggactg agacacggcc cag 323

SEQ ID NO: 23 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 note = Lachnospiraceae
 organism = unidentified

SEQUENCE: 23
 gatgaacgct ggcggcgctgc ttaacacatg caagtcgagc gaagcaattt aagtgaagtt 60
 ttccgatgga tcttaaattg actgagcggc ggacgggtga gtaacgcgtg gataacctgc 120
 ctccacacagg gggataacag ttgaaatga ctgctaatac cgcataagcg cacggtaaccg 180
 catggtacag tgtgaaaaac tccggtggtg tgagatggat ccgctgtcta ttaggtagtt 240
 ggtgaggttaa cggcccacca agccgacgat cagtagccga cctgagaggg tgaccggcca 300
 cattgggact gagacacggc ccaa 324

SEQ ID NO: 24 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 organism = Clostridium sp.

SEQUENCE: 24
 gatgaacgct ggcggcgctgc ctaacacatg caagtcgagc gaagcggttt taaggaaagt 60
 ttccgatgga attaaaactg actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc 120
 ctccatacagg gggataacag ttgaaatgg ctgctaatac cgcataagca cacagcttcg 180
 catggagcag tgtgaaaaac tccggtggtg tgagatggac ccgctgtcta ttagctagtt 240
 ggtaaggtaa cggcttacca aggcgacgat cagtagccga cctgagaggg tgaccggcca 300
 cattgggact gagacacggc ccaa 324

SEQ ID NO: 25 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 note = Clostridiaceae
 organism = unidentified

SEQUENCE: 25
 gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcaatct aacggaagt 60
 ttccgatgga agctagattg actgagtgcc ggacgggtga gtaacgcgtg ggtaacctgc 120
 ctccactcgg gggacaacag ttgaaatga ctgctaatac cgcataagcg cacaggaccg 180
 catggtccgg tgtgaaaaac tccggtggtg tgagatggac ccgctttga ttagctagtt 240
 ggtggggttaa cggcctacca aggcgacgat caatagccga cctgagaggg tgaccggcca 300
 cattgggact gagacacggc ccaa 324

SEQ ID NO: 26 moltype = DNA length = 311
 FEATURE Location/Qualifiers
 source 1..311
 mol_type = genomic DNA
 organism = Faecalibacterium sp.

SEQUENCE: 26
 gacgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggagcttaga gagcttgctt 60
 tttaagctta gtggcggaac ggtgagtaac gcgtgagtaa cctgccctag agtgggggac 120
 aacagttgga aacgactgct aataccgcat aagcccacgg taccgatgg tactgaggga 180
 aaaggattta ttgccttag gatggactcg cgtccaatta gctagttggt gaggtaacgg 240
 cccaccaagg ccacgattgg tagccggact gagaggttga acggccacat tgggactgag 300
 acacggccca g 311

SEQ ID NO: 27 moltype = DNA length = 316
 FEATURE Location/Qualifiers
 source 1..316
 mol_type = genomic DNA
 organism = Streptococcus sp.

SEQUENCE: 27

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gatgaacgct ggcggcggtgc ctaatacatg caagtagaac gctgaagctt ggtgcttgca 60
ccgagcggat gagttgcgaa cgggtgagta acgcgtaggt aacctgcctg gtacgggggg 120
ataactattg gaaacgatat ctaataccgc ataacagtag atgtcgcatg acctttgctt 180
gaaaggtgca attgcaccac taccagatgg acctgcgttg tattagctag ttggtgaggt 240
aacggctcac caaggcgacg atacatagcc gacctgagag ggtgatcggc cacactggga 300
ctgagacacg gccacg                                     316

```

```

SEQ ID NO: 28          moltype = DNA length = 322
FEATURE               Location/Qualifiers
source                1..322
                      mol_type = genomic DNA
                      note = Lachnospiraceae
                      organism = unidentified

```

```

SEQUENCE: 28
gatgaacgct ggcggcggtgc ttaacacatg caagtcgagc gaagcgattt aaatgagact 60
tcggtggatt ttaaattgac tgagcggcgg acgggtgagt aacgcgtgga taacctgcct 120
cacacagggg gataaacagt agaaatgact gctaataacc cataagcgca cggtaccgca 180
tggtacagtg tgaaaaactc cgggtggtgt agatggatcc gcgtctgatt aggtagtgtg 240
tgaggtaacg gccaccacag ccgacgatca gtagccgacc tgagagggtg accggccaca 300
ttgggactga gacacggccc aa                                     322

```

```

SEQ ID NO: 29          moltype = DNA length = 310
FEATURE               Location/Qualifiers
source                1..310
                      mol_type = genomic DNA
                      organism = Parabacteroides sp.

```

```

SEQUENCE: 29
gatgaacgct agcgacaggg ttaacacatg caagtcgagg ggcagcgggg gtagcaatac 60
ccgcggcgca ccggcgacag ggtgagtaac gcgtatgcaa cttacctatc agagggggat 120
aaccggcgca taagtcggact aataccgcat gaagcagggg ccccgcatgg ggatatttgc 180
taaagattca tcgctgatag ataggcatgc gttccattag tttgttggcg gggtaacggc 240
ccaccaagac cacgatggat aggggttctg agaggaaggt ccccccacatt ggaactgaga 300
cacggtccaa                                     310

```

```

SEQ ID NO: 30          moltype = DNA length = 326
FEATURE               Location/Qualifiers
source                1..326
                      mol_type = genomic DNA
                      organism = Lactobacillus sp.

```

```

SEQUENCE: 30
gatgaacgct ggcggcggtgc ttaacacatg caagtcgaac gaagcattta agacagatta 60
cttcggtttg aagtcttcta tgactgagtg gcggacgggt gagtaacggc tgggtaacct 120
gcctcataca ggggggatagc agctggaaac ggctggtaat accgcataag cgcacagtac 180
cacatgggtac agtgtgaaaa actccgggtg tatgagatgg acccgctctc gattagcttg 240
ttggcggggg aacggcccac caaggcgacg atcagtagcc gacctgagag ggtgacgggc 300
cacattggga ctgagacacg gccacg                                     326

```

```

SEQ ID NO: 31          moltype = DNA length = 325
FEATURE               Location/Qualifiers
source                1..325
                      mol_type = genomic DNA
                      organism = Clostridium sp.

```

```

SEQUENCE: 31
gatgaacgct ggcggcggtgc ctaacacatg caagtcgaac ggagtcgttt tggaaatctc 60
ttcggagatg gaattctcga cttagtggcg gacgggtgag taacgcgtga gcaatctgcc 120
tttaagaggg ggataaacagt cggaaacggc tgctaatacc gcataaagca ttaaattcgc 180
atgtttttga tgccaaagga gcaatccgct tttagatgag ctgcgctctg attagctagt 240
tggcgggggt acggccacac aaggcgacga tcagtagccg gactgagagg ttgaacggcc 300
acattgggac tgagacacg gccacg                                     325

```

```

SEQ ID NO: 32          moltype = DNA length = 307
FEATURE               Location/Qualifiers
source                1..307
                      mol_type = genomic DNA
                      organism = Roseburia sp.

```

```

SEQUENCE: 32
gataaacgct ggcggcggtgc ttaacacatg caagtcgaac gaagtttttc ttccgggagg 60
aacttagtgg cggacgggtg agtaacgcgt gggtaacctg cctgtacag ggggacaaca 120
gctggaaacg gctgctaata ccgcataagc ccttagcact gcatgggtga tagggaaaag 180
gagcaatccg gtacaggatg gacccgcgtc tgattagcca gttggcaggg taacggccta 240
ccaaagcgac gatcagtagc cgatctgaga ggatgtacgg ccacattggg actgagacac 300
ggcccag                                     307

```

```

SEQ ID NO: 33          moltype = DNA length = 316
FEATURE               Location/Qualifiers
source                1..316

```

-continued

```

mol_type = genomic DNA
organism = Bacteroides sp.

SEQUENCE: 33
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcattt tagtttgctt 60
gcaaaactgaa gatggcgacc ggcgcacggg tgagtaacac gtatccaacc tgccgataac 120
tcgggaatag cctttcgaaa gaaagattaa taccggatag catacgaata tcgcatgata 180
tttttattaa agaatttcgg ttatcgatgg ggatgcgttc cattagtttg ttggcgggggt 240
aacggccccc caagactacg atggataggg gttctgagag gaagggtccc cacattggaa 300
ctgagacacg gtccaa 316

SEQ ID NO: 34      moltype = DNA length = 308
FEATURE           Location/Qualifiers
source            1..308
                  mol_type = genomic DNA
                  organism = Bifidobacterium sp.

SEQUENCE: 34
gatgaacgct ggccggcgtgc ttaacacatg caagtcgaac gggatccatc aggccttgct 60
tggtgggtgag agtggcgaaac gggtagtaaa tgcgtgaccg acctgcccca tacaccggaa 120
tagctctcgg aaacgggtggg taatgccgga tgctccagtt gatcgcatgg tcttctggga 180
aagctttcgc ggatgggatg ggggtcgcgt cctatcagct tgacggcggg gtaacggccc 240
accgtggctt cgacgggtag ccggcctgag agggcgaccg gccacattgg gactgagata 300
cggcccgag 308

SEQ ID NO: 35      moltype = DNA length = 329
FEATURE           Location/Qualifiers
source            1..329
                  mol_type = genomic DNA
                  organism = Veillonella sp.

SEQUENCE: 35
gacgaacgct ggccggcgtgc ttaacacatg caagtcgaac gaagagcgat ggaagcttgc 60
ttctatcaat cttagtggcg aacgggtgag taacgcgtaa tcaacctgcc cttcagaggg 120
ggacaacagt tggaaacgac tgctaatacc gcatacgatc taatctcggc atcgaggata 180
gatgaaagggt ggctcttaca tgtaagctat cactgaagga ggggattgag tctgattagc 240
tagttggagg ggtaacggcg caccgaaggc atgatcagta gccggtctga gaggatgaac 300
ggccacattg ggactgagac acggcccgag 329

SEQ ID NO: 36      moltype = DNA length = 309
FEATURE           Location/Qualifiers
source            1..309
                  mol_type = genomic DNA
                  note = Schaaliala
                  organism = unidentified

SEQUENCE: 36
gacgaacgct ggccggcgtgc ttaacacatg caagtcgaac gctgaagccc agcttgctgg 60
gtggatgagt ggccgaacggg tgagtaacac gtgagtaacc tgcccccttc ttggggataa 120
cgcccggaaa cgggtgctaa tactggatat tcactggcct tcgcatgggg gttggtggaa 180
aggttttttc tgggtggggg tgggctcgcg gcctatcagc ttgttggtgg ggtgatggcc 240
taccgaaggct ttgacgggta gccggcctga gagggtgacc ggtcacattg ggactgagat 300
acggcccgag 309

SEQ ID NO: 37      moltype = DNA length = 310
FEATURE           Location/Qualifiers
source            1..310
                  mol_type = genomic DNA
                  organism = Parabacteroides sp.

SEQUENCE: 37
gatgaacgct agcgacaggc ttaacacatg caagtcgagg ggcagcacag gtagcaatac 60
cgggtggcga ccggcgacg ggtgagtaac gcgtatgcaa cttacctatc agagggggat 120
aaccggcgga aagtcggact aataccgcat gaagcagggg ccccgcatgg ggataattgc 180
taaagattca tcgctgatag ataggcatgc gttccattag gcagttaggc gggtaacggc 240
ccaccaaaac gacgatggat aggggttctg agaggaaggt cccccacatt ggtactgaga 300
cacggaccaa 310

SEQ ID NO: 38      moltype = DNA length = 316
FEATURE           Location/Qualifiers
source            1..316
                  mol_type = genomic DNA
                  organism = Bacteroides sp.

SEQUENCE: 38
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcattc cagtttgctt 60
gcaaaactgga gatggcgacc ggcgcacggg tgagtaacac gtatccaacc tgccgataac 120
tcggggatag cctttcgaaa gaaagattaa taccggatgg cataatagaa ccgcatgggt 180
tgattattaa agaatttcgg ttatcgatgg ggatgcgttc cattagcgag ttggtgggggt 240
aacggccccc caaaccttcg atggataggg gttctgagag gaagggtccc cacattggaa 300
ctgagacacg gtccaa 316

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-continued

SEQ ID NO: 39 moltype = DNA length = 314
 FEATURE Location/Qualifiers
 source 1..314
 mol_type = genomic DNA
 organism = Bacteroides sp.

SEQUENCE: 39

gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcacagga agaaagcttg	60
ctttctttgc tggcgacggg cgcacgggtg agtaacacgt atccaacctg ccctttactc	120
ggggatagcc ttctgaaaga aagattaata cccgatagca taatgattcc gcatgggttc	180
attattaaag gattccggta aaggatgggg atgctgtcca ttaggttgtt ggtgaggtaa	240
cggcccacca agccttcgat ggataggggt tctgagagga aggtcccca cattggaact	300
gagacacggt ccaa	314

SEQ ID NO: 40 moltype = DNA length = 309
 FEATURE Location/Qualifiers
 source 1..309
 mol_type = genomic DNA
 organism = Bifidobacterium sp.

SEQUENCE: 40

gatgaacgct ggcggcggtgc ttaacacatg caagtcgaac gggatccatc aggctttgct	60
tggtggtgag agtggcgaaac gggtagtaaa tgcgtgaccg acctgcccc tacaccggaa	120
tagctctcgg aaacgggtgg taatgccgga tgcctcggact cctcgcatgg ggtgtcggga	180
aagatttcat cggtatggga tggggtcgcg tcctatcagg tagtcggcgg ggtaacggcc	240
caccgagcct acgacgggta gccggcctga gagggcgacc ggccacattg ggactgagat	300
acggcccag	309

SEQ ID NO: 41 moltype = DNA length = 311
 FEATURE Location/Qualifiers
 source 1..311
 mol_type = genomic DNA
 organism = Faecalibacterium sp.

SEQUENCE: 41

gacgaacgct ggcggcgcgcg ctaacacatg caagtcgaac gagcgagaga gagcttgctt	60
tctcaagcga gtggcgaaacg ggtgagtaac gcgtgaggaa cctgcctcaa agagggggac	120
aacagttgga aacgactgct aataccgcat aagcccacga cccggcatcg ggtagaggga	180
aaaggagcaa tccgctttga gatggcctcg cgtccgatta gctagttggt gaggtaatgg	240
cccaccaagg cgcagatcgg tagccggact gagaggttga acggccacat tgggactgag	300
acacggccca g	311

SEQ ID NO: 42 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 note = Firmicutes
 organism = unidentified

SEQUENCE: 42

gatgaacgct ggcggcggtgc ttaacacatg caagtcgagc gaagcagtta agttgatttc	60
ttcggattga ttcttaactg actgagcggc ggacgggtga gtaacgcgtg ggtgacctgc	120
cccataccgg gggataacag ctggaaacgg ctgctaatac cgcataagcg cacagagctg	180
catggctcgg tgtgaaaaac tccggtggta tgggatgggc ccgctgtctg ttaggcagt	240
ggcggggtaa cggcccacca aaccgacgat cagtagccgg cctgagaggg cgaccggcca	300
cattgggact gagacacggc ccaa	324

SEQ ID NO: 43 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 organism = Blautia sp.

SEQUENCE: 43

gatgaacgct ggcggcggtgc ttaacacatg caagtcgagc gaagcacctt gacggatttc	60
ttcggattga agccttggtg actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc	120
ctcataccgg gggataacag ttggaaacgg ctgctaatac cgcataagcg cacagatccg	180
catgttacgg tgtgaaaaac tccggtggta tgagatggac ccgctgtctg ttagctagt	240
ggtggggtaa cggcctacca agcggatgat cagtagccgg tctgagagga tgaacggcca	300
cattgggact gagacacggc ccag	324

SEQ ID NO: 44 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 note = Lachnoclostridium
 organism = unidentified

SEQUENCE: 44

gatgaacgct ggcggcggtgc ctaacacatg caagtcgaac gaagcaatta aaatgaagtt	60
ttcggatgga tttttgattg actgagtgcc ggacgggtga gtaacgcgtg gataacctgc	120
ctcacactgg gggataacag ttgaaatga ctgctaatac cgcataagcg cacagtaccg	180

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catggtagcgg tgtgaaaaac tccgggtggtg tgagatggat ccgcgtctga ttagccagtt 240
ggcgggggtaa cggccacca aagcgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa 324

```

```

SEQ ID NO: 45      moltype = DNA length = 321
FEATURE           Location/Qualifiers
source            1..321
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 45
gatgaacgct ggcggcatgc ctaatacatg caagtcgaac gaagtttcga ggaagcttgc 60
ttccaaagag acttagtggc gaacgggtga gtaacacgta ggtaacctgc ccatgtgtcc 120
gggataactg ctggaaacgg tagctaaaac cggataggtg tacagagcgc atgctcagta 180
tattaaagcg cccatcaagg cgtgaacatg gatggacctg cggcgcatga gctagtgtgt 240
gaggtaacgg cccaccaagg cgatgatgcg tagccggcct gagagggtaa acggccacat 300
tgggactgag acacggccca a 321

```

```

SEQ ID NO: 46      moltype = DNA length = 326
FEATURE           Location/Qualifiers
source            1..326
                  mol_type = genomic DNA
                  organism = Anaerostipes sp.

```

```

SEQUENCE: 46
gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac gaaacacctt atttgatttt 60
cttcggaact gaagatttgg tgattgagtg gcggacgggt gagtaacgcg tgggtaacct 120
gcccctgtaca gggggataaac agtcagaaat gactgctaata accgcataag accacagcac 180
cgcatggtgc aggggtaaaa actccgggtg tacaggatgg acccgctct gattagctgg 240
ttggtgaggt aacggctcac caaggcgacg atcagtagcc ggcttgagag agtgaacggc 300
cacattggga ctgagacacg gcccga 326

```

```

SEQ ID NO: 47      moltype = DNA length = 324
FEATURE           Location/Qualifiers
source            1..324
                  mol_type = genomic DNA
                  note = Lachnospiraceae
                  organism = unidentified

```

```

SEQUENCE: 47
gatgaacgct ggcggcgtgc ctaacacatg caagtcgagc gaagcacttt gcttagattc 60
ttcggatgaa gaggattgtg actgagcggc ggcgggggtga gtaacgcgtg ggtaacctgc 120
ctcatacagg ggggataacag ttagaatatg ctgctaatac cgcataagac cacagacagg 180
catggtgcgg gggtaaaaaac tccgggtggt tgagatggac ccgcgtctga ttagctagtt 240
ggtaaggtaa cggcttacca agcgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa 324

```

```

SEQ ID NO: 48      moltype = DNA length = 311
FEATURE           Location/Qualifiers
source            1..311
                  mol_type = genomic DNA
                  organism = Faecalibacterium sp.

```

```

SEQUENCE: 48
gacgaacgct ggcggcgcgc ctaacacatg caagtcgaac gagagagagg aagcttgctt 60
cctcaatcga gtggcgaaac ggtgagtaac gcgtgaggaa cctgcctcaa agagggggac 120
aacagttgga aacgactgct aataccgcat aagcccacag gtcggcatcg accagaggga 180
aaaggagcaa tccgctttga gatggcctcg cgtccgatta gctagtgtgt gaggtaacgg 240
cccaccaagg caacgatcgg tagccggact gagaggttga acggccacat tgggactgag 300
acacggccca g 311

```

```

SEQ ID NO: 49      moltype = DNA length = 323
FEATURE           Location/Qualifiers
source            1..323
                  mol_type = genomic DNA
                  organism = Blautia sp.

```

```

SEQUENCE: 49
gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac gggaaacctt ttattgaagc 60
ttcggcagat ttagctgggt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatacaggg ggataacaac cagaaatggt tgctaatacc gcataagcgc acaggaccgc 180
atggtccggg gtgaaaaaact ccgggtggat aagatggacc cgcgttggat tagctagtgt 240
gcagggtaac ggctaccac ggcgacgat catagccggc ctgagagggg gaacggccac 300
attgggactg agacacggcc cag 323

```

```

SEQ ID NO: 50      moltype = DNA length = 317
FEATURE           Location/Qualifiers
source            1..317
                  mol_type = genomic DNA
                  organism = Barnesiella sp.

```

```

SEQUENCE: 50

```

-continued

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gatgaacgct agcgacaggg ctaacacatg caagtcgagg ggcagcgaag aggtagcaat 60
acctttgtcg gcgaccggcg cacgggtgag taacacgtat gcaatccacc tgtaacaggg 120
ggataacccg gagaaatccg gactaatacc ccataatatg ggcgctccgc atggagagcc 180
cattaagag agcaattttg gttacagacg agcatgcgct ccattagcca gttggcgggg 240
taacggccca ccaagcgac gatggatagg ggttctgaga ggaagggtccc ccacattgga 300
actgagacac ggtccaa 317

```

```

SEQ ID NO: 51      moltype = DNA length = 324
FEATURE           Location/Qualifiers
source            1..324
                  mol_type = genomic DNA
                  organism = Roseburia sp.

```

```

SEQUENCE: 51
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcacttt gatcgatttc 60
ttcggattga aatttttagtg actgagtggtg ggcgggtgga gtaacgcgtg ggtaacctgc 120
ctcacacagg gggataacag ttggaacagg ctgctaatac cgcataagcg cacagtaccg 180
catggtacag tgtgaaaaac tccggtgggt tgagatggac ccgctctgga ttagctagtt 240
ggcagggcaa cggcctacca aggcgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa 324

```

```

SEQ ID NO: 52      moltype = DNA length = 316
FEATURE           Location/Qualifiers
source            1..316
                  mol_type = genomic DNA
                  organism = Bilophila sp.

```

```

SEQUENCE: 52
attgaacgct ggcggcgctgc ttaacacatg caagtcgaac gtgaaagtcc ttcgggatga 60
gtaaagtggc gcacgggtgga gtaacgcgtg gataatctac ccttaagatg gggataacgg 120
ctggaaacgg tcgctaatac cgaatacgct cccgatttta tcgttggggg gaaagatggc 180
ctctgcttgc aagctatcgc ttaaggatga gtccgcgtcc cattagctag ttggcggggt 240
aacggcccac caaggcgacg atgggtagcc ggtctgagag gatgaccggc cacactggaa 300
ctggaacacg gtccag 316

```

```

SEQ ID NO: 53      moltype = DNA length = 311
FEATURE           Location/Qualifiers
source            1..311
                  mol_type = genomic DNA
                  organism = Actinomyces sp.

```

```

SEQUENCE: 53
gacgaacgct ggcggcgctgc ttaacacatg caagtcgaac gctgaaggct cagcttgctg 60
ggctggatga gtggcgaaac ggtgagtaac acgtgagtaa cctgccctct tctttgggat 120
aacggtcggg aacgggtgct aataccgggt attcactgtc cttcgcatgg ggggtgggtg 180
aaagggtttt tctgggtggg gatgggctcg cggcctatca gcttgttggg ggggtgatgg 240
cttaccgaag ctttgacggg taaccggcct gagagggtga ccggtcacat tgggactgag 300
atacggccca g 311

```

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SEQ ID NO: 54      moltype = DNA length = 323
FEATURE           Location/Qualifiers
source            1..323
                  mol_type = genomic DNA
                  organism = Ruminococcus sp.

```

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SEQUENCE: 54
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggaaatact ttattgaaac 60
ttcggtggtt ttaatttatt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatactggg ggataacacg cagaaatgac tgctaatacc gcataagcgc acagaaccgc 180
atggttcggt gtgaaaaact ccggtgggtat aagatggacc cgcgttggat tagctagttg 240
gcagggcagc ggcctaccaa ggcgacgatc catagccggc ctgagagggg gaacggccac 300
attgggactg agacacggcc cag 323

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SEQ ID NO: 55      moltype = DNA length = 325
FEATURE           Location/Qualifiers
source            1..325
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 55
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggagtcgttt tggaaaaatcc 60
ttcgggattg gaattctcga cttagtggcg gacgggtgag taacgcgtga gcaatctgcc 120
tttaagaggg ggataacagt cggaaacggc tgctaatacc gcataaagca atgaattcgc 180
atgttttcat tgccaaagga gcaatccgct tttagatgag ctgcgctctg attagctagt 240
tggcggggta acggccacc aaggcgacga tcagtagccg gactgagagg ttgaacggcc 300
acattgggac tgagacacgg cccag 325

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SEQ ID NO: 56      moltype = DNA length = 325
FEATURE           Location/Qualifiers
source            1..325
                  mol_type = genomic DNA

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                                organism = Butyrivicoccus sp.
SEQUENCE: 56
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggaactatgt tggaaactcc 60
ttcgggagcg gaattcttag tttagtggtg gacgggtgag taacgcgtga gcaatctgcc 120
tttaagaggg ggataaacagt cggaaacggc tgctaatacc gcataaagca tcaaattcgc 180
atgtttttga tgcacaagga gcaatccgct tttagatgag ctccgctctg attagctggt 240
tggcggggta acggccacc aaggcgacga tcagtagccg gactgagagg ttgaacggcc 300
acattgggac tgagacacgg cccag 325

SEQ ID NO: 57      moltype = DNA length = 311
FEATURE           Location/Qualifiers
source            1..311
                  mol_type = genomic DNA
                  organism = Faecalibacterium sp.

SEQUENCE: 57
gacgaacgct ggcggcgctgc ctaacacatg caagtcgaac gagcgagaga gagcttgctt 60
tctcaagcga gtggcgaaacg ggtgagtaac gcgtgaggaa cctgcctcaa agagggggac 120
aacagttgga aacgactgct aataccgcat aagccacggc gtcggcatcg accagaggga 180
aaaggagtga tccgctttga gatggcctcg cgtccgatta gctagttggt gaagtaacgg 240
cccaccaagg caacgatcgg tagccggact gagagggtga acggccacat tgggactgag 300
atacggccca g 311

SEQ ID NO: 58      moltype = DNA length = 326
FEATURE           Location/Qualifiers
source            1..326
                  mol_type = genomic DNA
                  organism = Megaspheera sp.

SEQUENCE: 58
gacgaacgct ggcggcgctgc ttaacacatg caagtcgaac gagaagagat gagaagcttg 60
cttcttatca attcgagtgg caaacgggtg agtaacgcgt aagcaacctg cccttcagat 120
ggggacaaca gctggaaacg gctgctaata ccgaatacgt tctttccgcc gcattgacggg 180
atgaagaaag gggcctttcg tggaggaggg gcttgctctc gattagctag 240
ttggaggggg aacggccacc caaggcgacg atcagtagcc ggtctgagag gatgaacggc 300
cacattggga ctgagacacg gccag 326

SEQ ID NO: 59      moltype = DNA length = 315
FEATURE           Location/Qualifiers
source            1..315
                  mol_type = genomic DNA
                  organism = Bacteroides sp.

SEQUENCE: 59
gatgaacgct agctacaggg ttaacacatg caagtcgagg ggcagcatgg tcttagcttg 60
ctaaggccga tggcgaccgg cgcacgggtg agtaacgcgt atccaacctg ccttacactc 120
ttggacagcc ttctgaaagg gagattaata caagatgta tcatgagtaa gcattttcgc 180
atgattaaag gtttaccggt gtaagatggg gatgcgttcc attagatagt aggcggggta 240
acggccacc tagtcttcga tggatagggg ttctgagagg aaggtcccc acattggaac 300
tgagacacgg tccaa 315

SEQ ID NO: 60      moltype = DNA length = 315
FEATURE           Location/Qualifiers
source            1..315
                  mol_type = genomic DNA
                  organism = Bacteroides sp.

SEQUENCE: 60
gatgaacgct agctacaggg ttaacacatg caagtcgagg ggcagcggga ttgaagcttg 60
cttcaattgc cggcgaccgg cgcacgggtg agtaacgcgt atccaacctt ccgtacactc 120
agggatagcc ttctgaaagg aagattaata cctgatgta tcttaagcac acatgtaatt 180
aagattaaag atttatcggt gtacgatggg gatgcgttcc attaggtagt aggcggggta 240
acggccacc tagccaacga tggatagggg ttctgagagg aaggtcccc acattggaac 300
tgagacacgg tccaa 315

SEQ ID NO: 61      moltype = DNA length = 313
FEATURE           Location/Qualifiers
source            1..313
                  mol_type = genomic DNA
                  organism = Prevotella sp.

SEQUENCE: 61
gatgaacgct ggcggcgctgc ttaacacatg caagtcgagg ggcagcatgt tggttgcttg 60
caaccgatga tggcgaccgg cgcacgggtg agtaacgcgt atccaacctt cccttgctca 120
tcggataacc cgtcgaaagg cggcctaaca cgatatcggg ttcacagcgg gcattcaacg 180
tgaacgaaat gtgaaggaga aggatgggga tgcgtctgat tagcttggtg gcggggtaac 240
ggccaccaca ggcaacgatc agtaggggtt ctgagaggaa gggtccccac attggaactg 300
agacacggtc caa 313

SEQ ID NO: 62      moltype = DNA length = 326
FEATURE           Location/Qualifiers

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source                1..326
                      mol_type = genomic DNA
                      organism = Anaerostipes sp.

SEQUENCE: 62
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagctgctt aactgatctt 60
cttcggaatt gacgttttgt agactgagtg gcggacgggt gagtaacgcg tgggtaacct 120
gccctgtaca gggggataac agtcagaaat gactgctaata accgcataag accacagcac 180
cgcatgggtgc aggggtaaaa actccgggtg tacaggatgg acccggtctt gattagctgg 240
ttggtgaggt aacggctcac caaggcgacg atcagtagcc ggcttgagag agtgaacggc 300
cacattggga ctgagacacg gcccaa 326

SEQ ID NO: 63         moltype = DNA length = 313
FEATURE              Location/Qualifiers
source               1..313
                      mol_type = genomic DNA
                      organism = Alistipes sp.

SEQUENCE: 63
gatgaacgct agcggcgaggc ttaacacatg caagtcgagg ggcagcataa tggatagcaa 60
tatctatggt ggcgacgggc gcacgggtgc gtaacgcgta tgcaacctac ctttaacagg 120
gggataaacac tgagaaattg gtactaatac ccataaatat catagaaggc atcttttatg 180
gttgaataatt ccgatgggtta gagatgggca tgcgttgat tagctagtgg gtggggtaac 240
ggctcaccaa ggcgacgata cataggggga ctgagaggtt aacccccac actggtactg 300
agacacggac cag 313

SEQ ID NO: 64         moltype = DNA length = 310
FEATURE              Location/Qualifiers
source               1..310
                      mol_type = genomic DNA
                      organism = Parabacteroides sp.

SEQUENCE: 64
gatgaacgct ggcggcgctgc ttaacacatg caagtcgagg ggcagcgggg gtagcaatac 60
ccgcggcgga cgcggcgacg ggtgagtaac gcgtatgcaa cttacctatc agagggggat 120
aaccggcgga aagtcggact aataccgcat gaagcaggga tcccgcatgg gaatatattg 180
taaagattca tcgctgatat ataggcatgc gttccattag gcagttggcg gggtaacggc 240
ccaccaaacc gacgatggat agggggttctg agaggaaggt cccccacatt ggtactgaga 300
cacggaccaa 310

SEQ ID NO: 65         moltype = DNA length = 323
FEATURE              Location/Qualifiers
source               1..323
                      mol_type = genomic DNA
                      note = Lachnospiraceae
                      organism = unidentified

SEQUENCE: 65
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggaaatact tcattgagac 60
ttcgggtggat ttgatttatt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatacagggg ggataacagt cagaaatgac tgctaatacc gcataagcgc acaggaccgc 180
atggtccggt gtgaaaaact ccggtgggtat aagatggacc cgcgttggtat tagctgggtg 240
gtggggtaac gggccaccaa ggcgacgata catagccggc ctgagagggg gaacggccac 300
attgggactg agacacggcc cag 323

SEQ ID NO: 66         moltype = DNA length = 314
FEATURE              Location/Qualifiers
source               1..314
                      mol_type = genomic DNA
                      organism = Bacteroides sp.

SEQUENCE: 66
gatgaacgct gacagaatgc ttaacacatg caagtcgagg ggcatcagga agaaagcttg 60
ctttctttgc tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccctttactc 120
ggggatagcc ttctgaaaga aagattaata cccgatggta taatgattcc gcattggttg 180
attattaaag gattccggtg aaggatgggg atgcgttcca ttaggttggt ggtgaggtaa 240
cggctcacca agccttcgat ggataggggt tctgagagga aggtccccc cattggaact 300
gagacacggt ccaa 314

SEQ ID NO: 67         moltype = DNA length = 325
FEATURE              Location/Qualifiers
source               1..325
                      mol_type = genomic DNA
                      organism = Clostridium sp.

SEQUENCE: 67
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac ggggattatt ttgacagaga 60
cttcgggttg agtcgttata atcctagtgg cggacgggtg agtaacgcgt gggtaacctg 120
cctcacactg ggggataaca gtcagaaatg actgctaata ccgcataagc gcacgggatt 180
gcatgatcca gtgtgaaaaa ctccgggtgt gtgagatgga cccgcgttgg attagccagt 240
tggcagggtg acggcctacc aaagcgacga tccatagccg gcctgagagg gtggacggcc 300
acattgggac tgagacacgg cccag 325

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SEQ ID NO: 68 moltype = DNA length = 325
 FEATURE Location/Qualifiers
 source 1..325
 mol_type = genomic DNA
 note = Flavonifractor
 organism = unidentified

SEQUENCE: 68

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgaac	ggggtgctca	tgacggagga	60
ttcgtccaac	ggattgagtt	acctagtggc	ggacgggtga	gtaacgcgtg	aggaacctgc	120
cttgagaggg	ggaataaacac	tccgaaagga	gtgctaatac	cgcataagc	agttgggtcg	180
catggctctg	actgccaaag	atttatcgct	ctgagatggc	ctcgcgtctg	attagctagt	240
agggggggtg	acggcccacc	taggcgacga	tcagtagccg	gactgagagg	ttgaccggcc	300
acattgggac	tgagacacgg	cccg				325

SEQ ID NO: 69 moltype = DNA length = 326
 FEATURE Location/Qualifiers
 source 1..326
 mol_type = genomic DNA
 organism = Clostridium sp.

SEQUENCE: 69

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgaac	gaagcacttt	acctgatttc	60
cttcggggat	gattgttctg	tgactgagtg	gcggacgggt	gagtaacgcg	tgataaacct	120
gcctcacaca	gggggataac	agttggaaac	gactgctaata	accgcataag	cgcacagcac	180
cgcattggtg	agtgtgaaaa	actccggtgg	tgtgagatgg	atccgcgtct	gattagctgg	240
ttggcggggg	aacggcccac	caaggcgacg	atcagtagcc	ggcctgagag	ggcgaccggc	300
cacattggga	ctgagacacg	gcccag				326

SEQ ID NO: 70 moltype = DNA length = 326
 FEATURE Location/Qualifiers
 source 1..326
 mol_type = genomic DNA
 organism = Eubacterium sp.

SEQUENCE: 70

gatgaacgct	ggcggcgctgc	ttaatacatg	caagtcgaac	gaagcacctt	ggacagaatc	60
cttcggggag	aagaccattg	tgactgagtg	gcggacgggt	gagtaacgcg	tggttaacct	120
gccttgtaga	gggggataac	agttggaaac	gactgctaata	accgcataag	cgcacagtac	180
cgcattggtg	agtgtgaaaa	actccggtgg	tacaagatgg	accgcgtctc	gattagctgg	240
ttggtgaggt	aacggcccac	caaggcgacg	atcagtagcc	gacttgagag	agtgatcggc	300
cacattggga	ctgagacacg	gcccac				326

SEQ ID NO: 71 moltype = DNA length = 309
 FEATURE Location/Qualifiers
 source 1..309
 mol_type = genomic DNA
 organism = Clostridium sp.

SEQUENCE: 71

gacgaacgct	ggcggcgacgc	ttaacacatg	caagtcgaac	gagagaagag	aagcttgctt	60
ttctgatcta	gtggcgagcg	ggtgagtaac	acgtgagcaa	tctgcctttc	agagggggat	120
accgattgga	aacgatcggt	aataccgcac	aacataattg	aaccgcacga	tttgattatc	180
aaagatttat	cgtgaaaaga	tgagctcgcg	tctgattagc	tagttggtaa	ggtaacggct	240
taccaaggcg	acgatcagta	gccggactga	gaggttgatc	ggccacattg	ggactgagac	300
acggcccag						309

SEQ ID NO: 72 moltype = DNA length = 323
 FEATURE Location/Qualifiers
 source 1..323
 mol_type = genomic DNA
 note = Lachnospiraceae
 organism = unidentified

SEQUENCE: 72

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgaac	gggaacatt	tcattgaggg	60
ttcggcggtg	ttgatctggt	tctagtggcg	gacgggtgag	taacgcgtgg	gtaacctgcc	120
ttatactggg	ggataacagc	cagaaatgac	tgctaatacc	gcataagcgc	acagaaccgc	180
atggttcggt	gtgaaaaact	ccggtgggtg	aagatggacc	cgcgttggtg	tagctgggtg	240
gcagggcagc	ggcctaccac	ggcgacgac	catagccggc	ctgagagggg	gaacggccac	300
attgggactg	agacacggcc	cag				323

SEQ ID NO: 73 moltype = DNA length = 323
 FEATURE Location/Qualifiers
 source 1..323
 mol_type = genomic DNA
 organism = Ruminococcus sp.

SEQUENCE: 73

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgaac	gggaattact	ttattgaaac	60
ttcgggtcgat	ttattttaat	tccagtggcg	gacgggtgag	taacgcgtgg	gtaacctgcc	120

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ttatacaggg ggataacagt tagaaatggc tgctaatacc gcataagcgc acagagctgc 180
atggctcagc gcgaaaaact cgggtgggtat aagatggacc cgcgttggtat tagctgggtg 240
gtggggtaac ggccccacaa ggcgacgatc catagccggc ctgagagggt gaacggccac 300
attgggactg agacacggcc cag                                     323

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```

SEQ ID NO: 74      moltype = DNA length = 309
FEATURE           Location/Qualifiers
source            1..309
                  mol_type = genomic DNA
                  organism = Coprococcus sp.

```

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SEQUENCE: 74
gatgaacgct ggcgggcgcg ctaacacatg caagtgcgaac ggacgatgaa gagcttgctt 60
ttcaaagtta gtggcgagcg ggtgagtaac gcgtgggtaa cctgcctcat acaggggaat 120
agcagctgga aacggctgat aaaaccgcat aagcgcacag tgccgcatgg cacagtgtga 180
aaaactccgg tggatgaga tggaccgcg tctgattagc ttgttggcgg ggtaacggcc 240
caccaaggcg acgatcagta gccggcctga gagggtgacc ggccacattg ggactgagac 300
acggcccaa                                     309

```

```

SEQ ID NO: 75      moltype = DNA length = 310
FEATURE           Location/Qualifiers
source            1..310
                  mol_type = genomic DNA
                  organism = Bacteroides sp.

```

```

SEQUENCE: 75
gatgaacgct agctacaggc ttaacacatg caagtgcagg ggcagcatga cctagcaata 60
ggttgatggc gaccggcgca cgggtgagta acacgtatcc aacctaccgg ttattccggg 120
atagccttcc gaaagaaaga ttaataccgg atagtataac gagaaggcat cttttgttta 180
ttaagaattc tcgataaccg atgggggatgc gttccattag ttgttggcg gggtaacggc 240
ccaccaagac atcgatggat aggggttctg agaggaaggt cccccacatt ggaactgaga 300
cacggtccaa                                     310

```

```

SEQ ID NO: 76      moltype = DNA length = 316
FEATURE           Location/Qualifiers
source            1..316
                  mol_type = genomic DNA
                  organism = Parabacteroides sp.

```

```

SEQUENCE: 76
gatgaacgct agcgacaggc ttaacacatg caagtgcagg ggcagcatag taggtagcaa 60
tacttattgg tggcgaccgg cgcacgggtg agtaacgcgt atgcaactta cctatcagag 120
gggggatgcc cggcgaaaagt cggattaata ctccataaaa cagggggtccc gcatgggact 180
atttgtaaaa gattcatcgc tgatagatag gcatgcgttc cattagcgac ttggcggggg 240
aacggcccac caaaccgacg atggataggg gttctgagag gaagggtccc caccattggga 300
ctgagacacg gaccaa                                     316

```

```

SEQ ID NO: 77      moltype = DNA length = 316
FEATURE           Location/Qualifiers
source            1..316
                  mol_type = genomic DNA
                  organism = Bacteroides sp.

```

```

SEQUENCE: 77
gatgaacgct agctacaggc ttaacacatg caagtgcagg ggcagcattt tagtttgctt 60
gcaaaactaaa gagttgcgaa cggcgacggg tgagtaaacac gtatccaacc tgccgataac 120
tcggggatag cctttcgaaa gaaagattaa tatccgatag tatattaaaa ccgcatgggt 180
ttactattaa agaatttcgg ttatcgatgg ggatgcgttc cattagtttg ttggcggggg 240
aacggcccac caagactacg atggataggg gttctgagag gaagggtccc caccattggaa 300
ctgagacacg gtccaa                                     316

```

```

SEQ ID NO: 78      moltype = DNA length = 316
FEATURE           Location/Qualifiers
source            1..316
                  mol_type = genomic DNA
                  organism = Streptococcus sp.

```

```

SEQUENCE: 78
gatgaacgct ggcgggcgct ctaatacatg caagtagaac gctgaagctt ggtgcttgca 60
ccgagcggat gagttgcgaa cgggtgagta acgctaggtt aacctgcctc ttacgggggg 120
ataactattg gaaacgatag ctaataccgc ataaaagtgc atatcgcatg atattgattt 180
gaaagggtgc attgcatcac taagagatgg acctgcgttg tattagctag ttggtgaggt 240
aacggctcac caaggcgacg atacatagcc gacctgagag ggtgatcgcc cacactggga 300
ctgagacacg gcccag                                     316

```

```

SEQ ID NO: 79      moltype = DNA length = 313
FEATURE           Location/Qualifiers
source            1..313
                  mol_type = genomic DNA
                  organism = Prevotella sp.

```

```

SEQUENCE: 79

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gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatgt tggttgcttg 60
caaccaatga tggcgacggg cgcacgggtg agtaacgcgt atccaacctt cccttggtcca 120
tcggataaac cgtcgaaagg cggcctaaca cgatatcgcg ttctcagcag gcatctaacy 180
agaacgaaat gtgaaggaga aggatgggga tgcgtctgat tagcttggtg gcggggtaac 240
ggcccaccaa ggcgacgatc agtaggggtt ctgagaggaa ggccccccac attggaactg 300
agacacggctc caa 313

```

```

SEQ ID NO: 80      moltype = DNA length = 315
FEATURE           Location/Qualifiers
source            1..315
                  mol_type = genomic DNA
                  organism = Bacteroides sp.

```

```

SEQUENCE: 80
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcgga ttgaagcttg 60
cttcaattgc cggcgacggg cgcacgggtg agtaacgcgt atccaacctt ccgtacactc 120
agggatagcc ttctgaaaga aagattaata cctgatggta tgatgagatt gcatgatagc 180
atcattaaag atttcgggtt gtacgatggg gatgcgttcc attaggtagt aggcggggta 240
acggcccccac tagccgacga tggatagggg ttctgagagg aaggtccccc acattggaac 300
tgagacacgg tccaa 315

```

```

SEQ ID NO: 81      moltype = DNA length = 324
FEATURE           Location/Qualifiers
source            1..324
                  mol_type = genomic DNA
                  note = Lachnospiraceae
                  organism = unidentified

```

```

SEQUENCE: 81
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcgctta aatggatttc 60
ttcggattga agtttttctg actgagtggtc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcatcacagg gggataacag ttgaaatga ctgctaatac cgcataagcg cacagtgtcg 180
catggcacag tgtgaaaaaac tccggtggta tgagatggac ccgcgtctga ttagctagtt 240
ggtaggggtaa cggcctacca aggcgacgat cagtagccgg cctgagaggg tgaacggcca 300
cattgggact gagacacggc ccaa 324

```

```

SEQ ID NO: 82      moltype = DNA length = 307
FEATURE           Location/Qualifiers
source            1..307
                  mol_type = genomic DNA
                  organism = Bifidobacterium sp.

```

```

SEQUENCE: 82
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggatcccag gagcttgctc 60
ctgggtgaga gtggcgaaac ggtgagtaat gcgtgaccga cctgccccat acaccggaat 120
agctcctgga aacgggtggg aatgccggat gctccgacat gacgcatgtc gtgtcgggaa 180
agattcatcg gtatgggatg gggtcgcgtc ctatcagctt gatggcgggg taacggccca 240
ccatgggttc gacgggtagc cggcctgaga gggcgaccgg ccacattggg actgagatac 300
ggcccag 307

```

```

SEQ ID NO: 83      moltype = DNA length = 322
FEATURE           Location/Qualifiers
source            1..322
                  mol_type = genomic DNA
                  note = Lachnospiraceae
                  organism = unidentified

```

```

SEQUENCE: 83
gatgaacgct ggcggcgctgc ttaacacatg caagtcgagc gaagcggtcg aattgattct 60
tcggatgatt ttcagttgac tgagcggcgg acgggtgagt aacgcgtggg tgacctgccc 120
cataccgggg gataacagct ggaacaggct gctaataccg cataagcgca cagagctgca 180
tggctcggtg tgaaaaactc cgggtggtat ggatggggcc gcgtctgatt aggcagtttg 240
cggggtaacg gcccacaaa cggacgatca gtagccggcc tgagagggcg accggccaca 300
ttgggactga gacacggccc aa 322

```

```

SEQ ID NO: 84      moltype = DNA length = 308
FEATURE           Location/Qualifiers
source            1..308
                  mol_type = genomic DNA
                  organism = Sutterella sp.

```

```

SEQUENCE: 84
attgaacgct ggcggcgtgc tttacacatg caagtcgaac ggcagcgcgg ggagcttgct 60
ccctggcgcg gagtggcgca cgggtgagta atacatcgga acgtgtcttc tagtggggga 120
taactgcccc aaagggcagc taataccgca tgagacctga gggtgaaagc gggggatcgc 180
aagacctcgc gctggaagag cggccgatgt ccgattagct agttggtgag gtaaaggctc 240
accaaggcga cgatcggtag ctggtctgag aggcagacca gccacactgg gactgagaca 300
cgcccag 308

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SEQ ID NO: 85      moltype = DNA length = 324
FEATURE           Location/Qualifiers

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-continued

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source                1..324
                      mol_type = genomic DNA
                      organism = Blautia sp.

SEQUENCE: 85
gatgaacgct ggcggcgtgc ttaacacatg caagtcgagc gaagcacctt gacggatttc 60
ttcggattga agccttggtg actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcgtacagg gggataacag ttggaacagg ctgctaatac cgcataagcg cacagtaccg 180
catggtagcg tgtgaaaaac tccggtggta tgagatggac ccgcgtctga ttaggtagtt 240
ggtggggtaa cggcctacca agccgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa                                     324

SEQ ID NO: 86         moltype = DNA length = 311
FEATURE              Location/Qualifiers
source               1..311
                      mol_type = genomic DNA
                      organism = Faecalibacterium sp.

SEQUENCE: 86
gacgaacgct ggcggcgcgc ctaacacatg caagtcgaac gacgcgagaga aagcttgctt 60
tctcgagcga gtggcgaaacg ggtgagtaac gcgtgaggaa cctgcctcaa agagggggac 120
aacagttgga aacgactgct aataccgcgt aagccacagg tgccgcattg cacagaggga 180
aaaggagcaa tccgctttga gatggcctcg cgtccgatta gctagttggt gaggtaatgg 240
cccaccaagg cgacgatcgg tagccggact gagaggttga acggccacat tgggactgag 300
acacggccca g                                     311

SEQ ID NO: 87         moltype = DNA length = 324
FEATURE              Location/Qualifiers
source               1..324
                      mol_type = genomic DNA
                      organism = Clostridium sp.

SEQUENCE: 87
gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac gaagcacttt ggacagattc 60
ttcggatgaa gtctcttagtg actgagtggc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcatacagg gggataacag ttagaatagg ctgctaatac cgcataagcg caccgtaccg 180
catggtacag tgtgaaaaac tccggtggta tgagatggac ccgcgttga ttagctagtt 240
ggcagggtaa cggcctacca aggcgacgat ccatagccgg cctgagaggg tggacggcca 300
cattgggact gagacacggc ccag                                     324

SEQ ID NO: 88         moltype = DNA length = 325
FEATURE              Location/Qualifiers
source               1..325
                      mol_type = genomic DNA
                      organism = Clostridium sp.

SEQUENCE: 88
gacgaacgct ggcggcgtgc ctaacacatg caagtcgaac ggaacttttt taaaggattt 60
cttcggaatg aatttgatta agtttagtgg cggacgggtg agtaacgcgt gagtaacctg 120
cctctaagag gggataaaca ttctgaaaag aatgctaata ccgcataaca tatatttatc 180
gcatggtaga tatataaag atttatcgct tagagatgga ctgcgctccg attagttagt 240
tggtgaggtg acggtcacc aagaccgcga tcggtagccg gactgagagg ttgaacggcc 300
acattgggac tgagacacgg cccag                                     325

SEQ ID NO: 89         moltype = DNA length = 316
FEATURE              Location/Qualifiers
source               1..316
                      mol_type = genomic DNA
                      organism = Parasutterella sp.

SEQUENCE: 89
attgaacgct ggcggaacgc tttacacatg caagtcgaac ggtaacgcgg agagaagctt 60
gcttctctcc ggcgacgagt ggcgaacggg tgagtaatac atcggaacgt gtccgctcgt 120
gggggacaac cagccgaaag gttggctaata accgcattgag ttctacggaa gaaagagggg 180
gaccgcgaag ggcctctcgc gagcggagcg gccgatgact gattagcctg ttggtgaggt 240
aacggctcac caaagcaacg atcagtagct ggtctgagag gacgaccagc cactactggga 300
ctgagacacg gcccag                                     316

SEQ ID NO: 90         moltype = DNA length = 322
FEATURE              Location/Qualifiers
source               1..322
                      mol_type = genomic DNA
                      note = Erysipelotrichales
                      organism = unidentified

SEQUENCE: 90
gatgaacgct ggcggcatgc ctaatacatg caagtcgaac ggagcgaata tggaagcttg 60
cttccgtaag agctcagtgg cgaacgggtg agtaaacagt aggtaacctg cccatgtgcc 120
cgggataact gctggaaacg gtatgtaaaa ccggataggt gaacaggagg catctcttat 180
tcattaaagg acctgtaaag gtgcgaacat ggatggacct gcggcgcatt agctggttgg 240
agtgttaacg gcacaccaag gcgacgatgc gtatgccgacc tgagaggggc aacggccaca 300
ttgggactga gacacggccc aa                                     322

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-continued

SEQ ID NO: 91 moltype = DNA length = 326
 FEATURE Location/Qualifiers
 source 1..326
 mol_type = genomic DNA
 organism = Eubacterium sp.

SEQUENCE: 91

gatgaacgct	ggcggcggtgc	ttaacacatg	caagtcgaac	gaagcattta	gaacagatta	60
tttcggtatg	aagtttttta	tgactgagtg	gcggaacggg	gagtaacgcg	tgggtaacct	120
gccttgtagt	gggggatagc	agctggaaac	ggctggtaat	accgcataag	cgcacaatgt	180
tgcatgacat	ggtgtgaaaa	actccggtgg	tataagatgg	accgcgtctc	gattagctag	240
ttggtgagat	aacagcccac	caaggcgacg	atcagtagcc	gacctgagag	ggtgaccggc	300
cacattggga	ctgagacacg	gcccag				326

SEQ ID NO: 92 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 organism = Coprococcus sp.

SEQUENCE: 92

gatgaacgct	ggcggcggtgc	ttaacacatg	caagtcgaac	gaagcacttt	aacctgattc	60
ttcggatgaa	ggtttttgtg	actgagtggc	ggacgggtga	gtaacgcgtg	ggtaacctgc	120
ctcacacagg	gggataacag	ttagaaatga	ctgctaatac	cgcataagac	cacggagccg	180
catggctcag	tgggaaaaac	tccggtggta	tgagatggac	ccgcgtctga	ttaggtagtt	240
ggtggggtaa	cggcctacca	agccaacgat	cagtagccga	cctgagaggg	tgaccggcca	300
cattgggact	gagacacggc	ccaa				324

SEQ ID NO: 93 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 note = Clostridiaceae
 organism = unidentified

SEQUENCE: 93

gatgaacgct	ggcggcggtgc	ttaacacatg	caagtcgaac	gaagcagttt	agcgggaagt	60
ttcggatgga	agttaaactg	actgagtggc	ggacgggtga	gtaacgcgtg	ggtaacctgc	120
ctcacacagg	gggacaacag	ttagaaatga	ctgctaatac	cgcataagcg	cacagcttcg	180
catgaagcag	tgtgaaaaac	tccggtggta	tgagatggac	ccgcgtctga	ttagctagtt	240
ggtgaggtaa	cggctcacca	agcgcacgat	cagtagccga	cctgagaggg	tgaccggcca	300
cattgggact	gagacacggc	ccaa				324

SEQ ID NO: 94 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 organism = Dorea sp.

SEQUENCE: 94

gatgaacgct	ggcggcggtgc	ttaacacatg	caagtcgagc	gaagcgcttt	gaaagattc	60
ttcggatgat	ttcctttgtg	actgagcggc	ggacgggtga	gtaacgcgtg	ggtaacctgc	120
ctcacacagg	gggataacag	ttagaaatga	ctgctaatac	cgcataagac	cacggtagcc	180
catggtacag	tggtaaaaac	tccggtggta	tgagatggac	ccgcgtctga	ttaggtagtt	240
ggtggggtaa	cggcctacca	agcgcacgat	cagtagccga	cctgagaggg	tgaccggcca	300
cattgggact	gagacacggc	ccag				324

SEQ ID NO: 95 moltype = DNA length = 323
 FEATURE Location/Qualifiers
 source 1..323
 mol_type = genomic DNA
 organism = Clostridium sp.

SEQUENCE: 95

gatgaacgct	ggcggcggtgc	ctaacacatg	caagtcgaac	ggagctgctt	tgatgaagtt	60
ttcggatgga	ttttaaaccag	cttagtggtg	gacgggtgag	taacgcgtgg	gtaacctgcc	120
tcacactggg	ggataaacagt	tagaaatagc	tgctaatacc	gcataagcgc	acggttcctc	180
atggaacagt	gtgaaaaaac	cgggtgggtg	gagatggacc	cgctctctga	tagccagttg	240
gcggggtaac	ggcccaccaa	agcgacgatc	agtagccggc	ctgagagggg	gaacggccac	300
attgggactg	agacacggcc	caa				323

SEQ ID NO: 96 moltype = DNA length = 309
 FEATURE Location/Qualifiers
 source 1..309
 mol_type = genomic DNA
 organism = Clostridium sp.

SEQUENCE: 96

gatgaacgct	ggcggcggtgc	ttaacacatg	caagtcgaac	ggacgatgaa	gagcttgctt	60
ttcagagtta	gtggcggacg	ggtgagtaac	gcgtgggtaa	cctgcctcat	acagggggat	120
agcagctgga	aacggctgat	aaaaccgcat	aagcgcacag	catcgcatga	tcagtggtga	180

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aaaaactccgg  tggatatgaga  tggaccgccg  tctgattagc  tggttggtga  ggtaacggcc  240
caccaaggcg  acgatcagta  gccggcctga  gaggggtgacc  ggccacattg  ggactgagac  300
acggcccaa  309

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```

SEQ ID NO: 97      moltype = DNA  length = 313
FEATURE           Location/Qualifiers
source            1..313
                  mol_type = genomic DNA
                  organism = Clostridium sp.

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SEQUENCE: 97
gatgaacgct  ggcggcgctgc  ctaacacatg  caagtcgaac  ggacttacat  tgaagcctag  60
cgatatgtaa  gttagtggcg  gacgggtgag  taacgcgtgg  gtaacctgcc  ttgtactggg  120
ggacaacagt  tggaaacgac  tgctaatacc  gcataagcgc  acagcttcgc  atgaagcagt  180
gtgaaaaact  ccggtgggtac  aagatggacc  cgcgtctgat  tagctagtgg  gtgaggtaac  240
ggcccaccaa  ggcgacgatc  agtagccggc  ctgagagggt  gaacggccac  attgggactg  300
agacacggcc  caa  313

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```

SEQ ID NO: 98      moltype = DNA  length = 324
FEATURE           Location/Qualifiers
source            1..324
                  mol_type = genomic DNA
                  note = Lachnospiraceae
                  organism = unidentified

```

```

SEQUENCE: 98
gatgaacgct  ggcggcgctgc  ttaacacatg  caagtcgagc  gaagcactta  ggaaagattc  60
ttcggatgat  ttcctatttg  actgagcggc  ggacgggtga  gtaacgcgtg  ggtaacctgc  120
ctcacacagg  gggataacag  ttgaaatga  ctgctaatac  cgcataagac  cacagcaccg  180
catggtgcag  gggtaaaaac  tccggtggta  tgagatggac  ccgcgtctga  ttagttagtt  240
ggtggggtaa  cggcctacca  aggcgacgat  cagtagccga  cctgagaggg  tgaccggcca  300
cattgggact  gagacacggc  ccaa  324

```

```

SEQ ID NO: 99      moltype = DNA  length = 317
FEATURE           Location/Qualifiers
source            1..317
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 99
gatgaacgct  ggcggcgctgc  ttaacacatg  caagtcgagc  ggagataaat  tgaaagcttg  60
cttttaattt  atcttagcgg  cggacgggtg  agtaacgtgt  gggcaacctg  cctcatacag  120
agggataatc  atgtgaaaac  gtgactaata  ccgcatgtcg  tttcgggagg  gcacccctct  180
gaaagaaaaa  gagcaatccg  gtatgagatg  ggcgcgcac  tgattagcta  gttggtgaga  240
taacagccca  ccaaggcgac  gatcagtagc  cgacctgaga  gggtagatcg  ccacattggg  300
actgagacac  ggcccaa  317

```

```

SEQ ID NO: 100     moltype = DNA  length = 318
FEATURE           Location/Qualifiers
source            1..318
                  mol_type = genomic DNA
                  note = Faecalitalea
                  organism = unidentified

```

```

SEQUENCE: 100
gatgaacgct  ggcggcgctgc  ctaatacatg  caagtcgaac  ggaccgaagg  gaaagcttgc  60
tttcctaaaa  gtcagtggcg  aacgggtgag  taacacgtag  gtaacctgcc  catgtgcccc  120
ggacaacgcc  tggaaacggg  tgctaaaacc  ggataggtag  agagatcgca  tgaactttct  180
attaaagggg  ctgcggccct  gaacatggat  ggacctgcgg  tgcattagct  agttggtgag  240
gtaacggccc  accaaggcga  cgatgcatag  ccgacctgag  agggcgaaac  gccacattgg  300
gactgagaca  cggcccaa  318

```

```

SEQ ID NO: 101     moltype = DNA  length = 324
FEATURE           Location/Qualifiers
source            1..324
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 101
gatgaacgct  ggcggcgctgc  ttaacacatg  caagtcgagc  gaagcgctta  cgacagaacc  60
ttcgggggaa  gatgtaaggg  actgagcggc  ggacgggtga  gtaacgcgtg  ggtaacctgc  120
ctcacacagg  gggataacag  ttgaaacgg  ctgctaatac  cgcataagcg  cacagtaccg  180
catggtacgg  tgtgaaaaac  tccggtggta  tgagatggac  ccgcgtctga  ttagctagtt  240
ggaggggtaa  cggccacca  aggcgacgat  cagtagccgg  cctgagaggg  tgaacggcca  300
cattgggact  gagacacggc  ccag  324

```

```

SEQ ID NO: 102     moltype = DNA  length = 323
FEATURE           Location/Qualifiers
source            1..323
                  mol_type = genomic DNA
                  organism = Blautia sp.

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SEQUENCE: 102
 gatgaacgct ggcggcgctgc ttaacacatg caagtogaac ggggaacatt ttatggaagc 60
 ttcggtggaa atagcttgtt cctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
 tcacactggg ggataacagt cagaaatgac tgctaatacc gcataagcgc acgggattgc 180
 atgatccagt gtgaaaaact ccgggtggtg gagatggacc cgcgttgat tagccagttg 240
 gcagggtaac ggccatacaa agcgcgcatc catagccggc ctgagagggg ggacggccac 300
 attgggactg agacacggcc cag 323

SEQ ID NO: 103 moltype = DNA length = 303
 FEATURE Location/Qualifiers
 source 1..303
 mol_type = genomic DNA
 organism = Collinsella sp.

SEQUENCE: 103
 gatgaacgct agctacaggc ttaacacatg caagtogaac ggcacctgcc ctgggacaga 60
 agcgagtggc gaacgggtga gtaacacgtg gagaacctgc cccctcccc gggatagccg 120
 cccgaaagga cgggtaatac cggatacccc ggggtgccgc atggcacccc ggctaaagcc 180
 ccgacgggag gggatggctc cgcggcccat caggtagacg gcgggggtgac ggcccaccgt 240
 gccgacaacg ggtagccggg ttgagagacc gaccggccag attgggactg agacacggcc 300
 cag 303

SEQ ID NO: 104 moltype = DNA length = 326
 FEATURE Location/Qualifiers
 source 1..326
 mol_type = genomic DNA
 organism = Blautia sp.

SEQUENCE: 104
 gatgaacgct ggcggcgctgc ttaacacatg caagtogaac gaagcacctt attctgattt 60
 ctctggaatg aaaaatttgg tgactgagtg gcggacgggt gagtaacgcg tgggtaacct 120
 gcctcataca gggggataac agttagaat gactgctaata accgcataag accacagcac 180
 tgcatggtgc aggggtgaaa actccgggtg tatgagatgg acccgctct gattaggtgg 240
 ttggtggggt aacggcctac caagccgacg atcagtagcc gacctgagag ggtgaccggc 300
 cacattggga ctgagacacg gcccaa 326

SEQ ID NO: 105 moltype = DNA length = 309
 FEATURE Location/Qualifiers
 source 1..309
 mol_type = genomic DNA
 organism = Bifidobacterium sp.

SEQUENCE: 105
 gatgaacgct ggcggcgctgc ttaacacatg caagtogaac gggatcggct ggagcttgct 60
 ccggccgtga gagtggcgaa cgggtgagta atgcgtgacc gacctgccc atacaccgga 120
 atagctcctg gaaacgggtg gtaatgccgg atgctccagt tgactgcatg gtctctggg 180
 aaagcttttg gggatgggga tggggtcgcg tcttatcagc ttgatggcgg ggtaacggcc 240
 caccatggct tcaacgggta gccggcctga gagggcgacc ggccacattg ggactgagac 300
 acggcccaa 309

SEQ ID NO: 106 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 organism = Ruminococcus sp.

SEQUENCE: 106
 gatgaacgct ggcggcgctgc ttaacacatg caagtogaac gaagcactta agtttgattc 60
 ttcggatgaa gacttttgtg actgagtggc ggacgggtga gtaacgcgtg ggtaacctgc 120
 ctcatacagg gggataacag ttagaaatga ctgctaatac cgcataagac cacagcaccg 180
 catggtgcag gggtaaaaac tccgggtgta tgagatggac ccgctctga ttagtgtgtt 240
 ggtggggtaa cggcctacca aggcgacgat cagtagccga cctgagaggg tgaccggcca 300
 cattgggact gagacacggc ccaa 324

SEQ ID NO: 107 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 note = Lachnospiraceae
 organism = unidentified

SEQUENCE: 107
 gatgaacgct agctacaggc ttaacacatg caagtgcagc gaagcactta agaattgattc 60
 ttcggatgaa ttcttttgtg actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc 120
 ctcatacagg gggataacag ttagaaatgg ctgctaatac cgcataagac cacagtactg 180
 catggtacag tggtaaaaac tccgggtgta tgagatggac ccgctctga ttaggtagtt 240
 ggtgaggtaa cggccacca agccgacgat cagtagccga cctgagaggg tgaccggcca 300
 cattgggact gagacacggc ccag 324

SEQ ID NO: 108 moltype = DNA length = 323
 FEATURE Location/Qualifiers

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source                1..323
                      mol_type = genomic DNA
                      organism = Clostridium sp.

SEQUENCE: 108
gatgaacgct ggcggcgtgc ctaacacatg caagtcgaac gaggtatttt gattgaagtt 60
ttcggatgga tttcaagata ccgagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
tcatacaggg ggataacggt tagaaatgac tgctaatacc gcataagcgc acagtaccgc 180
atgggtacgt gtgaaaaact ccggtgggat gagatggacc cgcgtctgat tagctagtgt 240
gtggggtaac ggcccaccaa ggcgacgatc agtagccgac ctgagagggg gaccggccac 300
attgggactg agacacggcc cag                                     323

SEQ ID NO: 109        moltype = DNA length = 325
FEATURE              Location/Qualifiers
source                1..325
                      mol_type = genomic DNA
                      organism = Clostridium sp.

SEQUENCE: 109
gacgaacgct ggcggcgtgc ttaacacatg caagtcgaac ggagcaccct tgactgaggt 60
ttcggccaaa tgataggaat gcttagtggc ggactgtgta gtaacgcgtg aggaacctgc 120
cttcacagag ggagacaacag ttggaacga ctgctaatac cgcagtacgc atgaccgggg 180
catcccgggc atgtcaaaga ttttatcgct ggaagatggc ctgcgctctg attagctaga 240
tggtggggta acggccacac atggcgacga tcagtagccg gactgagagg ttgaccggcc 300
acattggggc tgagatacgg ccacg                                     325

SEQ ID NO: 110        moltype = DNA length = 322
FEATURE              Location/Qualifiers
source                1..322
                      mol_type = genomic DNA
                      organism = Clostridium sp.

SEQUENCE: 110
gatgaacgct ggcggcgtgc ctaacacatg caagtcgaac gaagcacctt acttgattct 60
tcggatgaag gtctgggtgac tgagtggcgg acgggtgagt aacgcgtggg taacctgccc 120
tgtacagggg gataacagct ggaacaggct gctaataacc cataagcgca cgaggagaca 180
ttctcttggt tgaaaaaact ccgtgtgtaca ggatggggcc gcgtctgatt agctggttgg 240
cagggtaacg gcctaccaag gcaacgatca gtagccggtc tgagaggatg aacggccaca 300
ttggaactga gacacggtcc aa                                     322

SEQ ID NO: 111        moltype = DNA length = 313
FEATURE              Location/Qualifiers
source                1..313
                      mol_type = genomic DNA
                      organism = Clostridium sp.

SEQUENCE: 111
gatgaacgct ggcggcgtgc ctaacacatg caagtcgaac ggaatttaca tgaagcctag 60
cgattgtaaa tttagtggcg gacgggtgag taacgcgtgg gtaacctgcc ttgtactggg 120
ggacaacagt tggaaacgac tgctaatacc gcataagcgc acagcttcgc atgaagcagt 180
gtgaaaaact ccggtgggtac aagatggacc cgcgtctgat tagctggttg gtgaggtaac 240
ggcccaccaa ggcgacgatc agtagccggc ctgagagggg gaacggccac attgggactg 300
agacacggcc caa                                     313

SEQ ID NO: 112        moltype = DNA length = 313
FEATURE              Location/Qualifiers
source                1..313
                      mol_type = genomic DNA
                      note = Clostridiaceae
                      organism = unidentified

SEQUENCE: 112
gacgaacgct ggcggcgtgc ctaacacatg caagtcgaac ggagttgaaa ggagcttgct 60
tttttcaact cagttggcga cgggtgagta acgcgtgagt aacctgcctt tcagaggggg 120
ataacgcttc gaaaagaacg ctaataccgc ataagattgt agtttcgcat ggagcagcaa 180
tcaaaggagc aatccgctga aagatggact cgcgtccgat tagatagttg gcggggtaac 240
ggcccaccaa gtcgacgatc ggtagccgga ctgagagggt gatcggccac attgggactg 300
agacacggcc cag                                     313

SEQ ID NO: 113        moltype = DNA length = 310
FEATURE              Location/Qualifiers
source                1..310
                      mol_type = genomic DNA
                      organism = Bacteroides sp.

SEQUENCE: 113
gatgaacgct agctacaggg ttaacacatg caagtcgagg ggcatacagg tntagcaata 60
cattgctggc gaccggcgca cgggtgagta acacgtatcc aacctgcctt ttactcgggg 120
ataggcttcc gaaagaaaga ttaatacccg atggtataat atttcgcat gggattatta 180
ttaaagaatt tcggtaaaag atgggggatgc gttccattag gcagttggcg gggtaacggc 240
ccaccaaac ccgatggat aggggttctg agaggaaagt cccccaactt ggaactgaga 300
cacggtccaa                                     310

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SEQ ID NO: 114 moltype = DNA length = 312
 FEATURE Location/Qualifiers
 source 1..312
 mol_type = genomic DNA
 organism = Bacteroides sp.

SEQUENCE: 114

gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcattt cgctagcaat	60
agtgaagatg gcgaccggcg cacgggtgag taacacgtat ccaacctgcc gataactcgg	120
ggatagcctt tcgaaagaaa gattaatatc cgatagtata ttaaaaccgc atggttttac	180
tattaaagaa tttcggttat cgatggggat gcgttcocatt agtttggtgg cggggtaacg	240
gcccaccaag actacgatgg ataggggttc tgagagggaag gtccccca tgggaactga	300
gacacggtcc aa	312

SEQ ID NO: 115 moltype = DNA length = 326
 FEATURE Location/Qualifiers
 source 1..326
 mol_type = genomic DNA
 organism = Ruminococcus sp.

SEQUENCE: 115

gatgaacgct ggcggcgctgc ttaacacatg caagtcgagc gaagcactct gctttgattt	60
cttcgggatg aagagcttag tgactgagcg gcggacgggt gagtaacgcg tgggtaacct	120
gctcatatac gggggataac agttagaaat gactgctaata accgcataag gccacagcac	180
cgcatgggtc aggggtaaaa actccgggtg tatgagatgg acccgctctc gattaggtag	240
ttggcggggt aacggcccac caagccgacg atcagtagcc gacctgagag ggtgaccggc	300
cacattggga ctgagacacg gcccac	326

SEQ ID NO: 116 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 organism = Clostridium sp.

SEQUENCE: 116

gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcactta tctttgattc	60
ttcggatgaa gaggtttgtg actgagtgcc ggacgggtga gtaacgcgtg ggtaacctac	120
ctcatacagg gggataacag ttgaaatga ctgctaatac cgcataagac caccgagctg	180
catggctcag tgggaaaaac tccggtggtg tgagatggac ccgcgtctga ttaggtagtt	240
ggtggggtaa cggcctacca agccaacgat cagtagccga cctgagaggg tgaccggcca	300
cattgggact gagacacggc ccaa	324

SEQ ID NO: 117 moltype = DNA length = 323
 FEATURE Location/Qualifiers
 source 1..323
 mol_type = genomic DNA
 note = Lachnospiraceae
 organism = unidentified

SEQUENCE: 117

gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggaaacctt ttattgaagc	60
ttcggcagat ttagtgggtt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc	120
ttatacaggg ggataacagt cagaaatggc tgctaatacc gcataagcgc acagagctgc	180
atggctcagt gtgaaaaact ccggtgggtat aagatggacc cgcgttggtat tagctgggtg	240
gtggggtaac agcccaccaa ggcgacgatc catagccggc ctgagagggt gaacggccac	300
attgggactg gagacacggc cag	323

SEQ ID NO: 118 moltype = DNA length = 314
 FEATURE Location/Qualifiers
 source 1..314
 mol_type = genomic DNA
 organism = Bacteroides sp.

SEQUENCE: 118

gatgaacgct ggctacaggc ttaacacatg caagtcgagg ggcacgggga agaaagcttg	60
ctttctttgc tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccctttactc	120
ggggatagcc tttcgaaaga aagattaata cccgatggga taattattcc gcatggtttg	180
attattaaag gattccggtg aaggatgggg atgctgtcca ttaggcagtt ggtgaggtaa	240
cggtcacca aaccttcgat ggataggggt tctgagagga aggtcccca cattggaact	300
gagacacggt ccaa	314

SEQ ID NO: 119 moltype = DNA length = 303
 FEATURE Location/Qualifiers
 source 1..303
 mol_type = genomic DNA
 organism = Collinsella sp.

SEQUENCE: 119

gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggcacctgcc ctcgggcaga	60
agcgagtgcc gaacggctga gtaacacgtg gagaacctgc cccctcccc gggatagccg	120
cccgaaagga cgggtaatac cggatacccc cgggcgccgc atggcgcccc ggctaaagcc	180

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ccgacgggag gggatggctc cgcggcccat caggtagacg gcgggggtgac ggcccaccgt 240
gccgacaacg ggtagccggg ttgagagacc gaccggccag attgggactg agacacggcc 300
cag 303

```

```

SEQ ID NO: 120      moltype = DNA length = 311
FEATURE            Location/Qualifiers
source             1..311
                   mol_type = genomic DNA
                   organism = Clostridium sp.

```

```

SEQUENCE: 120
gacgaacgct ggcggcgctc ctaacacatg caagtcgaac ggacgaggag gagcttgctt 60
ctccaagtca gtggcggaac ggtgagtaac gcgtgagtaa cctgcccttc agagggggat 120
aacgtctgga aacggacgct aataccgcat gacatatttc catcgcatgg tggagatc 180
aaaggagcaa tccgtgaa gattggactcg cgtccgatta gatagttggc ggggtaacgg 240
cccaccaagt cgacgatcgg tagccggact gagagggtga acggccacat tgggactgag 300
acacggccca g 311

```

```

SEQ ID NO: 121      moltype = DNA length = 314
FEATURE            Location/Qualifiers
source             1..314
                   mol_type = genomic DNA
                   organism = Bacteroides sp.

```

```

SEQUENCE: 121
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatga acttagcttg 60
ctaagtttga tggcgaccgg cgcacgggtg agtaaacagt atccaacctg ccgatgactc 120
ggggatagcc ttctgaaaga aagattaata cccgatggca tagttcttcc gcattgtaga 180
actattaaag aatttcggtc atcgatgggg atgctgtcca ttaggttggt ggcggggtaa 240
cgggccacca agccttcgat ggataggggt tctgagagga aggtccccc cattggaact 300
gagacacggt ccaa 314

```

```

SEQ ID NO: 122      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source             1..316
                   mol_type = genomic DNA
                   organism = Bacteroides sp.

```

```

SEQUENCE: 122
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatgt cagtgtgctt 60
gcaaacctga gatggcgacc ggcgcacggg tgagtaacac gtatccaacc tgcgataac 120
tcggggatag cctttcgaaa gaaagattaa taccgatggg tataattaga ccgatggctc 180
ttgttattaa agaatttcgg ttatcgatgg ggatgcgttc cattaggcag ttggtgaggt 240
aacggctcac caaaccttcg atggataggg gttctgagag gaaggtcccc cacattggaa 300
ctgagacacg gtccaa 316

```

```

SEQ ID NO: 123      moltype = DNA length = 317
FEATURE            Location/Qualifiers
source             1..317
                   mol_type = genomic DNA
                   organism = Bacteroides sp.

```

```

SEQUENCE: 123
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatgg tcttagcttg 60
ctaaggctga tggcgaccgg cgcacgggtg agtaaacagt atccaacctg ccgtctactc 120
ttggccagcc ttctgaaagg aagattaatc caggatggga tcatgagttc acatgtccgc 180
atgattaaag gtattttccg gtacacgatg gggatgcgtt ccattagata gtaggcgggg 240
taacggccca cctagtcaac gatggatagg ggttctgaga ggaaggtccc ccacattgga 300
actgagacac ggtccaa 317

```

```

SEQ ID NO: 124      moltype = DNA length = 315
FEATURE            Location/Qualifiers
source             1..315
                   mol_type = genomic DNA
                   organism = Bacteroides sp.

```

```

SEQUENCE: 124
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatgg tcttagcttg 60
ctaaggccga tggcgaccgg cgcacgggtg agtaaacagt atccaacctg ccgtctactc 120
ttggacagcc ttctgaaagg aagattaata caagatggca tcatgagttc gcattgtcac 180
atgattaaag gtatttcggg agacgatggg gatgcgttcc attagatagt aggcggggta 240
acggccccc tagtcttcga tggatagggg ttctgagagg aaggtcccc acattggaac 300
tgagacacgg tccaa 315

```

```

SEQ ID NO: 125      moltype = DNA length = 314
FEATURE            Location/Qualifiers
source             1..314
                   mol_type = genomic DNA
                   organism = Bacteroides sp.

```

```

SEQUENCE: 125
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatca tcaaagcttg 60

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ctttgatgga tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccgacaacac 120
tgggatagcc tttcgaaaga aagattaata ccggatggca tagttttccc gcatgggatg 180
attattaaag aatttcggtt gtcgatgggg atgcgttcca ttaggcagtt ggcggggtaa 240
cgccccacca aaccaacgat ggataggggt tctgagagga aggtccccc cATTggaact 300
gagacacggg ccaa                                     314

```

```

SEQ ID NO: 126      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source             1..323
                  mol_type = genomic DNA
                  organism = Blautia sp.

```

```

SEQUENCE: 126
gatgaacgct ggcggcggtgc ttaacacatg caagtcgaac gggaaatatt tcattgagac 60
ttcgggtgat ttgatctatt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatacaggg ggataacagt cagaaatggc tgctaatacc gcataagcgc acagagctgc 180
atggctcagt gtgaaaaact ccggtgggtat aagatggacc cgcgttggat tagcttgttg 240
gtggggtaac ggcccaccaa ggcgacgatc catagccggc ctgagagggg gaacggccac 300
attgggactg agacacggcc cag                                     323

```

```

SEQ ID NO: 127      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source             1..323
                  mol_type = genomic DNA
                  note = Phascolarctobacterium
                  organism = unidentified

```

```

SEQUENCE: 127
gacgaacgct ggcggcggtgc ctaacacatg caagtcgaac ggagaatttt atttcggtag 60
aattcttagt ggcgaacggg tgagtaacgc gtaggcaacc tgccctttag acggggacaa 120
cattccgaaa ggagtgtctaa taccggatgt gatcatcgtg ccgcattggc ggatgaagaa 180
agatggcctc tacaagtaag ctatcgctaa aggatggggc tgcgtctgat tagctagtgt 240
gtagtgtaac ggactaccaa ggcgatgatc agtagccggt ctgagagggg gaacggccac 300
attgggactg agacacggcc caa                                     323

```

```

SEQ ID NO: 128      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                  mol_type = genomic DNA
                  organism = Dorea sp.

```

```

SEQUENCE: 128
gatgaacgct ggcggcggtgc ttaacacatg caagtcgagc gaagcactta agtttgattc 60
ttcggatgaa gacttttgtg actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcatacagg gggataacag ttgaaatga ctgctaatac cgcataagac caccgtaccg 180
catggtacag tggtaaaaac tctggtggta tgagatggac ccgcgtctga ttaggtagtt 240
ggtagggtaa cggcctacca agccgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccag                                     324

```

```

SEQ ID NO: 129      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source             1..323
                  mol_type = genomic DNA
                  organism = Blautia sp.

```

```

SEQUENCE: 129
gatgaacgct ggcggcggtgc ttaacacatg caagtcgaac gggaattact ttattgaaac 60
ttcggtcgat atgatttaat tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatacaggg ggataacagt cagaaatgac tgctaatacc gcataagcgc acagagctgc 180
atggctcggg gtgaaaaact ccggtgggtat aagatggacc cgcgttggat tagctagtgt 240
gtgaggtaac ggcccaccaa ggcgacgatc catagccggc ctgagagggg gaacggccac 300
attgggactg agacacggcc cag                                     323

```

```

SEQ ID NO: 130      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source             1..316
                  mol_type = genomic DNA
                  organism = Bacteroides sp.

```

```

SEQUENCE: 130
gatgaacgct agtacaggc ttaacacatg caagtcgagg ggcagcattt tagtttgctt 60
gcaaaactaaa gatggcgacc ggcgcacggg tgagtaacac gtatccaacc tgccgataac 120
tcggggtag cctttcgaaa gaaagattaa tatccgatgg tatattttct ccgcattggg 180
agaatattaa agaatttcgg ttatcgatgg ggatgcgttc cattagtttg ttggcgggg 240
aacggccac caagaccagc atggataggg gttctgagag gaaggtcccc cacattggaa 300
ctgagacacg gtccaa                                     316

```

```

SEQ ID NO: 131      moltype = DNA length = 322
FEATURE            Location/Qualifiers
source             1..322
                  mol_type = genomic DNA

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                                organism = Megamonas sp.
SEQUENCE: 131
gacgaacgct ggcggcgctgc ttaacacatg caagtcgaac ggggtgttta ttccggtaaa 60
caccaagtgg cgaacgggtg agtaacgcgt aagcaatcta ccttcaagat ggggacaaca 120
cttcgaaaagg ggtgctaata ccgaatgaat gtaagagtat cgcgatgagac acttactaaa 180
ggaggcctct gacataagct tccgcttgaa gatgagcttg cgtctgatta gctagtgtgt 240
gagggtaaa gcccaccaag gcgacgatca gtagccggctc tgagaggatg aacggccaca 300
ttgggactga gacacggccc ag                                     322

SEQ ID NO: 132      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source              1..316
                    mol_type = genomic DNA
                    organism = Staphylococcus sp.
SEQUENCE: 132
gacgaacgct ggcggcgctgc ctaatacatg caagtagaac gctgaagaga ggagcttgct 60
cttcttggat gagttgcgaa cgggtgagta acgcgtaggt aacctgcctt gtagcggggg 120
ataactattg gaaacgatag ctaataccgc ataacaatgg atgacacatg tcatttattt 180
gaaaggggca attgctccac tacaagatgg acctgcgttg tattagctag taggtgaggt 240
aacggctcac ctaggcgacg atacatagcc gacctgagag ggtgatcggc cacactggga 300
ctgagacacg gcccag                                     316

SEQ ID NO: 133      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source              1..324
                    mol_type = genomic DNA
                    organism = Blautia sp.
SEQUENCE: 133
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcacctt gatttgattc 60
ttcggatgaa gatcttggtg actgagtggtc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcatacagg gggataacag ttagaatatg ctgctaatac cgcataagac cacagcaccg 180
catggtgcag gggtaaaaac tccggtggtg tgagatggac ccgcgtctga ttaggtagtt 240
ggtggggtaa cggcctacca agccgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa                                     324

SEQ ID NO: 134      moltype = DNA length = 307
FEATURE            Location/Qualifiers
source              1..307
                    mol_type = genomic DNA
                    organism = Bifidobacterium sp.
SEQUENCE: 134
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggatcccag gagcttgctc 60
ctgggtgaga gtggcgaaacg ggtgagtaat gcgtgaccga cctgccccat acaccggaat 120
agctcctgga aacgggtggt aatgcgggat gctccagttg gatgcgatgc cttctgggaa 180
agattcatcg gtatgggatg gggtcgcgtc ctatcagctt gatggcgggg taacggccca 240
ccatggcttc gacgggtagc cggcctgaga gggcgaccgg ccacattggg actgagatac 300
ggcccag                                     307

SEQ ID NO: 135      moltype = DNA length = 303
FEATURE            Location/Qualifiers
source              1..303
                    mol_type = genomic DNA
                    organism = Collinsella sp.
SEQUENCE: 135
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggcacccctc tccggaggga 60
agcgagtggc gaacgggtga gtaaacctgc gagaacctgc cccctccccg gggatagccg 120
cccgaaggga cgggtaatac cggatacccc ggggtgccgc atggcaccgc ggctaagacc 180
ccgacgggag gggatggctc cgcggcccat caggtagacg gcgggggtgac ggcccaccgt 240
gccgacaacg ggtagccggg ttgagagacc gaccggccag attgggactg agacacggcc 300
cag                                     303

SEQ ID NO: 136      moltype = DNA length = 326
FEATURE            Location/Qualifiers
source              1..326
                    mol_type = genomic DNA
                    organism = Clostridium sp.
SEQUENCE: 136
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcacttt atttgatttc 60
cttcgggact gattattttg tgactgagtg gcggacgggt gagtaacgcg tgggtaacct 120
gccttgatga gggggataac agttggaaac ggctgctaata accgcataag cgcacagcat 180
cgcatgatgc agtggtgaaa actccgggtg tataagatgg accgcgttg gattagctag 240
ttggtgaggt aacggccccc caaggcgacg atccatagcc gacctgagag ggtgacggcg 300
cacattggga ctgagacacg gcccaa                                     326

SEQ ID NO: 137      moltype = DNA length = 316
FEATURE            Location/Qualifiers

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source                1..316
                      mol_type = genomic DNA
                      organism = Parabacteroides sp.

SEQUENCE: 137
gatgaacgct agcgacaggc ttaacacatg caagtcgagg ggcagcatga ttgtagcaa 60
tacagattga tggcgaccgg cgcacgggtg agtaacgcgt atgcaactta cctatcagag 120
ggggatagcc cggcgaaagt cggattaata cccataaaa caggggtccc gcatgggaat 180
at ttgttaaa gattcatcgc tgatagatag gcatgcggtc cattaggcag ttggcggggt 240
aacggcccac caaacgcacg atggataggg gttctgagag gaaggtcccc cacattggta 300
ctgagacacg gaccaa                                     316

SEQ ID NO: 138        moltype = DNA length = 307
FEATURE              Location/Qualifiers
source                1..307
                      mol_type = genomic DNA
                      organism = Bifidobacterium sp.

SEQUENCE: 138
gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac gggatcccag gagcttgctc 60
ctgggtgaga gtggcgaacg ggtgagtaat gcgtgaccga cctgccccat acaccggaat 120
agctcctgga aacgggtggt aatgccggat gctccagttg accgcatggt cctctgggaa 180
agcttttgcg gtatgggatg gggtcgcgtc ctatcagctt gatggcgggg taacggccca 240
ccatggcttc gacgggtagc cggcctgaga gggcgaccgg ccacattggg actgagatac 300
ggcccag                                     307

SEQ ID NO: 139        moltype = DNA length = 302
FEATURE              Location/Qualifiers
source                1..302
                      mol_type = genomic DNA
                      organism = Clostridium sp.

SEQUENCE: 139
gatgaacgct ggcggcgtgc ctaatacatg caagtcgaac gcttcacttc ggtgaagagt 60
ggcgaacggg tgagtaatac ataagtaacc tggcatctac agggggataa ctgatggaaa 120
cgtcagctaa gaccgcatag gtgtagagat cgcatagaac ctatatgaaa agtgctacgg 180
gactggtaga tgatggactt atggcgcatc agctgggttg tagggtaacg gcctaccaag 240
ggcagcatgc gtagccgacc tgagaggggtg accggccaca ctgggactga gacacggccc 300
ag                                     302

SEQ ID NO: 140        moltype = DNA length = 315
FEATURE              Location/Qualifiers
source                1..315
                      mol_type = genomic DNA
                      organism = Bacteroides sp.

SEQUENCE: 140
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatga acttagcttg 60
ctaagtttga tggcgaccgg cgcacgggtg agtaacacgt atccaacctt ccgtttactc 120
agggatagcc tttcgaaaga aagattaata cctgatagta tggtagattg gcatgatagc 180
accattaaag atttattggt aaacgatggg gatgcggtcc attaggtagt aggcggggta 240
acggccaccc tagccgacga tggatagggg ttctgagagg aaggtccccc acattggaac 300
tgagacacgg tccaa                                     315

SEQ ID NO: 141        moltype = DNA length = 326
FEATURE              Location/Qualifiers
source                1..326
                      mol_type = genomic DNA
                      organism = Roseburia sp.

SEQUENCE: 141
gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac gaagcactct atttgatttt 60
cttcggaaat gaagattttg tgactgagtg gcggacgggt gagtaacgcg tgggtaacct 120
gcctcataca gggggataac agttggaaac gactgctaata accgcataag cgcacaggat 180
cgcagatgcc ggtgtgaaaa actccgggtg tatgggatgg acccgctctc gattagccag 240
ttggcagggt aacggcctac caaagcgacg atcagtagcc gacctgagag ggtgaccggc 300
cacattggga ctgagacacg gcccacaa                                     326

SEQ ID NO: 142        moltype = DNA length = 323
FEATURE              Location/Qualifiers
source                1..323
                      mol_type = genomic DNA
                      note = Lachnospiraceae
                      organism = unidentified

SEQUENCE: 142
gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac gggaaacctt ttattgaagc 60
ttcggcagat ttagattggt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatacaggg ggataacagt cagaaatgac tgctaatacc gcataagcgc acaggaccgc 180
atggtccggt gtgaaaaact cgggtgggat aagatggacc cgcgttggat tagctagtgt 240
gtgaggtaac ggcccaccaa ggcgacgacg catagccggc ctgagagggt gaacggccac 300
attgggactg agacacggcc cag                                     323

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-continued

SEQ ID NO: 143 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 note = Lachnospiraceae
 organism = unidentified

SEQUENCE: 143

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgagc	gaagcaattt	aagtgaagtt	60
ttcggatgga	tcttaaattg	actgagcggc	ggacgggtga	gtaacgcgtg	gataacctgc	120
ctcacacagg	gggataacag	ttagaaatga	ctgctaatac	cgcataagcg	cacggtaaccg	180
catggtacag	tgtgaaaaac	tccggtggtg	tgagatggat	ccgcgtctga	ttaggtagtt	240
ggtgaggtaa	cggcccacca	agccgacgat	cagtagccga	cctgagaggg	tgaccggcca	300
cattgggact	gagacacggc	ccaa				324

SEQ ID NO: 144 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 organism = Clostridium sp.

SEQUENCE: 144

gatgaacgct	ggcggcgctgc	ctaacacatg	caagtcgagc	gaagcggttt	taagggaagtt	60
ttcggatgga	attaaaactg	actgagcggc	ggacgggtga	gtaacgcgtg	ggtaacctgc	120
ctcacacagg	gggataacag	ttagaaatgg	ctgctaatac	cgcataagca	cacagcttcg	180
catggagcag	tgtgaaaaac	tccggtggtg	tgagatggac	ccgcgtctga	ttagctagtt	240
ggtaaggtaa	cggcttacca	agccgacgat	cagtagccga	cctgagaggg	tgaccggcca	300
cattgggact	gagacacggc	ccaa				324

SEQ ID NO: 145 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 note = Clostridiaceae
 organism = unidentified

SEQUENCE: 145

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgaac	gaagcaatct	aacgggaagtt	60
ttcggatgga	agctagattg	actgagtggc	ggacgggtga	gtaacgcgtg	ggtaacctgc	120
ctcacactgg	gggacaacag	ttagaaatga	ctgctaatac	cgcataagcg	cacaggaccg	180
catggtccgg	tgtgaaaaac	tccggtggtg	tgagatggac	ccgcgtttga	ttagctagtt	240
ggtggggtaa	cggcctacca	agccgacgat	caatagccga	cctgagaggg	tgaccggcca	300
cattgggact	gagacacggc	ccaa				324

SEQ ID NO: 146 moltype = DNA length = 311
 FEATURE Location/Qualifiers
 source 1..311
 mol_type = genomic DNA
 organism = Faecalibacterium sp.

SEQUENCE: 146

gacgaacgct	ggcggcgcgcg	ctaacacatg	caagtcgaac	ggagcttaga	gagcttgctt	60
tttaagctta	gtggcgaaacg	ggtgagtaac	gcgtgagtaa	cctgccctag	agtgggggac	120
aacagttgga	aacgactgct	aataccgcat	aagcccacgg	taccgcatgg	tactgaggga	180
aaaggattta	ttcgctttag	gatggactcg	cgctccaatta	gctagtgtgt	gaggtaacgg	240
cccaccaagg	cgacgattgg	tagccggact	gagaggttga	acggccacat	tgggactgag	300
acacggccca	g					311

SEQ ID NO: 147 moltype = DNA length = 316
 FEATURE Location/Qualifiers
 source 1..316
 mol_type = genomic DNA
 organism = Streptococcus sp.

SEQUENCE: 147

gatgaacgct	ggcggcgctgc	ctaatacatg	caagtagaac	gctgaagctt	ggtgcttgca	60
ccgagcggat	gagttgcgaa	cgggtgagta	acgcgtaggt	aacctgcctg	gtagcggggg	120
ataactattg	gaaacgatag	ctaataccgc	ataacagtag	atgtcgcata	acctttgctt	180
gaaagggtga	attgcaccac	taccagatgg	acctgcgttg	tattagctag	ttggtgaggt	240
aacggctcac	caaggcgacg	atacatagcc	gacctgagag	ggtgatcgcc	cacactggga	300
ctgagacacg	gcccag					316

SEQ ID NO: 148 moltype = DNA length = 322
 FEATURE Location/Qualifiers
 source 1..322
 mol_type = genomic DNA
 note = Lachnospiraceae
 organism = unidentified

SEQUENCE: 148

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgagc	gaagcgattt	aatgagact	60
------------	-------------	------------	------------	------------	-----------	----

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tcggtggatt ttaaattgac tgagcggcgg acgggtgagt aacgcgtgga taacctgcct 120
cacacagggg gataacagtt agaaatgact gctaataccg cataagcgca cggtaaccgca 180
tggtacagtg tgaanaactc cgggtggtgtg agatggatcc gcgtctgatt aggtagtgtg 240
tgaggtaacg gcccaccaag ccgacgatca gtagccgacc tgagagggtg accggccaca 300
ttgggactga gacacggccc aa 322

```

```

SEQ ID NO: 149      moltype = DNA length = 310
FEATURE            Location/Qualifiers
source             1..310
                  mol_type = genomic DNA
                  organism = Parabacteroides sp.

```

```

SEQUENCE: 149
gatgaacgct agcgacaggc ttaacacatg caagtcgagg ggcagcgggg gtagcaatac 60
cgcggcgca cgcggcgacg ggtgagtaac gcgtatgcaa cttacctatc agagggggat 120
aaccggcgca aagtcggact aataccgcat gaagcagggg ccccgcatgg ggatatttgc 180
taaagattca tcgctgatag ataggcatgc gttccattag tttgttggcg gggtaacggc 240
ccaccaagac cacgatggat aggggttctg agaggaaagt cccccacatt ggaactgaga 300
cacggtccaa 310

```

```

SEQ ID NO: 150      moltype = DNA length = 326
FEATURE            Location/Qualifiers
source             1..326
                  mol_type = genomic DNA
                  organism = Lactobacillus sp.

```

```

SEQUENCE: 150
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcattta agacagatta 60
cttcgggttg aagtcttcta tgactgagtg gcggacgggt gagtaacgcg tgggtaacct 120
gcctcataca gggggatagc agctggaaac ggctggtaac accgcataag cgcacagtac 180
cacatgggtac agtgtgaaaa actccgggtg tatgagatgg acccgctctc gattagcttg 240
ttggcggggt aacggcccac caaggcgacg atcagtagcc gacctgagag ggtgacggcg 300
cacattggga ctgagacacg gccccag 326

```

```

SEQ ID NO: 151      moltype = DNA length = 325
FEATURE            Location/Qualifiers
source             1..325
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 151
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggagtcgttt tggaaatctc 60
ttcggagatg gaattctcga cttagtggcg gacgggtgag taacgcgtga gcaatctgcc 120
tttaagaggg ggataacagt cggaaacggc tgctaatacc gcataaagca ttaaatctgc 180
atgtttttga tgccaaagga gcaatccgct tttagatgag ctccgctctg attagctagt 240
tggcggggta acggcccaac aaggcgacga tcagtagccg gactgagagg ttgaacggcc 300
acattgggac tgagacacgg cccag 325

```

```

SEQ ID NO: 152      moltype = DNA length = 307
FEATURE            Location/Qualifiers
source             1..307
                  mol_type = genomic DNA
                  organism = Roseburia sp.

```

```

SEQUENCE: 152
gataaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagttttct ttcggggagg 60
aacttagtgg cggacggggtg agtaacgcgt gggtaacctg ccctgtacag ggggacaaca 120
gctggaaacg gctgctaata ccgcataaag ccttagcact gcattggtgca tagggaaaag 180
gagcaatccg gtacaggatg gacccgcgtc tgattagcca gttggcaggg taacggccta 240
ccaaagcgac gatcagtagc cgatctgaga ggatgtacgg ccacattggg actgagacac 300
ggcccag 307

```

```

SEQ ID NO: 153      moltype = DNA length = 314
FEATURE            Location/Qualifiers
source             1..314
                  mol_type = genomic DNA
                  organism = Bacteroides sp.

```

```

SEQUENCE: 153
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatga acttagcttg 60
ctaagtttga tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccgatgactc 120
ggggatagcc tttcgaaaga aagattaata cccgatggca tagttcttcc gcattgtaga 180
actattaaag aatttcggtc atcgatgggg atgcgttcca ttaggttggt ggcggggtaa 240
cgcccccaca agccttcgat ggataggggt tctgagagga aggtccccc cattggaact 300
gagacacggt ccaa 314

```

```

SEQ ID NO: 154      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source             1..316
                  mol_type = genomic DNA
                  organism = Bacteroides sp.

```

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SEQUENCE: 154
 gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcattt cagtttgctt 60
 gcaaaactgga gatggcgacc ggcgcacggg tgagtaacac gtatccaacc tgccgataac 120
 tcggggatag cctttcgaaa gaaagattaa taccgatgg tataattaga ccgatgggtc 180
 ttgttattaa agaatttcgg ttatcgatgg ggatgcgttc cattaggcag ttggtgaggt 240
 aacggctcac caaaccttcg atggataggg gttctgagag gaagggtccc cacattggaa 300
 ctgagacacg gtccaa 316

SEQ ID NO: 155 moltype = DNA length = 317
 FEATURE Location/Qualifiers
 source 1..317
 mol_type = genomic DNA
 organism = Bacteroides sp.

SEQUENCE: 155
 gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatgg tcttagcttg 60
 ctaaggctga tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccgtctactc 120
 ttggccagcc ttctgaaagg aagattaatc caggatggga tcatgagttc acatgtccgc 180
 atgattaaag gtattttccg gtagacgatg gggatgcgtt ccattagata gtaggcgggg 240
 taacggccca cctagtcaac gatggatagg ggttctgaga ggaagggtccc ccacattgga 300
 actgagacac ggtccaa 317

SEQ ID NO: 156 moltype = DNA length = 315
 FEATURE Location/Qualifiers
 source 1..315
 mol_type = genomic DNA
 organism = Bacteroides sp.

SEQUENCE: 156
 gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatgg tcttagcttg 60
 ctaaggcga tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccgtctactc 120
 ttggacagcc ttctgaaagg aagattaata caagatggca tcatgagttc gcatgttcac 180
 atgattaaag gtatttcggg agacgatggg gatgcgttcc attagatagt aggcggggta 240
 acggccccc tagtctcga ttgatagggg ttctgagagg aaggtccccc acattggaa 300
 tgagacacgg tccaa 315

SEQ ID NO: 157 moltype = DNA length = 316
 FEATURE Location/Qualifiers
 source 1..316
 mol_type = genomic DNA
 organism = Bacteroides sp.

SEQUENCE: 157
 gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcattt tagtttgctt 60
 gcaaaactgaa gatggcgacc ggcgcacggg tgagtaacac gtatccaacc tgccgataac 120
 tccggaatag cctttcgaaa gaaagattaa taccggatag catacgaata tcgcatgata 180
 tttttattaa agaatttcgg ttatcgatgg ggatgcgttc cattagtttg ttggcggggg 240
 aacggccccc caagactacg atggataggg gttctgagag gaagggtccc cacattggaa 300
 ctgagacacg gtccaa 316

SEQ ID NO: 158 moltype = DNA length = 314
 FEATURE Location/Qualifiers
 source 1..314
 mol_type = genomic DNA
 organism = Bacteroides sp.

SEQUENCE: 158
 gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatca tcaaagcttg 60
 ctttgatgga tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccgacaacac 120
 tgggatagcc ttctgaaaga aagattaata ccggatggca tagttttccc gcatgggatg 180
 attattaaag aatttcgggt gtcgatgggg atgcgttcca ttaggcagtt ggcggggtaa 240
 cggcccccac aaccaacgat ggataggggt tctgagagga aggtccccc cattggaact 300
 gagacacggt ccaa 314

SEQ ID NO: 159 moltype = DNA length = 308
 FEATURE Location/Qualifiers
 source 1..308
 mol_type = genomic DNA
 organism = Bifidobacterium sp.

SEQUENCE: 159
 gatgaacgct ggcggcggtg ttaacacatg caagtcgaac gggatccatc aggccttgct 60
 tgggtggtg agtggcgaac gggtagtaaa tgcgtgaccg acctgcccc tacaccggaa 120
 tagctcctgg aaacgggtgg taatgccgga tgctccagtt gatcgcatgg tcttctggga 180
 aagctttcgc ggtatgggat ggggtcgcgt cctatcagct tgacggcggg gtaacggccc 240
 accgtggctt cgacgggtag ccggcctgag agggcgaccg gccacattgg gactgagata 300
 cggcccag 308

SEQ ID NO: 160 moltype = DNA length = 316
 FEATURE Location/Qualifiers
 source 1..316

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mol_type = genomic DNA
organism = Parabacteroides sp.

SEQUENCE: 160
gatgaacgct agcgacaggc ttaacacatg caagtcgagg ggcagcatga ttgttagcaa 60
tacagattga tggcgacggg cgcacgggtg agtaacgcgt atgcaactta cctatcagag 120
ggggatagcc cggcgaaagt cggattaata ccccataaaa caggggtccc gcatgggaat 180
atttgttaaa gattcatgcg ttagatagat gcatgcgttc cattaggcag ttggcggggg 240
aacggccccc caaacggacg atggataggg gttctgagag gaagggtccc cacattggta 300
ctgagacacg gaccaa 316

SEQ ID NO: 161      moltype = DNA length = 329
FEATURE            Location/Qualifiers
source             1..329
                   mol_type = genomic DNA
                   organism = Veillonella sp.

SEQUENCE: 161
gacgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagagcgat ggaagcttgc 60
ttctatcaat cttagtggcg aacgggtgag taacgcgtaa tcaacctgcc cttcagaggg 120
ggacaacagt tggaaacgac tgctaatacc gcatacgatc taatctcggc atcgaggata 180
gatgaaagggt ggctcttaca tgtaagctat cactgaagga ggggattgag tctgattagc 240
tagttggagg ggtaacggcc caccaaggcg atgatcagta gccggtctga gaggatgaac 300
ggccacattg ggactgagac acggcccag 329

SEQ ID NO: 162      moltype = DNA length = 309
FEATURE            Location/Qualifiers
source             1..309
                   mol_type = genomic DNA
                   note = Schaalialia
                   organism = unidentified

SEQUENCE: 162
gacgaacgct ggcggcgctgc ttaacacatg caagtcgaac gctgaagccc agcttgcgtg 60
gtggatgagt gggaacgggg tgagtaaacac gtgagtaacc tgcccccttc ttggggataa 120
cgcccggaac cgggtgctaa tactggatat tcaactggcct tcgcatgggg gttggtggaa 180
aggttttttc ttgtggggga tgggctcgcg gcctatcagc ttgttggtgg ggtgatggcc 240
taccagggtt ttgacgggta gccggcctga gaggggtgacc ggtcacattg ggactgagat 300
acggcccag 309

SEQ ID NO: 163      moltype = DNA length = 310
FEATURE            Location/Qualifiers
source             1..310
                   mol_type = genomic DNA
                   organism = Parabacteroides sp.

SEQUENCE: 163
gatgaacgct agcgacaggc ttaacacatg caagtcgagg ggcagcacag gtagcaatac 60
cgggtggcga cgcgcgcacg ggtgagtaac gcgtatgcaa cttacctatc agagggggat 120
aaaccggcga aagtcggact aataccgcat gaagcagggg ccccgcatgg ggatatttgc 180
taaaagattca tcgctgtag atagggcatgc gttccattag gcagttggcg gggtaacggc 240
ccaccaaacg gacgatggat aggggttctg agaggaaggt ccccccacatt ggtactgaga 300
cacggaccaa 310

SEQ ID NO: 164      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source             1..316
                   mol_type = genomic DNA
                   organism = Bacteroides sp.

SEQUENCE: 164
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcattc cagtttgctt 60
gcaaaactgga gatggcgacc ggcgcacggg tgagtaaacac gtatccaacc tgccgataac 120
tcggggatag cctttcgaaa gaaagattaa taccgatgg cataatagaa ccgcatgggt 180
tgattattaa agaatttcgg ttatcgatgg ggatgcgttc cattaggcag ttggtggggg 240
aacggccccc caaaccttcg atggataggg gttctgagag gaagggtccc cacattggaa 300
ctgagacacg gtccaa 316

SEQ ID NO: 165      moltype = DNA length = 314
FEATURE            Location/Qualifiers
source             1..314
                   mol_type = genomic DNA
                   organism = Bacteroides sp.

SEQUENCE: 165
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcacatcagga agaaagcttg 60
ctttctttgc tggcgacggg cgcacgggtg agtaacacgt atccaacctg ccctttactc 120
ggggatagcc cttcgaaaaga aagattaata cccgatagca taatgattcc gcatgggttc 180
attattaaag gattccggta aaggatgggg atgcgttcca ttaggttgtt ggtgaggtaa 240
cggcccccac agccttcgat ggataggggt tctgagagga aggtccccca cattggaact 300
gagacacggt ccaa 314

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-continued

SEQ ID NO: 166 moltype = DNA length = 309
 FEATURE Location/Qualifiers
 source 1..309
 mol_type = genomic DNA
 organism = Bifidobacterium sp.

SEQUENCE: 166

gatgaacgct	ggcggcggtgc	ttaacacatg	caagtcgaac	gggatccatc	aggetttgct	60
tgggtggtgag	agtggcggaac	gggtgagtaa	tgcgtgaccg	acctgcccc	tacaccggaa	120
tagctcctgg	aaacgggtgg	taatgccgga	tgctccgact	cctcgcatgg	ggtgtcggga	180
aagatttcat	cggatatggga	tggggtcgcg	tcctatcagg	tagtcggcgg	ggtaacggcc	240
caccgagcct	acgacgggta	gccggcctga	gagggcgacc	ggccacattg	ggactgagat	300
acggcccag						309

SEQ ID NO: 167 moltype = DNA length = 323
 FEATURE Location/Qualifiers
 source 1..323
 mol_type = genomic DNA
 organism = Blautia sp.

SEQUENCE: 167

gatgaacgct	ggcggcggtgc	ttaacacatg	caagtcgaac	gggaaatatt	tcattgagac	60
tccggtggat	ttgatctatt	tctagtggcg	gacgggtgag	taacgcgtgg	gtaacctgcc	120
ttatacaggg	ggataacagt	cagaaatggc	tgctaatacc	gcataagcgc	acagagctgc	180
atggctcagt	gtgaaaaact	cgggtgggtat	aagatggacc	cgcgttggat	tagcttgttg	240
gtggggtaac	ggcccaccaa	ggcgacgatc	catagccggc	ctgagagggg	gaacggccac	300
attgggactg	agacacggcc	cag				323

SEQ ID NO: 168 moltype = DNA length = 311
 FEATURE Location/Qualifiers
 source 1..311
 mol_type = genomic DNA
 organism = Faecalibacterium sp.

SEQUENCE: 168

gacgaacgct	ggcggcgcgcg	ctaacacatg	caagtcgaac	gagcgagaga	gagcttgctt	60
tctcaagcga	gtggcggaacg	ggtgagtaac	gcgtgaggaa	cctgcctcaa	agagggggac	120
aacagttgga	aacgactgct	aataccgcat	aagcccacga	cccggcatcg	ggtagaggga	180
aaaggagcaa	tccgctttga	gatggcctcg	cgtccgatta	gctagttggt	gaggtaatgg	240
cccaccaagg	cgacgatcgg	tagccggact	gagaggttga	acggccacat	tgggactgag	300
acacggccca	g					311

SEQ ID NO: 169 moltype = DNA length = 316
 FEATURE Location/Qualifiers
 source 1..316
 mol_type = genomic DNA
 organism = Staphylococcus sp.

SEQUENCE: 169

gacgaacgct	ggcggcggtgc	ctaatacatg	caagtagaac	gctgaagaga	ggagcttgct	60
cttcttggat	gagttgcgaa	cgggtgagta	acgcgtaggt	aacctgcctt	gtagcggggg	120
ataactattg	gaaacgatag	ctaataccgc	ataacaatgg	atgacacatg	tcatttattt	180
gaaaggggca	attgtccac	tacaagatgg	acctgcgttg	tattagctag	taggtgaggt	240
aacggctcac	ctagggcgacg	atacatagcc	gacctgagag	ggtgatcggc	cacactggga	300
ctgagacacg	gcccag					316

SEQ ID NO: 170 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 organism = Blautia sp.

SEQUENCE: 170

gatgaacgct	ggcggcggtgc	ttaacacatg	caagtcgaac	gaagcacctt	gatttgattc	60
ttcggatgaa	gatcttggtg	actgagtggc	ggacgggtga	gtaacgcgtg	ggtaacctgc	120
ctcatacagg	gggataacag	ttagaaatga	ctgctaatac	cgcataagac	cacagcaccg	180
catggtgcag	gggtaaaaac	tccggtggta	tgagatggac	ccgcgtctga	ttaggtagtt	240
ggtggggtaa	cggcctacca	agccgacgat	cagtagccga	cctgagaggg	tgaccggcca	300
cattgggact	gagacacggc	caa				324

SEQ ID NO: 171 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 note = Firmicutes
 organism = unidentified

SEQUENCE: 171

gatgaacgct	ggcggcggtgc	ttaacacatg	caagtcgagc	gaagcagtta	agttgatttc	60
ttcggattga	ttcttaactg	actgagcggc	ggacgggtga	gtaacgcgtg	ggtgacctgc	120
cccataccgg	gggataaacg	ctggaaaacg	ctgctaatac	cgcataagcg	cacagagctg	180
catggctcgg	tgtgaaaaac	tccggtggta	tgggatgggc	ccgcgtctga	ttaggcagtt	240

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ggcggggtaa cggcccaacca aaccgacgat cagtagccgg cctgagaggg cgaccggcca 300
cattgggact gagacacggc ccaa 324

```

```

SEQ ID NO: 172      moltype = DNA length = 307
FEATURE            Location/Qualifiers
source             1..307
                  mol_type = genomic DNA
                  organism = Bifidobacterium sp.

```

```

SEQUENCE: 172
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggatcccag gagcttgctc 60
ctgggtgaga gtggcgaacg ggtgagtaat gcgtgaccga cctgccccat acaccggaat 120
agctcctgga aacgggtggt aatgccggat gctccagttg gatgcattgc cttctgggaa 180
agattcatcg gtatgggatg ggttcgcgtc ctatcagctt gatggcgggg taacggccca 240
ccatgggttc gacgggtagc cggcctgaga gggcgaccgg ccacattggg actgagatac 300
ggccccag 307

```

```

SEQ ID NO: 173      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                  mol_type = genomic DNA
                  organism = Blautia sp.

```

```

SEQUENCE: 173
gatgaacgct ggcggcgctgc ttaacacatg caagtcgagc gaagcacctt gacggatttc 60
ttcggattga agccttggtg actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcatacagg gggataacag ttggaacagg ctgctaatac cgcataagcg cacagtaccg 180
catggtacgg tgtgaaaaac tccggtggta tgagatggac ccgcgtctga ttagctagtt 240
ggtggggtaa cggcctacca aggcgatgat cagtagccgg tctgagagga tgaacggcca 300
cattgggact gagacacggc ccag 324

```

```

SEQ ID NO: 174      moltype = DNA length = 303
FEATURE            Location/Qualifiers
source             1..303
                  mol_type = genomic DNA
                  organism = Collinsella sp.

```

```

SEQUENCE: 174
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggcacccctc tccggaggga 60
agcgaagtgc gaacggctga gtaacacgtg gagaacctgc cccctcccc gggatagccg 120
ccgaaaggga cgggtaatac cggatacccc ggggtgccgc atggcacccc ggetaaagcc 180
ccgacgggag gggatggctc cgcggcccat caggtagacg gcggggtgac ggcccaccgt 240
gccgacaacg ggtagccggg ttgagagacc gaccggccag attgggactg agacacggcc 300
cag 303

```

```

SEQ ID NO: 175      moltype = DNA length = 326
FEATURE            Location/Qualifiers
source             1..326
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 175
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcacttt atttgatttc 60
cttcgggact gattattttg tgactgagtg gcggacgggt gagtaacgag tgggtaacct 120
gccttgatac gggggataac agttggaaac ggctgctaata accgcataag cgcacagcat 180
cgcatgatgc agtgtgaaaa actccgggtg tataaagatgg acccgcttg gattagctag 240
ttggtgaggt aacggccccc caaggcgagc atccatagcc gacctgagag ggtgaccggc 300
cacattggga ctgagacagc gcccaa 326

```

```

SEQ ID NO: 176      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                  mol_type = genomic DNA
                  note = Lachnoclostridium
                  organism = unidentified

```

```

SEQUENCE: 176
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac gaagcaatta aaatgaagtt 60
ttcggatgga tttttgattg actgagtggc ggacgggtga gtaacgcgtg gataacctgc 120
ctcacactgg gggataaacg ttgaaatga ctgctaatac cgcataagcg cacagtaccg 180
catggtacgg tgtgaaaaac tccggtgggt tgagatggat ccgcgtctga ttagccagtt 240
ggcggggtaa cggccacca aagcgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa 324

```

```

SEQ ID NO: 177      moltype = DNA length = 307
FEATURE            Location/Qualifiers
source             1..307
                  mol_type = genomic DNA
                  organism = Bifidobacterium sp.

```

```

SEQUENCE: 177
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggatcccag gagcttgctc 60

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ctgggtgaga gtggcgaacg ggtgagtaat gcgtgaccga cctgccccat acaccggaat 120
agctcctgga aacgggtggg aatgccggat gctccagttg accgcattgg cctctgggaa 180
agcttttgcg gtatgggatg gggctgcgtc ctatcagctt gatggcgggg taacggccca 240
ccatggcttc gacgggtagc cggcctgaga gggcgaccgg ccacattggg actgagatac 300
ggcccag                                     307

```

```

SEQ ID NO: 178      moltype = DNA length = 302
FEATURE            Location/Qualifiers
source             1..302
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 178
gatgaacgct ggcggcgtgc ctaatacatg caagtcgaac gcttcaacttc ggtgaagagt 60
ggcgaacggg tgagtaatac ataagtaacc tggcatctac aggggggataa ctgatggaaa 120
cgtcagctaa gaccgcatag gtgtagagat cgcatagaact ctatatgaaa agtgctacgg 180
gactggtaga tgatggactt atggcgcatc agctgggttg tagggtaacg gcctaccaag 240
gcgacgatgc gtagccgacc tgagaggggt accggccaca ctgggactga gacacggccc 300
ag                                     302

```

```

SEQ ID NO: 179      moltype = DNA length = 321
FEATURE            Location/Qualifiers
source             1..321
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 179
gatgaacgct ggcggcgtgc ctaatacatg caagtcgaac gaagtcttga ggaagcttgc 60
ttccaaagag acttagtggt gaacgggtga gtaaacacgta ggtaacctgc ccatgtgtcc 120
gggataactg ctggaaacgg tagctaaaac cggataggta tacagagcgc atgctcagta 180
tattaaagcg cccatcaagg cgtgaacatg gatggacctg cggcgcatga gctagtgtgt 240
gaggtaacgg cccaccaagg cgatgatgct tagccggcct gagagggtaa acggccacat 300
tgggactgag acacggccca a                                     321

```

```

SEQ ID NO: 180      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                  mol_type = genomic DNA
                  organism = Dorea sp.

```

```

SEQUENCE: 180
gatgaacgct ggcggcgtgc ttaacacatg caagtcgagc gaagcactta agtttgattc 60
ttcggatgaa gacttttgtg actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcatcacag gggataacag ttagaataag ctgctaatac cgcataagac caccgtaccg 180
catgttacag ttgtaaaaac tctggttggt tagatggac ccgcgtctga ttaggtagt 240
gggtgggtta agccctacca agccgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact ggacacggcg ccag                                     324

```

```

SEQ ID NO: 181      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source             1..323
                  mol_type = genomic DNA
                  organism = Blautia sp.

```

```

SEQUENCE: 181
gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac gggaattact ttattgaaac 60
ttcggtcgat atgatttaat tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatacaggg ggataacagt cagaaatgac tgctaatacc gcataagcgc acagagctgc 180
atggctcggg gtgaaaaaac ccgggtggtat aagatggacc cgcgttggat tagctagt 240
gtgaggtaac ggcccaccaa ggcgacgac catagccggc ctgagagggg gaacggccac 300
attgggactg agacacggcg cag                                     323

```

```

SEQ ID NO: 182      moltype = DNA length = 326
FEATURE            Location/Qualifiers
source             1..326
                  mol_type = genomic DNA
                  organism = Anaerostipes sp.

```

```

SEQUENCE: 182
gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac gaaacacctt atttgatttt 60
cttcggaact gaagatttgg tgattgagtg gcggacgggt gagtaacgcg tgggtaacct 120
gcccgtgata gggggataac agtcagaaat gactgctaata accgcataag accacagcac 180
cgcatgggtg aggggtaaaa actccgggtg tacaggatgg acccgctctc gattagctgg 240
ttggtgaggt aacggctcac caaggcgacg atcagtagcc ggcttgagag agtgaacggc 300
cacattggga ctgagacacg gcccac                                     326

```

```

SEQ ID NO: 183      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                  mol_type = genomic DNA
                  note = Lachnospiraceae

```

-continued

```

                                organism = unidentified
SEQUENCE: 183
gatgaacgct ggcggcgctgc ctaacacatg caagtcgagc gaagcacttt gcttagattc 60
ttcggatgaa gaggattgtg actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcatacagg gggataaacg ttgaaatga ctgctaatac cgcataagac cacagcaccg 180
catggtgcgg gggtaaaaac tccggtggta tgagatggac ccgcgtctga ttagctagtt 240
ggtaaggtaa cggcttacca aggcgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa                                     324

SEQ ID NO: 184      moltype = DNA length = 311
FEATURE            Location/Qualifiers
source             1..311
                   mol_type = genomic DNA
                   organism = Faecalibacterium sp.

SEQUENCE: 184
gacgaacgct ggcggcgcgcc ctaacacatg caagtcgaac gagagagagg aagcttgctt 60
cctcaatcga gtggcggaacg ggtgagtaac gcgtgaggaa cctgcctcaa agagggggac 120
aacagttgga aacgactgct aataccgcat aagcccacag gtcggcatcg accagaggga 180
aaaggagcaa tccgctttga gatggcctcg cgtccgatta gctagttggt gaggtaacgg 240
cccaccaagg caacgatcgg tagccggact gagaggttga acggccacat tgggactgag 300
acacggccca g                                     311

SEQ ID NO: 185      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source             1..323
                   mol_type = genomic DNA
                   organism = Blautia sp.

SEQUENCE: 185
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggaaacctt ttattgaagc 60
ttcggcagat ttagctgggt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatacaggg ggataacaac cagaaatggt tgctaatacc gcataagcgc acaggaccgc 180
atggtccggg gtgaaaaaact ccggtgggtat aagatggacc cgcgttggtg ttagctagtt 240
gcagggtaac ggcctacca ggcgacgatc catagccggc ctgagagggg gaacggccac 300
attgggactg agacacggcc cag                                     323

SEQ ID NO: 186      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                   mol_type = genomic DNA
                   note = Lachnospiraceae
                   organism = unidentified

SEQUENCE: 186
gatgaacgct ggcggcgctgc ttaacacatg caagtcgagc gaagcaattt aagtgaagtt 60
ttcggatgga tcttaaatgt actgagcggc ggacgggtga gtaacgcgtg gataacctgc 120
ctcacacagg gggataaacg ttgaaatga ctgctaatac cgcataagcg cacggtaccg 180
catggtacag ggtgaaaaac tccggtgggt tgagatggat ccgcgtctga ttaggtagtt 240
ggtgaggtaa cggcccaacca agccgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa                                     324

SEQ ID NO: 187      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                   mol_type = genomic DNA
                   organism = Clostridium sp.

SEQUENCE: 187
gatgaacgct ggcggcgctgc ctaacacatg caagtcgagc gaagcggttt taagggaagtt 60
ttcggatgga attaaaactg actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcatacagg gggataaacg ttgaaatga ctgctaatac cgcataagca cacagcttcg 180
catggagcag tgtgaaaaac tccggtgggt tgagatggac ccgcgtctga ttagctagtt 240
ggtaaggtaa cggcttacca aggcgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa                                     324

SEQ ID NO: 188      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                   mol_type = genomic DNA
                   note = Clostridiaceae
                   organism = unidentified

SEQUENCE: 188
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcaatct aacgggaagtt 60
ttcggatgga agctagattg actgagtggt ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcacactgg gggacaacag ttgaaatga ctgctaatac cgcataagcg cacaggaccg 180
catggtccgg tgtgaaaaac tccggtgggt tgagatggac ccgcgtttga ttagctagtt 240
ggtggggtaa cggcctacca aggcgacgat caatagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa                                     324

```

-continued

SEQ ID NO: 189 moltype = DNA length = 311
 FEATURE Location/Qualifiers
 source 1..311
 mol_type = genomic DNA
 organism = *Faecalibacterium* sp.

SEQUENCE: 189

gacgaacgct	ggcggcgcg	ctaacacatg	caagtcgaac	ggagcttaga	gagcttgctt	60
tttaagctta	gtggcggaac	ggtgagtaac	gcgtgagtaa	cctgccctag	agtgggggac	120
aacagttgga	aacgactgct	aataccgcat	aagcccacgg	taccgcatgg	tactgagggg	180
aaaggattta	ttcgctttag	gatggactcg	cgtccaatta	gctagttggt	gaggtaacgg	240
cccaccaagg	cgacgattgg	tagccgggact	gagaggttga	acggccacat	tgggactgag	300
acacggccca	g					311

SEQ ID NO: 190 moltype = DNA length = 317
 FEATURE Location/Qualifiers
 source 1..317
 mol_type = genomic DNA
 organism = *Barnesiella* sp.

SEQUENCE: 190

gatgaacgct	agcgacaggc	ctaacacatg	caagtcgagg	ggcagcgaag	aggtagcaat	60
acctttgtcg	gcgaccggcg	caagggtgag	taacacgtat	gcaatccacc	tgtaacaggg	120
ggataaccgg	gagaaatccg	gactaatacc	ccataatatg	ggcgtcccg	atggagagcc	180
cattaaagag	agcaattttg	gttacagacg	agcatgcgct	ccattagcca	gttggcgggg	240
taacggccca	ccaaagcgac	gatggatagg	ggttctgaga	ggaaggtccc	ccacattgga	300
actgagacac	ggtccaa					317

SEQ ID NO: 191 moltype = DNA length = 322
 FEATURE Location/Qualifiers
 source 1..322
 mol_type = genomic DNA
 note = *Lachnospiraceae*
 organism = unidentified

SEQUENCE: 191

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgagc	gaagcgattt	aaatgagact	60
tcggtggatt	ttaaattgac	tgagcggcg	acgggtgagt	aacgcgtgga	taacctgcct	120
cacacagggg	gataacagtt	agaaatgact	gctaataacc	cataagcgca	cggtaccgca	180
tggtacagtg	tgaaaaactc	cggtggtgtg	agatggatcc	gcgtctgatt	aggtagtgtg	240
tgaggtaacg	gccccaccaag	ccgacgatca	gtagccgacc	tgagaggggt	accggccaca	300
ttgggactga	gacacggccc	aa				322

SEQ ID NO: 192 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 organism = *Roseburia* sp.

SEQUENCE: 192

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgaac	gaagcacttt	gatcgatttc	60
ttcgatttga	aatttttagtg	actgagtggc	ggacgggtga	gtaacgcgtg	ggtaacctgc	120
ctcacacagg	gggataaacg	ttggaaacgg	ctgctaatac	cgcataagcg	cacagtaccg	180
catggtacag	tgtgaaaaac	tcgggtggtg	tgagatggac	ccgcgtctga	ttagctagtt	240
ggcagggcaa	cggcctacca	aggcgacgat	cagtagccga	cctgagaggg	tgaccggcca	300
cattgggact	gagacacggc	caa				324

SEQ ID NO: 193 moltype = DNA length = 316
 FEATURE Location/Qualifiers
 source 1..316
 mol_type = genomic DNA
 organism = *Bilophila* sp.

SEQUENCE: 193

attgaacgct	ggcggcgctgc	ttaacacatg	caagtcgaac	gtgaaagtcc	ttcgggatga	60
gtaaagtggc	gcacgggtga	gtaacgcgtg	gataatctac	ccttaagatg	gggataacgg	120
ctggaaacgg	tcgctaatac	cgaatacgct	cccgatTTta	tcgttggggg	gaaagatggc	180
ctctgcttgc	aagctatcgc	ttaaggatga	gtccgcgtcc	cattagctag	ttggcggggg	240
aacggcccac	caaggcgacg	atgggtagcc	ggtctgagag	gatgaccggc	cacactggaa	300
ctggaaacag	gtccag					316

SEQ ID NO: 194 moltype = DNA length = 311
 FEATURE Location/Qualifiers
 source 1..311
 mol_type = genomic DNA
 organism = *Actinomyces* sp.

SEQUENCE: 194

gacgaacgct	ggcggcgctgc	ttaacacatg	caagtcgaac	gctgaaggct	cagcttgctg	60
ggctggatga	gtggcggaac	ggtgagtaac	acgtgagtaa	cctgccctct	tctttgggat	120
aacggtcgga	aacggctgct	aataccgggt	attcactgtc	cttcgcatgg	ggggtggtgg	180
aaaggttttt	tctggtgggg	gatgggctcg	cggcctatca	gcttggtggt	ggggtgatgg	240

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cttaccaagg ctttgacggg taaccggcct gagagggtga cgggtcacat tgggactgag 300
atacggccca g                                     311

```

```

SEQ ID NO: 195      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source             1..323
                  mol_type = genomic DNA
                  organism = Ruminococcus sp.

```

```

SEQUENCE: 195
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggaaatact ttattgaaac 60
ttcgggtggat ttaatttatt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatactggg ggataacagc cagaaatgac tgctaatacc gcataagcgc acagaaccgc 180
atggttcggt gtgaaaaact ccggtgggat aagatggacc cgcgttggat tagctagttg 240
gcagggcagc ggcctaccaa ggcgacgatc catagccggc ctgagagggt gaacggccca 300
attgggactg agacacggcc cag                                     323

```

```

SEQ ID NO: 196      moltype = DNA length = 325
FEATURE            Location/Qualifiers
source             1..325
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 196
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggagtcgttt tggaaaatcc 60
ttcgggattg gaattctcga cttagtggcg gacgggtgag taacgcgtga gcaatctgcc 120
tttaagaggg ggataacagt cggaaacggc tgctaatacc gcataagca atgaattcgc 180
atgttttcat tgccaaagga gcaatccgct tttagatgag ctccgctctg attagctagt 240
tgccggggta acggcccacc aaggcgacga tcagtagccg gactgagagg ttgaacggcc 300
acattgggac tgagacacgg cccag                                     325

```

```

SEQ ID NO: 197      moltype = DNA length = 325
FEATURE            Location/Qualifiers
source             1..325
                  mol_type = genomic DNA
                  organism = Butyricicoccus sp.

```

```

SEQUENCE: 197
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggaactatct tggaaaatcc 60
ttcgggagcg gaattctttag tttagtggcg gacgggtgag taacgcgtga gcaatctgcc 120
tttaagaggg ggataacagt cggaaacggc tgctaatacc gcataagca tcaaatctgc 180
atgttttga tgccaaagga gcaatccgct tttagatgag ctccgctctg attagctagt 240
tgccggggta acggcccacc aaggcgacga tcagtagccg gactgagagg ttgaacggcc 300
acattgggac tgagacacgg cccag                                     325

```

```

SEQ ID NO: 198      moltype = DNA length = 311
FEATURE            Location/Qualifiers
source             1..311
                  mol_type = genomic DNA
                  organism = Faecalibacterium sp.

```

```

SEQUENCE: 198
gacgaacgct ggcggcgcgcg ctaacacatg caagtcgaac gagcgagaga gagcttgctt 60
tctcaagcga gtggcggaacg ggtgagtaac gcgtgaggaa cctgcctcaa agagggggac 120
aacagttgga aacgactgct aataccgcat aagcccacgg gtcggcatcg accagaggga 180
aaaggagtga tccgctttga gatggcctcg cgtccgatta gctagttggt gaagtaacgg 240
cccaccaagg caacgatcgg tagccggact gagagggtga acggccacat tgggactgag 300
atacggccca g                                     311

```

```

SEQ ID NO: 199      moltype = DNA length = 314
FEATURE            Location/Qualifiers
source             1..314
                  mol_type = genomic DNA
                  organism = Bacteroides sp.

```

```

SEQUENCE: 199
gatgaacgct agctacaggg ttaacacatg caagtcgagg ggcagcatga acttagcttg 60
ctaagtttga tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccgatgactc 120
ggggatagcc tttcgaaaga aagattaata cccgatggca tagttcttcc gcatggtaga 180
actattaaag aatttcggtc atcgatgggg atgcgttcca ttaggttggt ggcggggtaa 240
cggccccaac agccttcgat ggataggggt tctgagagga aggtcccca cattggaact 300
gagacacggt ccaa                                     314

```

```

SEQ ID NO: 200      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source             1..316
                  mol_type = genomic DNA
                  organism = Bacteroides sp.

```

```

SEQUENCE: 200
gatgaacgct agctacaggg ttaacacatg caagtcgagg ggcagcatct cagtttgctt 60
gcaaactgga gatggcgacc ggcgcacggg tgagtaacac gtatccaacc tgccgataac 120

```

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```

tcggggatag cctttcgaaa gaaagattaa tacccgatgg tataattaga ccgcatggtc 180
ttgttattaa agaatttcgg ttatcgatgg ggatgcgttc cattaggcag ttggtgaggt 240
aacggctcac caaaccttcg atggataggg gttctgagag gaaggtcccc cacattggaa 300
ctgagacacg gtccaa 316

```

```

SEQ ID NO: 201      moltype = DNA length = 317
FEATURE            Location/Qualifiers
source              1..317
                    mol_type = genomic DNA
                    organism = Bacteroides sp.

```

```

SEQUENCE: 201
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatgg tcttagcttg 60
ctaaggctga tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccgtctactc 120
tggccagcc  ttctgaaagg aagattaatc caggatggga tcatgagttc acatgtccgc 180
atgattaaag gtattttccg gtagacgatg gggatgcgtt ccattagata gtaggcgggg 240
taacggccca cctagtcaac gatggatagg ggttctgaga ggaaggtccc ccacattgga 300
actgagacac ggtccaa 317

```

```

SEQ ID NO: 202      moltype = DNA length = 315
FEATURE            Location/Qualifiers
source              1..315
                    mol_type = genomic DNA
                    organism = Bacteroides sp.

```

```

SEQUENCE: 202
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatgg tcttagcttg 60
ctaaggcga  tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccgtctactc 120
tggacagcc  ttctgaaagg aagattaata caagatggca tcatgagttc gcatgttcac 180
atgattaaag gtattccggt agacgatggg gatgcgttcc attagatagt aggcggggta 240
acggcccaac tagtcttcga tggatagggg ttctgagagg aaggtccccc acattggaaac 300
tgagacacgg tccaa 315

```

```

SEQ ID NO: 203      moltype = DNA length = 314
FEATURE            Location/Qualifiers
source              1..314
                    mol_type = genomic DNA
                    organism = Bacteroides sp.

```

```

SEQUENCE: 203
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatca tcaaagcttg 60
ctttgatgga tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccgacaacac 120
tgggatagcc ttctgaaaga aagattaata ccgcatggca tagttttccc gcatgggatg 180
attattaaag aatttcggtt gtcatggggg atgcgttcca ttaggcagtt ggccggggtaa 240
cggccccaac aaccaacgat ggataggggt tctgagagga aggtccccc cattggaact 300
gagacacggt ccaa 314

```

```

SEQ ID NO: 204      moltype = DNA length = 314
FEATURE            Location/Qualifiers
source              1..314
                    mol_type = genomic DNA
                    organism = Bacteroides sp.

```

```

SEQUENCE: 204
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatga acttagcttg 60
ctaagttga  tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccgatgactc 120
ggggatagcc ttctgaaaga aagattaata cccgatggca tagttcttcc gcatggtaga 180
actattaaag aatttcggtc atcgatgggg atgcgttcca ttaggttgtt ggccggggtaa 240
cggccccaac agccttcgat ggataggggt tctgagagga aggtccccc cattggaact 300
gagacacggt ccaa 314

```

```

SEQ ID NO: 205      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source              1..323
                    mol_type = genomic DNA
                    organism = Blautia sp.

```

```

SEQUENCE: 205
gatgaacgct ggcggcggtgc ttaacacatg caagtcgaac gggaaatatt tcattgagac 60
ttcggtggat ttgatctatt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatacaggg ggataaacagt cagaaatggc tgctaatacc gcataagcgc acagagctgc 180
atggctcagt gtgaaaaact ccggtgggat aagatggacc cgcgttggat tagcttgttg 240
gtggggtaac ggcccaccaa ggcgacgatc catagccggc ctgagagggt gaacggccac 300
attgggactg agacacggcc cag 323

```

```

SEQ ID NO: 206      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source              1..316
                    mol_type = genomic DNA
                    organism = Bacteroides sp.

```

```

SEQUENCE: 206

```

-continued

```

gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcattt cagtttgctt 60
gcaaaactgga gatggcgacc ggcgcacggg tgagtaaacac gtatccaacc tgccgataac 120
tcgggggatag cctttcgaaa gaaagattaa tacccgatgg tataattaga ccgcatgggtc 180
ttgttattaa agaatttcgg ttatcgatgg ggatgcgttc cattaggcag ttggtgaggt 240
aacggctcac caaaccttcg atggataggg gttctgagag gaagggtccc cacattggaa 300
ctgagacacg gtccaa 316

```

```

SEQ ID NO: 207      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source             1..323
                   mol_type = genomic DNA
                   note = Phascolarctobacterium
                   organism = unidentified

```

```

SEQUENCE: 207
gacgaacgct ggcggcggtgc ctaacacatg caagtcgaac ggagaatttt atttcggtag 60
aattcttagt ggcgaacggg tgagtaaacg gttaggcaacc tgccctttag acggggacaa 120
cattccgaaa ggagtgcata taccggatgt gatcatcggt ccgcataagca ggatgaagaa 180
agatggcctc tacaagtaag ctatcgctaa aggatggggc tcgctctgat tagctagttg 240
gtagtgtaac ggactaccaa ggcgatgatc agtagccggt ctgagaggat gaacggccac 300
attgggactg agacacggcc caa 323

```

```

SEQ ID NO: 208      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                   mol_type = genomic DNA
                   organism = Dorea sp.

```

```

SEQUENCE: 208
gatgaacgct ggcggcggtgc ttaacacatg caagtcgagc gaagcactta agtttgattc 60
ttcggatgaa gactttttgt actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcatacagg gggataacag ttgaaatga ctgctaatac cgcataagac caccgtaccg 180
catggtacag tggtaaaaac tctggtggtg tgagatggac ccgctctgga ttaggtagtt 240
ggtagggtaa cggcctacca agccgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccag 324

```

```

SEQ ID NO: 209      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source             1..323
                   mol_type = genomic DNA
                   organism = Blautia sp.

```

```

SEQUENCE: 209
gatgaacgct ggcggcggtgc ttaacacatg caagtcgaac gggaattact ttattgaaac 60
ttcggtcgat atgatttaat tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatacaggg ggataacagt cagaaatgac tgctaatacc gcataagcgc acagagctgc 180
atggctcggg gtgaaaaact ccgggtggtat aagatggacc cgcgttggat tagctagttg 240
gtgaggtaac ggcccaccaa ggcgacgatc catagccggc ctgagagggt gaacggccac 300
attgggactg agacacggcc cag 323

```

```

SEQ ID NO: 210      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source             1..316
                   mol_type = genomic DNA
                   organism = Bacteroides sp.

```

```

SEQUENCE: 210
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcattt tagtttgctt 60
gcaaaactaaa gatggcgacc ggcgcacggg tgagtaaacac gtatccaacc tgccgataac 120
tcgggggatag cctttcgaaa gaaagattaa tatccgatgg tatattttct ccgcatgggtg 180
agaatattaa agaatttcgg ttatcgatgg ggatgcgttc cattagtttg ttggcgggggt 240
aacggccccc caagaccacg atggataggg gttctgagag gaagggtccc cacattggaa 300
ctgagacacg gtccaa 316

```

```

SEQ ID NO: 211      moltype = DNA length = 322
FEATURE            Location/Qualifiers
source             1..322
                   mol_type = genomic DNA
                   organism = Megamonas sp.

```

```

SEQUENCE: 211
gacgaacgct ggcggcggtgc ttaacacatg caagtcgaac ggggtgttta ttccggtaaa 60
caccgaagtgg cgaacgggtg agtaacgcgt aagcaatcta ccttcaagat ggggacaaca 120
cttcgaaagg ggtgctaata ccgaatgaat gtaagagtat cgcatgagac acttactaaa 180
ggaggcctct gacataagct tccgcttgaa gatgagcttg cgtctgatta gctagttggt 240
gagggtaaaag gccaccacag gcgacgatca gtagccggtc tgagaggatg aacggccaca 300
ttgggactga gacacggccc ag 322

```

```

SEQ ID NO: 212      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source             1..316

```

-continued

```

mol_type = genomic DNA
organism = Staphylococcus sp.

SEQUENCE: 212
gacgaacgct ggcggcgctgc ctaatacatg caagtagaac gctgaagaga ggagcttgct 60
cttcttggat gagttgacgaa cgggtgagta acgcgtaggt aacctgcctt gtagcggggg 120
ataactattg gaaacgatag ctaataccgc ataacaatgg atgacacatg tcatttattt 180
gaaaggggca attgctccac tacaagatgg acctgcgttg tattagctag taggtgaggt 240
aacggctcac ctaggcgacg atacatagcc gacctgagag ggtgatcggc cacactggga 300
ctgagacacg gccacg 316

SEQ ID NO: 213      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source              1..324
                    mol_type = genomic DNA
                    organism = Blautia sp.

SEQUENCE: 213
gatgaacgct ggcggcgctgc ttaacacatg caagtogaac gaagcacctt gatttgattc 60
ttcggatgaa gatcttggtg actgagtggt ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcatcacag gggataaacg ttagaatga ctgctaatac cgcataagac cacagcaccg 180
catggtgcag gggtaaaaaa tccggtggtg tgagatggac ccgcgtctga ttaggtagtt 240
ggtggggtaa cggcctacca agccgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa 324

SEQ ID NO: 214      moltype = DNA length = 307
FEATURE            Location/Qualifiers
source              1..307
                    mol_type = genomic DNA
                    organism = Bifidobacterium sp.

SEQUENCE: 214
gatgaacgct ggcggcgctgc ttaacacatg caagtogaac gggatcccag gagcttgctc 60
ctgggtgaga gtggcgaaac ggtgagtaat gcgtgaccga cctgccccat acaccggaat 120
agctcctgga aacgggtggt aatgcgggat gctccagttg gatgcattgc cttctgggaa 180
agattcatcg gtatgggatg gggtcgcgtc ctatcagctt gatggcgggg taacggccca 240
ccatggcttc gacgggtagc cggcctgaga gggcgaccgg ccacattggg actgagatac 300
ggcccag 307

SEQ ID NO: 215      moltype = DNA length = 303
FEATURE            Location/Qualifiers
source              1..303
                    mol_type = genomic DNA
                    organism = Collinsella sp.

SEQUENCE: 215
gatgaacgct ggcggcgctgc ctaacacatg caagtogaac ggcacccttc tccggaggga 60
agcgagtggc gaaacgggtga gtaacacgtg gagaacctgc cccctcccc gggatagccg 120
ccgaaaggga cgggtaatac cggatacccc ggggtgccgc atggcaccgc ggctaagacc 180
ccgacgggag gggatggctc cgcggcccat caggtagacg gcgggggtgac ggcccaccgt 240
gcccacaacg ggtagccggg ttgagagacc gaccggccag attgggactg agacacggcc 300
cag 303

SEQ ID NO: 216      moltype = DNA length = 326
FEATURE            Location/Qualifiers
source              1..326
                    mol_type = genomic DNA
                    organism = Clostridium sp.

SEQUENCE: 216
gatgaacgct ggcggcgctgc ttaacacatg caagtogaac gaagcacttt atttgatttc 60
cttcgggact gattattttg tgactgagtg gcggacgggt gagtaacgcg tgggtaacct 120
gccttgtaga gggggataac agttgaaac ggctgctaata accgcataag cgcacagcat 180
cgcatgatgc agtgtgaaaa actccggtgg tataagatgg acccgcgttg gattagctag 240
ttggtgaggt aacggcccac caaggcgacg atccatagcc gacctgagag ggtgaccggc 300
cacattggga ctgagacacg gcccaa 326

SEQ ID NO: 217      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source              1..316
                    mol_type = genomic DNA
                    organism = Parabacteroides sp.

SEQUENCE: 217
gatgaacgct agcgacaggg ttaacacatg caagtcgagg ggcagcatga ttgtagcaa 60
tacagattga tggcgaccgg cgcacgggtg agtaacgcgt atgcaactta cctatcagag 120
ggggatagcc cggcgaaaag cggattaata cccataaaa cagggggtccc gcatgggaat 180
atttgtaaaa agttcatcgc tgatagatag gcatgcgttc cattagcagc ttggcggggg 240
aacggcccac caaaccgacg atggataggg gttctgagag gaaggtcccc cacattggta 300
ctgagacacg gaccaa 316

SEQ ID NO: 218      moltype = DNA length = 307

```

-continued

FEATURE
source Location/Qualifiers
 1..307
 mol_type = genomic DNA
 organism = Bifidobacterium sp.

SEQUENCE: 218
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggatcccag gagcttgctc 60
ctgggtgaga gtggcgaaacg ggtgagtaat gcgtgaccga cctgcccacat acaccggaat 120
agctcctgga aacgggtgggt aatgccggat gctccagttg accgcatggt cctctgggaa 180
agcttttgcg gtatgggatg gggtcgcgctc ctatcagctt gatggcgggg taacggccca 240
ccatggcttc gacgggtagc cggcctgaga gggcgaccgg ccacattggg actgagatac 300
ggcccag 307

SEQ ID NO: 219 moltype = DNA length = 302
FEATURE Location/Qualifiers
source 1..302
 mol_type = genomic DNA
 organism = Clostridium sp.

SEQUENCE: 219
gatgaacgct ggcggcgctgc ctaatacatg caagtcgaac gcttcacttc ggtgaagagt 60
ggcgaaacggg tgagtaatac ataagtaacc tggcatctac aggggggataa ctgatggaaa 120
cgtcagctaa gaccgcatag gtgtagagat cgcatagaact ctatatgaaa agtgctacgg 180
gactggtaga tgatggactt atggcgcat agctgggttg tagggtaacg gcctaccaag 240
gcgacgatgc gtagccgacc tgagaggggt accggccaca ctgggactga gacacgggcc 300
ag 302

SEQ ID NO: 220 moltype = DNA length = 315
FEATURE Location/Qualifiers
source 1..315
 mol_type = genomic DNA
 organism = Bacteroides sp.

SEQUENCE: 220
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatga acttagcttg 60
ctaagtttga tggcgaccgg cgcacgggtg agtaaacagt atccaacctt ccgtttactc 120
agggatagcc tttcgaaaga aagattaata cctgatagta tggtagagatt gcatgatagc 180
accattaaag atttattggt aaacgatggg gatgcgttcc attaggtagt aggcggggta 240
acggccacc tagccgacga tggatagggg ttctgagagg aaggtcccc acattggaac 300
tgagacacgg tccaa 315

SEQ ID NO: 221 moltype = DNA length = 326
FEATURE Location/Qualifiers
source 1..326
 mol_type = genomic DNA
 organism = Roseburia sp.

SEQUENCE: 221
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcactct atttgatttt 60
cttcggaaat gaagattttg tgactgagtg gcggacgggt gagtaacgcg tgggtaacct 120
gcctcataca gggggataac agttggaaac gactgctaat accgcataag cgcacaggat 180
cgcatgatcc ggtgtgaaaa actccgggtg tatgggatgg acccgctct gattagccag 240
ttggcagggt aacggcctac caaagcgacg atcagtagcc gacctgagag ggtgaccggc 300
cacattggga ctgagacacg gcccac 326

SEQ ID NO: 222 moltype = DNA length = 323
FEATURE Location/Qualifiers
source 1..323
 mol_type = genomic DNA
 note = Lachnospiraceae
 organism = unidentified

SEQUENCE: 222
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggaaacctt ttattgaagc 60
ttcggcagat ttatagttgt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatacaggg ggataacagt cagaaatgac tgctaatacc gcataagcgc acaggaccgc 180
atggtccgggt gtgaaaaact ccggtgggtat aagatggacc cgcgttggat tagctagttg 240
gtgaggtaac ggcccaccaa ggcgacgatc catagccggc ctgagagggg gaacggccac 300
attgggactg agacacggcc cag 323

SEQ ID NO: 223 moltype = DNA length = 324
FEATURE Location/Qualifiers
source 1..324
 mol_type = genomic DNA
 note = Lachnospiraceae
 organism = unidentified

SEQUENCE: 223
gatgaacgct ggcggcgctgc ttaacacatg caagtcgagc gaagcaattt aagtgaagtt 60
ttcggatgga tcttaaatgt actgagcggc ggacgggtga gtaacgcgtg gataacctgc 120
ctcacacagg ggataaacag ttgaaatga ctgctaatac cgcataagcg cagggtaccg 180
catggtacag tgtgaaaaac tccggtgggt tgagatggat ccgcgtctga ttaggtagtt 240

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ggtgaggttaa cggcccccacga agccgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa 324

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SEQ ID NO: 224      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 224
gatgaacgct ggcggcgctgc ctaacacatg caagtcgagc gaagcggttt taaggaagtt 60
ttcggatgga attaaaaatg actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcatcacgg gggataacag ttgaaatgg ctgctaatac cgcataagca cacagcttcg 180
catggagcag tgtgaaaaac tccggtggtg tgagatggac ccgcgtctga ttagctagtt 240
ggtaaggtaa cggcttacca agccgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa 324

```

```

SEQ ID NO: 225      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                  mol_type = genomic DNA
                  note = Clostridiaceae
                  organism = unidentified

```

```

SEQUENCE: 225
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcaatct aacggaagtt 60
ttcggatgga agctagattg actgagtggc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcacactgg gggacaacag ttgaaatga ctgctaatac cgcataagcg cacaggaccg 180
catggtccgg tgtgaaaaac tccggtggtg tgagatggac ccgcgtttga ttagctagtt 240
ggtggggtaa cggcctacca agccgacgat caatagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa 324

```

```

SEQ ID NO: 226      moltype = DNA length = 311
FEATURE            Location/Qualifiers
source             1..311
                  mol_type = genomic DNA
                  organism = Faecalibacterium sp.

```

```

SEQUENCE: 226
gacgaacgct ggcggcgcgcc ctaacacatg caagtcgaac ggagcttaga gagcttgctt 60
tttaagctta gtggcggaac ggtgagtaac gcgtgagtaa cctgccctag agtgggggac 120
aacagttgga aacgactgct aataccgcat aagccacggg taccgcatgg tactgaggga 180
aaaggattta ttcgctttag gatggactcg cgtccaatta gctagttggt gaggtaacgg 240
cccaccaagg cgaagattgg tagccggact gagaggttga acggccacat tgggactgag 300
acacggccca g 311

```

```

SEQ ID NO: 227      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source             1..316
                  mol_type = genomic DNA
                  organism = Streptococcus sp.

```

```

SEQUENCE: 227
gatgaacgct ggcggcgctgc ctaatacatg caagtagaac gctgaagctt ggtgcttgca 60
cagagcggat gagttgcgaa cgggtgagta acgcgtaggt aacctgcctg gtacgggggg 120
ataactattg gaaacgatag ctaataccgc ataacagtag atgtcgcatg acctttgctt 180
gaaagggtgca attgcaccac taccagatgg acctgcgttg tattagctag ttggtgaggt 240
aacggctcac caaggcgacg atacatagcc gacctgagag ggtgatcggc cacactggga 300
ctgagacacg gccacag 316

```

```

SEQ ID NO: 228      moltype = DNA length = 322
FEATURE            Location/Qualifiers
source             1..322
                  mol_type = genomic DNA
                  note = Lachnospiraceae
                  organism = unidentified

```

```

SEQUENCE: 228
gatgaacgct ggcggcgctgc ttaacacatg caagtcgagc gaagcgattt aaatgagact 60
tcgggtgatt ttaaatggac tgagcggcgg acgggtgagt aacgcgtgga taacctgcct 120
cacacagggg gataacagtt agaaatgact gctaataacc cataagcgca cggtaaccga 180
tggtacagtg tgaaaaactc cgggtggttg agatggatcc gcgtctgatt aggtagttgg 240
tgaggtaacg gccacacaag ccgacgatca gtagccgacc tgagagggtg accggccaca 300
ttgggactga gacacggccc aa 322

```

```

SEQ ID NO: 229      moltype = DNA length = 310
FEATURE            Location/Qualifiers
source             1..310
                  mol_type = genomic DNA
                  organism = Parabacteroides sp.

```

```

SEQUENCE: 229

```

-continued

```

gatgaacgct agcgacaggc ttaacacatg caagtcgagg ggcagcgggg gtagcaatac 60
ccgccggcga ccggcgacag ggtgagtaac gcgtatgcaa cttacctatc agagggggat 120
aaccggcgca aagtcggact aataccgcat gaagcagggg ccccgcatgg ggatatattgc 180
taaagattca tcgctgatag ataggcatgc gttccattag tttgttggcg gggtaacggc 240
ccaccaagac cacgatggat aggggttctg agaggaaggt cccccaactt ggaactgaga 300
cacggtccaa                                     310

```

```

SEQ ID NO: 230      moltype = DNA length = 326
FEATURE            Location/Qualifiers
source             1..326
                   mol_type = genomic DNA
                   organism = Lactobacillus sp.

```

```

SEQUENCE: 230
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcattta agacagatta 60
cttcgggtttg aagtctttta tgactgagtg gcggacgggt gagtaacgcg tgggtaacct 120
gcctcataka gggggatagc agctggaaac ggctggtaat accgcataag cgcacagtac 180
cacatgggtac agtgtgaaaa actccggtag tatgagatgg acccgctctc gattagcttg 240
ttggcgggggt aacggcccac caaggcgacg atcagtagcc gacctgagag ggtgaccggc 300
cacattggga ctgagacacg gcccgag                                     326

```

```

SEQ ID NO: 231      moltype = DNA length = 325
FEATURE            Location/Qualifiers
source             1..325
                   mol_type = genomic DNA
                   organism = Clostridium sp.

```

```

SEQUENCE: 231
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggagtcgttt tggaaatctc 60
ttcggagatg gaattctcga cttagtggcg gacgggtgag taacgcgtga gcaatctgcc 120
tttaagagggg ggataaacgt cggaaaacggc tgctaatacc gcataaagca ttaaattcgc 180
atgtttttga tgcctaaagga gcaatccgct tttagatgag ctccgctctg attagctagt 240
tgggcggggtg acggcccaacc aaggcgacga tcagtagccg gactgagagg ttgaacggcc 300
acattggggac tgagacacgg cccag                                     325

```

```

SEQ ID NO: 232      moltype = DNA length = 307
FEATURE            Location/Qualifiers
source             1..307
                   mol_type = genomic DNA
                   organism = Roseburia sp.

```

```

SEQUENCE: 232
gataaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagtttttc ttccgggagg 60
aaacttagtgg cggacgggtg agtaacgctg gggtaacctg ccctgtacag ggggacaaca 120
gctggaaacg gctgctaata ccgcataaag ccttagcact gcattggtgca tagggaaaag 180
gagcaatccg gtacaggatg gacccgcgtc tgattagcca gttggcaggg taacggccta 240
ccaaagcgac gatcagtagc cgatctgaga ggatgtacgg ccacattggg actgagacac 300
ggcccag                                     307

```

```

SEQ ID NO: 233      moltype = DNA length = 317
FEATURE            Location/Qualifiers
source             1..317
                   mol_type = genomic DNA
                   organism = Bacteroides sp.

```

```

SEQUENCE: 233
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatgg tcttagcttg 60
ctaagggtga tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccgtctactc 120
ttggccagcc ttctgaaaag aagattaatc caggatggga tcatgagttc acatgtccgc 180
atgattaaag gtattttccg gtagacgatg gggatgctgt ccattagata gtaggcgggg 240
taacggccca cctagtcaac gatggatagg ggttctgaga ggaaggtccc ccacattgga 300
actgagacac ggtccaa                                     317

```

```

SEQ ID NO: 234      moltype = DNA length = 309
FEATURE            Location/Qualifiers
source             1..309
                   mol_type = genomic DNA
                   organism = Bifidobacterium sp.

```

```

SEQUENCE: 234
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggatccatc aggctttgct 60
tgggtggtgag agtggcgaaac ggggtgagtaa tgcgtgacgg acctgcccca tacacgggaa 120
tagctccttg aaacgggtgg taatgccgga tgctccgact cctcgcatgg ggtgtcgggg 180
aagatttcat cggtatggga tggggtcgcg tcctatcagg tagtcggcgg ggtaacggcc 240
caccgagcct acgacgggta gccggcctga gagggcgacc ggccacattg ggactgagat 300
acggccag                                     309

```

```

SEQ ID NO: 235      moltype = DNA length = 311
FEATURE            Location/Qualifiers
source             1..311
                   mol_type = genomic DNA

```

-continued

```

                                organism = Faecalibacterium sp.
SEQUENCE: 235
gacgaacgct ggcggcgctgc ctaacacatg caagtcgaac gagcgagaga gagcttgctt 60
tctcaagcga gtggcgaaacg ggtgagtaac gcgtgaggaa cctgcctcaa agagggggac 120
aacagtgtga aacgactgct aataccgcat aagcccacga cccggcatcg ggtagaggga 180
aaaggagcaa tccgctttga gatggcctcg cgtccgatta gctagttggt gaggtaatgg 240
cccaccaagg cgacgatcgg tagccggact gagagggtga acggccacat tgggactgag 300
acacgggcca g                                     311

SEQ ID NO: 236          moltype = DNA length = 326
FEATURE                Location/Qualifiers
source                 1..326
                        mol_type = genomic DNA
                        organism = Megasphaera sp.
SEQUENCE: 236
gacgaacgct ggcggcgctgc ttaacacatg caagtcgaac gagaagagat gagaagcttg 60
cttcttatca attcgagtgg caaacgggtg agtaacgcgt aagcaacctg cccttcagat 120
ggggacaaca gctggaaacg gctgctaata ccgaatacgt tctttccgcc gcatgacggg 180
atgaagaaag ggaggccttc gggctttcgc tggaggaggg gcttgctgtc gattagctag 240
ttggaggggt aacggcccac caaggcgacg atcagtagcc ggtctgagag gatgaacggc 300
cacattggga ctgagacacg gccacg                                     326

SEQ ID NO: 237          moltype = DNA length = 315
FEATURE                Location/Qualifiers
source                 1..315
                        mol_type = genomic DNA
                        organism = Bacteroides sp.
SEQUENCE: 237
gatgaacgct agctacaggg ttaacacatg caagtcgagg ggcagcatgg tcttagcttg 60
ctaagggcga tggcgaccgg cgcacgggtg agtaacgcgt atccaacctg ccttacactc 120
ttggacagcc ttctgaaagg gagattaata caagatgta tcatgagtaa gcattttcgc 180
atgattaaag gtttaccggt gtaagatggg gatgcgttcc attagatagt aggcggggta 240
acggcccacc tagtcttcga ttgatagggt ttctgagagg aaggtcccc acattggaac 300
tgagacacgg tccaa                                     315

SEQ ID NO: 238          moltype = DNA length = 315
FEATURE                Location/Qualifiers
source                 1..315
                        mol_type = genomic DNA
                        organism = Bacteroides sp.
SEQUENCE: 238
gatgaacgct agctacaggg ttaacacatg caagtcgagg ggcagcggga ttgaagcttg 60
cttcaattgc cggcgaccgg cgcacgggtg agtaacgcgt atccaacctt ccgtacactc 120
agggatagcc ttctgaaaga aagattaata cctgatgta tcttaagcac acatgtaatt 180
aagattaaag atttatcggt gtacgatggg gatgcgttcc attagtagt aggcggggta 240
acggcccacc tagccaacga ttgatagggt ttctgagagg aaggtcccc acattggaac 300
tgagacacgg tccaa                                     315

SEQ ID NO: 239          moltype = DNA length = 326
FEATURE                Location/Qualifiers
source                 1..326
                        mol_type = genomic DNA
                        organism = Anaerostipes sp.
SEQUENCE: 239
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaaacacctt atttgatttt 60
cttcggaact gaagatttgg tgattgagtg gcggacgggt gagtaacgcg tgggtaacct 120
gccctgtaca gggggtaaac agtcagaaat gactgctaata accgcataag accacagcac 180
cgcatggtgc aggggtaaaa actccgggtg tacaggatgg acccgctct gattagctgg 240
ttggtgaggt aacggctcac caaggcgacg atcagtagcc ggcttgagag agtgaacggc 300
cacattggga ctgagacacg gcccaa                                     326

SEQ ID NO: 240          moltype = DNA length = 313
FEATURE                Location/Qualifiers
source                 1..313
                        mol_type = genomic DNA
                        organism = Prevotella sp.
SEQUENCE: 240
gatgaacgct ggcggcgctgc ttaacacatg caagtcgagg ggcagcatgt tggttgcttg 60
caaccgatga tggcgaccgg cgcacgggtg agtaacgcgt atccaacctt cccttgctca 120
tcggataacc cgtcgaaagg cggcctaaca cgatatcgcg ttcacacggy gcatctaacy 180
tgaacgaaat gtgaaggaga aggatgggga tgcgtctgat tagcttgatt gcggggtaac 240
ggcccaccaa ggcaacgatc agtaggggtt ctgagaggaa ggteccccac attggaactg 300
agacacggtc caa                                     313

SEQ ID NO: 241          moltype = DNA length = 326
FEATURE                Location/Qualifiers

```


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```

source                1..326
                      mol_type = genomic DNA
                      organism = Anaerostipes sp.

SEQUENCE: 241
gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac gaagctgctt aactgatctt 60
cttcggaatt gacgttttgt agactgagtg gcggacgggt gagtaacgcg tgggtaacct 120
gccctgtaca gggggataac agtcagaaat gactgctaata accgcataag accacagcac 180
cgcatgggtg aggggtaaaa actccgggtg tacaggatgg acccggtctt gattagctgg 240
ttggtgaggt aacggctcac caaggcgacg atcagtagcc ggcttgagag agtgaacggc 300
cacattggga ctgagacacg gcccaa                                     326

SEQ ID NO: 242        moltype = DNA length = 313
FEATURE              Location/Qualifiers
source                1..313
                      mol_type = genomic DNA
                      organism = Alistipes sp.

SEQUENCE: 242
gatgaacgct agcggcaggc ttaacacatg caagtcgagg ggcagcataa tggatagcaa 60
tatctatggt ggcgaccggc gcacgggtgc gtaacgcgta tgcaacctac ctttaacagg 120
gggataaacac tgagaaattg gtactaatac ccataaatat catagaaggc atcttttatg 180
gttgaatttc ccgatgggta gagatgggca tgcgttgat tagctagttg gtggggtaac 240
ggctcaccaa ggcgacgata cataggggga ctgagaggtt aacccccac actggtactg 300
agacacggac cag                                     313

SEQ ID NO: 243        moltype = DNA length = 310
FEATURE              Location/Qualifiers
source                1..310
                      mol_type = genomic DNA
                      organism = Parabacteroides sp.

SEQUENCE: 243
gatgaacgct ggcggcgtgc ttaacacatg caagtcgagg ggcagcgggg gtagcaatac 60
cgccggcgca ccggcgacag ggtgagtaac gcgtatgcaa cttacctatc agagggggat 120
aaccggcgca aagtcggact aataccgcat gaagcaggga tcccgcatgg gaatatattg 180
taaagattca tcgctgatat ataggcatgc gttccattag gcagtggcg gggtaacggc 240
ccaccaaacc gacgatggat agggggtctg agaggaaggt cccccacatt ggtactgaga 300
cacggaccaa                                     310

SEQ ID NO: 244        moltype = DNA length = 323
FEATURE              Location/Qualifiers
source                1..323
                      mol_type = genomic DNA
                      note = Lachnospiraceae
                      organism = unidentified

SEQUENCE: 244
gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac gggaaatact tcattgagac 60
ttcgggtgat ttgatttatt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatacaggg ggataacagt cagaaatgac tgctaatacc gcataagcgc acaggaccgc 180
atggtccggg gtgaaaaact ccggtgggat aagatggacc cgcgttgat tagctgggtg 240
gtggggtaac gggccaccaa ggcgacgata catagccggc ctgagagggg gaacggccac 300
attgggactg agacacggcc cag                                     323

SEQ ID NO: 245        moltype = DNA length = 324
FEATURE              Location/Qualifiers
source                1..324
                      mol_type = genomic DNA
                      organism = Roseburia sp.

SEQUENCE: 245
gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac gaagcacttt gatcgatttc 60
ttcggattga aatttttagtg actgagtggt ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcacacagg gggataacag ttggaacggc ctgctaatac cgcataagcg cacagtaccg 180
catggtacag tgtgaaaaaac tccgggtggt tgagatggac ccgcgtctga ttagctagtt 240
ggcagggcaa cggcctacca aggcgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa                                     324

SEQ ID NO: 246        moltype = DNA length = 314
FEATURE              Location/Qualifiers
source                1..314
                      mol_type = genomic DNA
                      organism = Bacteroides sp.

SEQUENCE: 246
gatgaacgct gacagaatgc ttaacacatg caagtcgagg ggcatacagg agaaagcttg 60
ctttctttgc tggcgacggc cgcacgggtg agtaacacgt atccaaacctg ccctttactc 120
ggggatagcc ttgcgaaaga aagattaata ccgatggta taatgattcc gcatggtttg 180
attattaaag gattccggta aaggatgggg atgcgttcca ttaggttgtt ggtgaggtaa 240
cggtcacca agccttcgat ggataggggt tctgagagga aggtocccca cattggaact 300
gagacacggc ccaa                                     314

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-continued

SEQ ID NO: 247 moltype = DNA length = 316
 FEATURE Location/Qualifiers
 source 1..316
 mol_type = genomic DNA
 organism = Bilophila sp.

SEQUENCE: 247

attgaacgct	ggcggcggtgc	ttaacacatg	caagtcgaac	gtgaaagtcc	ttcgggatga	60
gtaaagtggc	gcacgggtga	gtaacgcgtg	gataatctac	ccttaagatg	gggataacgg	120
ctggaacggc	tcgctaatac	cgaatacgct	cccgatttta	tcgttggggg	gaaagatggc	180
ctctgcttgc	aagctatcgc	ttaaggatga	gtccgcgtcc	cattagctag	ttggcggggg	240
aacggcccac	caaggcgacg	atgggtagcc	ggtctgagag	gatgaccggc	cacactggaa	300
ctggaacacg	gtccag					316

SEQ ID NO: 248 moltype = DNA length = 325
 FEATURE Location/Qualifiers
 source 1..325
 mol_type = genomic DNA
 organism = Clostridium sp.

SEQUENCE: 248

gatgaacgct	ggcggcggtgc	ttaacacatg	caagtcgaac	ggggattatt	ttgacagaga	60
cttcgggtga	agtcgttata	atcctagtgg	cggacgggtg	agtaacgcgt	gggtaacctg	120
cctcacactg	ggggataaac	gtcagaaatg	actgctaata	ccgcataagc	gcacgggatt	180
gcatgatcca	gtgtgaaaaa	ctccggtggt	gtgagatgga	cccgcttggg	attagccagt	240
tggcagggta	acggcctacc	aaagcgacga	tccatagccg	gcctgagagg	gtggacggcc	300
acattggggc	tgagacacgg	cccag				325

SEQ ID NO: 249 moltype = DNA length = 325
 FEATURE Location/Qualifiers
 source 1..325
 mol_type = genomic DNA
 note = Flavonifractor
 organism = unidentified

SEQUENCE: 249

gatgaacgct	ggcggcggtgc	ttaacacatg	caagtcgaac	gggggtgctca	tgacggagga	60
ttcgtccaac	ggattgagtt	acctagtggc	ggacgggtga	gtaacgcgtg	aggaaacctg	120
cttgagaggg	ggaataaacac	tccgaaagga	gtgctaatac	cgcataagc	agttgggtcg	180
catggctctg	actgccaaag	atctatcgct	ctgagatggc	ctcgcgtctg	attagctagt	240
agggggggta	acggccacc	tagggcagca	tcagtagccg	gactgagagg	ttgaccggcc	300
acattggggc	tgagacacgg	cccag				325

SEQ ID NO: 250 moltype = DNA length = 326
 FEATURE Location/Qualifiers
 source 1..326
 mol_type = genomic DNA
 organism = Clostridium sp.

SEQUENCE: 250

gatgaacgct	ggcggcggtgc	ttaacacatg	caagtcgaac	gaagcacttt	acctgatttc	60
cttcggggat	gattgttctg	tgactgagtg	gcggacgggt	gagtaacgcg	tgataaacct	120
gcctcacaca	gggggataac	agttggaaac	gactgctaata	accgcataag	cgacacagc	180
cgcattggtg	agtgtagaaa	actccgggtg	tgtgagatgg	atccgcgtct	gattagctgg	240
ttggcggggg	aacggccacc	caaggcgacg	atcagtagcc	ggcctgagag	ggcgacggcc	300
cacattggga	ctgagacacg	gccccag				326

SEQ ID NO: 251 moltype = DNA length = 326
 FEATURE Location/Qualifiers
 source 1..326
 mol_type = genomic DNA
 organism = Eubacterium sp.

SEQUENCE: 251

gatgaacgct	ggcggcggtgc	ttaatacatg	caagtcgaac	gaagcacctt	ggacagaatc	60
cttcggggag	aagaccattg	tgactgagtg	gcggacgggt	gagtaacgcg	tggttaacct	120
gcctgtgaca	gggggataac	agttggaaac	gactgctaata	accgcataag	cgacacagtc	180
cgcattggtg	gggtgtaaaa	actccgggtg	tacaagatgg	acccgcgtct	gattagctgg	240
ttggtagggt	aacggccacc	caaggcgacg	atcagtagcc	gacttgagag	agtgatcgcc	300
cacattggga	ctgagacacg	gccccaa				326

SEQ ID NO: 252 moltype = DNA length = 309
 FEATURE Location/Qualifiers
 source 1..309
 mol_type = genomic DNA
 organism = Clostridium sp.

SEQUENCE: 252

gacgaacgct	ggcggcacgc	ttaacacatg	caagtcgaac	gagagaagag	aagcttgctt	60
ttctgatcta	gtggcggaag	ggtgagtaac	acgtgagcaa	tctgcctttc	agagggggat	120
accgattgga	aacgatcggt	aataccgcat	aacataattg	aaccgcataga	tttgattatc	180

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aaagatttat cgctgaaaga tgagctcgcg tctgattagc tagttggtaa ggtaacggct 240
taccaaggcg acgatcagta gccggactga gaggttgatc ggccacattg ggactgagac 300
acggcccag 309

```

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SEQ ID NO: 253      moltype = DNA length = 325
FEATURE            Location/Qualifiers
source              1..325
                    mol_type = genomic DNA
                    organism = Clostridium sp.

```

```

SEQUENCE: 253
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggagtcgttt tggaaaatcc 60
ttcgggattg gaattctcga cttagtggcg gacgggtgag taacgcgtga gcaatctgcc 120
ttaagagggg ggataacagt cggaaacggc tgctaatacc gcataaagca atgaattcgc 180
atgttttcat tgccaaaagga gcaatccgct tttagatgag ctgcgctctg attagctagt 240
tggcggggta acggcccacc aaggcgacga tcagtagccg gactgagagg ttgaacggcc 300
acattgggac tgagacacgg cccag 325

```

```

SEQ ID NO: 254      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source              1..323
                    mol_type = genomic DNA
                    note = Lachnospiraceae
                    organism = unidentified

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SEQUENCE: 254
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggaaacatt tcattgaggc 60
ttcggcggtt ttgatctggt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatactggg ggataacagc cagaaatgac tgctaatacc gcataaagcg acagaaccgc 180
atggttcggt gtgaaaaaact ccggtgggtat aagatggacc cgcgttggat tagctgggtg 240
gcagggcagc ggctaccaa gccgacgatc catagccggc ctgagagggg gaacggccac 300
attgggactg agacacggcc cag 323

```

```

SEQ ID NO: 255      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source              1..323
                    mol_type = genomic DNA
                    organism = Ruminococcus sp.

```

```

SEQUENCE: 255
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggaattact ttattgaaac 60
ttcggtcgat ttaatttaat tccagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatacaggg ggataacagt tagaaatggc tgctaatacc gcataaagcg acagagctgc 180
atggctcagc gcgaaaaaact ccggtgggtat aagatggacc cgcgttggat tagctgggtg 240
gtggggtaac ggcccaccaa gccgacgatc catagccggc ctgagagggg gaacggccac 300
attgggactg agacacggcc cag 323

```

```

SEQ ID NO: 256      moltype = DNA length = 325
FEATURE            Location/Qualifiers
source              1..325
                    mol_type = genomic DNA
                    organism = Butyrivicoccus sp.

```

```

SEQUENCE: 256
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggaactatct tggaaaatcc 60
ttcgggagcg gaattcttag tttagtggcg gacgggtgag taacgcgtga gcaatctgcc 120
tttaagaggg ggataacagt cggaaacggc tgctaatacc gcataaagca tcaaattcgc 180
atgttttga tgccaaaagga gcaatccgct tttagatgag ctgcgctctg attagctggt 240
tggcggggta acggcccacc aaggcgacga tcagtagccg gactgagagg ttgaacggcc 300
acattgggac tgagacacgg cccag 325

```

```

SEQ ID NO: 257      moltype = DNA length = 309
FEATURE            Location/Qualifiers
source              1..309
                    mol_type = genomic DNA
                    organism = Coprococcus sp.

```

```

SEQUENCE: 257
gatgaacgct ggcggcgcgcg ctaacacatg caagtcgaac ggacgatgaa gagcttgctt 60
ttcaaagtta gtggcgagcg ggtgagtaac gcgtgggtaa cctgcctcat acaggggaat 120
agcagctgga aacggctgat aaaaccgcat aagcgcacag tgccgcatgg cacagtgtga 180
aaaaatccgg tggtatgaga tggaccgcg tctgattagc ttgttggcgg ggtaacggcc 240
caccaaggcg acgatcagta gccggcctga gaggggtgacc ggccacattg ggactgagac 300
acggcccaa 309

```

```

SEQ ID NO: 258      moltype = DNA length = 310
FEATURE            Location/Qualifiers
source              1..310
                    mol_type = genomic DNA
                    organism = Bacteroides sp.

```

```

SEQUENCE: 258

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gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatga cctagcaata 60
ggttgatggc gaccggcgca cgggtgagta acacgtatcc aacctacggg ttattccggg 120
atagccttcc gaaagaaaga ttaataccgg atagtataac gagaaggcat ctttttgtaa 180
ttaaagaatt tcgataaccg atgggggatgc gttccattag tttgttggcg gggtaacggc 240
ccaccaagac atcgatggat aggggttctg agaggaaggc ccccccattt ggaactgaga 300
cacggtccaa                                     310

```

```

SEQ ID NO: 259      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source             1..316
                  mol_type = genomic DNA
                  organism = Parabacteroides sp.

```

```

SEQUENCE: 259
gatgaacgct agcgacaggc ttaacacatg caagtcgagg ggcagcatag taggtagcaa 60
tacttattgg tggcgaccgg cgcacgggtg agtaacgcgt atgcaactta cctatcagag 120
ggggatagcc cggcgaaagt cggattaata ctccataaaa cagggggtccc gcattgggact 180
atttggtaaa gattcatcgc tgatagatag gcatgcgttc cattaggcag ttggcggggg 240
aacggccccc caaaccgacg atggataggg gttctgagag gaagggtccc cacttgggaa 300
ctgagacacg gaccaa                                     316

```

```

SEQ ID NO: 260      moltype = DNA length = 310
FEATURE            Location/Qualifiers
source             1..310
                  mol_type = genomic DNA
                  organism = Parabacteroides sp.

```

```

SEQUENCE: 260
gatgaacgct agcgacaggc ttaacacatg caagtcgagg ggcagcacag gtagcaatac 60
cgggtggcga cggcgccacg ggtgagtaac gcgtatgcaa cttacctatc agaggggggat 120
aaccggcgga aagtcggact aataccgcgt gaagcagggg ccccgcatgg ggatatttgc 180
taaagattca tcgctgtag ataggcatgc gttccattag gcagtgtggc gggtaacggc 240
ccaccaaacg gacgatggat aggggttctg agaggaaggc ccccccattt ggtactgaga 300
cacggaccaa                                     310

```

```

SEQ ID NO: 261      moltype = DNA length = 308
FEATURE            Location/Qualifiers
source             1..308
                  mol_type = genomic DNA
                  organism = Bifidobacterium sp.

```

```

SEQUENCE: 261
gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac gggatccatc aggccttgct 60
tggtggtgag agtggcgaaac ggggtgagtaa tgcgtgaccg acctgcccc aacacgggaa 120
tagctcctgg aaacgggtgg taatgccgga tgctccagtt gatcgcatgg tcttctggga 180
aagctttcgc ggtatgggat ggggtcgcgt cctatcagct tgacggcggg gtaacggccc 240
accgtggctt cgacgggtag cgggcctgag agggcgaccg gccacattgg gactgagata 300
cggccag                                     308

```

```

SEQ ID NO: 262      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source             1..316
                  mol_type = genomic DNA
                  organism = Bacteroides sp.

```

```

SEQUENCE: 262
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcattt tagtttgctt 60
gcaaaactaaa gatggcgacc ggcgcacggg tgagtaaacac gtatccaaac tgccgataac 120
tcggggatag cctttcgaaa gaaagattaa tatccgatat tatattaaaa ccgcattggt 180
ttactattaa agaatttcgg ttatcgatgg ggatgcgttc cattagtttg ttggcggggg 240
aacggccccc caagactacg atggataggg gttctgagag gaagggtccc cacttggaa 300
ctgagacacg gtccaa                                     316

```

```

SEQ ID NO: 263      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source             1..316
                  mol_type = genomic DNA
                  organism = Streptococcus sp.

```

```

SEQUENCE: 263
gatgaacgct ggcggcgtgc ctaatacatg caagtagaac gctgaagctt ggtgcttgca 60
ccgagcggat gagttgcgaa cgggtgagta acgcgtaggt aacctgcctc ttgacggggg 120
ataactattg gaaacgatag ctaataccgc ataaaagtcg atatcgcatg atattgattt 180
gaaagggtga attgcatcac taagagatgg acctgcgttg tattagctag ttggtgaggt 240
aacggctcac caaggcgacg atacatagcc gacctgagag ggtgatcggc cacactggga 300
ctgagacacg gcccag                                     316

```

```

SEQ ID NO: 264      moltype = DNA length = 313
FEATURE            Location/Qualifiers
source             1..313
                  mol_type = genomic DNA

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                                organism = Prevotella sp.
SEQUENCE: 264
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatgt tggttgcttg 60
caaccaatga tggcgacggg cgcacgggtg agtaacgcgt atccaacctt cccttgctca 120
tcggataacc cgtcgaaagg cggcctaaca cgatatgcgg ttctcagcag gcattctaag 180
agaacgaaat gtgaaggaga aggatgggga tgcgtctgat tagcttggtg gcggggtaac 240
ggccccacaa ggcgacgacg agtaggggtt ctgagaggaa ggtccccac attggaactg 300
agacacggtc caa 313

SEQ ID NO: 265      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                   mol_type = genomic DNA
                   note = Firmicutes
                   organism = unidentified
SEQUENCE: 265
gatgaacgct ggcggcggtgc ttaacacatg caagtcgagc gaagcagtta agttgatttc 60
ttcggattga ttcttaactg actgagcggc ggacgggtga gtaacgcgtg ggtgacctgc 120
cccataccgg gggataaacg ctggaacggg ctgctaatac cgcataagcg cacagagctg 180
catggctcgg tgtgaaaaac tccgggtgga tgggatgggc ccgctctga ttaggcagtt 240
ggcggggtaa cggccccaca aaccgacgat cagtagccgg cctgagaggg cgaccggcca 300
cattgggact gagacacggc ccaa 324

SEQ ID NO: 266      moltype = DNA length = 315
FEATURE            Location/Qualifiers
source             1..315
                   mol_type = genomic DNA
                   organism = Bacteroides sp.
SEQUENCE: 266
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcggga ttgaagcttg 60
cttcaattgc cggcgaccgg cgcacgggtg agtaacgcgt atccaacctt ccgtacactc 120
agggatagcc ttctgaaaga aagattaata cctgatggta tgatgagatt gcattgatgc 180
atcattaaag atttatcggt gtacgatggg gatgcgttcc attaggtagt aggcggggta 240
acggccccac tagccgacga tggatagggg ttctgagagg aaggtcccc acattggaac 300
tgagacacgg tccaa 315

SEQ ID NO: 267      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                   mol_type = genomic DNA
                   note = Lachnospiraceae
                   organism = unidentified
SEQUENCE: 267
gatgaacgct ggcggcggtgc ttaacacatg caagtcgaac gaagcgctta aatggatttc 60
ttcggattga agtttttggg actgagtggt ggacgggtga gtaacgcgtg ggttaacctgc 120
ctcatacagg gggataaacg ttagaatatg ctgctaatac cgcataagcg cacagtgcgtg 180
catggcacag tgtgaaaaac tccgggtgga tgagatggac ccgctctga ttagctagtt 240
ggtggggtaa cggcctacca aggcgacgat cagtagccgg cctgagaggg tgaacggcca 300
cattgggact gagacacggc ccaa 324

SEQ ID NO: 268      moltype = DNA length = 307
FEATURE            Location/Qualifiers
source             1..307
                   mol_type = genomic DNA
                   organism = Bifidobacterium sp.
SEQUENCE: 268
gatgaacgct ggcggcggtgc ttaacacatg caagtcgaac gggatccag gagcttgctc 60
ctgggtgaga gtggcgaaag ggtgagtaat gcgtgaccga cctgccccat acaccggaat 120
agctcctgga aacgggtggt aatgcccgat gctccgacat gacgcattgc gtgtcgggaa 180
agattcatcg gtatgggatg gggtcgcgtc ctatcagctt gatggcgggg taacggccca 240
ccatggcttc gacgggtagc cggcctgaga gggcgaccgg ccacattggg actgagatac 300
ggcccag 307

SEQ ID NO: 269      moltype = DNA length = 322
FEATURE            Location/Qualifiers
source             1..322
                   mol_type = genomic DNA
                   note = Lachnospiraceae
                   organism = unidentified
SEQUENCE: 269
gatgaacgct ggcggcggtgc ttaacacatg caagtcgagc gaagcggttg aattgattct 60
tcggatgatt ttcagttgac tgagcggcgg acgggtgagt aacgcgtggg tgacctgccc 120
cataccgggg gataacagct ggaacgggct gctaataacc cataagcgca cagagctgca 180
tggctcgggt tgaaaaactc cgggtggtat ggtggggccc gcgtctgatt aggcagttgg 240
cggggtaacg gccacacaaa ccgacgatca gtagecggcc tgagaggggc accggccaca 300
ttgggactga gacacggccc aa 322

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SEQ ID NO: 270 moltype = DNA length = 308
 FEATURE Location/Qualifiers
 source 1..308
 mol_type = genomic DNA
 organism = Sutterella sp.

SEQUENCE: 270

attgaacgct	ggcggcgatgc	tttacacatg	caagtcgaac	ggcagcgcg	ggagcttgct	60
cctggcgcg	gagtgggcgca	cgggtgagta	atacatcgga	acgtgtcttc	tagtggggga	120
taactgccc	aaagggcagc	taataccgca	tgagacctga	gggtgaaagc	gggggatcgc	180
aagacctcg	gctggaagag	cggccgatgt	cgcattagct	agttggtgag	gtaaagggtc	240
accaaggcga	cgatcggtag	ctggtctgag	aggacgacca	gccacactgg	gactgagaca	300
cggcccgag						308

SEQ ID NO: 271 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 organism = Blautia sp.

SEQUENCE: 271

gatgaacgct	ggcggcgatgc	ttaacacatg	caagtcgagc	gaagcacctt	gacggatttc	60
ttcggattga	agccttggtg	actgagcggc	ggacgggtga	gtaacgcgtg	ggtaacctgc	120
ctcgtacagg	gggataacag	ttggaacagg	ctgctaatac	cgcataagcg	cacagtaccg	180
catggtacgg	tgtgaaaaac	tccggtggta	tgagatggac	ccgcgtctga	ttaggtagtt	240
ggtggggtaa	cggcctacca	agccgacgat	cagtagccga	cctgagaggg	tgaccggcca	300
cattgggact	gagacacggc	ccaa				324

SEQ ID NO: 272 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 note = Lachnospiraceae
 organism = unidentified

SEQUENCE: 272

gatgaacgct	ggcggcgatgc	ctaacacatg	caagtcgagc	gaagcacctt	gcttagattc	60
ttcggatgaa	gaggattgtg	actgagcggc	ggacgggtga	gtaacgcgtg	ggtaacctgc	120
ctcatacagg	gggataacag	ttagaaatga	ctgctaatac	cgcataagac	cacagcaccg	180
catggtcgcg	gggtaaaaac	tccggtggta	tgagatggac	ccgcgtctga	ttagctagtt	240
ggtaaggtaa	cggcttacca	agccgacgat	cagtagccga	cctgagaggg	tgaccggcca	300
cattgggact	gagacacggc	ccaa				324

SEQ ID NO: 273 moltype = DNA length = 311
 FEATURE Location/Qualifiers
 source 1..311
 mol_type = genomic DNA
 organism = Faecalibacterium sp.

SEQUENCE: 273

gacgaacgct	ggcggcgcg	ctaacacatg	caagtcgaac	gagagagagg	aagcttgctt	60
cctcaatcga	gtggcgaaac	ggtgagtaac	gcgtgaggaa	cctgcctcaa	agagggggac	120
aacagttgga	aacgactgct	aataccgcat	aagcccacag	gtcggcatcg	accagaggga	180
aaaggagcaa	tccgctttga	gatggcctcg	cgtccgatta	gctagttggt	gaggtaacgg	240
cccaccaagg	caacgatcgg	tagccggact	gagaggttga	acggccacat	tgggactgag	300
acacggccca	g					311

SEQ ID NO: 274 moltype = DNA length = 323
 FEATURE Location/Qualifiers
 source 1..323
 mol_type = genomic DNA
 organism = Blautia sp.

SEQUENCE: 274

gatgaacgct	ggcggcgatgc	ttaacacatg	caagtcgaac	gggaaacctt	ttattgaagc	60
ttcggcagat	ttagctgggt	tctagtggcg	gacgggtgag	taacgcgtgg	gtaacctgcc	120
ttatacagg	ggataacaac	cagaaatggt	tgctaatacc	gcataagcgc	acaggaccgc	180
atggtccggt	gtgaaaaact	cgggtggat	aagatggacc	cgcttggat	tagctagttg	240
gcagggtaac	ggcctaccaa	ggcgacgatc	catagccggc	ctgagagggt	gaacggccac	300
attgggactg	agacacggcc	cag				323

SEQ ID NO: 275 moltype = DNA length = 311
 FEATURE Location/Qualifiers
 source 1..311
 mol_type = genomic DNA
 organism = Faecalibacterium sp.

SEQUENCE: 275

gacgaacgct	ggcggcgcg	ctaacacatg	caagtcgaac	gagcgagaga	aagcttgctt	60
tctcgagcga	gtggcgaaac	ggtgagtaac	gcgtgaggaa	cctgcctcaa	agagggggac	120
aacagttgga	aacgactgct	aataccgcat	aagcccacgg	tgccgcatgg	cacagaggga	180

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aaaggagcaa tccgctttga gatggcctcg cgtccgatta gctagtgtgt gaggtaattg 240
cccaccaagg cgacgatcgg tagccggact gagagggtga acggccacat tgggactgag 300
acacggccca g                                     311

```

```

SEQ ID NO: 276      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 276
gatgaacgct ggcggcgtgc ttaacacatg caagtogaac gaagcacttt ggacagattc 60
ttcggatgaa gtccttagtg actgagtggc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcatacagg gggataacag ttagaaatgg ctgctaatac cgcataagcg cacgggtaccg 180
catggtagag tgtgaaaaac tccgggtgga tgagatggac ccggttgga ttagctagtt 240
ggcagggtaa cggcctacca aggcgacgat ccatagccgg cctgagaggg tggacggcca 300
cattgggact gagacacggc ccag                                     324

```

```

SEQ ID NO: 277      moltype = DNA length = 325
FEATURE            Location/Qualifiers
source             1..325
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 277
gacgaacgct ggcggcgtgc ctaacacatg caagtogaac ggaacttttt taaaggattt 60
cttcggaatg aatttgatta agtttagtgg cggacgggtg agtaacgcgt gagtaacctg 120
cctctaagag gggaataaca ttctgaaaag aatgctaata ccgcataaca tatatttacc 180
gcatggtaga tatatcaaa atttatcgct tagagatgga ctgcgctccg attagttagt 240
tggtgaggtg acggctcacc aagaccgcga tcggtagccg gactgagagg ttgaacggcc 300
acattgggac tgagacacgg ccacg                                     325

```

```

SEQ ID NO: 278      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source             1..316
                  mol_type = genomic DNA
                  organism = Parasutterella sp.

```

```

SEQUENCE: 278
attgaacgct ggcgggaacgc tttacacatg caagtogaac ggtaacgcgg agagaagctt 60
gtttctctcc ggcgacgagt ggcgaacggg tgagtaatac atcggaacgt gtccgctcgt 120
gggggacaac cagccgaagg gttggctaata accgcatgag ttctacggaa gaaagagggg 180
gacccgcaag ggcctctcgc gagcggagcg gccgatgact gattagcctg ttggtgaggt 240
aacggctcac caagcaaacg atcagtagct ggtctgagag gacgaccagc cacactggga 300
ctgagacacg gccacg                                     316

```

```

SEQ ID NO: 279      moltype = DNA length = 322
FEATURE            Location/Qualifiers
source             1..322
                  mol_type = genomic DNA
                  note = Erysipelotrichales
                  organism = unidentified

```

```

SEQUENCE: 279
gatgaacgct ggcggcgtgc ctaatacatg caagtogaac ggagcgaata tggaagcttg 60
cttcgctaag agctcagtg cgaacgggtg agtaacacgt aggaacctg cccatgtgcc 120
cgggataact gctggaaacg gtagctaaaa cgggataagg gaacaggagg catctcttat 180
tcattaaagg acctgtaaag gtgcgaacat ggatggacct gcggcgcat agctggttgg 240
agtggtaacg gcacaccaag gcgacgatgc gtagccgacc tgagaggggc aacggccaca 300
ttgggactga gacacggccc aa                                     322

```

```

SEQ ID NO: 280      moltype = DNA length = 317
FEATURE            Location/Qualifiers
source             1..317
                  mol_type = genomic DNA
                  organism = Barnesiella sp.

```

```

SEQUENCE: 280
gatgaacgct agcgacaggg ctaacacatg caagtcgagg ggcagcgaag aggtagcaat 60
acctttgtcg gcgaccggcg cacgggtgag taacacgtat gcaatccacc tgtaacaggg 120
ggataaaccg gagaaatccg gactaatacc ccataatatg ggcgctccgc atggagagcc 180
cattaagag agcaattttg gttacagacg agcatgcgct ccattagcca gttggcgggg 240
taacggccca ccaaagcgac gatggatagg ggttctgaga ggaaggtccc ccacattgga 300
actgagacac ggtccaa                                     317

```

```

SEQ ID NO: 281      moltype = DNA length = 326
FEATURE            Location/Qualifiers
source             1..326
                  mol_type = genomic DNA
                  organism = Eubacterium sp.

```

```

SEQUENCE: 281

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gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcattta gaacagatta 60
tttcgggatg aagtttttta tgactgagtg gcggacgggt gagtaacgcg tgggtaacct 120
gccttgtagt gggggatagc agctggaaac ggctggtaat accgcataag cgcacaatgt 180
tgcatgacat ggtgtgaaaa actccgggtg tataagatgg acccgctctt gattagctag 240
ttggtgagat aacagcccac caaggcgacg atcagtagcc gacctgagag ggtgaccggc 300
cacattggga ctgagacacg gccacg 326

```

```

SEQ ID NO: 282      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source              1..324
                    mol_type = genomic DNA
                    organism = Coprococcus sp.

```

```

SEQUENCE: 282
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcatttt aacctgattc 60
tttcgggatg aagtttttta tgactgagtg gcggacgggt gagtaacgctg ggtaacctgc 120
ctcacaacag ggggataacag ttgaaatga ctgctaatac cgcataagac cacggagacc 180
catggctcag tgggaaaaaac tccggtggta tgagatggac ccgcgtctga ttaggtagtt 240
ggtggggtaa cggcctacca agccaacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa 324

```

```

SEQ ID NO: 283      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source              1..324
                    mol_type = genomic DNA
                    note = Clostridiaceae
                    organism = unidentified

```

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SEQUENCE: 283
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcagttt agcgggaagt 60
tttcgggatg aagtttttta tgactgagtg gcggacgggt gagtaacgctg ggtaacctgc 120
ctcacaacag ggggataacag ttgaaatga ctgctaatac cgcataagac cacagcttcg 180
catgaagcag tgtgaaaaaac tccggtggta tgagatggac ccgcgtctga ttaggtagtt 240
ggtgaggtaa cggctacca agcgcacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa 324

```

```

SEQ ID NO: 284      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source              1..324
                    mol_type = genomic DNA
                    organism = Dorea sp.

```

```

SEQUENCE: 284
gatgaacgct ggcggcgctgc ttaacacatg caagtcgagc gaagcgcttt ggaaagattc 60
tttcgggatg ttcctttgtg actgagcggc ggacgggtga gtaacgctg ggtaacctgc 120
ctcacaacag ggggataacag ttgaaatga ctgctaatac cgcataagac cacggtaacc 180
catggtacag tggtaaaaaa tccggtggta tgagatggac ccgcgtctga ttaggtagtt 240
ggtggggtaa cggcctacca agcgcacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccag 324

```

```

SEQ ID NO: 285      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source              1..323
                    mol_type = genomic DNA
                    organism = Clostridium sp.

```

```

SEQUENCE: 285
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggagctgctt tgatgaagt 60
tttcgggatg tttaaaaacag cttagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
tcacactggg ggataaacagt tagaaatagc tgctaatacc gcataagcgc acgggtccgc 180
atggaacagt gtgaaaaaact ccggtgggtg gagatggacc cgcgtctgat tagccagttg 240
gcggggtaac ggcccaccaa agcgcacgat agtagccgcg ctgagagggg gaacggccac 300
attgggactg agacacggcc caa 323

```

```

SEQ ID NO: 286      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source              1..323
                    mol_type = genomic DNA
                    organism = Ruminococcus sp.

```

```

SEQUENCE: 286
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggaaatact ttattgaaac 60
tttcggtgat ttaatttatt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatactggg ggataacagc cagaaatgac tgctaatacc gcataagcgc acagaaccgc 180
atggttcggt gtgaaaaaact ccggtgggtg aagatggacc cgcgttgat tagctagttg 240
gcagggcagc ggcctacca ggcgacgat catagccgcg ctgagagggg gaacggccac 300
attgggactg agacacggcc cag 323

```

```

SEQ ID NO: 287      moltype = DNA length = 309
FEATURE            Location/Qualifiers
source              1..309

```


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                                mol_type = genomic DNA
                                organism = Clostridium sp.

SEQUENCE: 287
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac ggacgatgaa gagcttgctt 60
ttcagagtta gtggcggacg ggtgagtaac gcgtgggtaa cctgcctcat acagggggat 120
agcagctgga aacggctgat aaaaccgcat aagcgacacg catcgcatga tgcagtgtga 180
aaaaactcgg tggatagaga tggaccgcg tctgattagc tgggtggtga ggtaacggcc 240
caccaaggcg acgatcagta gccggcctga gagggtgacc ggccacattg ggactgagac 300
acggcccaa                                     309

SEQ ID NO: 288                moltype = DNA length = 313
FEATURE                      Location/Qualifiers
source                       1..313
                                mol_type = genomic DNA
                                organism = Clostridium sp.

SEQUENCE: 288
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggacttacat tgaagcctag 60
cgatatgtaa gttagtggcg gacgggtgag taacgcgtgg gtaacctgcc ttgtactggg 120
ggacaacagt tggaaacgac tgctaatacc gcataagcgc acagcttcgc atgaagcagt 180
gtgaaaaact ccggtgggtac aagatggacc cgcgtctgat tagctagtgt gtgaggtaac 240
ggcccaccaa ggcgacgatc agtagccggc ctgagagggt gaacggccac attgggactg 300
agacacggcc caa                                     313

SEQ ID NO: 289                moltype = DNA length = 324
FEATURE                      Location/Qualifiers
source                       1..324
                                mol_type = genomic DNA
                                note = Lachnospiraceae
                                organism = unidentified

SEQUENCE: 289
gatgaacgct ggcggcgctgc ttaacacatg caagtcgagc gaagcactta ggaaagattc 60
ttcggatgat ttctattttg actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcatacagg gggataacag ttgaaatga ctgctaatac cgcataagac cacagcaccg 180
catggtgcag gggtaaaaac tccggtggtg tgagatggac ccgcgtctga ttagttagtt 240
ggtggggtaa cggcctacca aggcgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa                                     324

SEQ ID NO: 290                moltype = DNA length = 317
FEATURE                      Location/Qualifiers
source                       1..317
                                mol_type = genomic DNA
                                organism = Clostridium sp.

SEQUENCE: 290
gatgaacgct ggcggcgctgc ttaacacatg caagtcgagc ggagataaat tgaaagcttg 60
cttttaattt atcttagcgg cggacgggtg agtaacgtgt gggcaacctg cctcatcacg 120
agggataatc atgtgaaaac gtgactaata ccgcatgtcg ttccgggagg gcatcctcct 180
gaaagaaaag gagcaatccg gtagatgatg gggccgcgac tgattagcta gttggtgaga 240
taacagccca ccaaggcgac gatcagtagc cgacctgaga gggtagatcg ccacattggg 300
actgagacac ggccccaa                                     317

SEQ ID NO: 291                moltype = DNA length = 318
FEATURE                      Location/Qualifiers
source                       1..318
                                mol_type = genomic DNA
                                note = Faecalitalea
                                organism = unidentified

SEQUENCE: 291
gatgaacgct ggcggcgctgc ctaatacatg caagtcgaac ggaccgaagg gaaagcttgc 60
tttcctaaaa gtcagtggcg aacgggtgag taacacgtag gtaacctgcc catgtgcccg 120
ggacaacgcc tggaaacggg tgctaaaacc ggataggtag agagatcgca tgaactttct 180
attaaagggg ctgcggccct gaacatggat ggacctgcgg tgcattagct agttggtgag 240
gtaacggccc accaaggcga cgatgcatag ccgacctgag agggcgaacg gccacattgg 300
gactgagaca cggcccaa                                     318

SEQ ID NO: 292                moltype = DNA length = 324
FEATURE                      Location/Qualifiers
source                       1..324
                                mol_type = genomic DNA
                                organism = Clostridium sp.

SEQUENCE: 292
gatgaacgct ggcggcgctgc ttaacacatg caagtcgagc gaagcgctta cgacagaacc 60
ttcgggggaa gatgtaaggg actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcatacagg gggataacag ttgaaacagg ctgctaatac cgcataagcg cacagtaccg 180
catggtacgg tgtgaaaaac tccggtggtg tgagatggac ccgcgtctga ttagctagtt 240
ggaggggtaa cggcccacca aggcgacgat cagtagccgg cctgagaggg tgaacggcca 300
cattgggact gagacacggc ccag                                     324

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-continued

SEQ ID NO: 293 moltype = DNA length = 323
 FEATURE Location/Qualifiers
 source 1..323
 mol_type = genomic DNA
 organism = *Blautia* sp.

SEQUENCE: 293

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgaac	ggggaacatt	ttatggaagc	60
ttcgggtgaa	atagcttgtt	cctagtggcg	gacgggtgag	taacgcgtgg	gtaacctgcc	120
tcacactggg	ggataacagt	cagaaatgac	tgctaatacc	gcataagcgc	acgggattgc	180
atgatccagt	gtgaaaaaact	ccgggtggtg	gagatggacc	cgcgttggat	tagccagttg	240
gcagggtaac	ggcctaccac	agcgacgatc	catagccggc	ctgagagggg	ggacggccac	300
attgggactg	agacacggcc	cag				323

SEQ ID NO: 294 moltype = DNA length = 303
 FEATURE Location/Qualifiers
 source 1..303
 mol_type = genomic DNA
 organism = *Collinsella* sp.

SEQUENCE: 294

gatgaacgct	agctacaggc	ttaacacatg	caagtcgaac	ggcacctgcc	ctcgggcaga	60
agcgaagtgg	gaacgggtga	gtaacacgtg	gagaacctgc	cccctcccc	gggatatgcc	120
cccgaaggga	cgggtaatac	cggatacccc	ggggtgccgc	atggcacccc	gggtaaacgc	180
ccgacgggag	gggatggctc	cgcggcccat	caggtagacg	gcgggggtgac	ggcccaccgt	240
gccgacaacg	ggtagccggg	ttgagagacc	gaccggccag	attgggactg	agacacggcc	300
cag						303

SEQ ID NO: 295 moltype = DNA length = 326
 FEATURE Location/Qualifiers
 source 1..326
 mol_type = genomic DNA
 organism = *Blautia* sp.

SEQUENCE: 295

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgaac	gaagcacctt	attctgattt	60
cttcggaatg	aaaaatttgg	tgactgagtg	gcggacgggt	gagtaacgcg	tgggtaacct	120
gcctcataca	gggggataac	agttagaaat	gactgctaat	accgcataag	accacagcac	180
tgcatggtgc	aggggtgaaa	actccgggtg	tatgagatgg	accgcgtct	gattaggtgg	240
ttggtggggg	aacgggcctac	caagccgacg	atcagtagcc	gacctgagag	ggtgaccggc	300
cacattggga	ctgagacacg	gcccaa				326

SEQ ID NO: 296 moltype = DNA length = 309
 FEATURE Location/Qualifiers
 source 1..309
 mol_type = genomic DNA
 organism = *Bifidobacterium* sp.

SEQUENCE: 296

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgaac	gggatcggct	ggagcttgct	60
ccggccgtga	gagtgggcga	cgggtgagta	atgcgtgacc	gacctgcccc	atacacccga	120
atagctcctg	gaaacgggtg	gtaatgcccg	atgctccagt	tgactgcata	gtcctctggg	180
aaagcttttg	gggtatggga	tggggtccgc	tctatcacgc	ttgatggcgg	ggtaacggcc	240
caccatggct	tcaacgggta	gccggcctga	gagggcgacc	ggccacattg	ggactgagac	300
acggcccaa						309

SEQ ID NO: 297 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 organism = *Ruminococcus* sp.

SEQUENCE: 297

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgaac	gaagcactta	agtttgattc	60
ttcggatgaa	gacttttgtg	actgagtgcc	ggacgggtga	gtaacgcgtg	ggtaacctgc	120
ctcatcacag	gggataacag	ttagaaatga	ctgctaatac	cgcataagac	cacagaccgc	180
catggtgcag	gggtaaaaac	tccggtggtg	tgagatggac	ccgcgtctga	ttagctgggt	240
ggtggggtaa	cggcctacca	aggcgacgat	cagtagccga	cctgagaggg	tgaccggcca	300
cattgggact	gagacacggc	ccaa				324

SEQ ID NO: 298 moltype = DNA length = 311
 FEATURE Location/Qualifiers
 source 1..311
 mol_type = genomic DNA
 organism = *Faecalibacterium* sp.

SEQUENCE: 298

gacgaacgct	ggcggcgctgc	ctaacacatg	caagtcgaac	gagcgagaga	gagcttgctt	60
tctcaagcga	gtggcgaacg	ggtgagtaac	gcgtgaggaa	cctgcctcaa	agagggggac	120
aacagtttga	aacgactgct	aataccgcac	aagccacagg	gtcggcatcg	accagaggga	180
aaaggagtga	tccgctttga	gatggcctcg	cgtccgatta	gctagttggt	gaagtaacgg	240

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cccaccaagg caacgatcgg tagccgggact gagaggttga acggccacat tgggactgag 300
atacggccca g 311

```

```

SEQ ID NO: 299      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                  mol_type = genomic DNA
                  note = Lachnospiraceae
                  organism = unidentified

```

```

SEQUENCE: 299
gatgaacgct agctacaggc ttaacacatg caagtcgagc gaagcactta agaatgattc 60
ttcggatgaa ttcttttttg actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcatcacagg gggataaacg ttagaaatgg ctgctaatac cgcataagac cacagtactg 180
catggtagacg tggtaaaaaa tccgggtgga tgagatggac ccgcgtctga ttaggtagtt 240
ggtgaggttaa cggccacca agccgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccag 324

```

```

SEQ ID NO: 300      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source             1..323
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 300
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac gaggtatattt gattgaagtt 60
ttcggatgga ttccaagata ccgagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
tcatacaggg ggataaacggt tagaaatgac tgctaatacc gcataagcgc acagtaccgc 180
atggtacggt gtgaaaaact ccggtgggtat gagatggacc cgcgtctgat tagctagttg 240
gtgggggtaac ggcccaccaa ggcgacgac agtagccgac ctgagagggg gaccggccac 300
attgggactg agacacggcc cag 323

```

```

SEQ ID NO: 301      moltype = DNA length = 325
FEATURE            Location/Qualifiers
source             1..325
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 301
gacgaacgct ggcggcgctgc ttaacacatg caagtcgaac ggagcaccct tgactgaggt 60
ttcggccaaa ttgtaggaat gcttagtggtc ggactgggtga gtaacgcgtg aggaacctgc 120
cttcagagg gggacaacag ttggaacga ctgctaatac cgcataagcgc atgaccgggg 180
catccggggc atgtcaaaga ttttatcgct ggaagatggc ctgcgtctg attagctaga 240
tggtggggta acggccacc atggcgacga tcagtagccg gactgagagg ttgacgggcc 300
acattgggac tgagatacgg ccag 325

```

```

SEQ ID NO: 302      moltype = DNA length = 322
FEATURE            Location/Qualifiers
source             1..322
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 302
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac gaagcacctt acttgattct 60
tcggatgaag gtctgggtgac tgagtggcgg acgggtgagt aacgcgtggg taacctgccc 120
tgtacagggg gataacagct ggaacgggct gctaataacc cataagcgca cgaggagaca 180
ttctctttgt tgaaaaaact cggtggtaca ggatggggccc gcgtctgatt agctggttgg 240
cagggtaacg gcctaccaag gcaacgatca gtagccggtc tgagaggatg aacggccaca 300
ttggaactga gacacggtcc aa 322

```

```

SEQ ID NO: 303      moltype = DNA length = 313
FEATURE            Location/Qualifiers
source             1..313
                  mol_type = genomic DNA
                  organism = Clostridium sp.

```

```

SEQUENCE: 303
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggaatttaca tgaagcctag 60
cgattgtaaa tttagtggcg gacgggtgag taacgcgtgg gtaacctgcc ttgtactggg 120
ggacaacagt tggaaacgac tgctaatacc gcataagcgc acagcttcgc atgaagcagt 180
gtgaaaaact ccggtgttac aagatggacc cgcgtctgat tagctggttg gtgaggtaac 240
ggcccaccaa ggcgacgac agtagccggc ctgagagggg gaacggccac attgggactg 300
agacacggcc caa 313

```

```

SEQ ID NO: 304      moltype = DNA length = 313
FEATURE            Location/Qualifiers
source             1..313
                  mol_type = genomic DNA
                  note = Clostridiaceae
                  organism = unidentified

```

```

SEQUENCE: 304

```

-continued

```

gacgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggagttgaaa ggagcttgct 60
ttttcaact cagtggcgga cgggtgagta acgctgagta aacctgcctt tcagaggggg 120
ataacgttct gaaaagaacg ctaataccgc ataagattgt agtttcgcat ggagcagcaa 180
tcaaaggagc aatccgctga aagatggact cgcgtccgat tagatagttg gcggggtaac 240
ggcccaccaa gtgcacgatc ggtagccgga ctgagaggtt gatcggccac attgggactg 300
agacacggcc cag 313

```

```

SEQ ID NO: 305      moltype = DNA length = 310
FEATURE            Location/Qualifiers
source              1..310
                    mol_type = genomic DNA
                    organism = Bacteroides sp.

```

```

SEQUENCE: 305
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcacagga ttagcaata 60
cattgctggc gaccggcgca cgggtgagta acacgtatcc aacctgcctt ttactcgggg 120
ataggctttc gaaagaaaga ttaatacccg atggtataat attccgcat gggattatta 180
ttaaagaatt tcggtaaaag atggggatgc gttccattag gcagttggcg gggtaacggc 240
ccaccaaac ttcgatggat aggggttctg agaggaaggt cccccacatt ggaactgaga 300
cacggtccaa 310

```

```

SEQ ID NO: 306      moltype = DNA length = 312
FEATURE            Location/Qualifiers
source              1..312
                    mol_type = genomic DNA
                    organism = Bacteroides sp.

```

```

SEQUENCE: 306
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcattt cgctagcaat 60
agtgaagatg gcgaccggcg caccgggtgag taacacgtat ccaacctgcc gataactcgg 120
ggatagcctt tcgaaagaaa gattaatatc cgatagata ttaaaacgcg atgggtttac 180
tattaaagaa tttcggttat cgatggggat gcgttccatt agtttgttgg cggggtaacg 240
gccaccaag actacgatgg ataggggttc tgagaggaag gtcccccaca ttggaactga 300
gacacggtcc aa 312

```

```

SEQ ID NO: 307      moltype = DNA length = 326
FEATURE            Location/Qualifiers
source              1..326
                    mol_type = genomic DNA
                    organism = Ruminococcus sp.

```

```

SEQUENCE: 307
gatgaacgct ggcggcgctgc ttaacacatg caagtcgagc gaagcaactct gctttgattt 60
cttcgggatg aagagcttag tgactgagcg gcggacgggt gagtaacgcg tgggtaacct 120
gcctcataca gggggataac agttagaaat gactgctaata accgcataag gccacagcac 180
cgcatggtgc aggggtaaaa actccgggtg tatgagatgg accgcgctc gattaggtag 240
ttggcggggt aacggccccc caagccgacg atcagtagcc gacctgagag ggtgaccggc 300
cacattggga ctgagacacg gcccaa 326

```

```

SEQ ID NO: 308      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source              1..324
                    mol_type = genomic DNA
                    organism = Clostridium sp.

```

```

SEQUENCE: 308
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcaactta tctttgattc 60
ttcggatgaa gagttttgtg actgagtgcc ggcgggtgga gtaacgcgtg ggtaacctac 120
ctcatacagg gggataacag ttagaatatg ctgctaatac gcataagac cacggagctg 180
catggctcag tgggaaaaaac tccggtggta tgagatggac ccgctgtctg ttaggtagtt 240
ggtggggtaa cggcctacca agccaacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa 324

```

```

SEQ ID NO: 309      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source              1..323
                    mol_type = genomic DNA
                    note = Lachnospiraceae
                    organism = unidentified

```

```

SEQUENCE: 309
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggaaacctt ttattgaagc 60
ttcggcagat ttagctgggt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatacaggg ggataacagt cagaaatggc tgctaatacc gcataagcgc acagagctgc 180
atggctcagt gtgaaaaact ccgggtggat aagatggacc cgcgttggat tagctgggtg 240
gtggggtaac ggcccaccaa ggcgacgatc catagccggc ctgagagggg gaacggccac 300
attgggactg agacacggcc cag 323

```

```

SEQ ID NO: 310      moltype = DNA length = 314
FEATURE            Location/Qualifiers
source              1..314

```

-continued

mol_type = genomic DNA
organism = Bacteroides sp.

SEQUENCE: 310
gatgaacgct ggctacagcg ttaacacatg caagtcgagg ggcacgga agaaagcttg 60
ctttctttgc tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccctttactc 120
ggggatagcc ttctgaaaga aagattaata cccgatggga taattattcc gcatgggttg 180
attattaaag gattccggta aaggatgggg atgctgtcca ttaggcagtt ggtgaggtaa 240
cggctcacca aaccttcgat ggataggggt tctgagagga aggtccccc cattggaact 300
gagacacggt ccaa 314

SEQ ID NO: 311 moltype = DNA length = 303
FEATURE Location/Qualifiers
source 1..303
mol_type = genomic DNA
organism = Collinsella sp.

SEQUENCE: 311
gatgaacgct ggccggcgcc ctaacacatg caagtcgaac ggcacctgcc ctccggcgaga 60
agcgaagtggc gaacgggtga gtaacacgtg gagaacctgc cccctccccc gggatagccg 120
cccgaaagga cgggtaatac cggatacccc cggcgccgcg atggcgcccg ggctaaagcc 180
ccgacgggag gggatgggtc cgcggcccat caggtagacg gcgggggtgac ggcccaccgt 240
gcccacaacg ggtagccggg ttgagagacc gaccggccag attgggactg agacacggcc 300
cag 303

SEQ ID NO: 312 moltype = DNA length = 311
FEATURE Location/Qualifiers
source 1..311
mol_type = genomic DNA
organism = Clostridium sp.

SEQUENCE: 312
gacgaacgct ggccggcggt ctaacacatg caagtcgaac ggacgaggag gagcttgctt 60
ctccaagtca gtggcgagcg ggtgagtaac gcgtgagtaa cctgccttc agagggggat 120
aacgtctgga aacggagcgt aataccgcat gacatatttc catcgcatgg tggagatc 180
aaaggagcaa tccgctgaag gatggactcg cgtccgatta gatagttggc ggggtaacgg 240
cccaccaagt gcagcatcgg tagccggact gagaggttga acggccacat tgggactgag 300
acacggccca g 311

SEQ ID NO: 313 moltype = DNA length = 315
FEATURE Location/Qualifiers
source 1..315
mol_type = genomic DNA
organism = Bacteroides sp.

SEQUENCE: 313
gatgaacgct agctacagcg ttaacacatg caagtcgagg ggcagcatgg tcttagcttg 60
ctaagggcga tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccgtctactc 120
ttggacagcc ttctgaaagg aagattaata caagatggca tcatgagtcg gcatgttcac 180
atgattaaag gtattccggt agacgatggg gatgctgtcc attagatagt aggcggggta 240
acggccacc tagtcttcca tggatagggg ttctgagagg aaggtccccc acattggaac 300
tgagacacgg tccaa 315

SEQ ID NO: 314 moltype = DNA length = 314
FEATURE Location/Qualifiers
source 1..314
mol_type = genomic DNA
organism = Bacteroides sp.

SEQUENCE: 314
gatgaacgct agctacagcg ttaacacatg caagtcgagg ggcagcatga tcaaagcttg 60
ctttgatgga tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccgacaacac 120
tgggatagcc ttctgaaaga aagattaata ccggatggca tagttttccc gcatgggatg 180
attattaaag aatttcggtt gtcgatgggg atgctgtcca ttaggcagtt ggcggggtaa 240
cggccacca aaccaacgat ggataggggt tctgagagga aggtccccc cattggaact 300
gagacacggt ccaa 314

SEQ ID NO: 315 moltype = DNA length = 314
FEATURE Location/Qualifiers
source 1..314
mol_type = genomic DNA
organism = Bacteroides sp.

SEQUENCE: 315
gatgaacgct agctacagcg ttaacacatg caagtcgagg ggcagcatga acttagcttg 60
ctaagtttga tggcgaccgg cgcacgggtg agtaacacgt atccaacctg ccgatgactc 120
ggggatagcc ttctgaaaga aagattaata ccggatggca tagttcttc gcatggtaga 180
actattaaag aatttcggtc atcgatgggg atgctgtcca ttaggttgtt ggcggggtaa 240
cggccacca agccttcgat ggataggggt tctgagagga aggtccccc cattggaact 300
gagacacggt ccaa 314

SEQ ID NO: 316 moltype = DNA length = 323

-continued

FEATURE
source Location/Qualifiers
 1..323
 mol_type = genomic DNA
 organism = Blautia sp.

SEQUENCE: 316

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgaac	gggaaatatt	tcattgagac	60
ttcgggtgat	ttgatctatt	tctagtggcg	gacgggtgag	taacgcgtgg	gtaacctgcc	120
ttatacaggg	ggataacagt	cagaaatggc	tgctaatacc	gcataagcgc	acagagctgc	180
atggctcagt	gtgaaaaact	ccggtgggat	aagatggacc	cgcgttggat	tagcttgttg	240
ttggggtaac	ggcccaccaa	ggcgacgac	catagccggc	ctgagagggg	gaacggccac	300
attgggactg	agacacggcc	cag				323

SEQ ID NO: 317 moltype = DNA length = 316

FEATURE Location/Qualifiers
source 1..316
 mol_type = genomic DNA
 organism = Bacteroides sp.

SEQUENCE: 317

gatgaacgct	agctacaggg	ttaacacatg	caagtcgagg	ggcagcattt	cagtttgctt	60
gcaaaactgga	gatggcgacc	ggcgacggg	tgagtaaac	gtatccaacc	tgccgataac	120
tcggggatag	cctttcgaaa	gaaagattaa	taccgatgg	tataattaga	ccgcattggtc	180
ttgttattaa	agaatttcgg	ttatcgatgg	ggatgcgttc	cattaggcag	ttggtgaggt	240
aacggctcac	caaaccttcg	atggataggg	gttctgagag	gaaggtcccc	cacattggaa	300
ctgagacacg	gtccaa					316

SEQ ID NO: 318 moltype = DNA length = 323

FEATURE Location/Qualifiers
source 1..323
 mol_type = genomic DNA
 note = Phascolarctobacterium
 organism = unidentified

SEQUENCE: 318

gacgaacgct	ggcggcgctgc	ctaacacatg	caagtcgaac	ggagaatttt	atttcggtag	60
aattcttagt	ggcgaacggg	tgagtaacgc	gtaggcaacc	tgccctttag	acggggacaa	120
cattccgaaa	ggagtgcata	taccgatgt	gatcatcgtg	ccgcattggca	ggatgaagaa	180
agatggcctc	tacaagtaag	ctatcgctaa	aggatggggc	tgcgctctgat	tagctagtgtg	240
gtagtgtaac	ggactaccaa	ggcgatgac	agtagccggt	ctgagaggat	gaacggccac	300
attgggactg	agacacggcc	caa				323

SEQ ID NO: 319 moltype = DNA length = 324

FEATURE Location/Qualifiers
source 1..324
 mol_type = genomic DNA
 organism = Dorea sp.

SEQUENCE: 319

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgagc	gaagcactta	agtttgattc	60
ttcgggatgaa	gacttttgtg	actgagcggc	ggacgggtga	gtaacgcgtg	ggtaacctgc	120
ctcatcacagg	gggataacag	ttagaaatga	ctgctaatac	cgcataagac	cacggtagccg	180
catggtacag	tggtaaaaac	tctggtggta	tgagatggac	ccgcgtctga	ttaggtagtt	240
ggtggggtaa	cggcctacca	agccgacgat	cagtagccga	cctgagaggg	tgaccggcca	300
cattgggact	gagacacggc	ccag				324

SEQ ID NO: 320 moltype = DNA length = 323

FEATURE Location/Qualifiers
source 1..323
 mol_type = genomic DNA
 organism = Blautia sp.

SEQUENCE: 320

gatgaacgct	ggcggcgctgc	ttaacacatg	caagtcgaac	gggaattact	ttattgaaac	60
ttcgggtcgat	atgatttaat	tctagtggcg	gacgggtgag	taacgcgtgg	gtaacctgcc	120
ttatacaggg	ggataacagt	cagaaatgac	tgctaatacc	gcataagcgc	acagagctgc	180
atggctcggg	gtgaaaaact	ccggtgggat	aagatggacc	cgcgttggat	tagctagtgtg	240
gtgaggtaac	ggcccaccaa	ggcgacgac	catagccggc	ctgagagggg	gaacggccac	300
attgggactg	agacacggcc	cag				323

SEQ ID NO: 321 moltype = DNA length = 316

FEATURE Location/Qualifiers
source 1..316
 mol_type = genomic DNA
 organism = Bacteroides sp.

SEQUENCE: 321

gatgaacgct	agctacaggg	ttaacacatg	caagtcgagg	ggcagcattt	tagtttgctt	60
gcaaaactaaa	gatggcgacc	ggcgacggg	tgagtaaac	gtatccaacc	tgccgataac	120
tcggggatag	cctttcgaaa	gaaagattaa	tatccgatgg	tatatattct	ccgcattggtg	180
agaatattaa	agaatttcgg	ttatcgatgg	ggatgcgttc	cattagtttg	ttggcggggg	240
aacggccccc	caagaccacg	atggataggg	gttctgagag	gaaggtcccc	cacattggaa	300

-continued

ctgagacacg gtccaa 316

SEQ ID NO: 322 moltype = DNA length = 322
 FEATURE Location/Qualifiers
 source 1..322
 mol_type = genomic DNA
 organism = Megamonas sp.

SEQUENCE: 322
 gacgaacgct ggcggcgctgc ttaacacatg caagtcgaac ggggtgttta ttccggtaaa 60
 caccaagtgg cgaacgggtg agtaacgctg aagcaatcta ccttcaagat ggggacaaca 120
 cttcgaaaagg ggtgctaata ccgaatgaat gtaagagtat cgcagagac acttactaaa 180
 ggaggcctct gacataagct tccgcttgaa gatgagcttg cgtctgatta gctagtgtgt 240
 gagggtaaag gccccaagc gcgacgatca gtagccggctc tgagaggatg aacggccaca 300
 ttgggactga gacacggccc ag 322

SEQ ID NO: 323 moltype = DNA length = 316
 FEATURE Location/Qualifiers
 source 1..316
 mol_type = genomic DNA
 organism = Staphylococcus sp.

SEQUENCE: 323
 gacgaacgct ggcggcgctgc ctaatacatg caagtagaac gctgaagaga ggagcttgct 60
 cttcttggat gagttgcgaa cgggtgagta acgctaggt aacctgcctt gtagcggggg 120
 ataactattg gaaacgtag ctaataccgc ataacaatgg atgacacatg tcatttattt 180
 gaaaggggca attgctccac tacaagatgg acctgcgttg tattagctag taggtgaggt 240
 aacggctcac ctaggcgacg atacatagcc gacctgagag ggtgatcggc cacactggga 300
 ctgagacacg gccacg 316

SEQ ID NO: 324 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 organism = Blautia sp.

SEQUENCE: 324
 gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcacctt gatttgattc 60
 ttcggatgaa gatcttggtg actgagtggt ggacgggtga gtaacgcgtg ggtaacctgc 120
 ctcatacagg gggataacag ttagaatga ctgctaatac cgcataagac cacagcaccg 180
 catggtgcag gggtaaaaac tccggtggta tgagatggac ccgcgtctga ttaggtagtt 240
 ggtggggtaa cggcctacca agccgacgat cagtagccga cctgagaggg tgaccggcca 300
 cattgggact gagacacggc ccaa 324

SEQ ID NO: 325 moltype = DNA length = 307
 FEATURE Location/Qualifiers
 source 1..307
 mol_type = genomic DNA
 organism = Bifidobacterium sp.

SEQUENCE: 325
 gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggatcccag gagcttgctc 60
 ctgggtgaga gtggcgaaac ggtgagtaat gcgtgaccga cctgccccat acaccggaat 120
 agctcctgga aacgggtggt aatgcccgat gctccagttg gatgcatgtc cttctgggaa 180
 agattcatcg gtatgggatg gggtcgcgtc ctatcagctt gatggcgggg taacggccca 240
 ccatggcttc gacgggtagc cggcctgaga gggcgaccgg ccacattggg actgagatac 300
 ggcccag 307

SEQ ID NO: 326 moltype = DNA length = 303
 FEATURE Location/Qualifiers
 source 1..303
 mol_type = genomic DNA
 organism = Collinsella sp.

SEQUENCE: 326
 gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggcacccttc tccggaggga 60
 agcgagtggc gaacggctga gtaacacgtg gagaacctgc cccctcccc gggatagccg 120
 cccgaaagga cgggtaatac cggatacccc ggggtgccgc atggcaccgc ggctaagacc 180
 ccgacgggag gggatggctc cgcggcccat caggtagacg gcgggggtgac ggcccacgct 240
 gccgacaacg ggtagccggg ttgagagacc gaccggccag attgggactg agacacggcc 300
 cag 303

SEQ ID NO: 327 moltype = DNA length = 326
 FEATURE Location/Qualifiers
 source 1..326
 mol_type = genomic DNA
 organism = Clostridium sp.

SEQUENCE: 327
 gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcacttt atttgatttc 60
 cttcgggact gattattttg tgactgagtg gcggacgggt gagtaacgcg tgggtaacct 120
 gccttgatga gggggataac agttggaaac ggctgctaata accgcataag cgcacagcat 180

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cgcatgatgc agtgtgaaaa actccggtgg tataagatgg acccgcggtg gattagctag 240
ttgggtgaggt aacggccccc caaggcgacg atccatagcc gacctgagag ggtgaccggc 300
cacattggga ctgagacacg gcccaa 326

```

```

SEQ ID NO: 328      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source             1..316
                   mol_type = genomic DNA
                   organism = Parabacteroides sp.

```

```

SEQUENCE: 328
gatgaacgct agcgacaggc ttaacacatg caagtcgagg ggcagcatga ttgtagcaa 60
tacagattga tggcgaccgg cgcacgggtg agtaacgctg atgcaactta cctatcagag 120
ggggatagcc cggcgaaagt cggattaata cccataaaa caggggtccc gcatgggaat 180
atgtgttaaa gattcatcgc tgatagatag gcatgcgttc cattaggcag ttggcggggt 240
aacggccccc caaacggacg atggataggg gttctgagag gaaggtcccc cacattggta 300
ctgagacacg gaccaa 316

```

```

SEQ ID NO: 329      moltype = DNA length = 307
FEATURE            Location/Qualifiers
source             1..307
                   mol_type = genomic DNA
                   organism = Bifidobacterium sp.

```

```

SEQUENCE: 329
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggatcccgag gagcttgctc 60
ctgggtgaga gtggcgaaacg ggtgagtaat gcgtgaccga cctgccccat acaccggaat 120
agctcctgga aacggggggt aatgcgggat gctccagttg acccgatggt cctctgggaa 180
agcttttgcg gtatgggatg gggtcgcgtc ctatcagctt gatggcgggg taacggccca 240
ccatgggttc gacgggtagc cggcctgaga gggcgaccgg ccacattggg actgagatac 300
ggcccag 307

```

```

SEQ ID NO: 330      moltype = DNA length = 302
FEATURE            Location/Qualifiers
source             1..302
                   mol_type = genomic DNA
                   organism = Clostridium sp.

```

```

SEQUENCE: 330
gatgaacgct ggcggcgctgc ctaatacatg caagtcgaac gcttcacttc ggtgaagagt 60
ggcgaaacggg tgagtaatac ataagtaacc tggcatctac aggggggataa ctgatggaaa 120
cgtcagctaa gaccgcatag gtgtagagat cgcatagaact ctatatgaaa agtgctacgg 180
gactggtaga tgatgggactt atggcgcatg agctgggttg tagggtaacg gcctaccaag 240
gcgacgatgc gtagccgacc tgagaggggtg accggccaca ctgggactga gacacggccc 300
ag 302

```

```

SEQ ID NO: 331      moltype = DNA length = 315
FEATURE            Location/Qualifiers
source             1..315
                   mol_type = genomic DNA
                   organism = Bacteroides sp.

```

```

SEQUENCE: 331
gatgaacgct agctacaggc ttaacacatg caagtcgagg ggcagcatga acttagcttg 60
ctaagtttga tggcgaccgg cgcacgggtg agtaacacgt atccaacctt ccgtttactc 120
agggatagcc tttcgaaaaga aagattaata cctgatagta tggtagagatt gcatgatagc 180
accattaaag atttattggt aaacgatggg gatgcgttcc attaggtagt aggcggggta 240
acggccccc tagccgacga ttgatagggg ttctgagagg aaggtcccc acattggaac 300
tgagacacgg tccaa 315

```

```

SEQ ID NO: 332      moltype = DNA length = 326
FEATURE            Location/Qualifiers
source             1..326
                   mol_type = genomic DNA
                   organism = Roseburia sp.

```

```

SEQUENCE: 332
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcactct atttgatttt 60
cttcggaaat gaagattttg tgactgagtg gcggacgggt gagtaacgcg tgggtaacct 120
gcctcataca ggggggataac agttgaaaac gactgctaata accgcataag cgcacaggat 180
cgcatgatcc ggtgtgaaaa actccggtgg tatgggatgg acccgctctt gattagccag 240
ttggcagggt aacggcctac caaagcgacg atcagtagcc gacctgagag ggtgaccggc 300
cacattggga ctgagacacg gcccaa 326

```

```

SEQ ID NO: 333      moltype = DNA length = 323
FEATURE            Location/Qualifiers
source             1..323
                   mol_type = genomic DNA
                   note = Lachnospiraceae
                   organism = unidentified

```

```

SEQUENCE: 333

```


-continued

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gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gggaaacctt ttattgaagc 60
ttcggcagat ttagattgtt tctagtggcg gacgggtgag taacgcgtgg gtaacctgcc 120
ttatacaggg ggataaacagt cagaaatgac tgctaatacc gcataagcgc acaggaccgc 180
atggtccggt gtgaaaaaac ccggtgggtat aagatggacc cgcgttggat tagctagtgt 240
gtgagggtaac gggccaccaa ggcgacgatc catagccggc ctgagagggt gaacggccac 300
attgggactg agacacggcc cag                                     323

```

```

SEQ ID NO: 334      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                   mol_type = genomic DNA
                   note = Lachnospiraceae
                   organism = unidentified

```

```

SEQUENCE: 334
gatgaacgct ggcggcgctgc ttaacacatg caagtcgagc gaagcaattt aagtgaagtt 60
ttcggatgga tcttaaattg actgagcggc ggacgggtga gtaacgcgtg gataacctgc 120
ctcacacagg gggataaacag ttagaaatga ctgctaatac cgcataagcg caccgtaccg 180
catggtacag tgtgaaaaac tccggtgggtg tgagatggat ccgcgtctga ttaggtagtt 240
ggtgaggtaa cggccaccaa agccgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa                                     324

```

```

SEQ ID NO: 335      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                   mol_type = genomic DNA
                   organism = Clostridium sp.

```

```

SEQUENCE: 335
gatgaacgct ggcggcgctgc ctaacacatg caagtcgagc gaagcggttt taaggaagtt 60
ttcggatgga attaaaactg actgagcggc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcacacagg gggataaacag ttagaaatga ctgctaatac cgcataagca cacagcttcg 180
catggagcag tgtgaaaaac tccggtggta tgagatggac ccgcgtctga ttagctagtt 240
ggtaaggtaa cggtttacca agccgacgat cagtagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa                                     324

```

```

SEQ ID NO: 336      moltype = DNA length = 324
FEATURE            Location/Qualifiers
source             1..324
                   mol_type = genomic DNA
                   note = Clostridiaceae
                   organism = unidentified

```

```

SEQUENCE: 336
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcaatct aacggaagtt 60
ttcggatgga actagatttg actgagtgcc ggacgggtga gtaacgcgtg ggtaacctgc 120
ctcacactgg gggacaacag ttagaaatga ctgctaatac cgcataagcg cacaggaccg 180
catggtccgg tgtgaaaaac tccggtgggtg tgagatggac ccgcgtttga ttagctagtt 240
ggtggggtaa cggcctacca agccgacgat caatagccga cctgagaggg tgaccggcca 300
cattgggact gagacacggc ccaa                                     324

```

```

SEQ ID NO: 337      moltype = DNA length = 311
FEATURE            Location/Qualifiers
source             1..311
                   mol_type = genomic DNA
                   organism = Faecalibacterium sp.

```

```

SEQUENCE: 337
gacgaacgct ggcggcgcgcc ctaacacatg caagtcgaac ggagcttaga gagcttgctt 60
tttaagctta gtggcgaaac ggtgagtaac gcgtgagtaa cctgccctag agtgggggac 120
aacagtttga aacgactgct aataccgcat aagccacagg taccgcatgg tactgaggga 180
aaaggattta ttcgcttttag gatggactcg cgtccaatta gctagttggt gaggtaacgg 240
cccaccaagg cgacgatttg tagccggact gagaggttga acggccacat tgggactgag 300
acacggccca g                                     311

```

```

SEQ ID NO: 338      moltype = DNA length = 316
FEATURE            Location/Qualifiers
source             1..316
                   mol_type = genomic DNA
                   organism = Streptococcus sp.

```

```

SEQUENCE: 338
gatgaacgct ggcggcgctgc ctaatacatg caagtagaac gctgaagctt ggtgcttgca 60
ccgagcggat gagttgcgaa cgggtgagta acgcgtaggt aacctgcctg gtagcggggg 120
ataactattg gaaacgatag ctaataccgc ataacagtag atgtcgcagt acctttgctt 180
gaaaggtgca attgccaccac taccagatgg acctgcgttg tattagctag ttggtgaggt 240
aacggctcac caagggcgacg atacatagcc gacctgagag ggtgatcggc cacactggga 300
ctgagacacg gccacg                                     316

```

```

SEQ ID NO: 339      moltype = DNA length = 322
FEATURE            Location/Qualifiers

```

-continued

```

source                1..322
                      mol_type = genomic DNA
                      note = Lachnospiraceae
                      organism = unidentified

SEQUENCE: 339
gatgaacgct ggcggcgctgc ttaacacatg caagtcgagc gaagcgattt aaatgagact 60
tcggtggatt ttaaattgac tgagcggcgg acgggtgagt aacgcgtgga taacctgcct 120
cacacagggg gataacagtt agaaatgact gctaataccg cataagcgca cggtaaccga 180
tggtacagtg tgaaaaactc cgggtggtgt agatggatcc gcgtctgatt aggtagtgtg 240
tgaggtaacg gccccaacag ccgacgatca gtagccgacc tgagagggtg accgggccaca 300
ttgggactga gacacggccc aa                                     322

SEQ ID NO: 340        moltype = DNA length = 310
FEATURE              Location/Qualifiers
source                1..310
                      mol_type = genomic DNA
                      organism = Parabacteroides sp.

SEQUENCE: 340
gatgaacgct agcgacagggc ttaacacatg caagtcgagg ggcagcgggg gtagcaatac 60
ccgcccggcg cccggcgacg ggtgagtaac gcgtatgcaa cttacctatc agagggggat 120
aaccggcgga aagtcggact aataccgcat gaagcagggg ccccgcatgg ggatatttgc 180
taaagattca tcgctgtagc ataggcatgc gtccattag tttgttggcg gggtaacggc 240
ccaccaagac cacgatggat aggggttctg agaggaaggt ccccccacatt ggaactgaga 300
cacggtccaa                                     310

SEQ ID NO: 341        moltype = DNA length = 326
FEATURE              Location/Qualifiers
source                1..326
                      mol_type = genomic DNA
                      organism = Lactobacillus sp.

SEQUENCE: 341
gatgaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagcattta agacagatta 60
cttcgggttg aagtctttta tgactgagtg cgggacgggt gagtaacgag tgggtaacct 120
gctcatatac ggggggatagc agctggaaac ggctggtaat accgcataag cgcacagtac 180
cacatggtag agtgtgaaaa actccgggtg tatgagatgg acccgctctc gattagcttg 240
ttggcggggg aacggccccc caaggcgacg atcagtagcc gacctgagag ggtgacggcg 300
cacattggga ctgagacacg gccccag                                     326

SEQ ID NO: 342        moltype = DNA length = 325
FEATURE              Location/Qualifiers
source                1..325
                      mol_type = genomic DNA
                      organism = Clostridium sp.

SEQUENCE: 342
gatgaacgct ggcggcgctgc ctaacacatg caagtcgaac ggagtcgttt tggaaatctc 60
ttcggagatg gaattctcga cttagtggcg gacgggtgag taacgcgtga gcaatctgcc 120
tttaagaggg ggataacagt cggaaacggc tgctaatacc gcataaagca ttaaattcgc 180
atgtttttga tgccaaagga gcaatccgct ttagatgagc ctccgctctg attagctagt 240
tggcgggggt acggcccacc aaggcgacga tcagtagccg gactgagagg ttgaacggcc 300
acattgggac tgagacacgg cccag                                     325

SEQ ID NO: 343        moltype = DNA length = 307
FEATURE              Location/Qualifiers
source                1..307
                      mol_type = genomic DNA
                      organism = Roseburia sp.

SEQUENCE: 343
gataaacgct ggcggcgctgc ttaacacatg caagtcgaac gaagtttttc ttccgggagg 60
aacttagtgg cggacgggtg agtaacgctg gggtaacctg ccctgtacag ggggacaaca 120
gctggaaacg gctgctaata ccgcataaag ccttagcact gcatgggtga tagggaaaag 180
gagcaatccg gtacaggatg gacccgcgtc tgattagcca gttggcaggg taacggccta 240
ccaaagcgac gatcagtagc cgatctgaga ggatgtacgg ccacattggg actgagacac 300
ggcccag                                     307

SEQ ID NO: 344        moltype = DNA length = 317
FEATURE              Location/Qualifiers
source                1..317
                      mol_type = genomic DNA
                      organism = Bacteroides sp.

SEQUENCE: 344
gatgaacgct agctacagggc ttaacacatg caagtcgagg ggcagcatgg tcttagcttg 60
ctaaggctga tggcgacggg cgcacgggtg agtaacacgt atccaacctg ccgtctactc 120
ttggccagcc ttctgaaagg aagattaatc caggatggga tcatgagttc acatgtccgc 180
atgattaaag gtattttccg gtacagcatg gggatgcgtt ccattagata gtaggcgggg 240
taacggccca cctagtcaac gatggatagg ggttctgaga ggaagggtccc ccacattgga 300
actgagacac ggtccaa                                     317

```

-continued

```

SEQ ID NO: 345      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
misc_feature        1..20
                    note = Synthetic DNA
source              1..20
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 345
agrgtttgat ymtggctcag                                20

SEQ ID NO: 346      moltype = DNA  length = 19
FEATURE            Location/Qualifiers
misc_feature        1..19
                    note = Synthetic DNA
source              1..19
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 346
tgctgctctcc cgtaggagat                                19

```

1. A method for treating graft-versus-host disease (GVHD) comprising:

orally administering a composition to a subject having GVHD, wherein the composition comprises an active ingredient in powder form and an excipient, which is formulated in a capsule formulation, and the active ingredient comprises *Bacteroides dorei*.

2. The method of claim 1, wherein the active ingredient further comprises at least one selected from the group consisting of:

Bacteroides vulgatus;
Bacteroides stercoris;
Bacteroides uniformis;
Blautia wexlerae;
Bacteroides sp. AR29;
Dorea longicatena;
Phascolarctobacterium faecium;
Blautia massiliensis;
Ruminococcus sp. K-1;
Bacteroides ovatus;
Faecalibacterium prausnitzii;
Megamonas funiformis;
Bifidobacterium adolescentis;
[Ruminococcus] torques;
Collinsella aerofaciens;
Parabacteroides merdae;
Butyrate-producing bacterium M104/1;
Bifidobacterium faecale;
Clostridium sp. 826;
Bacteroides coprocola;
Roseburia faecis;
Lachnospiraceae bacterium DJF_VP18k1;
Clostridium sp. AT4;
Lactobacillus rogosae;
Butyrate-producing bacterium A2-207; and
Roseburia sp. 1120.

3. The method of claim 1, wherein the active ingredient further comprises:

Bacteroides vulgatus;
Bacteroides stercoris;
Bacteroides uniformis;
Blautia wexlerae;
Bacteroides sp. AR29;
Dorea longicatena;
Phascolarctobacterium faecium;
Blautia massiliensis;
Ruminococcus sp. K-1;
Bacteroides ovatus;
Faecalibacterium prausnitzii;
Megamonas funiformis;
Bifidobacterium adolescentis;
[Ruminococcus] torques;
Collinsella aerofaciens;
Parabacteroides merdae;
Butyrate-producing bacterium M104/1;
Bifidobacterium faecale;
Clostridium sp. 826;
Bacteroides coprocola;
Roseburia faecis;
Lachnospiraceae bacterium DJF_VP18k1;
Clostridium sp. AT4;
Lactobacillus rogosae;
Butyrate-producing bacterium A2-207; and
Roseburia sp. 1120.

4. The method of claim 1, wherein the GVHD is steroid-refractory GVHD, steroid-dependent GVHD or acute GVHD.

5. The method of claim 2, wherein the GVHD is steroid-refractory GVHD, steroid-dependent GVHD or acute GVHD.

6. The method of claim 3, wherein the GVHD is steroid-refractory GVHD, steroid-dependent GVHD or acute GVHD.

* * * * *